

# Companion to Bird & de Moor, Algebra of Programming

## §0. Laws from Relation Algebra

### §0.1. Adjunctions

(0.1a)

|    | $F \dashv G$                                            | F's type                                                              | monad $FG$                            | $1 \sqsubseteq FG$                                                             | $GF \sqsubseteq 1$                                                             | F preserves $\cup$                                                | G preserves $\cap$                                                                     | $FGF=F$                                                         | $GFG=G$                                                           |
|----|---------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|
| 1  | $\triangleleft \dashv \triangleright$                   | $A \rightarrow A \otimes A$                                           | $\triangleleft \triangleright = 1$    | $1 \sqsubseteq \triangleleft \triangleright$                                   | $\triangleright \triangleleft \sqsubseteq 1$                                   | $(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$ | $(R \cap S) \triangleright = R \triangleright \cap S \triangleright$                   | $\triangleleft \triangleright \triangleleft = \triangleleft$    | $\triangleright \triangleleft \triangleright = \triangleright$    |
| 2  | $\multimap \dashv \multimap$                            | $A \rightarrow I$                                                     | $\multimap \multimap = T$             | $1 \sqsubseteq \multimap \multimap$                                            | $\multimap \multimap \sqsubseteq 1$                                            | $(R \cup S) \multimap = R \multimap \cup S \multimap$             | $(R \cap S) \multimap = R \multimap \cap S \multimap$                                  | $\multimap \multimap \multimap = \multimap$                     | $\multimap \multimap \multimap = \multimap$                       |
| 3  | $\circ \dashv \circ$                                    | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $R^{\circ \circ} = R$                 | $R^{\circ \circ} = R$                                                          | $R^{\circ \circ} = R$                                                          | $(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$                   | $(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$                                        | $R^{\circ \circ} = R$                                           | $R^{\circ \circ} = R$                                             |
| 4  | $\multimap \triangleleft \triangleright \multimap$      | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                                   | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $=$                                                               | $=$                                                                                    | $=$                                                             | $=$                                                               |
| 5  | $\triangleright \multimap \triangleleft \triangleright$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                                   | $(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $=$                                                               | $=$                                                                                    | $=$                                                             | $=$                                                               |
| 6  | $\Delta \dashv \cap$                                    | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \cap R = R$                        | $R \cap R = R$                                                                 | $R \cap S \sqsubseteq R$                                                       | $\Delta(R \cup S) = \Delta R \cup \Delta S$                       | $(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$                              | $R \cap R = R$                                                  | $R \cap R = R$                                                    |
| 7  | $\cup \dashv \Delta$                                    | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (R \cup S, R \cup S)$ | $R \sqsubseteq R \cup S$                                                       | $R \cup R = R$                                                                 | $(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$         | $\Delta(R \cap S) = \Delta R \cap \Delta S$                                            | $R \cup R = R$                                                  | $R \cup R = R$                                                    |
| 8  | $\emptyset \dashv !$                                    | $\{*\} \rightarrow (A \rightarrow B)$                                 |                                       |                                                                                | $\emptyset \sqsubseteq R$                                                      |                                                                   |                                                                                        |                                                                 |                                                                   |
| 9  | $\cdot S \dashv / S$                                    | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                                 | $R \sqsubseteq (RS)/S$                                                         | $(T/S)S \sqsubseteq T$                                                         | $(S \cup T)R = (SR) \cup (TR)$                                    | $(R \cap R') / S = (R / S) \cap (R' / S)$                                              | $((RS)/S)S = RS$                                                | $((T/S)S)/S = T/S$                                                |
| 10 | $S \cdot \dashv S \setminus$                            | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                       | $R \sqsubseteq S \setminus (SR)$                                               | $S(S \setminus T) \sqsubseteq T$                                               | $R(S \cup T) = (RS) \cup (RT)$                                    | $S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$                      | $S(S \setminus (SR)) = SR$                                      | $S \setminus (S(S \setminus T)) = S \setminus T$                  |
| 11 | $R \cap \dashv R \Rightarrow$                           | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (X \cap R)$  | $X \sqsubseteq R \Rightarrow (X \cap R)$                                       | $R \cap (R \Rightarrow Y) \sqsubseteq Y$                                       | $R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$                  | $R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$                    | $R \cap (R \Rightarrow (X \cap R)) = X \cap R$                  | $R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$      |
| 12 | $\mathcal{D} \dashv \cdot T$                            | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$           | $R \sqsubseteq (\mathcal{D}R)T$                                                | $\mathcal{D}(AT) \sqsubseteq A$                                                | $\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$          | $(A \cap B)T = AT \cap BT$                                                             | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$                   | $(\mathcal{D}(AT))T = AT$                                         |
| 13 | $\mathcal{R} \dashv T \cdot$                            | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$           | $R \sqsubseteq T(\mathcal{R}R)$                                                | $\mathcal{R}(TA) \sqsubseteq A$                                                | $\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$          | $T(A \cap B) = TA \cap TB$                                                             | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$                   | $T(\mathcal{R}(TA)) = TA$                                         |
| 14 | $\cdot f \dashv \cdot f^{\circ}$                        | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $ff^{\circ}$                          | $\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$                          | $f^{\circ} f \sqsubseteq 1$                                                    | $(S \cup T)R = (SR) \cup (TR)$                                    | $\text{Simple}(g) \Rightarrow (X \cap Y) g^{\circ} = (X g^{\circ}) \cap (Y g^{\circ})$ | $ff^{\circ}f = f$                                               | $f^{\circ}ff^{\circ} = f^{\circ}$                                 |
| 15 | $f^{\circ} \cdot \dashv f \cdot$                        | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $ff^{\circ}$                          | $\text{Map}(f) \Rightarrow 1 \sqsubseteq f f^{\circ}$                          | $f^{\circ} f \sqsubseteq 1$                                                    | $R(S \cup T) = (RS) \cup (RT)$                                    | $f(X \cap Y) = fX \cap fY$                                                             | $ff^{\circ}f = f$                                               | $f^{\circ}ff^{\circ} = f^{\circ}$                                 |
| 16 | $i \dashv E$                                            | $\text{Map} \multimap \text{Rel}$                                     | $E$                                   | $\frac{1}{\exists} : A \rightarrow PA$                                         | $\exists : PB \rightarrow B$                                                   | —                                                                 | —                                                                                      | $\frac{\exists}{\exists} = 1$                                   | $\frac{R}{\exists} \exists = R$                                   |
| 17 | $\cdot \exists \dashv \frac{\exists}{\exists}$          | $\text{Map}(A, PB) \rightarrow (A \rightarrow B)$                     | $1$                                   | $\frac{f \exists}{\exists} = f$                                                | $\frac{R}{\exists} \exists = R$                                                | —                                                                 | —                                                                                      | $\frac{\exists}{\exists} \frac{f \exists}{\exists} = f \exists$ | $\frac{R}{\exists} \frac{f \exists}{\exists} \exists = f \exists$ |
| 18 | $\frac{\exists}{\exists} \dashv \cdot \exists$          | $(A \rightarrow B) \rightarrow \text{Map}(A, PB)$                     | $1$                                   | $\frac{R}{\exists} \exists = R$                                                | $\frac{f \exists}{\exists} = f$                                                | —                                                                 | —                                                                                      | $\frac{\exists}{\exists} \frac{R}{\exists} = \frac{R}{\exists}$ | $\frac{f \exists}{\exists} \frac{R}{\exists} \exists = f \exists$ |

### §0.2. Composing adjunctions

(0.2a)

|   | $\bullet T$             | $T \bullet$                                                       | $\bullet g$                                                      | $g^\circ \bullet$                                      | $Tn$                                                          | $^\circ$                                                                                                                                  | $\bullet \triangleleft$                        | $\Delta$                                                                                        |
|---|-------------------------|-------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1 | $\bullet S$             | $R / (S_1 \ S_2)$<br>$= R / S_2 / S_1$                            | $S \setminus (R / W)$<br>$= (S \setminus R) / W$                 | $R / (Sg) =$<br>$(Rg^\circ) / S$                       | $\text{Map}(f) \Rightarrow f$<br>$(R / S) =$<br>$(f R) / S$   | $(Y/S)^\circ =$<br>$S^\circ \setminus (Y^\circ)$                                                                                          | $(RS)^\wedge =$<br>$R^\wedge (S \otimes 1)$    | $(R \cap R') /$<br>$S = (R / S)$<br>$\cap (R' / S)$                                             |
| 2 | $S \bullet$             | $S \setminus (R / W)$<br>$= (S \setminus R) / W$                  | $(S_1 \ S_2) \setminus R$<br>$= S_2 \setminus (S_1 \setminus R)$ | $S \setminus (Rg^\circ) =$<br>$(S \setminus R)g^\circ$ | $(g^\circ S) \setminus R =$<br>$S \setminus (gR)$             | $(S \setminus Y)^\circ =$<br>$Y^\circ / S^\circ$                                                                                          | $((S \otimes 1)R)^\wedge =$<br>$SR^\wedge$     | $S \setminus (R \cap$<br>$R') = (S \setminus$<br>$R) \cap (S \setminus$<br>$R')$                |
| 3 | $\bullet f$             | $\text{Map}(f) \Rightarrow$<br>$R / (f S) =$<br>$(R / S) f^\circ$ | $T \setminus (Rf^\circ) =$<br>$(T \setminus R) f^\circ$          | $(R S)^\circ = S^\circ$<br>$R^\circ$                   |                                                               | $(R S)^\circ = S^\circ$<br>$R^\circ$                                                                                                      |                                                | $\text{Simple}(g)$<br>$\Rightarrow (X \cap Y)$<br>$g^\circ = (X g^\circ)$<br>$\cap (Y g^\circ)$ |
| 4 | $f^\circ \bullet$       | $\text{Map}(f) \Rightarrow f$<br>$(R / S) =$<br>$(f R) / S$       | $(Tf^\circ) \setminus R =$<br>$f(T \setminus R)$                 | $(R S)^\circ = S^\circ$<br>$R^\circ$                   |                                                               | $(R S)^\circ = S^\circ$<br>$R^\circ$                                                                                                      |                                                | $f(T_1 \cap T_2) =$<br>$fT_1 \cap fT_2$                                                         |
| 5 | $Rn$                    | —                                                                 | —                                                                | —                                                      | $R \Rightarrow S \Rightarrow T =$<br>$R \cap S \Rightarrow T$ | $(R \Rightarrow Y)^\circ =$<br>$R^\circ \Rightarrow (Y^\circ)$                                                                            | —                                              | $R \Rightarrow S \cap T =$<br>$(R \Rightarrow S) \cap$<br>$(R \Rightarrow T)$                   |
| 6 | $^\circ$                | $(Y/T)^\circ =$<br>$T^\circ \setminus (Y^\circ)$                  | $(T \setminus Y)^\circ =$<br>$Y^\circ / T^\circ$                 | $(R S)^\circ = S^\circ$<br>$R^\circ$                   | $(R S)^\circ = S^\circ$<br>$R^\circ$                          | $(T \Rightarrow Y)^\circ =$<br>$T^\circ \Rightarrow (Y^\circ)$                                                                            | $R^{\circ\circ} = R$                           | $(T_1 \cap T_2)^\circ =$<br>$T_1^\circ \cap T_2^\circ$                                          |
| 7 | $\bullet \triangleleft$ | $(RT)^\wedge =$<br>$R^\wedge (T \otimes 1)$                       | $((T \otimes 1)R)^\wedge =$<br>$TR^\wedge$                       | —                                                      | —                                                             | $(R^\wedge)^\circ =$<br>$(R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \bullet)$<br>both bends: the<br>$^\circ$ section's<br>display | —                                              | —                                                                                               |
| 8 | $u$                     | $(S \cup T) R =$<br>$(S R) \cup (T R)$                            | $R (S \cup T) =$<br>$(R S) \cup (R T)$                           | $(S \cup T) R =$<br>$(S R) \cup (T R)$                 | $R (S \cup T) =$<br>$(R S) \cup (R T)$                        | $R \cap (S \cup T) =$<br>$R \cap S \cup R \cap T$                                                                                         | $(R \cup S)^\circ =$<br>$S^\circ \cup R^\circ$ |                                                                                                 |
| 9 | $\emptyset$             | $\emptyset R = \emptyset$                                         | $R \emptyset = \emptyset$                                        | $\emptyset R = \emptyset$                              | $R \emptyset = \emptyset$                                     | $R \cap \emptyset = \emptyset$                                                                                                            | $\emptyset^\circ = \emptyset$                  |                                                                                                 |

### §0.3. Adjoint triples

(0.3a)

$$\bullet f \dashv \bullet f^\circ \dashv / f^\circ$$

$$f^\circ \bullet \dashv f \bullet \dashv f \setminus$$

### §0.4. Frobenius

#### Frobenius.

(0.4a)

$$(n \otimes n) \otimes n = n \otimes (n \otimes n)$$

$\triangleright$  associative

Frobenius, one half — the other is its  $^\circ$

$$n \otimes n = n \otimes n$$

$\triangleright$  commutative

$$R \triangleleft \leq \triangleleft (R \otimes R)$$

$$n \otimes I = n$$

$\bullet$  is its unit

$$R \multimap \leq \multimap$$

$R \multimap \leq \multimap$

(0.2a)

$$\begin{array}{cccc}
 \begin{array}{c} \text{Diagram 1: } \Delta \triangleright \leq 1 \\ \text{Diagram 2: } 1 \leq \triangleleft \triangleright \\ \text{Diagram 3: } \circ - \circ \leq 1 \\ \text{Diagram 4: } 1 \leq \circ - \circ \end{array} & & & (0.4b)
 \end{array}$$

## §0.5. $\circ$ is a contravariant 2-functor

(0.5a)

$\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure **DRAWS** that — the picture and the formula below are  $R^\circ$ , not its definition:



$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circ)$$

where  $\circ \triangleleft : \mathbb{I} \rightarrow a \otimes a$  opens a pair of wires out of nothing and  $\triangleright \circ : b \otimes b \rightarrow \mathbb{I}$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\circ$  are  $\triangleright^\circ$  and  $\circ^\circ$ . The laws of  $\circ$  are that it is a contravariant 2-functor  $\circ : \mathcal{C}^\circ \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

## §0.6. Meet and composition

(0.6a)

| semi-distributivity, and what supplies it                                                                                                               | picture |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $R \ (S \cap T) \sqsubseteq RS \cap RT$ — the lax copy law. Equality exactly when $R$ is single valued: the Maps section's $F(R \cap S) = FR \cap FS$ . |         |

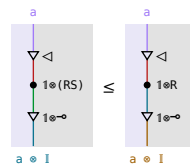
## §0.7. Domain and range

(0.7a)

|   |                                                                                                                                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{dom}(R) \triangleq 1 \cap (R \ R^\circ)$                                                                                                                                                                         |
| 2 | $\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X \ R$                                                                                                                                   |
| 3 | $\text{dom}(R \ S) \sqsubseteq \text{dom}(R)$                                                                                                                                                                           |
| 4 | $\text{dom}(R \cap S) = 1 \cap (S \ R^\circ)$                                                                                                                                                                           |
| 5 | $R \text{ entire } \text{Entire}(R) \triangleq \text{dom}(R) = 1$                                                                                                                                                       |
| 6 | $R \text{ simple } \text{Simple}(R) \triangleq R^\circ \ R \sqsubseteq 1$                                                                                                                                               |
| 7 | $R \text{ a map } \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$                                                                                                                                    |
| 8 | $\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R \ S)$<br>$\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f \ g)$<br>$\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f \ g)$ |
| 9 | $\text{Entire}(R \ S) \implies \text{Entire}(R)$                                                                                                                                                                        |

## §0.8. Sliding the discard

(0.8a)



$S \multimap \multimap \multimap$ , the lax axiom for  $\multimap$  in the first section  
— the discard slides back past  $S$

## §0.9. Division

(0.9a)

|   |                                                                                                                                                                                                                |  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 | $X \in R/S \iff X \in R$<br>$X$ is any $X$ -to- $Y$ pairing; one that only pairs $X$ with a $Y$ such that $X$ admires everyone $Y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                    |  |
| 2 | $X \in S \setminus R \iff S \setminus X \in R$<br>The mirror — divide on the left when $X$ comes first.                                                                                                        |  |
| 3 | $(R/S) \in R$<br>There is a $Y$ such that $X$ admires everyone $Y$ hates, and $p$ is one of the people $Y$ hates — then $X$ admires $p$ too. Strict at $S=\emptyset$ : $R/S$ is everyone, $(R/S)S=\emptyset$ . |  |
| 4 | $S \setminus (S \setminus R) \in R$<br>The mirror.                                                                                                                                                             |  |
| 5 | <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                    |  |
| 6 | $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                       |  |
| 7 | <b>maps:</b> $f \cdot (R/S) = (fR)/S$<br>Rename $X$ before or after dividing — the licence to write $fR/S$ .                                                                                                   |  |
| 8 | $R/(fS) = (R/S) f^\circ$<br>Rename $Y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                              |  |
| 9 | $(R/S) \cdot (S/W) \in R/W$<br>Someone who admires all of a hate-set that already covers everyone $Z$ works for admires those people too.                                                                      |  |

(0.8a)

|    |                                                                                                                                                                                                                                                      |                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 10 | <div><math>1 \sqsubseteq R/R</math><br/><math>R/R</math> runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only <b>a</b> and <b>b</b> admire each other's idols too, and still stay two people.</div>   |    |
| 11 | <div><math>(R/R) (R/R) = R/R</math><br/><math>R/R</math> is the preorder <b>admires at least as much as</b>, and a preorder is idempotent. Freyd writes <math>\sqsubseteq</math>; with <math>1 \sqsubseteq R/R</math> above it is an equality.</div> |    |
| 12 | <div><math>(R/R)R = R</math><br/>Reaching <b>p</b> through someone whose idols <b>x</b> fully admires is reaching <b>p</b> directly, since <b>x</b> admires their own idols.</div>                                                                   |    |
| 13 | <div><math>R/1 = R</math><br/>Dividing by <math>1</math>: <b>p</b>'s set is just <math>\{p\}</math>, so admiring all of it is admiring <b>p</b>.</div>                                                                                               |    |
| 14 | <div><math>R/(S_1 \cup S_2) = R/S_1 \cap R/S_2</math><br/>Admiring a combined hate-set is admiring each set in full.</div>                                                                                                                           |  |
| 15 | <div><math>S \setminus (R/W) = (S \setminus R)/W</math><br/>Which is why <math>S \setminus R/W</math> needs no bracket.</div>                                                                                                                        |  |

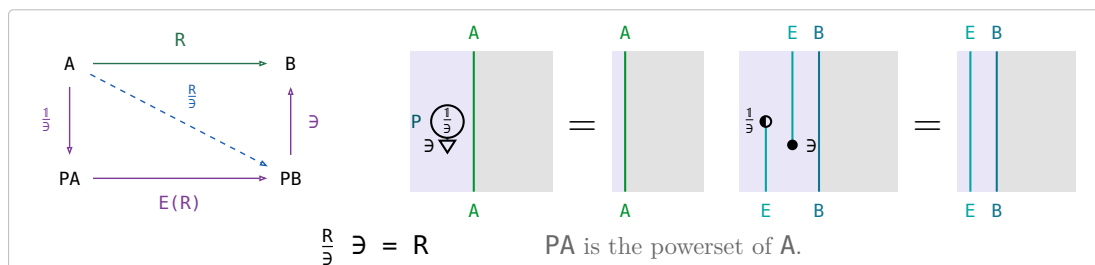
§0.10. Power allegories

(0.10a)

|    | law                                                                                                                                                        | the reading                                                                                                                                                                                      |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $\exists_R \square = R\square, \exists_R = \exists_R \square$                                                                                              | One $\exists$ per object, not per arrow.                                                                                                                                                         |
| 2  | $\exists$ is <b>thick</b>                                                                                                                                  | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                                                                                     |
| 3  | $\frac{R}{\exists} : A \rightarrow PB$ , for $R : A \rightarrow B$                                                                                         | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                                                                                       |
| 4  | $\text{Map}(\frac{R}{\exists})$                                                                                                                            |                                                                                                                                                                                                  |
| 5  | $\frac{R}{\exists} \exists = R$                                                                                                                            | reading the chosen set back through $\exists$ returns the relation.                                                                                                                              |
| 6  | $\text{Simple}(F) \Rightarrow F \sqsubseteq \frac{F \exists}{\exists}$                                                                                     | A partial choice of sets is inside the total one.                                                                                                                                                |
| 7  | the <b>singleton map</b> is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq 1$ two points with the same one-person set are the same point. | The one-person set.                                                                                                                                                                              |
| 8  | $\frac{\exists}{\exists} = 1$                                                                                                                              | Make the set of a set, then read it back one level down.                                                                                                                                         |
| 9  | <b>fusion:</b> $\text{Map}(f) \Rightarrow \frac{f R}{\exists} = f \frac{R}{\exists}$                                                                       | Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$ . renaming a point and then taking its one-person set is taking the set and renaming inside it. |
| 10 | $\text{Map}(f) \wedge 1\square = \exists\square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$                                                       | Rename first or take singletons first — the fusion row above at $R=1$ .                                                                                                                          |
| 11 | $R\square = \exists\square \Rightarrow R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$                                                              | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                                                                                          |
| 12 | $E(R) \triangleq \frac{\exists R}{\exists}$                                                                                                                | $E(R) : PA \rightarrow PB$ , image of a set of $A$                                                                                                                                               |
| 13 | $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                                                                               | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                                                                                            |
| 14 | $\frac{S}{\exists} E(R) = \frac{S R}{\exists}$                                                                                                             | absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$ , $\mathcal{A} \equiv \text{Kleisli}(E)$ in Relation Algebra                      |
| 15 | $E(R) = E(\frac{R}{\exists})$ union                                                                                                                        | the image of a set under $R$ is the union of the $R$ -sets of its points.                                                                                                                        |
| 16 | $\supseteq \triangleq \exists / \exists$                                                                                                                   | $xs \supseteq ys \Leftrightarrow \forall a. ys \exists a \rightarrow xs \exists a$                                                                                                               |
| 17 | $\subseteq = \supseteq^\circ$                                                                                                                              | $xs \subseteq ys \Leftrightarrow ys \supseteq xs$                                                                                                                                                |
| 18 | $T^\circ \frac{T}{\exists} \sqsubseteq E$                                                                                                                  | every $T$ -value of a point is a member of its $T$ -set: $\Lambda(T)$ is a map and $\Lambda(T)\exists=T$ .                                                                                       |

## §0.11. $\mathbf{i}\text{-E}$ : power transpose through singleton

(0.11a)



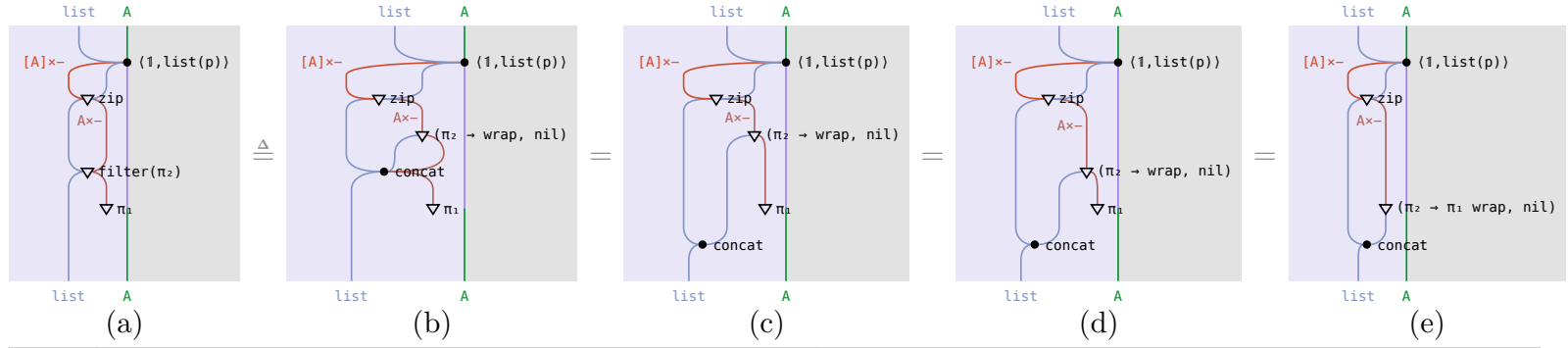
## §0.12. $\epsilon \setminus$ as a composite



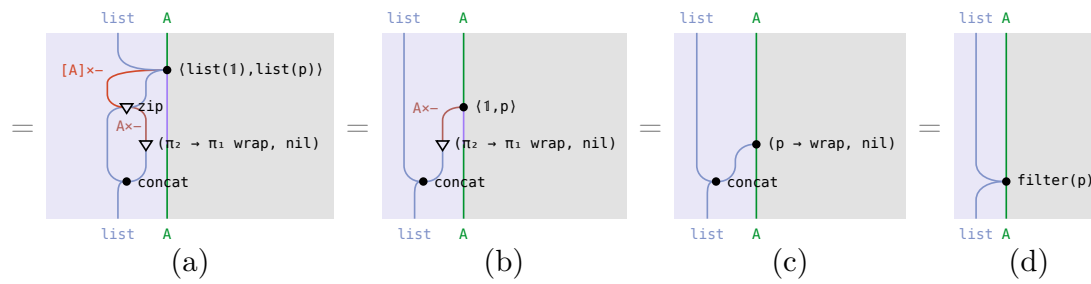
$$\langle 1, \text{list}(p) \rangle \text{ zip filter}(\pi_2) \text{ list}(\pi_1) = \text{filter}(p) \quad (1.7a)$$

pairing each element with its answer under  $p$  ( $\langle 1, \text{list}(p) \rangle \text{ zip}$ ), keeping the pairs answered **true** ( $\text{filter}(\pi_2)$ ) and dropping the answers ( $\text{list}(\pi_1)$ ) is **filter**( $p$ )

the starting composite



|     |                                                                                                                                                     |                                                                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle 1, \text{list}(p) \rangle \text{ zip filter}(\pi_2) \text{ list}(\pi_1)$                                                                   | definition of filter: $\text{filter}(p) \triangleq \text{list}((p \rightarrow \text{wrap}, \text{nil})) \text{ concat}$                             |
| (b) | $\triangleq \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil})) \text{ concat list}(\pi_1)$             | (1.6): $\text{concat list}(f) = \text{list}(\text{list}(f)) \text{ concat}$                                                                         |
| (c) | $= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil})) \text{ list}(\text{list}(\pi_1)) \text{ concat}$ | (1.8) backwards: $\text{list}(f) \text{ list}(g) = \text{list}(fg)$                                                                                 |
| (d) | $= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil}) \text{ list}(\pi_1)) \text{ concat}$              | (1.11): $(p \rightarrow f, g)h = (p \rightarrow fh, gh)$ ; (1.5): $\text{wrap list}(f) = f \text{ wrap}$ ; (1.4): $\text{nil list}(f) = \text{nil}$ |
| (e) | $= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$                           | (1.9) backwards: $1 = \text{list}(1)$                                                                                                               |



|     |                                                                                                                                        |                                                                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $= \langle \text{list}(1), \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$ | (1.7): $\langle \text{list}(f), \text{list}(g) \rangle \text{ zip} = \text{list}(\langle f, g \rangle)$<br>(1.8) backwards: $\text{list}(f) \text{ list}(g) = \text{list}(fg)$ |
| (b) | $= \text{list}(\langle 1, p \rangle (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$                                | (1.10): $h(p \rightarrow f, g) = (hp \rightarrow hf, hg)$ ; (1.1): $\langle f, g \rangle \pi_1 = f$ ; (1.3): $f \text{ nil} = \text{nil}$                                      |
| (c) | $= \text{list}((p \rightarrow \text{wrap}, \text{nil})) \text{ concat}$                                                                | (1.12): $1 f = f$                                                                                                                                                              |
| (d) | $= \text{filter}(p)$                                                                                                                   | definition of filter                                                                                                                                                           |

## §2. Functions and Categories

### §2.1. Categories

### §2.2. Functors

### §2.3. Natural transformations

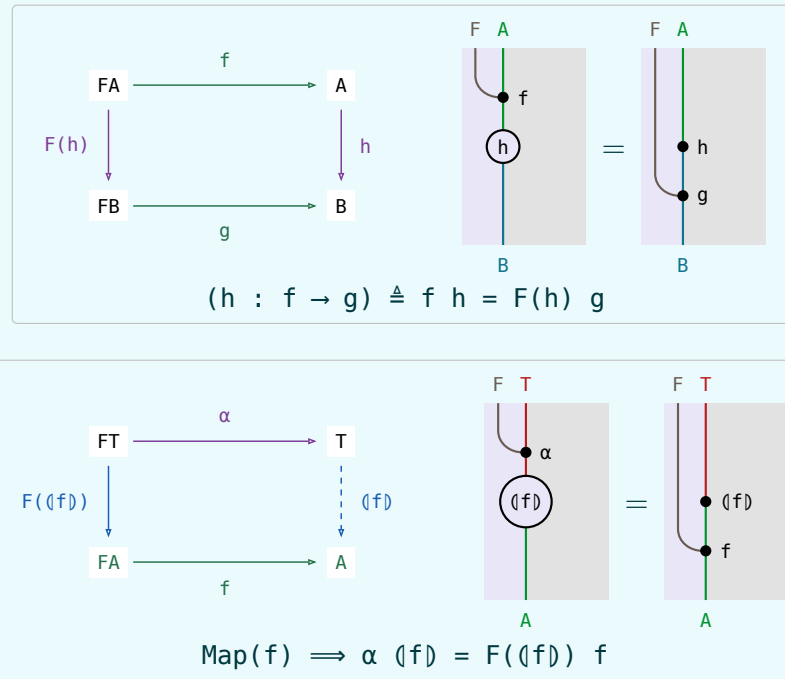
### §2.4. Constructing datatypes

### §2.5. Products and coproducts

### §2.6. Initial algebras

(2.6a)

An **F-algebra** is a map  $f : F(A) \rightarrow A$ ;  $A$  is its **carrier**. An **F-homomorphism** from  $f : F(A) \rightarrow A$  to  $g : F(B) \rightarrow B$  is a map  $h : A \rightarrow B$  with  $f \circ h = F(h) \circ g$ . The **initial algebra**  $\alpha : F(T) \rightarrow T$  is the F-algebra with exactly one F-homomorphism  $(f) : T \rightarrow A$  to every F-algebra  $f$ .

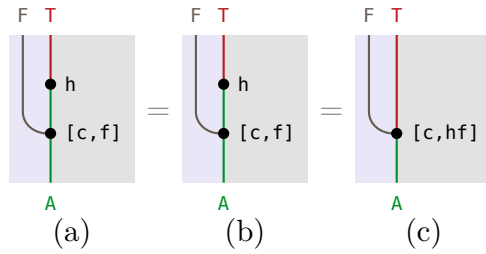


$$\alpha \circ h = F(h) \circ [c, f] \iff \text{zero } h = c \wedge \text{succ } h = h \circ f$$

(2.6b)

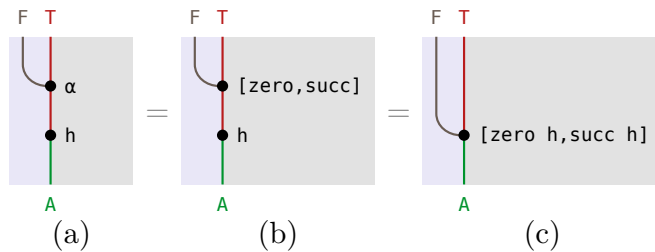
$h$  is a homomorphism from  $\alpha = [\text{zero}, \text{succ}]$  to  $[c, f]$  exactly when  $h$  sends **zero** to  $c$  and **succ** then  $h$  equals  $h$  then  $f$

the right side

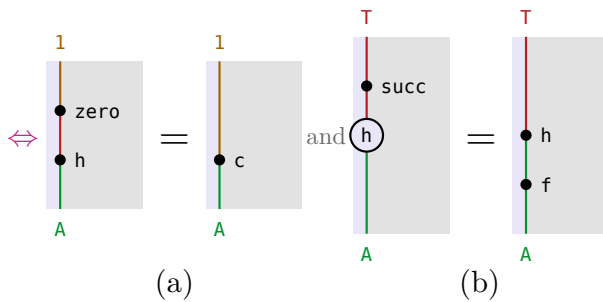


|     |                    |                   |
|-----|--------------------|-------------------|
| (a) | $F(h) [c, f]$      | definition of $F$ |
| (b) | $= (1 + h) [c, f]$ | coproduct         |
| (c) | $= [c, h \circ f]$ |                   |

the left side



|     |                                        |                                             |
|-----|----------------------------------------|---------------------------------------------|
| (a) | $\alpha \circ h$                       | since $\alpha = [\text{zero}, \text{succ}]$ |
| (b) | $= [\text{zero}, \text{succ}] \circ h$ | coproduct                                   |
| (c) | $= [\text{zero } h, \text{succ } h]$   | cancellation, <b>zero</b> arm               |



|     |                                                                                             |                               |
|-----|---------------------------------------------------------------------------------------------|-------------------------------|
| (a) | $\iff [\text{zero } h, \text{succ } h] = [c, h \circ f] \implies \text{zero } h = c$        | cancellation, <b>SUCC</b> arm |
| (b) | and $[\text{zero } h, \text{succ } h] = [c, h \circ f] \implies \text{succ } h = h \circ f$ |                               |

### §2.6.1. Reflection

(2.6a)

(2.6.1a)



Taking a value apart with  $\alpha$  and putting it straight back is doing nothing.

**§2.6.2.  $R S = F(S) Q \implies (R) S = (Q)$  — folding with  $R$ , then applying  $S$ , is folding with  $Q$**

When  $S : B \rightarrow C$  is an  $F$ -homomorphism from  $R : F(B) \rightarrow B$  to  $Q : F(C) \rightarrow C$ , folding with  $R$  and then applying  $S$  is folding with  $Q$ .

(2.6.2a)



## §2.7. Type functors

(2.7a)

Let  $F$  be a bifunctor taking both the parameter  $A$  and the recursive position  $TA$ , with an initial algebra  $\alpha_A : F(A, TA) \rightarrow TA$  for every object  $A$ . Then  $T$  is a functor, acting on a map  $f : A \rightarrow B$  by

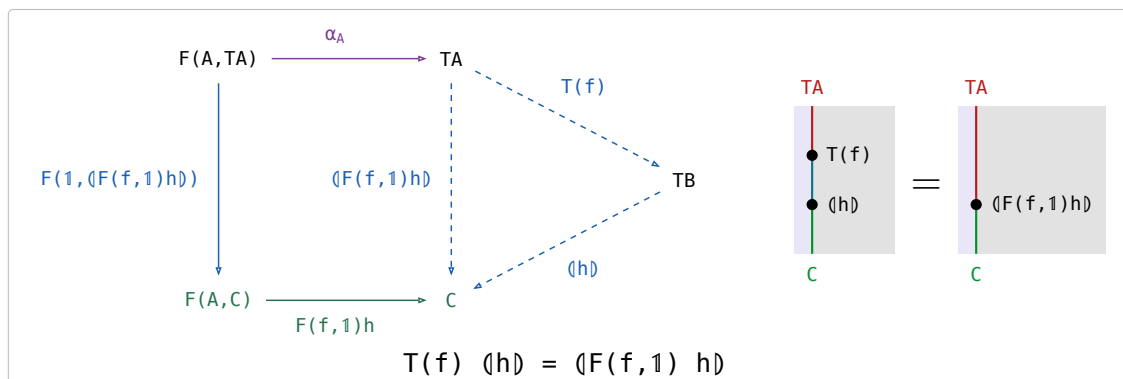
$$T(R) \triangleq (F(R, 1) \alpha)$$

(2.7b)



- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$  is a bifunctor and a wire is a unary functor, so the two arguments are packed first:  $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$  sends  $A$  to  $(A, TA)$ , and  $F(\langle 1, T \rangle(A))$  is  $F(A, TA)$ .
- The picture is three wires —  $F$ ,  $\langle 1, T \rangle$ , and the object — and the region between the first two is  $\mathcal{A} \times \mathcal{A}$ .
- $\alpha$  is then an ordinary natural transformation  $F \circ \langle 1, T \rangle \Rightarrow T$ : its bead eats the  $F$  and  $\langle 1, T \rangle$  wires, and the  $T$  wire is born under it.
- $F(f, T(f))$  costs no notation. It is the bead  $f$  on the object wire with  $\langle 1, T \rangle$  and  $F$  running past:  $\langle 1, T \rangle$  is what turns  $f$  into the pair  $(f, T(f))$ , and  $F$  is what applies it.
- The law is the naturality of  $\alpha$ , which is exactly the freedom to slide that  $f$  bead past it.

(2.7c)



(2.7d)

|   | name                   | law                                 | what it says                                                                                                                                                                                                |
|---|------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | functor                | $T(1) = 1$ and $T(R) T(S) = T(R S)$ | Acting by the identity changes nothing, and two actions in a row are one action.                                                                                                                            |
| 2 | type functor fusion    | $T(f) (fh) = (F(f, 1) h)$           | A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because $F$ is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$ . |
| 3 | naturality of $\alpha$ | $\alpha T(f) = F(f, T(f)) \alpha$   | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ .                                                                      |

## §3. Applications

### §3.1. Banana-split

(3.1a)

|   | definition                                                                                                                                                                           | type                                                    | note                                                                                                         |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1 | $\text{listr } A ::= \text{nil} \mid \text{cons}(A, \text{listr } A)$<br>$\text{list}(A) = [A]$                                                                                      | $\mathcal{A} \rightarrow \mathcal{A}$                   | The cons-lists over $A$ , the datatype every row below folds.                                                |
| 2 | $\text{sum} = \langle [\text{zero}, \text{plus}] \rangle$                                                                                                                            | $[A] \rightarrow A$                                     | $\text{plus}(a, b) = a + b$ .                                                                                |
| 3 | $\text{length} = \langle [\text{zero}, \pi_2 \text{ succ}] \rangle$                                                                                                                  | $[A] \rightarrow \mathbb{N}$                            | $\pi_2$ drops the head and keeps the count of the tail, $\text{succ}$ adds one for the head.                 |
| 4 | $\text{average} \triangleq \langle \text{sum}, \text{length} \rangle \text{ div}$                                                                                                    | $[\mathbb{N}] \rightarrow \mathbb{N}$                   | $\text{div}(m, n) = m/n$ , with $\text{div}(0, 0) = 0$ so <b>average</b> is total. Traverses the list twice. |
| 5 | banana-split law<br>$\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1) h, F(\pi_2) k \rangle \rangle$                                                  | $T \rightarrow A \times B$                              | Any fork of folds is a single fold, hence one traversal — $F$ the base functor.                              |
| 6 | what it reduces to<br>$\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\pi_1) h, F(\pi_2) k \rangle$ | $FT \rightarrow A \times B$                             | All that $\triangleq$ <b>reduce</b> leaves to check: the fork satisfies the defining equation.               |
| 7 | the instance<br>$\langle \text{sum}, \text{length} \rangle = \langle [\text{zeros}, \text{pluss}] \rangle$                                                                           | $[\mathbb{N}] \rightarrow \mathbb{N} \times \mathbb{N}$ | $\text{pluss}(a, (b, n)) = (a + b, n + 1)$ , so <b>average</b> runs in one pass.                             |
| 8 | $\text{preds } n = [n, n-1, \dots, 1]$                                                                                                                                               | $\mathbb{N} \rightarrow [\mathbb{N}]$                   | apply the earlier special case to write <b>preds</b> as $\langle k \rangle \pi_1$ .                          |

(3.1b)



### §3.1.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$  — the algebras and their homomorphisms — has binary products whenever  $\mathcal{A}$  does, and the forgetful  $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$  creates them: the product of  $h : FA \rightarrow A$  and  $k : FB \rightarrow B$  is carried by  $A \times B$ , with structure  $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$ .
- $\pi_1$  and  $\pi_2$  are homomorphisms out of it, and  $\langle p, q \rangle : X \rightarrow A \times B$  is a homomorphism **iff**  $p$  and  $q$  both are — substitute the structure and compare the two forks component by component.
- The banana-split law of  $\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$  IS that product's universal property read at the initial algebra:  $\langle h \rangle$  and  $\langle k \rangle$  are the unique homomorphisms to  $(A, h)$  and  $(B, k)$ , so their fork is the unique homomorphism into the product, which is  $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$ .
- Fokkinga's theorem is **strictly more general** and is not that product: in mutual recursion the two arrows leave  $F(A \times B)$ , not  $FA$  and  $FB$ , so each sees BOTH components. The product is the case where they factor as  $F(\pi_1)h$  and  $F(\pi_2)k$ . What it says: **any algebra on a product carrier is folded by a fork**.

(3.1.1a)



### §3.2. Ruby triangles and Horner's rule

(3.2a)

| stage                         | definition                                                                                                                            |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1 informally                  | $\text{tri}(f)[a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f(a_1), \dots, f^i(a_i), \dots, f^n(a_n)]$                                   |
| 2 for cons-lists              | $\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$                                               |
| 3 with the base functor named | $F(A, B) = 1 + A \times B$ , $\alpha = [\text{nil}, \text{cons}]$ , so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$ |
| 4 abstractly                  | $F$ a bifunctor with initial type $(\alpha, T)$ : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$                                 |

For the definition to make sense  $f : A \rightarrow A$  is required, and then  $\text{tri}(f) : TA \rightarrow TA$ .

(3.2b)



Horner's rule is Horner's rule, generalised from cons-lists to any initial type. Fusion reduces it to  $F(1, T(f)) \alpha (g) = F(1, (g)) F(1, f) g$ .

## §3.2.1. Depth of a tree

(3.2.1a)

|   |                  |                                                                                                                                                                                                                                  |
|---|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | the datatype     | $\text{tree } A ::= \text{tip}(A) \mid \text{bin}(\text{tree } A, \text{tree } A)$ , base functor $F(A,B)=A+B \times B$ ,<br>initial type $([\text{tip}, \text{bin}], \text{tree})$                                              |
| 2 | the fold         | $\langle [g, h] \rangle$ is the unique $f$ with $f(\text{tip}(a))=g(a)$ and $f(\text{bin}(x, y))=h(f(x), f(y))$                                                                                                                  |
| 3 | $\text{tree}(f)$ | $\text{tree}(f) = \langle F(f, 1) \mid [\text{tip}, \text{bin}] \rangle$ ; pointwise $\text{tree}(f)(\text{tip}(a))=\text{tip}(f(a))$ and<br>$\text{tree}(f)(\text{bin}(x, y))=\text{bin}(\text{tree}(f)(x), \text{tree}(f)(y))$ |
| 4 | $\text{max}$     | $\text{max} = \langle [1, \text{bmax}] \rangle$ , where $\text{bmax}(a, b)$ is the larger of $a$ and $b$                                                                                                                         |
| 5 | $\text{depths}$  | $\text{depths} = \text{tree}(\text{zero}) \text{ tri}(\text{succ})$ — replaces every tip by its depth in the tree, $\text{zero}$ the constant<br>function returning 0 and $\text{succ}$ the successor                            |
| 6 | $\text{depth}$   | $\text{depth} = \text{depths } \text{max}$                                                                                                                                                                                       |

### §3.3. The TeX problem

### §3.4. Conditions and conditionals

### §3.5. Concatenation and currying

If  $\varphi$  is natural, i.e.  $G(h \times id) \cdot \varphi = \varphi \cdot (F(h) \times id)$ , then  $h \cdot G(f) \cdot \varphi = f \iff f = apply \cdot ((h \cdot G(apply)) \cdot \varphi) \times id$  The unique solution of a two-argument recursion  $f(a, x) = h(G(f)(a, x))$  built from a natural  $\varphi$  is **apply** composed with a one-argument fold, which is how **cat** — not itself a fold — is derived from one. Theorem 3.1

## §4. Relations and Allegories

### §4.1. Allegories

### §4.2. Special properties of arrows

$R: A \rightarrow B$ ,  $S: B \rightarrow A$ ,  $1_A \in RS$ ,  $SR \in 1_B \implies S = R^\circ$  and  $R$  is a map A relation with a two-sided partial inverse under composition — identity below one order, identity above the other — is that inverse's converse, and  $R$  is itself a map. Proposition 4.1

### §4.3. Tabular allegories

$(f, g)$  a tabulation of  $R \implies k^\circ h \in R \iff \exists! \text{ map } m \text{ with } h = mf, k = mg$  Every pair below  $R$ 's tabulating pair factors, uniquely, through a single map into the tabulation's apex. Proposition 4.2

### §4.4. Locally complete allegories

### §4.5. Boolean allegories

### §4.6. Power allegories

## §5. Datatypes in Allegories

### §5.1. Relators

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \sqsubseteq S$ . A **relator**  $F: \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories: (5.1a)

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving  $^\circ$  is **not** asked for —  $^\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ \circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

#### the statement

- 1  $\text{Map}(f) \implies \text{Map}(F(f))$  and  $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$ . Lemma 5.1
- 2 Over a **tabular** allegory a functor is a relator  $\iff$  it preserves  $^\circ$ . Theorem 5.1 the book's reverse direction (preserving  $^\circ \implies$  relator) needs the source to be **Rel**: the counterexample over a tabular allegory that is not **Rel** shows it can fail
- 3  $F(R^\circ) = F(R)^\circ$  for every  $R$ , so  $F(R)^\circ$  needs no bracket.
- 4 Two relators agreeing on maps are equal. Corollary 5.1
- 5  $\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$  Ex 5.2
- 6  $F(R \cap S) \sqsubseteq F(R) \cap F(S)$ , and strictly. Ex 5.2
- 7  $\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$  Ex 5.5

The **power relator**  $P \text{ — } xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

### §5.2. Relational products

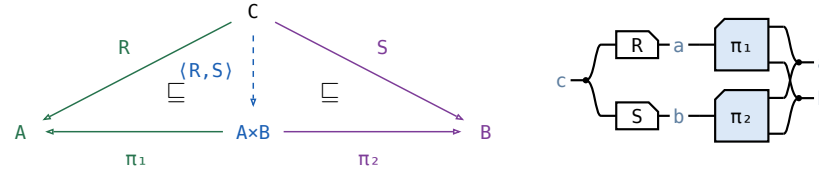
The **fork** of  $R: \mathcal{C} \rightarrow \mathcal{A}$  and  $S: \mathcal{C} \rightarrow \mathcal{B}$  is  $\langle R, S \rangle \triangleq R\pi_1^\circ \cap S\pi_2^\circ$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ . (5.1)

(5.2a)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) \circ R \quad (5.6)$$

$$\langle R, S \rangle \pi_2 = \text{dom}(R) \circ S \quad (5.7)$$

(5.2b)



A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \sqsubseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In  $\mathbf{Rel}$ ,  $c \langle R, S \rangle (a, b)$  iff  $c R a$  and  $c S b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft(R \otimes S)$  on the right.

No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \omega$  discards the second component, so  $\pi_1 \circ = 1 \otimes \omega$  **creates** one out of nothing, and  $\eta$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2c)

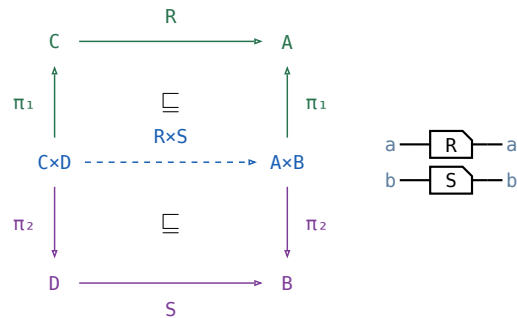


### §5.2.1. Relational product $R \times S$

(5.2)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.

(5.2.1a)

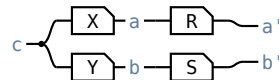


Right-then-up is  $(R \times S) \pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S) \pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) (R \times S) (a, b)$  iff  $c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §5.2.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:

(5.3)



(5.2.2a)

| the law                                                                                                                                                                                    | picture |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 $\langle X, Y \rangle \langle R, S \rangle^\circ = (X \circ R) \times (Y \circ S)$                                                                                                       |         |
| 2 $\langle R, S \rangle^\circ \langle U, V \rangle \subseteq (R^\circ U) \times (S^\circ V)$                                                                                               |         |
| 3 $\text{Map}(f) \Rightarrow f \langle R, S \rangle = \langle f \circ R, f \circ S \rangle$<br>f a map; it fails for an arbitrary arrow;                                                   |         |
| 4 $X \langle R, S \rangle \subseteq \langle X \circ R, X \circ S \rangle$<br>the half of the row above that holds for every X                                                              |         |
| 5 $F(R \times S) \text{unzip}(F) \subseteq \text{unzip}(F) (F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$ ; only $\subseteq$ for an arbitrary R, S |         |
| 6 $g = \text{curry}(f) \Leftrightarrow (g \times 1) \text{eval} = f$<br>reading $\times$ as the relational product, does $\text{Rel}$ have exponentials?                                   |         |

### §5.3. Relational coproducts

(5.3a)

| the statement<br>$R : a_1 \rightarrow c, S : a_2 \rightarrow c$                                                                                                                                              | picture |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 $[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$<br>(5.9) The tape is the union — a particle entering at $A+B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |
| 2 $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$                                                                                                                                                  |         |
| 3 $l [R, S] = R, r [R, S] = S$<br>$l T = R \wedge r T = S \Rightarrow T = [R, S],$                                                                                                                           |         |
| 4 $U : a_1 \rightarrow d, V : a_2 \rightarrow d$<br>$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S) \quad (5.11)$                                                                                        |         |
| 5 $Z : c \rightarrow d$<br>$[R, S] Z = [R Z, S Z]$                                                                                                                                                           |         |

$$1 \quad R + S \triangleq [R \text{ l}, S \text{ r}] \quad (5.10)$$
$$a_1 \text{---} \boxed{\text{L}} \text{---} a \text{---} \boxed{\text{L}} \text{---} a_1 = a_1 \text{---} a_1 \text{---} a_2 \text{---} \boxed{\text{R}} \text{---} a \text{---} \boxed{\text{R}} \text{---} a_2 = a_2 \text{---} a_2$$
$$a_1 \text{---} \boxed{\text{l}} \text{---} a \text{---} \boxed{\text{r}} \text{---} a_2 = a_1 \text{---} \boxed{\text{0}} \text{---} a_2$$

The universal-property row is not free:  $\mathbf{l}, \mathbf{r}$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\Xi$  sends an arrow  $\mathbf{A} \rightarrow \mathbf{C}$  to a map  $\mathbf{A} \rightarrow \mathbf{P}\mathbf{C}$  reversibly, so the map coproduct can be applied underneath it. For any  $\mathbf{T} : \mathbf{A} + \mathbf{B} \rightarrow \mathbf{C}$ ,

$$\begin{array}{ccccc}
 \begin{array}{|c|} \hline \mathbb{1}T \\ \hline R \\ \hline \end{array} & \xleftrightarrow{\cdot \partial \mathbb{1} \ominus} & \begin{array}{|c|} \hline \mathbb{1}T \\ \hline R \\ \hline \end{array} & \xleftrightarrow{\text{fusion}} & \begin{array}{|c|} \hline \mathbb{1}T \\ \hline R \\ \hline \end{array} \\
 \begin{array}{|c|} \hline rT \\ \hline S \\ \hline \end{array} & & \begin{array}{|c|} \hline rT \\ \hline S \\ \hline \end{array} & & \begin{array}{|c|} \hline rT \\ \hline S \\ \hline \end{array} \\
 \\ 
 \xleftrightarrow{\text{coproduct of maps}} & \begin{array}{|c|} \hline T \\ \hline [\frac{R}{\ominus}, \frac{S}{\ominus}] \\ \hline \end{array} & \xleftrightarrow{\cdot \partial \mathbb{1} \ominus} & \begin{array}{|c|} \hline T \\ \hline [\frac{R}{\ominus}, \frac{S}{\ominus}] \partial \\ \hline \end{array}
 \end{array}$$

A commutative diagram illustrating the relationship between objects  $A$ ,  $B$ ,  $A+B$ , and  $PC$ . The diagram shows the following components and mappings:

- Objects:  $A$  (top left),  $B$  (top right),  $A+B$  (middle left), and  $PC$  (middle right).
- Mappings:
  - $A \rightarrow A+B$  (labeled  $l$ )
  - $B \rightarrow A+B$  (labeled  $r$ )
  - $A \rightarrow PC$  (labeled  $R$ )
  - $B \rightarrow PC$  (labeled  $S$ )
  - $A+B \rightarrow PC$  (dashed blue arrow, labeled  $[R, S]$ )
  - $A \rightarrow PC$  (curved green arrow, labeled  $R$ )
  - $B \rightarrow PC$  (curved purple arrow, labeled  $S$ )

$$(5.3.1c)$$

1 prove (5.11), and say why duality  
does not carry it over from the  
product law

§5.4. The power relator  $P(R)$ 

(5.4a)

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq (\epsilon \setminus (R \epsilon)) \cap ((\exists R) / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

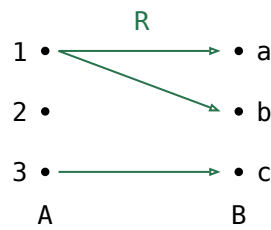
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(5.4b)

|                                         | in Rel                                                                                              | in words                                                                       | $\text{im}(xs)$              | partner                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| 1 $(\exists R) / \exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                     | $\forall y. \text{ some } xs \ R \ y$                                          | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| 2 $(\exists / (\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                     | every $y$ with some $xs \ R \ y$ is in $ys$                                    | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| 3 $E(R)$                                | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                       | $ys = \text{every } y \text{ with some } xs \ R \ y$                           | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| 4 $((\exists R^\circ) / \exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                     | $\forall x. x \ R \ \text{some } ys$                                           | —                            | every $x$ has a partner                       |
| 5 $P(R)$                                | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$ | —                            | every $x$ and every $y$ has a partner         |

(5.4c)



| $xs$            | $(\exists R) / \exists$ | $(\exists / (\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ) / \exists)^\circ$ | $P(R)$        |
|-----------------|-------------------------|---------------------------------|-------------|---------------------------------------|---------------|
| 1 $\emptyset$   | $\emptyset$             | all 8 subsets of $abc$          | $\emptyset$ | all 8 subsets of $abc$                | $\emptyset$   |
| 2 $\{1\}$       | $\emptyset, a, b, ab$   | $ab, abc$                       | $ab$        | $a, b, ab, ac, bc, abc$               | $a, b, ab$    |
| 3 $\{2\}$       | $\emptyset$             | all 8 subsets of $abc$          | $\emptyset$ | none                                  | none          |
| 4 $\{3\}$       | $\emptyset, c$          | $c, ac, bc, abc$                | $c$         | $c, ac, bc, abc$                      | $c$           |
| 5 $\{1, 2\}$    | $\emptyset, a, b, ab$   | $ab, abc$                       | $ab$        | none                                  | none          |
| 6 $\{1, 3\}$    | all 8 subsets of $abc$  | $abc$                           | $abc$       | $ac, bc, abc$                         | $ac, bc, abc$ |
| 7 $\{2, 3\}$    | $\emptyset, c$          | $c, ac, bc, abc$                | $c$         | none                                  | none          |
| 8 $\{1, 2, 3\}$ | all 8 subsets of $abc$  | $abc$                           | $abc$       | none                                  | none          |

(5.4d)

|   | law                                                                                                | the reading                                                                                                                                                                                                                                                                                                    |
|---|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $X \sqsubseteq P(R) \iff X \sqsubseteq \exists R \text{ and } X^\circ \sqsubseteq \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$ .                                                                                                                  |
| 2 | $P(1) = 1$                                                                                         | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                            |
| 3 | $\text{Map}(f) \implies P(f) = E(f)$                                                               | In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f(a)$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| 4 | $P(R \ S) = P(R) \ P(S)$                                                                           | $\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                    |

§5.4.1. Type relator

(5.4.1a)

Let  $F$  be a binary relator with initial type  $(\alpha, T)$ , so  $T$  is a type functor.  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the reduce section. For every object  $A$  the initial algebra is  $\alpha : F(A, TA) \rightarrow TA$ , among the maps.  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = \langle F(R, 1) \ \alpha \rangle$$

(5.4.1b)

|   |  | name                  | law                                       | what it says                                                                                         |
|---|--|-----------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1 |  | the defining equation | $T(R) = \langle F(R, 1) \ \alpha \rangle$ | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                      |
| 2 |  | type relator          | $T(R)^\circ = T(R^\circ)$                 | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map. |

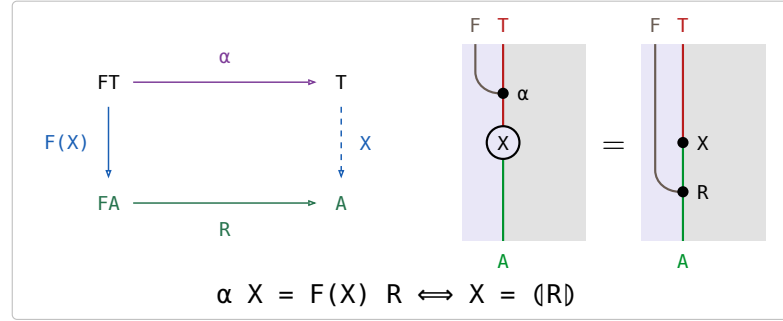
## §5.5. Relational catamorphisms

(5.5a)

let  $F$  be a relator and has **initial algebra**  $\alpha : F(T) \rightarrow T$  in the subcategory of functions.  $\alpha$  is also initial in the allegory:

### §5.5.1. The defining equation

(5.5.1a)



An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $A$ .

(5.5.1b)

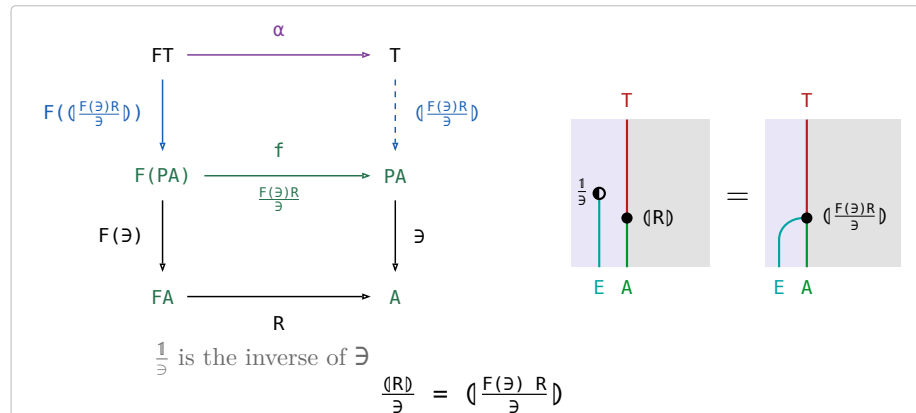
$$\langle [c, f] \rangle = \text{reduce}(c, f)$$

catamorphism: arrows of the form  $\langle f \rangle$  are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing

|   |                          |                                                                                                                      |                                                                                                                           |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>part</b>              | <b>Nat</b>                                                                                                           | <b>[A]</b>                                                                                                                |
| 2 | datatype                 | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                                                    | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                             |
| 3 | base functor $F$         | $F(X) = 1 + X$                                                                                                       | $F(X) = 1 + A \times X$                                                                                                   |
| 4 | initial algebra $\alpha$ | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$                                   | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$                                              |
| 5 | the fold, pointwise      | $\langle [c, f] \rangle(\text{zero}) = c$<br>$\langle [c, f] \rangle(\text{succ}(n)) = f(\langle [c, f] \rangle(n))$ | $\langle [c, f] \rangle(\text{nil}) = c$<br>$\langle [c, f] \rangle(\text{cons}(a, x)) = f(a, \langle [c, f] \rangle(x))$ |

### §5.5.2. $\langle R \rangle = \langle \frac{F(\exists) R}{\exists} \rangle \exists$ — folding with $R$ is a fold of maps into sets, then $\exists$

(5.5.2a)



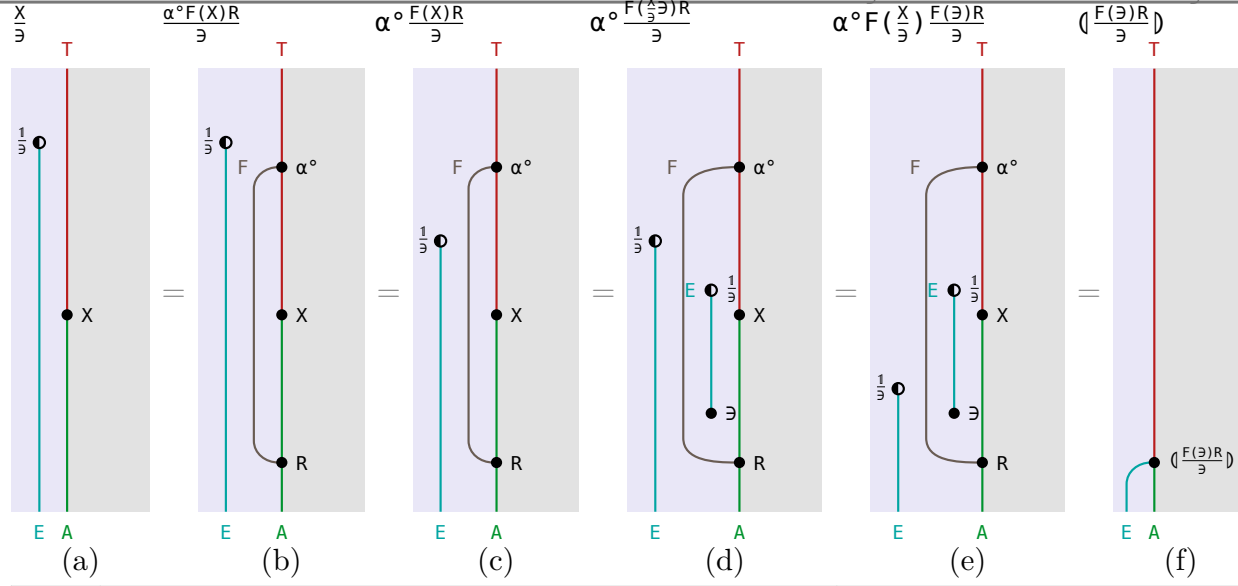
(5.12)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \boxed{\frac{\frac{\alpha}{\frac{X}{\exists}}}{\frac{F(\frac{X}{\exists})}{\frac{F(\exists)R}{\exists}}}} \xleftrightarrow{\text{reduce of maps}} \boxed{\frac{\frac{X}{\exists}}{\langle \frac{F(\exists)R}{\exists} \rangle}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{X}{\langle \frac{F(\exists)R}{\exists} \rangle \exists}}
 \end{array}$$

the last two rows at  $X := \langle R \rangle$ :

$$\alpha X = F(X) R \iff \frac{X}{\exists} = \langle \frac{F(\exists) R}{\exists} \rangle \quad (5.5.2b)$$

in a tabular allegory, with  $F$  a relator and  $\alpha : F(T) \rightarrow T$  its initial algebra: a relation  $X$  out of  $T$  satisfies the fold equation  $\alpha X = F(X) R$  of  $R$  exactly when  $\frac{X}{\exists}$  is the fold of the map  $\frac{F(\exists) R}{\exists}$



|     |                                                                |                                                                                                                                                                                                                                                                               |
|-----|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{X}{\exists}$                                            | $\alpha$ iso; $\alpha X = F(X) R$                                                                                                                                                                                                                                             |
| (b) | $= \frac{\alpha^0 F(X) R}{\exists}$                            | $\alpha^0$ is a map because $\alpha$ is an iso, so it leaves $\Lambda$ : $\frac{\alpha^0 F(X) R}{\exists} = \alpha^0$<br>$\frac{F(X) R}{\exists} \text{ --- Map}(f)$<br>$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$                                                  |
| (c) | $= \alpha^0 \frac{F(X) R}{\exists}$                            | $X = \frac{X}{\exists} \text{ --- } \frac{R}{\exists} \text{ --- } \exists = R$                                                                                                                                                                                               |
| (d) | $= \alpha^0 \frac{F(\frac{X}{\exists}) R}{\exists}$            | $F(\frac{X}{\exists}) = F(\frac{X}{\exists}) F(\exists)$ as $F$ is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves $\Lambda$ --- $\triangleq$ relator,<br>$\text{Map}(f)$<br>$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$ |
| (e) | $= \alpha^0 F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$ | fold uniqueness, $\frac{F(\exists) R}{\exists}$ being a map --- $(h : f \rightarrow g) \triangleq f h$<br>$= F(h) g, \text{Map}(f)$<br>$\implies \alpha \langle f \rangle = F(\langle f \rangle) f$                                                                           |
| (f) | $= \langle \frac{F(\exists) R}{\exists} \rangle$               |                                                                                                                                                                                                                                                                               |

## §5.6. Combinatorial functions

(5.6a)

|    | definition                                                                                                                                                                                                                                                                           | type                                                                                                                            | note                                                                                                                                                                                                                                |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$<br>$\text{list}(A) = [A]$                                                                                                                                                                                                              | $\mathcal{A} \rightarrow \mathcal{A}$                                                                                           | The list type, under the short name it keeps.                                                                                                                                                                                       |
| 2  | $\text{list}(R) = \langle [\text{nil}, (R \times 1) \text{ cons}] \rangle$                                                                                                                                                                                                           | $[A] \rightarrow [B]$                                                                                                           | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                                                                                                           |
| 3  | $\text{subseq} = \langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$                                                                                                                                                                                                               | $[A] \rightarrow [A]$                                                                                                           | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — $\text{cons}$ keeps the head, $\pi_2$ drops it.                                                                                                                      |
| 4  | $\text{prefix} = \langle [\text{nil}, \multimap \text{nil} \cup \text{cons}] \rangle$<br>$\text{prefix} = \text{cat}^\circ \pi_1 = \text{init}^*$                                                                                                                                    | $[A] \rightarrow [A]$                                                                                                           | $ys$ is an initial segment of $xs$ ; the first $\text{nil}$ is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$ .                                                                                             |
| 5  | $\text{suffix} = \text{cat}^\circ \pi_2 = \text{tail}^*$                                                                                                                                                                                                                             | $[A] \rightarrow [A]$                                                                                                           | The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$ ; as a reduce it needs $\text{snoc}$ -lists.                                                                                                                             |
| 6  | $\text{partition} = \text{concat}^\circ$                                                                                                                                                                                                                                             | $[A] \rightarrow [[A]]$                                                                                                         | This $\text{cat}$ is restricted to $[A]^* \times [A] \rightarrow [A]$ , so $ys$ is a list of non-empty segments of $xs$ .                                                                                                           |
| 7  | $\text{concat} = \langle [\text{nil}, \text{cat}] \rangle$                                                                                                                                                                                                                           | $[[A]] \rightarrow [A]$                                                                                                         | Joins the segments back up, which is why its converse splits a list.                                                                                                                                                                |
| 8  | $\text{inits}$                                                                                                                                                                                                                                                                       | $[A] \rightarrow [[A]]$                                                                                                         | Implements $\frac{\text{prefix}}{\exists}$ , listing the prefixes by increasing length.                                                                                                                                             |
| 9  | $\text{tails}$                                                                                                                                                                                                                                                                       | $[A] \rightarrow [[A]]$                                                                                                         | Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.                                                                                                                                               |
| 10 | $\text{cpp}, \text{cpl}, \text{cpr}$                                                                                                                                                                                                                                                 | $[A] \times [B] \rightarrow [A \times B]$<br>$[A] \times B \rightarrow [A \times B]$<br>$A \times [B] \rightarrow [A \times B]$ | Every pair $(a, b)$ with $a$ from the first list and $b$ from the second; $\text{cpl}$ , $\text{cpr}$ keep one side a single element.                                                                                               |
| 11 | $\text{cpp setify} = (\text{setify} \times \text{setify}) \frac{\exists \times \exists}{\exists}$<br>$\text{cpl setify} = (\text{setify} \times 1) \frac{\exists \times 1}{\exists}$<br>$\text{cpr setify} = (1 \times \text{setify}) \frac{1 \times \exists}{\exists}$              | $[A] \times [B] \rightarrow P(A \times B)$                                                                                      | Forgetting the order with $\text{setify}$ , $\text{cpp}$ is the transpose of $\exists \times \exists$ .                                                                                                                             |
| 12 | $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$                                                                                                                                                                                                                                 | $F(PA) \rightarrow PFA$                                                                                                         | $\text{cp}(F)$ , the transpose of a relator's action on $\exists$ ; $\text{cpp}$ implements it at $F(A) = A \times A$ .                                                                                                             |
| 13 | $\text{cp}(\text{list}) = \langle \frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} \rangle$<br>$\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{\exists \times \exists}{\exists}]$<br>$E(\text{cons})$ | $\text{list}(PA) \rightarrow P(\text{list}(A))$                                                                                 | $\text{cp}(\text{list})$ : (5.12) $\langle R \rangle = \langle \frac{F(\exists) R}{\exists} \rangle \exists$ at $\text{list}(\exists) = \langle [\text{nil}, (\exists \times 1) \text{ cons}] \rangle$ , then the algebra expanded. |
| 14 | $\text{cplist} = \langle [\text{nil} \text{ wrap}, \text{cpp list}(\text{cons})] \rangle$                                                                                                                                                                                            | $[[A]] \rightarrow [[A]]$                                                                                                       | $\text{cp}(\text{list})$ with every set a list: $\text{cpp}$ in place of $\frac{\exists \times \exists}{\exists}$ .                                                                                                                 |

**glue concat  $\sqsubseteq (1 \times \text{concat}) \text{ cons}$** 

(5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(1×concat) cons**)

the starting composite



|     |                                                                                                              |                                                    |
|-----|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| (a) | <b>glue concat</b>                                                                                           | definition of glue                                 |
| (b) | $\triangleq (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cons } \text{concat}$   | since $\text{concat} = ([\text{nil}, \text{cat}])$ |
| (c) | $= (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 (1 \times \text{concat}) \text{ cat}$   | naturality of $\text{assocl}$                      |
| (d) | $= (1 \times (\text{cons}^\circ (1 \times \text{concat}))) \text{ assocl } \text{cons} \times 1 \text{ cat}$ | since $\text{concat} = ([\text{nil}, \text{cat}])$ |



|     |                                                                                                                            |                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cat}$        | since $(\text{cons} \times 1) \text{ cat} = \text{assocr } (1 \times \text{cat}) \text{ cons}$ |
| (b) | $= (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{assocr } (1 \times \text{cat}) \text{ cons}$ | since $\text{assocl } \text{assocr} = 1$ and $\text{cat}^\circ \text{ cat} \sqsubseteq 1$      |
| (c) | $\sqsubseteq (1 \times \text{concat}) \text{ cons}$                                                                        |                                                                                                |

**§5.6.1.**  $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$  — the set of subsequences is a fold of maps into sets

(5.6.1a)

$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : \text{PA} \times \text{PA} \rightarrow \text{PA}$ , so  $\frac{R \cup S}{\exists} = (\frac{R}{\exists}, \frac{S}{\exists}) \text{ cup}$ .

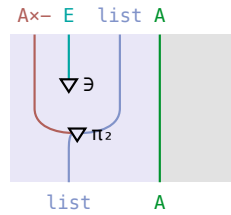
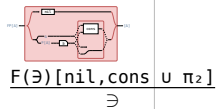
$$\frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} (\text{cons} \cup \pi_2))]$$

(5.6.1b)

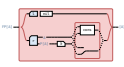
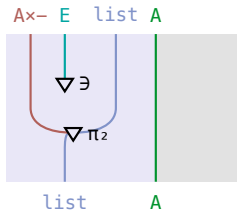
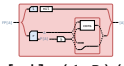
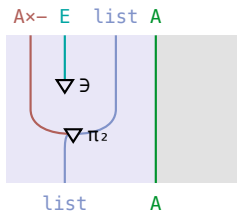
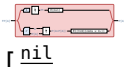
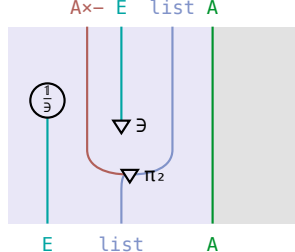
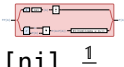
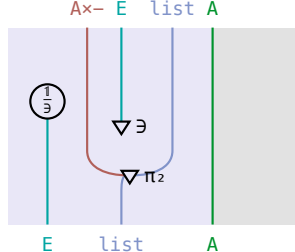
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12)  $(R) = (\frac{F(\exists)}{\exists} R) \exists$  at  $\text{subseq} = ([\text{nil}, \text{cons} \cup \pi_2])$ ,  $\text{subseq} = ([\text{nil}, \text{cons} \cup \pi_2])$ .

**circuit** —  
the fork is  
 $F([A]) = 1 + A \times [A]$ :  
nil above,  
the pair  
below

Hinze–Marsden



the  $\pi_2$  operand of  $\text{cons} \cup \pi_2$  under the  $1 \times \exists$  summand of  $F(\exists)$ , i.e.  $(1 \times \exists) \pi_2$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>circuit</b> —<br/>the fork is<br/><math>F([A]) = 1 + A \times [A]</math>:<br/>nil above,<br/>the pair<br/>below</p>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p style="text-align: center;"><b>Hinze–Marsden</b></p>                                                                                                                                                                                                                                                                                                                                                                                |
|  $\frac{(1 + 1 \times \exists)(\text{nil}, \text{cons } u \ \pi_2)}{\exists}$ $= F(X) = 1 + A \times X$ <p>—<br/><math>\text{list}(A)</math><br/><math>= [A]</math></p>                                                                                                                                                                                                                                                                                                            |  <p style="text-align: center;">the same operand under <math>1 + 1 \times \exists</math>, whose <math>1 \times \exists</math> summand it sits in</p>                                                                                                                                                                                                 |
|  $\frac{[\text{nil}, (1 \times \exists)(\text{cons } u \ \pi_2)]}{\exists}$ $R + S \triangleq [Rl, Sr],$ $l[R, S] = R,$ $r[R, S] = S$ <p>— <math>[R, S]</math><br/><math>\triangleq (l^\circ R)</math><br/><math>u(r^\circ</math><br/><math>= S), l</math><br/><math>[R, S] =</math><br/><math>R, r</math><br/><math>[R, S] =</math><br/><math>S, l \ T =</math><br/><math>R \wedge r \ T</math><br/><math>= S</math><br/><math>\Rightarrow T =</math><br/><math>[R, S]</math></p> |  <p style="text-align: center;">the <math>\pi_2</math> operand of the second arm <math>(1 \times \exists)(\text{cons } u \ \pi_2)</math></p>                                                                                                                                                                                                        |
|  $\frac{[\frac{\text{nil}}{\exists}, \frac{(1 \times \exists)(\text{cons } u \ \pi_2)}{\exists}]}{\exists}$ $[R, S] =$ $[\frac{R}{\exists}, \frac{S}{\exists}] \ \exists$ <p>at<br/><math>T := [\text{nil},</math><br/><math>(1 \times \exists)</math><br/><math>(\text{cons } u</math><br/><math>\pi_2)]</math></p>                                                                                                                                                             |  <p style="text-align: center;">the <math>\pi_2</math> operand under its <math>\frac{1}{\exists}</math>, the arm <math>\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2) \text{ cup}])</math> 's square rewrites, <math>(1 \times \exists)\pi_2 = \pi_2 \exists</math></p> |
|  $\frac{[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \ \pi_2)}{\exists}]}{\exists}$ $\text{Map}(f) \wedge$ $1 \square = \exists \square$ $\Rightarrow \frac{f}{\exists} =$ $f \frac{1}{\exists}, \frac{f}{\exists}$ $= f \frac{1}{\exists} \text{ for}$ <p><math>f</math> a map,<br/>at <math>f := \text{nil}</math></p>                                                                                                                              |  <p style="text-align: center;">the same operand; the two rows differ only in the <b>nil</b> arm</p>                                                                                                                                                                                                                                               |

(5.6.1c)

$$\begin{array}{ccc}
 A \times P[A] & \xrightarrow{\pi_2} & P[A] \\
 \downarrow 1 \times \exists & & \downarrow \exists \\
 A \times [A] & \xrightarrow{\pi_2} & [A]
 \end{array}$$



## §5.7. Lax natural transformations (LaT)

(5.7a)

For relators  $G, F : \mathcal{C} \rightarrow \mathcal{D}$  and components  $\varphi_A : GA \rightarrow FA$ ,  $\varphi$  is **lax at**  $R : A \rightarrow B$  when  $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$  — both sides  $GA \rightarrow FB$ , the left through the component at  $B$ , the right through the one at  $A$ .

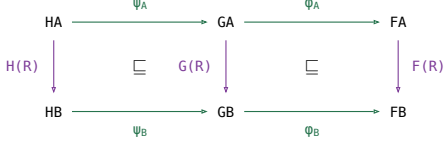
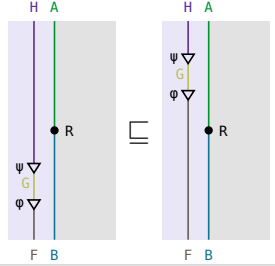
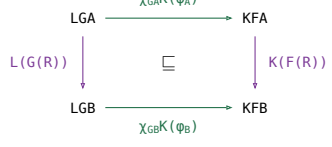
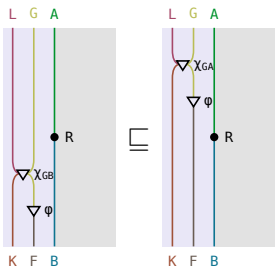
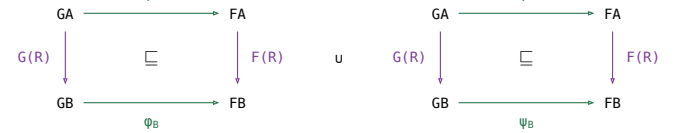
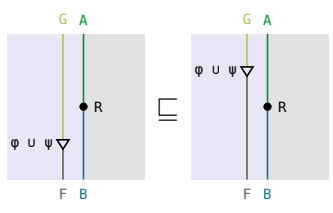
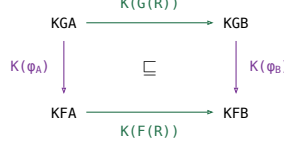
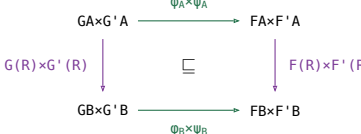
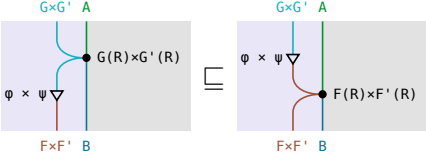
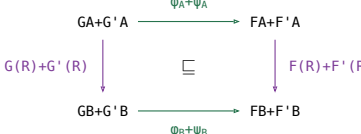
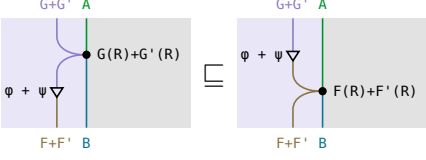
$\varphi : G \Rightarrow F$  is a **lax natural transformation** (LaT) when it is lax at **every**  $R$ . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality  $G(f)\varphi = \varphi F(f)$ : laxness is about relations only.

(5.7b)



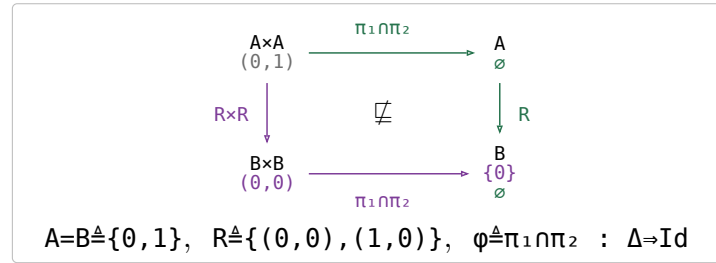
(5.7c)

| closed under                                                                                                                                                                                                                                                                                                                                                         | commutative diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | string diagram                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1<br>composition<br>$\psi\phi$                                                                                                                                                                                                                                                                                                                                       |  <p><math>H(R)\psi_B \sqsubseteq \psi_A G(R)</math> and <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> give <math>H(R)(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| 2<br>horizontal composition<br>$\chi\circ\phi$                                                                                                                                                                                                                                                                                                                       |  <p><math>\phi : G \Rightarrow F</math> with <math>G, F : \mathcal{C} \rightarrow \mathcal{D}</math> and <math>\chi : L \Rightarrow K</math> with <math>L, K : \mathcal{D} \rightarrow \mathcal{E}</math> give <math>\chi \circ \phi : L \circ G \Rightarrow K \circ F</math>, the family <math>A \mapsto \chi_{GA} K(\phi_A)</math><br/> <math>L(G(R))\chi_{GB} \sqsubseteq \chi_{GA} K(G(R))</math> is <math>\chi</math> lax at <math>G(R)</math>; then<br/> <math>K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))</math> is <math>K</math> applied to <math>\phi</math>'s own inequation;<br/> <math>\chi := \mathbb{1}_K</math> gives <math>K(\phi) : K \circ G \Rightarrow K \circ F</math>, <math>K</math> composed on the outside; <math>\phi := \mathbb{1}_G</math> gives <math>\chi G : L \circ G \Rightarrow K \circ G</math>, <math>G</math> composed on the inside<br/> two candidates, <math>\chi_{GA} K(\phi_A)</math> and <math>L(\phi_A)\chi_{FA}</math>, ordered by <math>\sqsubseteq</math>; this row is the first</p> |    |
| 3<br>union<br>$\phi \cup \psi$                                                                                                                                                                                                                                                                                                                                       |  <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> and <math>G(R)\psi_B \sqsubseteq \psi_A F(R)</math> give <math>G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| 4<br>a relator $K$<br>$K(\phi)$                                                                                                                                                                                                                                                                                                                                      |  <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> gives <math>K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                       |
| 5<br>product<br>$\phi \times \psi$                                                                                                                                                                                                                                                                                                                                   |  <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi \times \psi : G \times G' \Rightarrow F \times F'</math><br/> <math>(R \times S)(U \times V) = (RU) \times (SV)</math> and monotonicity in both slots — the row above's <math>K</math> applied to the inequation, run on a bifunctor; the fork is the derived case <math>\langle \phi, \psi \rangle = \triangleleft(\phi \times \psi)</math>, and its ONE cost is the lax copy law <math>R \triangleleft \sqsubseteq \triangleleft(R \times R) \rightarrow R \triangleleft \leq \triangleleft(R \otimes R), R \multimap \leq \multimap</math></p>                                                                                                                                                                                                                                                                                                                                                                             |  |
| 6<br>coproduct<br>$\phi + \psi$                                                                                                                                                                                                                                                                                                                                      |  <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi + \psi : G + G' \Rightarrow F + F'</math><br/> <math>(R + S)(U + V) = (RU) + (SV)</math> and monotonicity in both slots; the co-fork is the derived case <math>[\phi, \psi] = (\phi + \psi) \nabla</math>, and <math>\nabla</math> costs nothing<br/> the step it would need is <math>(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)</math>, the wrong direction of (0.6a)<br/> a counterexample: (5.7.1a)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| 7<br>meet — <b>fails</b><br>$\phi \cap \psi$                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |
| 8 Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$ , quantified over $R$ ; at $X' = X$ with $R_A, R_B$ endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |

## §5.7.1. meet is not closed in LaT

counterexample —  $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$  lax natural does NOT give  $\varphi \cap \psi$  lax natural

(5.7.1a)



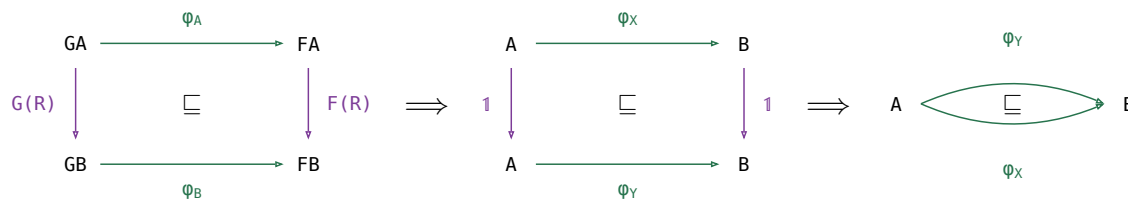
## §5.7.2. Two stacked towers

(5.7.2a)

|                | lower — inside ONE allegory      | upper — between allegories                                                                                                                                                                                                                                                                         |
|----------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 0-cell       | an object $A$                    | an allegory $\mathcal{C}$ — a whole lower tower                                                                                                                                                                                                                                                    |
| 2 1-cell       | a relation $R : A \rightarrow B$ | a relator $F : \mathcal{C} \rightarrow \mathcal{D}$                                                                                                                                                                                                                                                |
| 3 2-cell       | $R \sqsubseteq S$                | a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F} \triangleq \text{lax}$                                                                                                                                                                                                                          |
| 4 order        | $R \sqsubseteq S$                | $\varphi \sqsubseteq \psi$ componentwise                                                                                                                                                                                                                                                           |
| 5 union        | $R \cup S$                       | $\varphi \cup \psi$ , survives $G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$                                                                                                                                                                                             |
| 6 product      | $R \times S \triangleq \times$   | $\varphi \times \psi$ , survives; a TENSOR, not a categorical product; SYMMETRIC $\langle R, S \rangle$<br>$\pi_1 = \text{dom}(S) \ R, \langle R, S \rangle \ \pi_2 = \text{dom}(R) \ S \ (G(R) \times G'(R)) (\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A) (F(R) \times F'(R))$ |
| 7 coproduct    | $R + S$                          | $\varphi + \psi$ , survives; a BIPRODUCT $(G(R) + G'(R)) (\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A) (F(R) + F'(R))$                                                                                                                                                                     |
| 8 meet         | $R \cap S$                       | $\varphi \cap \psi$ componentwise, fails (5.7.1a)                                                                                                                                                                                                                                                  |
| 9 converse     | $R^\circ$                        | $\varphi^\circ$ , fails, oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$                                                                                                                                                                                                            |
| 10 zero object | $z$ with $1 \ z = 0$             | the constant relator at such a $z$ , initial and terminal                                                                                                                                                                                                                                          |

## §5.7.3. RES is a special case of LaT

(5.7.3a)



$G \triangleq \text{const } A, F \triangleq \text{const } B$  — the relators  $X \rightarrow A$  and  $X \rightarrow B$ , each sending every arrow to  $1 \triangleq \text{lax}$

with  $\mathcal{C}$  two objects and one non-identity arrow  $X \rightarrow Y$ , a LaT  $\text{const } A \Rightarrow \text{const } B$  is the pair  $\varphi_X, \varphi_Y : A \rightarrow B$  with  $\varphi_Y \sqsubseteq \varphi_X$  — the datum  $R \in S$

not every LaT is constant:  $\exists : \mathbf{P} \Rightarrow \text{Id}$  (5.4d.1)  $\pi_2 : (A \times -) \Rightarrow \text{Id} \xrightarrow{\text{subseq}} = ([\text{nil } \frac{1}{3}, (\frac{1}{3} \times \exists \ E(\text{cons}), \pi_2) \text{ cup}]) \triangleleft : \text{Id} \Rightarrow \Delta$

$$R \triangleleft \leq \triangleleft (R \otimes R), R \multimap \leq \multimap$$

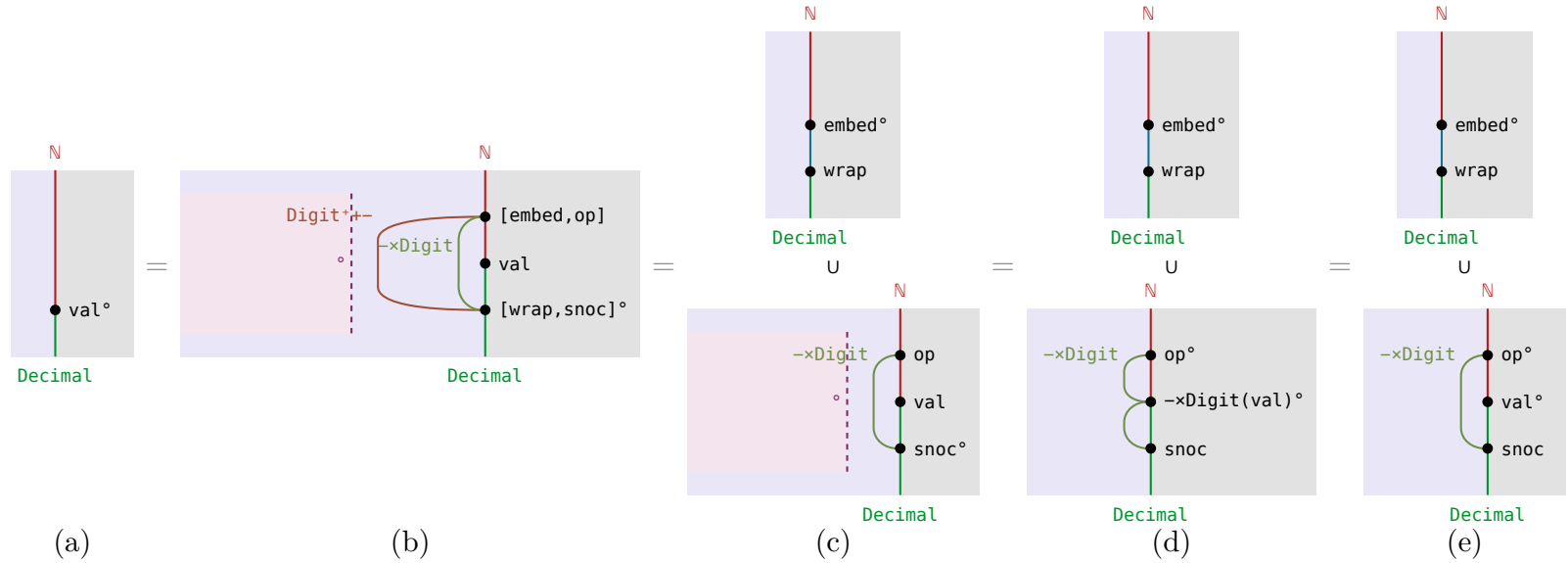
## §6. Recursive Programs

## §6.1. Digits of a number

$$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{ snoc}) \quad (6.1a)$$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

the converse of  $\text{val}$ , which reads digits as a number



|     |                                                                                                                        |                                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{val}^\circ$                                                                                                     | $\text{val}$ undoes $\text{wrap}$ or $\text{snoc}$ , reads the front with $\text{val}$ and combines by $\text{embed}$ or $\text{op}$ , so its converse runs that backwards |
| (b) | $= ([\text{wrap}, \text{snoc}]^\circ \text{Digit}^{++} \neg \text{Digit}(\text{val})) [\text{embed}, \text{op}]^\circ$ | undoing $\text{wrap}$ or $\text{snoc}$ first gives two cases: a decimal and its last digit read by $\text{val}$ then $\text{op}$ , or one digit read by $\text{embed}$     |
| (c) | $= (\text{wrap}^\circ \text{embed})^\circ \cup (\text{snoc}^\circ \neg \text{Digit}(\text{val}) \text{op})^\circ$      | the converse of each case is its steps undone in reverse: undo $\text{op}$ , undo reading the front, then $\text{snoc}$ ; or undo $\text{embed}$ , then $\text{wrap}$      |
| (d) | $= (\text{embed}^\circ \text{wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val})^\circ \text{snoc})$            | undoing the read of the front while keeping the last digit is $\text{val}^\circ$ on the front alone                                                                        |
| (e) | $= (\text{embed}^\circ \text{wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{snoc})$            |                                                                                                                                                                            |

(6.1b)

- 1  $\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$   
 $\text{op}^\circ$  splits a number into its quotient and remainder by 10
- 2  $(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$   
 $\text{op}^\circ$  gives a pair with a nonzero first component exactly at the numbers with two or more digits
- 3  $(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$   
 $\text{embed}^\circ$  gives a digit exactly at the one-digit numbers

## §6.2. Least fixed points

(6.2a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$  .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$  .

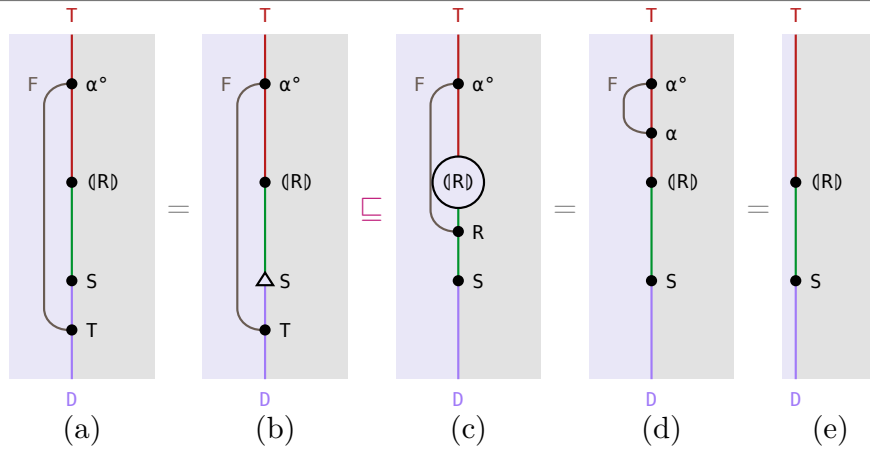
(6.2b)

| the law                                                                                      | what it says                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §6.3 and every chapter after it uses                                              |
| 2 $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |
| 3 $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$<br>(6.2)   | a relation $X$ that contains one unfolding of the fold's recursion at $X$ contains the fold                                                                                     |
| 4 $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$<br>(6.3)   | a relation $X$ contained in one unfolding of the fold's recursion at $X$ is contained in the fold                                                                               |

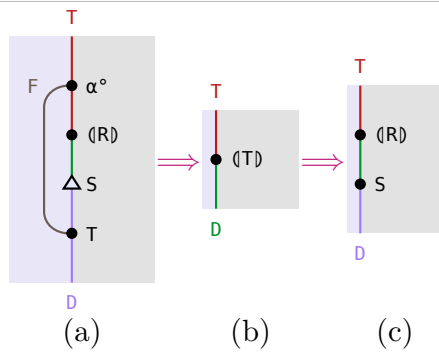
$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2c)

if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$



|     |                                                     |                                                     |
|-----|-----------------------------------------------------|-----------------------------------------------------|
| (a) | $\alpha^\circ F(\langle R \rangle S) T$             |                                                     |
| (b) | $= \alpha^\circ F(\langle R \rangle) F(S) T$        | $F(S) T \sqsubseteq R S$                            |
| (c) | $\sqsubseteq \alpha^\circ F(\langle R \rangle) R S$ | $\alpha \langle R \rangle = F(\langle R \rangle) R$ |
| (d) | $= \alpha^\circ \alpha \langle R \rangle S$         | $\alpha^\circ \alpha = 1$                           |
| (e) | $= \langle R \rangle S$                             |                                                     |

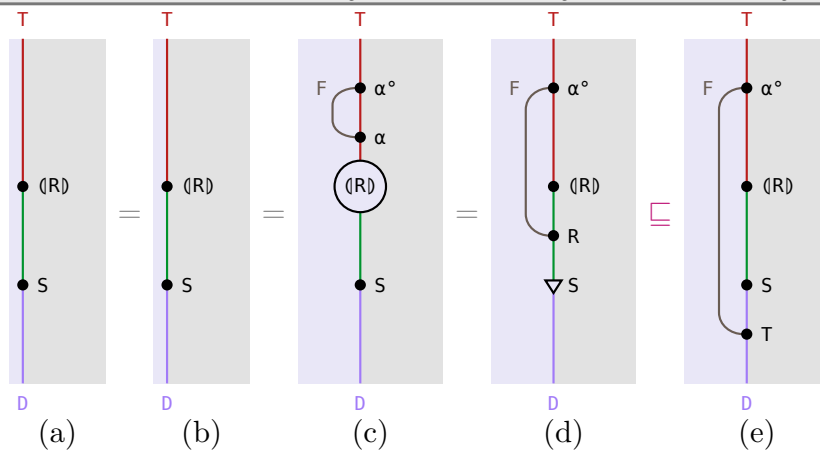


|     |                                                                                     |                                                                                                                 |
|-----|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| (a) | $F(S) T \sqsubseteq R S$                                                            | $F(S) T \sqsubseteq R S$<br>$\Rightarrow \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S$ |
| (b) | $\Rightarrow \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S$ | $\alpha^\circ F(X) R \sqsubseteq X$<br>$\Rightarrow \langle R \rangle \sqsubseteq X$                            |
| (c) | $\Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$                     |                                                                                                                 |

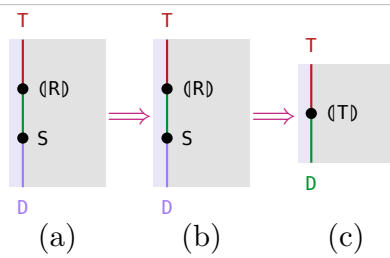
$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



|     |                                                       |                                                         |
|-----|-------------------------------------------------------|---------------------------------------------------------|
| (a) | $\langle R \rangle \ S$                               |                                                         |
| (b) | $= \langle R \rangle \ S$                             | $\alpha^0 \ \alpha = 1$                                 |
| (c) | $= \alpha^0 \ \alpha \ \langle R \rangle \ S$         | $\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$ |
| (d) | $= \alpha^0 \ F(\langle R \rangle) \ R \ S$           | $R \ S \sqsubseteq F(S) \ T$                            |
| (e) | $\sqsubseteq \alpha^0 \ F(\langle R \rangle \ S) \ T$ |                                                         |



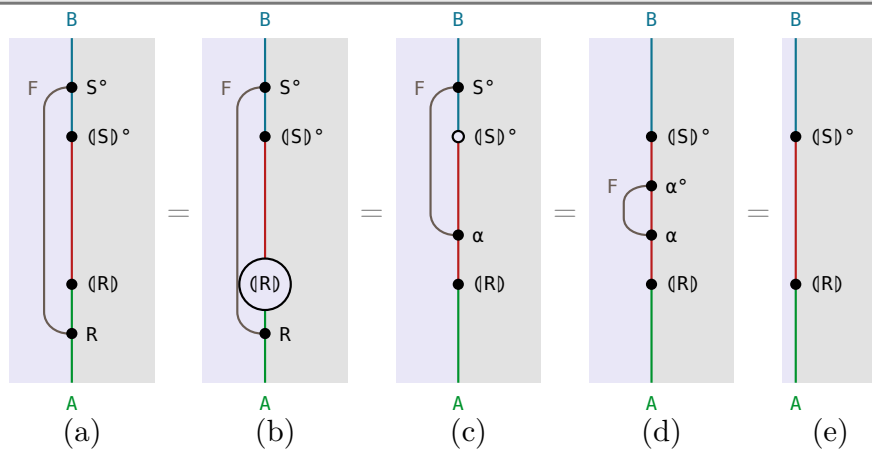
|     |                                                                                         |                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| (a) | $R \ S \sqsubseteq F(S) \ T$                                                            | $R \ S \sqsubseteq F(S) \ T$<br>$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^0 \ F(\langle R \rangle \ S) \ T$ |
| (b) | $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^0 \ F(\langle R \rangle \ S) \ T$ | $X \sqsubseteq \alpha^0 \ F(X) \ R$<br>$\Rightarrow X \sqsubseteq \langle R \rangle$                                    |
| (c) | $\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$                       |                                                                                                                         |

### §6.3. Hylomorphisms

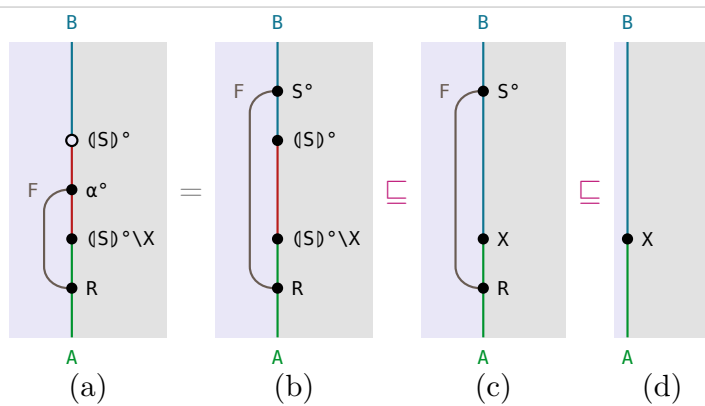
$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R)$$

(6.3a)

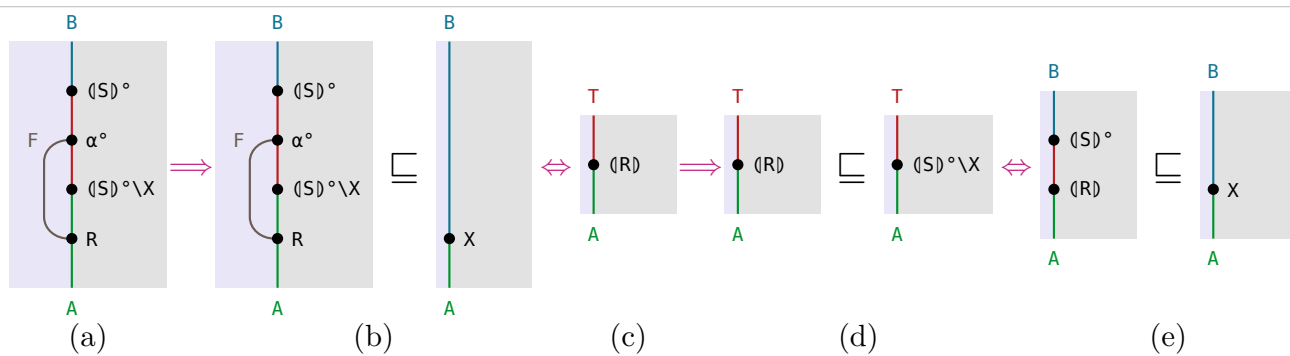
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



|     |                                                         |                                                                   |
|-----|---------------------------------------------------------|-------------------------------------------------------------------|
| (a) | $S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R$          |                                                                   |
| (b) | $= S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$     | $\alpha (\mathbb{R}) = F((\mathbb{R})) R$                         |
| (c) | $= S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$   | $(\mathbb{R})^\circ \alpha^\circ = R^\circ F((\mathbb{R})^\circ)$ |
| (d) | $= (\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$ | $\alpha^\circ \alpha = 1$                                         |
| (e) | $= (\mathbb{S})^\circ (\mathbb{R})$                     |                                                                   |



|     |                                                                       |                                                                   |
|-----|-----------------------------------------------------------------------|-------------------------------------------------------------------|
| (a) | $(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$ | $(\mathbb{R})^\circ \alpha^\circ = R^\circ F((\mathbb{R})^\circ)$ |
| (b) | $= S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$  | $S (S \setminus R) \subseteq R$                                   |
| (c) | $\sqsubseteq S^\circ F(X) R$                                          | $S^\circ F(X) R \subseteq X$                                      |
| (d) | $\sqsubseteq X$                                                       |                                                                   |

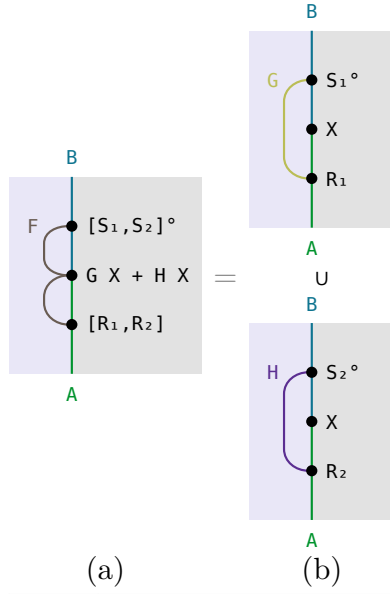


|     |                                                                                                             |                                                             |
|-----|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| (a) | $S^\circ F(X) R \subseteq X$                                                                                | $S^\circ F(X) R \subseteq X$                                |
| (b) | $\Rightarrow (\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \subseteq X$               | $X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$ |
| (c) | $\Leftrightarrow \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \subseteq (\mathbb{S})^\circ \setminus X$ | $\alpha^\circ F(X) R \subseteq X$                           |
| (d) | $\Rightarrow (\mathbb{R}) \subseteq (\mathbb{S})^\circ \setminus X$                                         | $\Rightarrow (\mathbb{R}) \subseteq X$                      |
| (e) | $\Leftrightarrow (\mathbb{S})^\circ (\mathbb{R}) \subseteq X$                                               | $X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$ |

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2))) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

the body of Theorem 6.2 at  $F(X)=G(X)+H(X)$

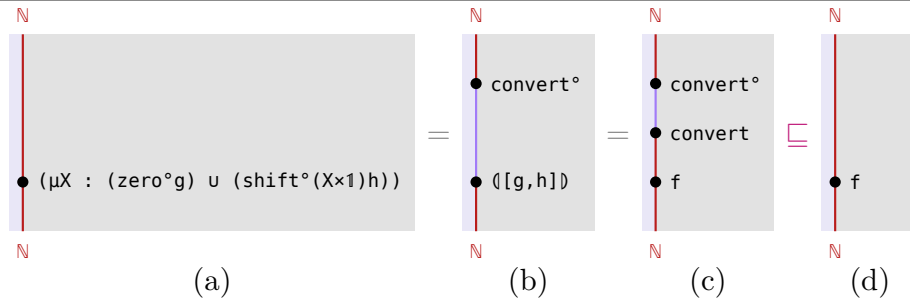


|     |                                                    |                                                                                                                                                                                            |
|-----|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$        | $[S, T] = (S+T)\nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$ ; the + stands for the codiagonals $\nabla^\circ, \nabla$ , which are not drawn |
| (b) | $= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$ | $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct                                                                                                                                         |

## §6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f \quad (6.4a)$$

once  $f$  sends **zero** to  $g$  and turns **shift** into  $h$ , the recursion that halves the argument at each step computes  $f$



|     |                                                                            |                                                                                                                                |
|-----|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$ | $\text{convert}^\circ \llbracket [g, h] \rrbracket = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$ |
| (b) | $= \text{convert}^\circ \llbracket [g, h] \rrbracket$                      | $\text{zero } f = g \wedge \text{shift } f = (f \times 1) h$                                                                   |
| (c) | $= \text{convert}^\circ \text{convert } f$                                 | $\Rightarrow \text{convert } f = \llbracket [g, h] \rrbracket$                                                                 |
| (d) | $\sqsubseteq f$                                                            | $\text{Simple}(\text{convert})$                                                                                                |

$$(\mu X : (\text{zero}^\circ \text{one}) \cup (\text{shift}^\circ (X \times 1) \text{op}(a))) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes  $a$  to the power  $b$

$\text{zero } \text{exp}(a) = \text{one}$

$a$  to the power  $0$  is  $1$

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

$a$  to the power  $2n+d$  is  $\text{op}(a)$  of  $a$  to the power  $n$  and  $d$

$$(\mu X : (\text{zero}^\circ \text{zero}) \cup (\text{shift}^\circ (X \times 1) \text{op}(b))) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves  $a$  at each step computes the remainder of  $a$  by  $b$

$\text{zero } \text{mod}(b) = \text{zero}$

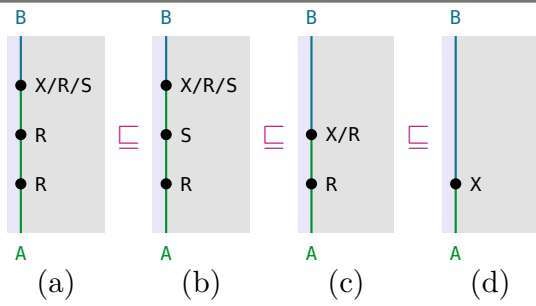
$0 \text{ mod } b$  is  $0$

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

$(2a+d) \text{ mod } b$  is  $\text{op}(b)$  of  $a \text{ mod } b$  and  $d$

## §6.5. Unique fixed points

$$\begin{aligned}
 & \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{X} & (6.5a) \\
 & ((\mathbf{X}/\mathbf{R})/\mathbf{S}) \text{ followed by } \mathbf{R}\mathbf{R} \text{ stays below } \mathbf{X} \text{ when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \\
 & \mathbf{inductive}(\mathbf{S}) \wedge \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow \mathbf{inductive}(\mathbf{R}) \\
 & \mathbf{R} \text{ is inductive when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \text{ for an inductive } \mathbf{S}
 \end{aligned}$$



|     |                                                                            |                                                               |
|-----|----------------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $(\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R}$             | $\mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R}$     |
| (b) | $\sqsubseteq (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{S} \mathbf{R}$ | $(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$ |
| (c) | $\sqsubseteq (\mathbf{X} / \mathbf{R}) \mathbf{R}$                         | $(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$ |
| (d) | $\sqsubseteq \mathbf{X}$                                                   |                                                               |

$$\mathbf{member} = \emptyset \quad (6.5b)$$

a constant relator records no elements, so its membership is empty

$$\mathbf{member} = [\mathbf{member}, \mathbf{member}]$$

a member of  $\mathbf{F}+\mathbf{G}$  is a member of whichever summand the value is in

$$\mathbf{member} = (\pi_1 \mathbf{member}) \cup (\pi_2 \mathbf{member})$$

a member of  $\mathbf{F} \times \mathbf{G}$  is a member of either component

$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{X} = \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \wedge \mathbf{Y} = \mathbf{S} \mathbf{F}(\mathbf{Y}) \mathbf{R} \Rightarrow \mathbf{X} = \mathbf{Y} & (6.5c) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive, two solutions } \mathbf{X}, \mathbf{Y} \text{ of } \mathbf{X} = \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ are equal}
 \end{aligned}$$

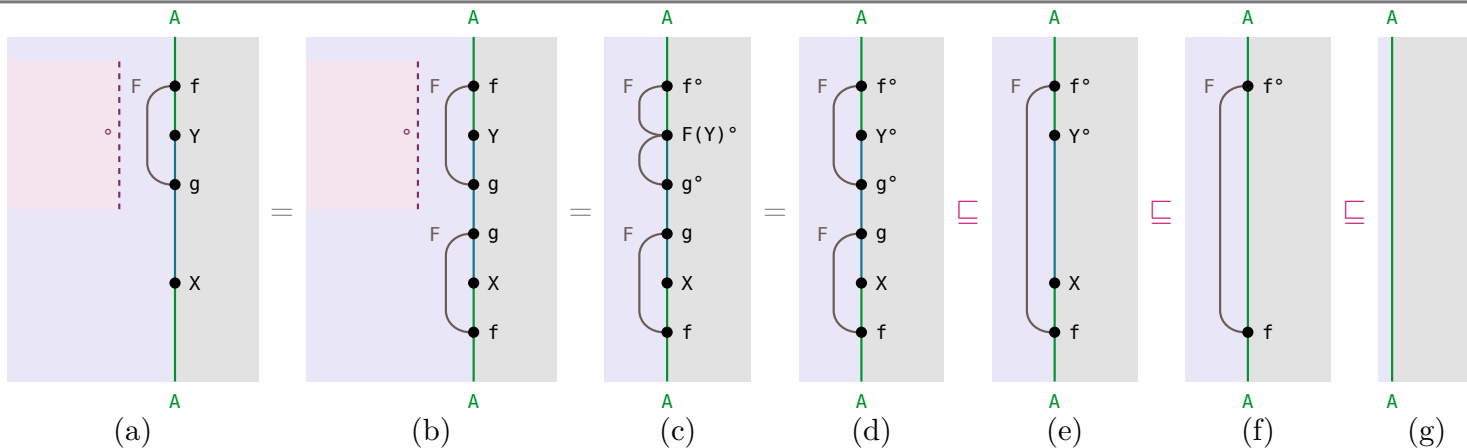
$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{Entire}(\mathbf{S}) \wedge \mathbf{Entire}(\mathbf{R}) \wedge \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \sqsubseteq \mathbf{X} \Rightarrow \mathbf{Entire}(\mathbf{X}) & (6.5d) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive and } \mathbf{S}, \mathbf{R} \text{ are entire, a pre-fixed point of } \mathbf{X} \mapsto \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ is entire}
 \end{aligned}$$

$$\mathbf{inductive}(\mathbf{g} \mathbf{member}) \wedge \mathbf{Map}(\mathbf{g}) \wedge \mathbf{Map}(\mathbf{f}) \wedge \mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \Rightarrow \mathbf{Map}(\mathbf{X}) \quad (6.5e)$$

when  $\mathbf{g} \mathbf{member}$  is inductive and  $\mathbf{f}, \mathbf{g}$  are maps, the solution of  $\mathbf{X} = \mathbf{g}\mathbf{F}(\mathbf{X})\mathbf{f}$  is a map

$$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \wedge \mathbf{Simple}(\mathbf{g}) \wedge \mathbf{Simple}(\mathbf{f}) \wedge \mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1} \Rightarrow (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X} \sqsubseteq \mathbf{1}$$

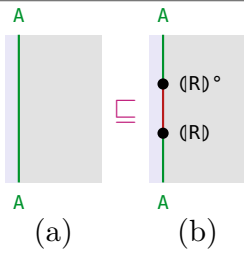
$(\mathbf{g}\mathbf{F}(\mathbf{Y})\mathbf{f})^\circ$  followed by  $\mathbf{X}$  stays below  $\mathbf{1}$  when  $\mathbf{g}, \mathbf{f}$  are simple and  $\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$



|     |                                                                                                                 |                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| (a) | $(\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X}$                                               | $\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$         |
| (b) | $= (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$           | $(\mathbf{R} \mathbf{S})^\circ = \mathbf{S}^\circ \mathbf{R}^\circ$ |
| (c) | $= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y})^\circ \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$ | $\mathbf{PreservesRecip}(\mathbf{F})$                               |
| (d) | $= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ) \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$ | $\mathbf{Simple}(\mathbf{g})$                                       |
| (e) | $\sqsubseteq \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ \mathbf{X}) \mathbf{f}$                               | $\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$                |
| (f) | $\sqsubseteq \mathbf{f}^\circ \mathbf{f}$                                                                       | $\mathbf{Simple}(\mathbf{f})$                                       |
| (g) | $\sqsubseteq \mathbf{1}$                                                                                        |                                                                     |

(6.5f)

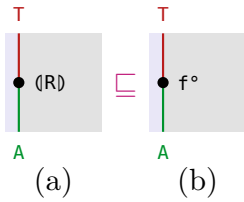
**inductive( $R^\circ$  member)  $\wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R)$**   
 when  $R^\circ$  member is inductive and  $R$  is surjective, the fold  $(R)$  is surjective



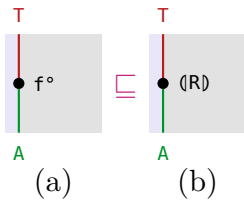
|     |                             |                                                                                                                                                                   |
|-----|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $1$                         | inductive( $S$ member) $\wedge$ Entire( $S$ ) $\wedge$ Entire( $R$ )<br>$\wedge S \sqsubseteq F(X) R \sqsubseteq X$<br>$\Rightarrow$ Entire( $X$ ) at $S=R^\circ$ |
| (b) | $\sqsubseteq (R)^\circ (R)$ |                                                                                                                                                                   |

(6.5g)

**Map( $f$ )  $\wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = (R)$**   
 a map  $f$  with  $Rf \sqsubseteq F(f) \alpha$ , for a surjective  $R$ , has the fold  $(R)$  as its converse



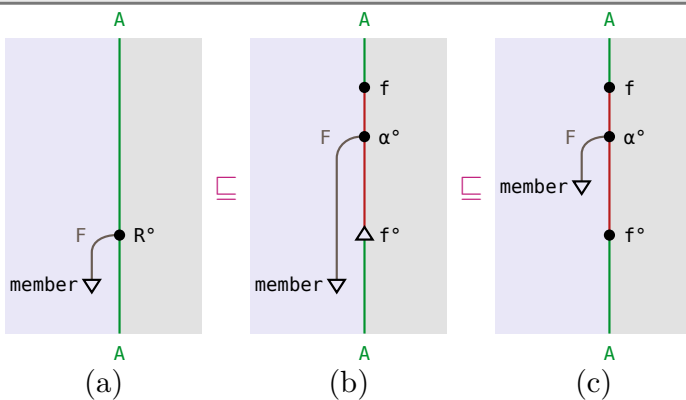
|     |                       |                                                     |
|-----|-----------------------|-----------------------------------------------------|
| (a) | $(R)$                 | shunting $f$ , fusion, $Rf \sqsubseteq F(f) \alpha$ |
| (b) | $\sqsubseteq f^\circ$ |                                                     |



|     |                   |                                                                                                                                      |
|-----|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $f^\circ$         | inductive( $R^\circ$ member) $\wedge 1 \sqsubseteq R^\circ R$<br>$\Rightarrow 1 \sqsubseteq (R)^\circ (R)$ and $(R) f \sqsubseteq 1$ |
| (b) | $\sqsubseteq (R)$ |                                                                                                                                      |

(6.5h)

**Map( $f$ )  $\wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ$  member  $\sqsubseteq f \alpha^\circ$  member  $f^\circ$**   
 $R^\circ$  member is below  $\alpha^\circ$  member conjugated by  $f$ , so it is inductive when  $\alpha^\circ$  member is



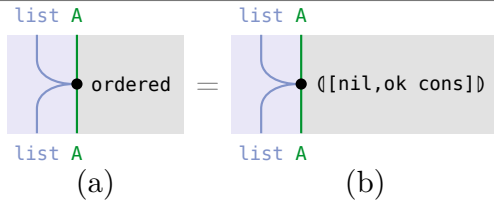
|     |                                                |                                                                                                                |
|-----|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| (a) | $R^\circ$ member                               | Map( $f$ ) $\wedge R f \sqsubseteq F(f) \alpha$<br>$\Rightarrow R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$ |
| (b) | $\sqsubseteq f \alpha^\circ F(f^\circ)$ member | $F(R)$ member $\sqsubseteq$ member $R$                                                                         |
| (c) | $\sqsubseteq f \alpha^\circ$ member $f^\circ$  |                                                                                                                |

## §6.6. Sorting by selection

$$\text{ordered} = ([\text{nil}, \text{ok cons}])$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

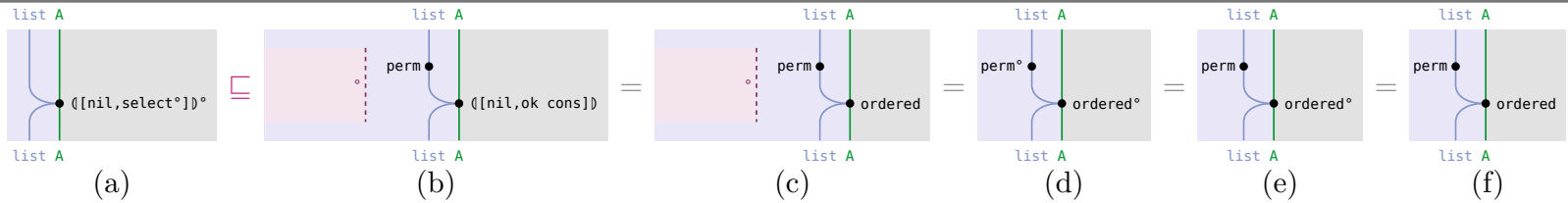


|     |                    |                 |
|-----|--------------------|-----------------|
| (a) | ordered            | fold uniqueness |
| (b) | = ([nil, ok cons]) |                 |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

every output of unfolding the input by **select** is a sorted permutation of it

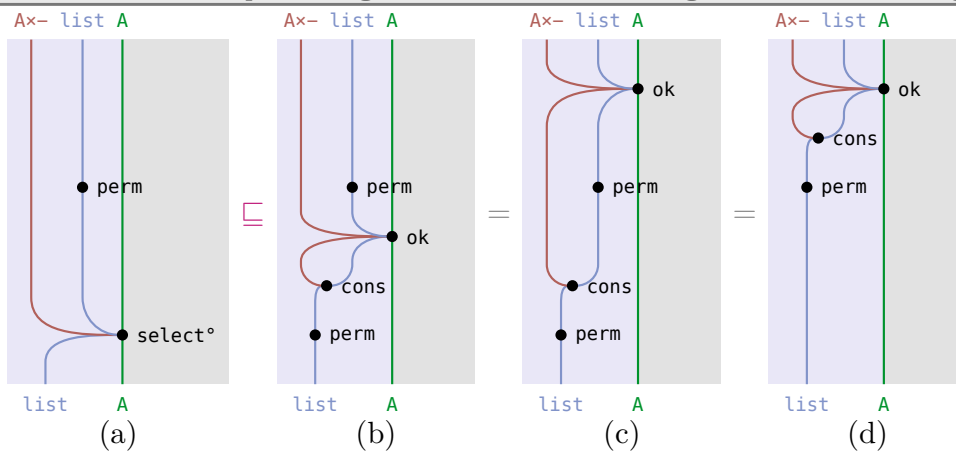


|     |                                                                       |                                                                                                                                                                                      |
|-----|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{select}^\circ]^\circ$                             | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$<br>$\Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq ([\text{nil}, \text{ok cons}])^\circ \text{ perm}^\circ$ |
| (b) | $\sqsubseteq ([\text{nil}, \text{ok cons}])^\circ \text{ perm}^\circ$ | (6.6a) $\text{ordered} = ([\text{nil}, \text{ok cons}])^\circ$                                                                                                                       |
| (c) | $= (\text{ordered perm})^\circ$                                       | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                      |
| (d) | $= \text{perm}^\circ \text{ ordered}^\circ$                           | $\text{perm}^\circ = \text{perm}$                                                                                                                                                    |
| (e) | $= \text{perm ordered}^\circ$                                         | $\text{ordered}^\circ = \text{ordered}$                                                                                                                                              |
| (f) | $= \text{perm ordered}$                                               |                                                                                                                                                                                      |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{ select}^\circ \sqsubseteq \text{ok cons perm}$$

(6.6c)

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation

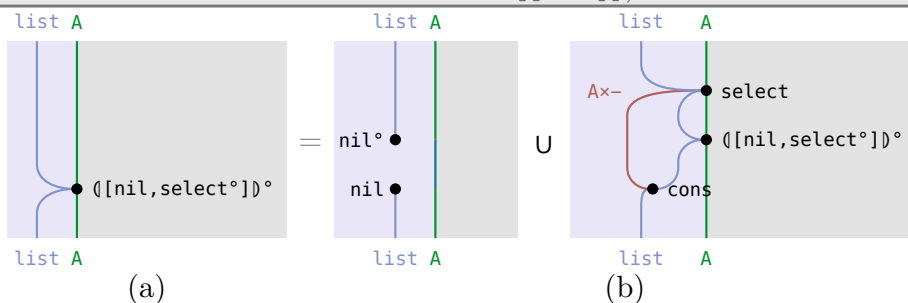


|     |                                                           |                                                                         |
|-----|-----------------------------------------------------------|-------------------------------------------------------------------------|
| (a) | $(1 \times \text{perm}) \text{ select}^\circ$             | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$                   |
| (b) | $\sqsubseteq (1 \times \text{perm}) \text{ ok cons perm}$ | $\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ ok}$ |
| (c) | $= \text{ok } (1 \times \text{perm}) \text{ cons perm}$   | $\text{cons perm} = (1 \times \text{perm}) \text{ cons perm}$           |
| (d) | $= \text{ok cons perm}$                                   |                                                                         |

$$([\text{nil}, \text{select}^\circ])^\circ = (\text{nil}^\circ \text{ nil}) \cup (\text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$$

(6.6d)

selection sort returns **[]** on **[]**, and otherwise selects **(a, y)**, sorts **y** and conses **a** back on

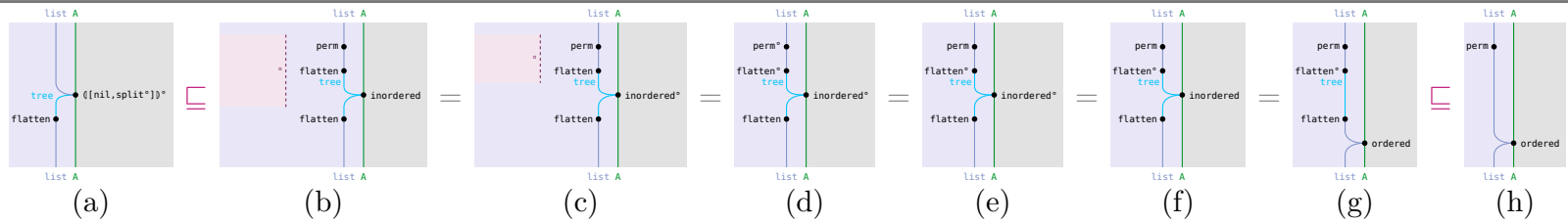


|     |                                                                                                                            |                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| (a) | $[\text{nil}, \text{select}^\circ]^\circ$                                                                                  | unfolding the converse of a fold |
| (b) | $= (\text{nil}^\circ \text{ nil}) \cup (\text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$ |                                  |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6e)$$

**flatten  $\sqsubseteq$  perm ordered**

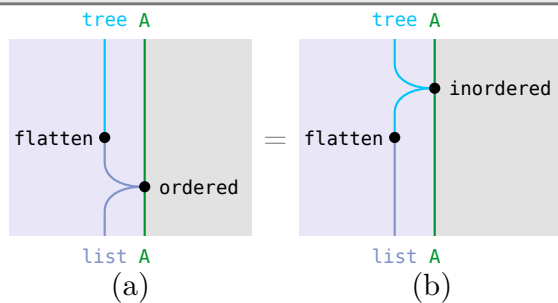
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



|     |                                                                                     |                                                                                                                                                                                                              |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[[\text{nil}, \text{split}^\circ]]^\circ \text{ flatten}$                          | $\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$<br>$\Rightarrow [[\text{nil}, \text{split}^\circ]]^\circ \text{ flatten} \sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$ |
| (b) | $\sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$                 | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                                              |
| (c) | $= (\text{flatten perm})^\circ \text{ inordered}^\circ \text{ flatten}$             | $(R S)^\circ = S^\circ R^\circ$                                                                                                                                                                              |
| (d) | $= \text{perm}^\circ \text{ flatten}^\circ \text{ inordered}^\circ \text{ flatten}$ | $\text{perm}^\circ = \text{perm}$                                                                                                                                                                            |
| (e) | $= \text{perm flatten}^\circ \text{ inordered}^\circ \text{ flatten}$               | $\text{inordered}^\circ = \text{inordered}$                                                                                                                                                                  |
| (f) | $= \text{perm flatten}^\circ \text{ inordered flatten}$                             | $(6.6f) (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c))$<br>$\Rightarrow \text{flatten ordered} = \text{inordered flatten}$                                                  |
| (g) | $= \text{perm flatten}^\circ \text{ flatten ordered}$                               | $\text{flatten}^\circ \text{ flatten} \sqsubseteq 1$                                                                                                                                                         |
| (h) | $\sqsubseteq \text{perm ordered}$                                                   |                                                                                                                                                                                                              |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

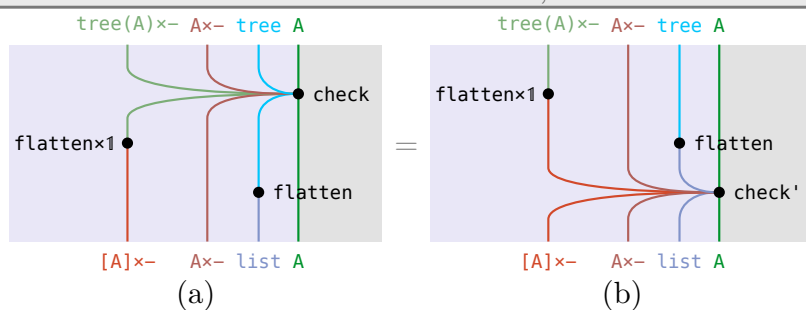
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



|     |                              |                                                                         |
|-----|------------------------------|-------------------------------------------------------------------------|
| (a) | $\text{flatten ordered}$     | $R$ transitive: a flattened tree is ordered iff every node passes check |
| (b) | $= \text{inordered flatten}$ |                                                                         |

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \quad (6.6g)$$

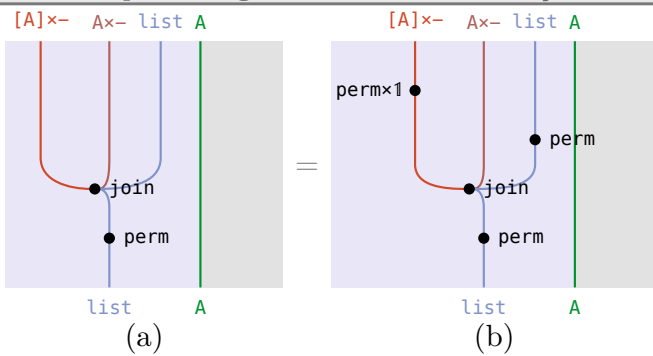
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



|     |                                                                      |                                                                |
|-----|----------------------------------------------------------------------|----------------------------------------------------------------|
| (a) | $\text{check} (\text{flatten} \times (1 \times \text{flatten}))$     | $b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$ |
| (b) | $= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$ |                                                                |

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6h)$$

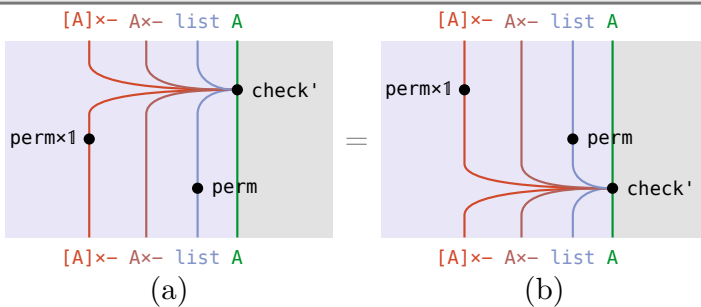
permuting the list  $x \mathrel{++} [a] \mathrel{++} y$  is the same as first permuting  $x$  and  $y$  and then permuting the joined list



|     |                                 |                                        |
|-----|---------------------------------|----------------------------------------|
| (a) | join perm                       | perm is transitive and closed under ++ |
| (b) | = (perm × (1 × perm)) join perm |                                        |

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}' \quad (6.6i)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

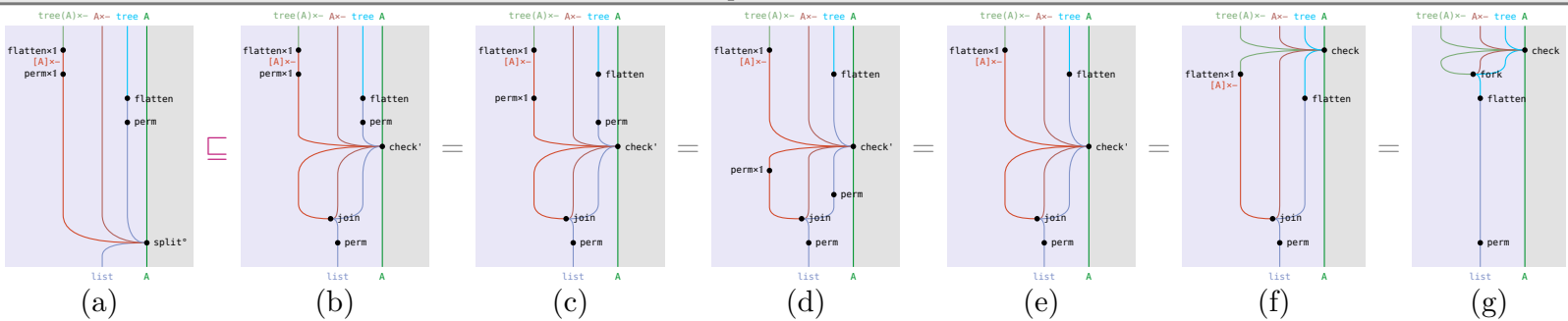


|     |                              |                                                     |
|-----|------------------------------|-----------------------------------------------------|
| (a) | check' (perm × (1 × perm))   | $b \text{ inlist } x \iff b \text{ inlist perm}(x)$ |
| (b) | = (perm × (1 × perm)) check' |                                                     |

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check fork} \quad (6.6j)$$

**flatten perm**

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



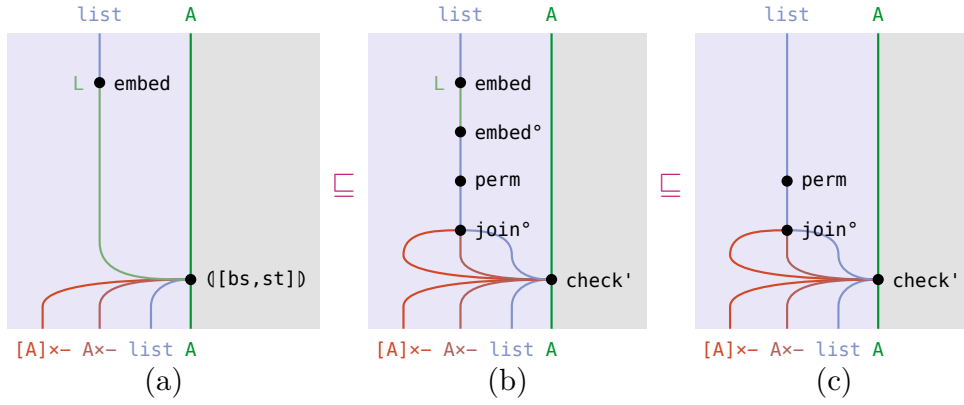
|     |                                                                                                                                    |                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ$                                              | $\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$                                                                           |
| (b) | $\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ check}' \text{ join perm}$                     |                                                                                                                                            |
| (c) | $= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{ check}' \text{ join perm}$ | (6.6i) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}'$            |
| (d) | $= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$ | (6.6h) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$                                                  |
| (e) | $= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \text{ join perm}$                                             | (6.6g) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$ |
| (f) | $= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{ join perm}$                                               | $\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{ join}$                                                     |
| (g) | $= \text{check fork flatten perm}$                                                                                                 |                                                                                                                                            |

$bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed}$  (6.6k)

$([bs, st]) \sqsubseteq \text{perm join}^\circ \text{check}'$

turning a list into a non-empty list and folding it with any  $[bs, st]$  meeting the two conditions below gives only

**checked triples**  $(x, a, y)$  whose  $\text{join } x \# [a] \# y$  is a permutation of the list

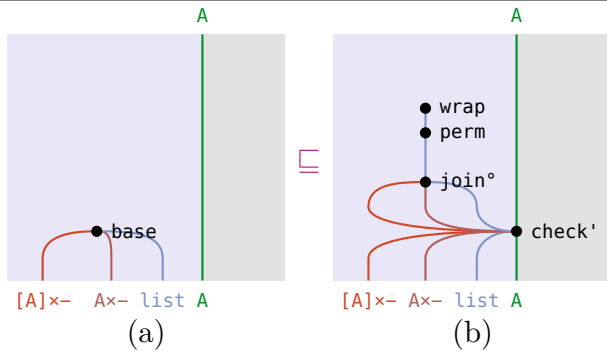


|     |                                                                             |                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{embed } ([bs, st])$                                                  | $bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed } ([bs, st]) \sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$ |
| (b) | $\sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$ | $\text{embed embed}^\circ \text{perm join}^\circ \text{check}' \sqsubseteq \text{perm join}^\circ \text{check}'$                                                                                                                                                                                    |
| (c) | $\sqsubseteq \text{perm join}^\circ \text{check}'$                          |                                                                                                                                                                                                                                                                                                     |

$\text{base} \sqsubseteq \text{wrap perm join}^\circ \text{check}'$

(6.6l)

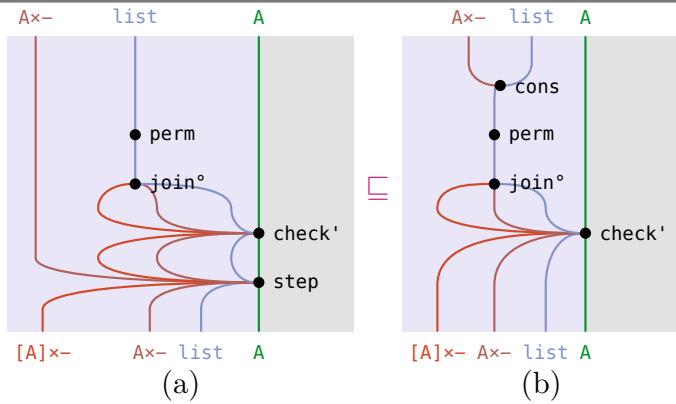
$\text{base}(a)$  is a **checked triple** whose  $\text{join}$  is a permutation of the one-element list  $[a]$



|     |                                                         |                                                                    |
|-----|---------------------------------------------------------|--------------------------------------------------------------------|
| (a) | $\text{base}$                                           | $[] \# [a] \# [] = [a]$ , and $\text{check}'$ holds on empty lists |
| (b) | $\sqsubseteq \text{wrap perm join}^\circ \text{check}'$ |                                                                    |

$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{check}'$  (6.6m)

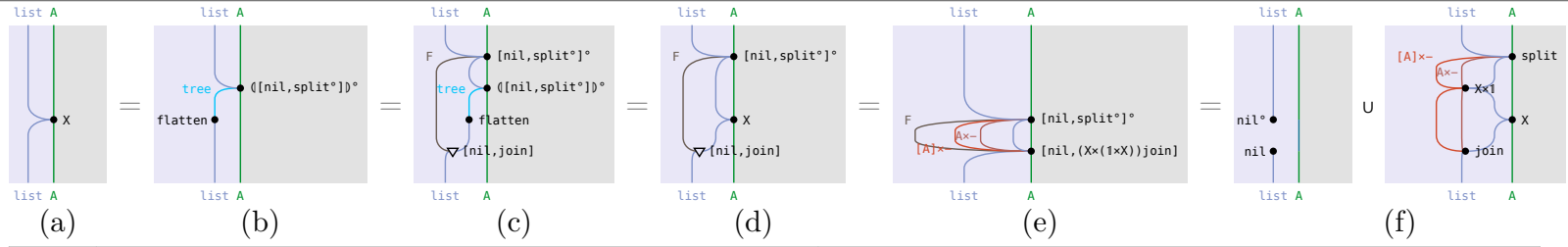
if  $(x, b, y)$  is a **checked triple** for a permutation of  $l$ , then  $\text{step}(a, (x, b, y))$  is a **checked triple** for a permutation of  $[a] \# l$



|     |                                                                  |                                                                               |
|-----|------------------------------------------------------------------|-------------------------------------------------------------------------------|
| (a) | $(1 \times (\text{perm join}^\circ \text{check}')) \text{ step}$ | $aRb$ puts $a$ in front of $x$ ; otherwise $bRa$ and $a$ goes in front of $y$ |
| (b) | $\sqsubseteq \text{cons perm join}^\circ \text{check}'$          |                                                                               |

$$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} \Rightarrow X = (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join}) \quad (6.6n)$$

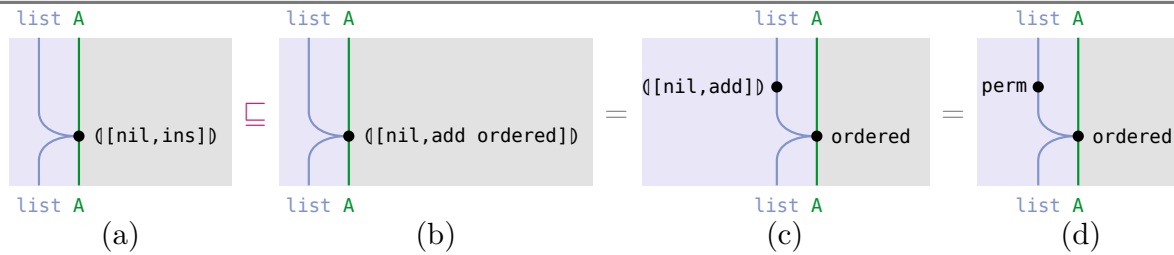
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



|     |                                                                                                                                |                                                                                          |
|-----|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| (a) | X                                                                                                                              | $X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$                           |
| (b) | $= ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$                                                                   | $S^\circ F([\text{nil}, \text{split}^\circ])^\circ R = [\text{nil}, \text{split}^\circ]$ |
| (c) | $= [\text{nil}, \text{split}^\circ]^\circ F([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} [\text{nil}, \text{join}]$ | $X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$                           |
| (d) | $= [\text{nil}, \text{split}^\circ]^\circ F(X) [\text{nil}, \text{join}]$                                                      | $F(S) [T, U] = [T, (S \times (1 \times S)) U]$                                           |
| (e) | $= [\text{nil}, \text{split}^\circ]^\circ [\text{nil}, (X \times (1 \times X)) \text{ join}]$                                  | $[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$                                     |
| (f) | $= (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join})$                                    |                                                                                          |

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}])^\circ \sqsubseteq \text{perm ordered} \quad (6.6o)$$

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations

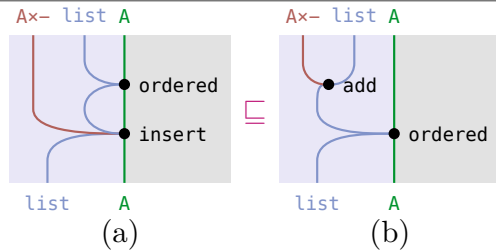


|     |                                                        |                                                                                                     |
|-----|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| (a) | $([\text{nil}, \text{ins}])^\circ$                     | $(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$                              |
| (b) | $\sqsubseteq ([\text{nil}, \text{add ordered}])^\circ$ | $\Rightarrow ([\text{nil}, \text{ins}])^\circ \sqsubseteq ([\text{nil}, \text{add ordered}])^\circ$ |
| (c) | $= ([\text{nil}, \text{add}])^\circ \text{ ordered}$   | $([\text{nil}, \text{add}])^\circ \text{ ordered} = ([\text{nil}, \text{add ordered}])^\circ$       |
| (d) | $= \text{perm ordered}$                                | $\text{perm} = ([\text{nil}, \text{add}])^\circ$                                                    |

$$(\forall a. \forall b. \text{leb}(a,b) = \text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b) = \text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. \quad (6.6p)$$

$$R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered}$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



|     |                                            |                                                 |
|-----|--------------------------------------------|-------------------------------------------------|
| (a) | $(1 \times \text{ordered}) \text{ insert}$ | insert splices a in, and R transitive and total |
| (b) | $\sqsubseteq \text{add ordered}$           |                                                 |

## §6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R\***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

**X** contains **R** exactly when it contains **R\***

$$1 \sqsubseteq R^* \quad (6.7c)$$

**R\*** is reflexive

|     |                              |                          |
|-----|------------------------------|--------------------------|
| (a) | 1                            | $R \sqsubseteq R \cup S$ |
| (b) | $\sqsubseteq 1 \cup (R R^*)$ | $1 \cup (R R^*) = R^*$   |
| (c) | $= R^*$                      |                          |

$$R \subseteq R^*$$

(6.7d)

 $R^*$  contains  $R$ 

|     |                     |                          |
|-----|---------------------|--------------------------|
| (a) | $R$                 |                          |
| (b) | $= R$               | (6.7c) $1 \subseteq R^*$ |
| (c) | $\sqsubseteq R R^*$ | $R R^* \subseteq R^*$    |
| (d) | $\sqsubseteq R^*$   |                          |

$$(1 \cup (R (R^* / R^*))) R^* \subseteq R^*$$

(6.7e)

 $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup RX$ , stated through division

$$R^* R^* \subseteq R^*$$

 $R^*$  is transitive, because  $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup RX$ 

|     |                                    |                                   |
|-----|------------------------------------|-----------------------------------|
| (a) | $(1 \cup (R (R^* / R^*))) R^*$     | $(S \cup T) R = (S R) \cup (T R)$ |
| (b) | $= (R^*) \cup (R (R^* / R^*) R^*)$ |                                   |
| (c) | $= R^* \cup (R (R^* / R^*) R^*)$   | $(R / R) R \subseteq R$           |
| (d) | $\sqsubseteq R^* \cup (R R^*)$     | $R R^* \subseteq R^*$             |
| (e) | $\sqsubseteq R^*$                  |                                   |

$$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow 1 \cup (R X) \subseteq X$$

(6.7f)

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $Y \mapsto 1 \cup RY$ 

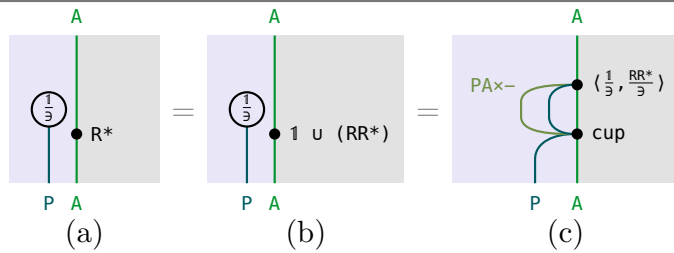
$$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow R^* \subseteq X$$

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $X \mapsto 1 \cup RX$ , so it contains  $R^*$ 

|     |                            |                 |
|-----|----------------------------|-----------------|
| (a) | $1 \cup (R X)$             | $1 \subseteq X$ |
| (b) | $\sqsubseteq X \cup (R X)$ | $R \subseteq X$ |
| (c) | $\sqsubseteq X \cup (X X)$ |                 |
| (d) | $\sqsubseteq X$            |                 |

$$\frac{R^*}{\exists} = \left( \frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$$

(6.7g)

the set of  $R^*$ -successors of  $a$  is  $a$  itself joined with the  $R^*$ -successors of its  $R$ -successors


|     |                                                                         |                                                                                              |
|-----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| (a) | $\frac{R^*}{\exists}$                                                   | $1 \cup (R R^*) = R^*$                                                                       |
| (b) | $= \frac{1 \cup (R R^*)}{\exists}$                                      | $\frac{R \cup S}{\exists} = \left( \frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$ |
| (c) | $= \left( \frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$ |                                                                                              |

$$R - \emptyset = R$$

(6.7h)

taking nothing away leaves  $R$ 

$$R \cup (S - R) = R \cup S$$

beside  $R$ , only the part of  $S$  outside  $R$  adds anything

$$R - (S \cup T) = R - S - T$$

taking away  $S$  and then  $T$  takes away  $S \cup T$ 

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking  $T$  away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \Rightarrow (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$$

(6.7i)

the least fixed point of  $\varphi$  after  $\psi$  is  $\varphi$  applied to the least fixed point of  $\psi$  after  $\varphi$ 

$$S R^* = (\mu X : S \cup (X R))$$

(6.7j)

 $S$  followed by any number of  $R$  steps is the least  $X$  containing  $S$  and closed under a further  $R$  step

$$R^* S = (\mu X : S \cup (R X))$$

any number of  $R$  steps followed by  $S$  is the least  $X$  containing  $S$  and closed under an  $R$  step in front

| $\theta(P, \emptyset) = P$ <span style="float: right;">(6.71)</span> |                                                            |                                                                 |
|----------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------|
| with nothing left to explore, $\theta$ returns what it has found     |                                                            |                                                                 |
| (a)                                                                  | $\theta(P, \emptyset)$                                     | $\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X \ R) - P))$ |
| (b)                                                                  | $\triangleq P \cup (\mu X : \emptyset \cup ((X \ R) - P))$ | $(\mu X : \emptyset \cup ((X \ R) - P)) = \emptyset$            |
| (c)                                                                  | $= P \cup \emptyset$                                       | $R \cup \emptyset = R$                                          |
| (d)                                                                  | $= P$                                                      |                                                                 |

### §7.1. Minimum and maximum

For  $R : A \rightarrow A$ ,  $\text{est}(R) \triangleq \exists n (\in R^\circ) : PA \rightarrow A \quad X \in \in R^\circ \iff EX \in R^\circ$ , and  $EX$  runs  $y \rightarrow xS \rightarrow x$ , member first, so its pair  $(y, x)$  is in  $R^\circ$  exactly when  $x R y$ ; at  $R \triangleq \leq$  that is  $x \leq y$  for every  $y \in xS$ , the least —  $\in \leq$  would give  $y \leq x$ , the greatest.

$$xs \text{ (est}(R)) \iff (x \text{ in } xs) \text{ and all } x \in R \setminus xs: xs \text{ in } q$$
$$E(R) \triangleq \frac{\exists R}{\exists} : PA \rightarrow PB, \quad xS \in E(R) \quad yS \iff yS = \{y \mid \exists x \in xS. \ x \ R \ y\} \quad \text{the image of } xS, \quad E(R) \triangleq \frac{\exists R}{\exists}$$

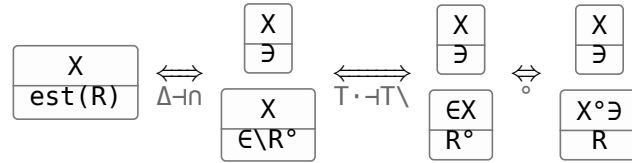
44

(7.1b)

|    | the law                                                                                                                      | what it says                                                                                                |
|----|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1  | $X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists \text{ and } X^\circ \sqsubseteq R$                                  | in the set, and below every element of it                                                                   |
| 2  | $\frac{1}{\exists} (\epsilon \setminus R) = R$                                                                               | bounding a singleton is bounding its element                                                                |
| 3  | $\frac{S}{\exists} (\epsilon \setminus R) = S^\circ \setminus R$                                                             | bound S's image without building the set                                                                    |
| 4  | $\text{union} \triangleq \frac{\exists \exists}{\exists} : \text{PPA} \rightarrow \text{PA}$                                 | flattens a set of sets                                                                                      |
| 5  | $\text{union} (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)$                                            | bound a union by bounding each member set                                                                   |
| 6  | $\frac{1}{\exists} \text{est}(R) = 1 \sqcap R^\circ$ (7.4)                                                                   | a singleton's minimum is its element, where R is reflexive<br>$\frac{S}{\exists} \text{est}(R)$ at $S := 1$ |
| 7  | $\frac{S}{\exists} \text{est}(R) = S \sqcap (S^\circ \setminus R^\circ)$                                                     | an S-value that points to every S-value                                                                     |
| 8  | $\frac{S}{\exists} \text{est}(R \sqcap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$                                       | only R between values S gives one argument counts — context                                                 |
| 9  | $E(S) \text{est}(R) = (\exists S) \sqcap ((\exists S)^\circ \setminus R^\circ)$ (7.7)                                        | the same for the image of a set<br>$\frac{S}{\exists} \text{est}(R)$ at $S := \exists S$                    |
| 10 | $\text{Map}(f) \implies P(f) \text{est}(R) = \text{est}(f R f^\circ) f$                                                      | shunt a function through a minimum                                                                          |
| 11 | $P(S) \text{est}(R) = (\exists S) \sqcap (\epsilon \setminus (SR^\circ))$ (7.9)<br>R reflexive                               | fusion with the power relator<br>$\exists$ is the only proof here that tabulates                            |
| 12 | $P(S) \text{est}(R) \sqsubseteq (\exists S) \sqcap (\epsilon \setminus (SR^\circ))$                                          | the half of the row above that costs nothing                                                                |
| 13 | $P(\text{est}(R)) \text{est}(R) \sqsubseteq \text{union} \text{est}(R)$<br>R transitive                                      | a minimum in each set, then a minimum of those                                                              |
| 14 | $R R \sqsubseteq R \implies P(\text{est}(R)) \text{est}(R) = P(\text{dom}(\text{est}(R))) \text{union} \text{est}(R)$ (7.12) | the same as an equality, once empty sets are dropped                                                        |

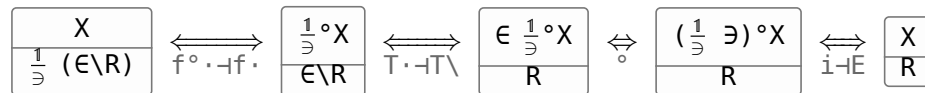
**§7.1.1.  $X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists \wedge \epsilon \in X \sqsubseteq R^\circ$  — X refines est(R) when it picks a member R-related to every member**

(7.1.1a)



**§7.1.2.  $\frac{1}{\exists} (\epsilon \setminus R) = R$  — all of {a} relates by R to c exactly when a does**

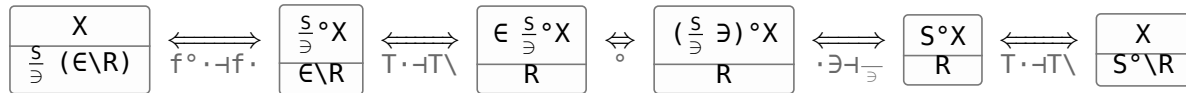
(7.1)



**§7.1.3.  $\frac{S}{\exists} (\epsilon \setminus R) = S^\circ \setminus R$  — gathering the S-values into one set first changes nothing**

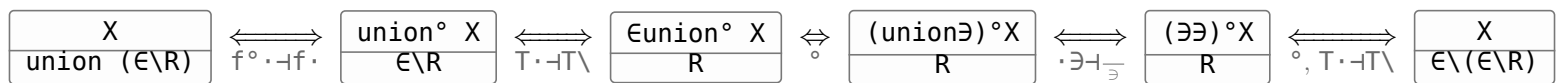
$\frac{S}{\exists}$  gathers the S-image of a point into one set and  $\epsilon \setminus R$  asks that every member of that set be R-related to the target, so the set cancels and  $S^\circ \setminus R$  asks it of the S-image directly.

(7.2)



**§7.1.4.  $\text{union} (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)$  — every member of the union relates by R iff every set's members do**

(7.3)



**§7.1.5.  $\frac{S}{\exists} \text{est}(R) = S \sqcap (S^\circ \setminus R^\circ)$  — the smallest S-value is an S-value R-related to every S-value**

$$\begin{array}{c}
 \boxed{\begin{array}{c} X \\ \exists \\ \text{est}(R) \end{array}} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \boxed{\begin{array}{c} S^\circ X \\ \exists \\ \text{est}(R) \end{array}} \xleftrightarrow{\Delta \neg n} \boxed{\begin{array}{c} S^\circ X \\ \exists \\ \exists \\ \epsilon \setminus R^\circ \end{array}} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \boxed{\begin{array}{c} X \\ \exists \\ \exists \\ \epsilon \setminus R^\circ \end{array}} \xleftrightarrow{\cdot \exists \neg \exists, (7.2) \frac{S}{\exists} (\epsilon \setminus R) = S^\circ} \boxed{\begin{array}{c} X \\ S \\ \exists \\ S^\circ \setminus R^\circ \end{array}} \xleftrightarrow{\Delta \neg n} \boxed{\begin{array}{c} X \\ S n(S^\circ \setminus R^\circ) \end{array}}
 \end{array}
 \quad (7.5)$$

**§7.1.6.**  $\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$  — only  $R$  between  $S$ -values of one argument matters for the smallest

$$\begin{array}{c}
 \boxed{\begin{array}{c} X \\ \exists \\ \text{est}(R n S^\circ S) \end{array}} \xleftrightarrow{(7.5) \frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \setminus R^\circ), \circ} \boxed{\begin{array}{c} X \\ S n(S^\circ \setminus (R^\circ n S^\circ S)) \end{array}} \xleftrightarrow{\Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} X \\ S \\ S^\circ X \\ R^\circ n S^\circ S \end{array}} \\
 \xleftrightarrow{\Delta \neg n, S^\circ \cdot \text{monotone}} \boxed{\begin{array}{c} X \\ S \\ S^\circ X \\ R^\circ \end{array}} \xleftrightarrow{T \cdot \neg T \setminus, \Delta \neg n} \boxed{\begin{array}{c} X \\ S n(S^\circ \setminus R^\circ) \end{array}} \xleftrightarrow{(7.5) \frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)} \boxed{\begin{array}{c} X \\ \exists \\ \text{est}(R) \end{array}}
 \end{array}
 \quad (7.6)$$

**§7.1.7.**  $\text{Map}(f) \Rightarrow P(f) \text{est}(R) = \text{est}(f R f^\circ) f$  — the smallest  $f$ -image is  $f$  of the smallest under  $f R f^\circ$

$$\begin{array}{c}
 \boxed{\begin{array}{c} P(f) \text{est}(R) \\ f \text{ a map} \end{array}} \xrightarrow{P=E \text{ on maps, } (7.5) \frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)} \boxed{(\exists f) n((\exists f)^\circ \setminus R^\circ)} \xrightarrow{\circ, T \cdot \neg T \setminus, f^\circ \cdot \neg f^\circ} \boxed{(\exists f) n(\epsilon \setminus (f R^\circ))} \\
 \xrightarrow{\text{modular law, } f \text{ simple}} \boxed{(\exists n((\epsilon \setminus (f R^\circ)) f^\circ)) f} \xrightarrow{\cdot f \neg f^\circ, \circ, \text{est}} \boxed{\text{est}(f R f^\circ) f}
 \end{array}
 \quad (7.8)$$

**§7.1.8.**  $P(S) \text{est}(R) \sqsubseteq (\exists S) \cap (\epsilon \setminus (S R^\circ))$  — the smallest of the  $S$ -image is  $R$ -below an  $S$ -value of every member

$$\begin{array}{c}
 \boxed{\begin{array}{c} P(S) \text{est}(R) \\ (\exists S) n(\epsilon \setminus (S R^\circ)) \end{array}} \xleftrightarrow{\Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(S) \text{est}(R) \\ \exists S \end{array}} \xleftrightarrow{\text{UP of est}} \boxed{\begin{array}{c} P(S) \exists \\ \exists S \end{array}} \\
 \boxed{\begin{array}{c} \epsilon P(S) \text{est}(R) \\ S R^\circ \end{array}} \xleftrightarrow{\text{UP of est}} \boxed{\begin{array}{c} \epsilon P(S) \\ S \epsilon \end{array}}
 \end{array}
 \quad (7.10)$$

**§7.1.9.**  $R R \sqsubseteq R \Rightarrow P(\text{est}(R)) \text{est}(R) \sqsubseteq \text{union est}(R)$  — the smallest of the smallests is a smallest of the union

$$\begin{array}{c}
 \boxed{\begin{array}{c} P(\text{est}(R)) \\ \text{est}(R) \\ \text{union est}(R) \\ R \text{ transitive} \end{array}} \xleftrightarrow{(7.5) \frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)} \boxed{\begin{array}{c} P(\text{est}(R)) \\ \text{est}(R) \\ (\exists \exists) n((\exists \exists)^\circ \setminus R^\circ) \end{array}} \xleftrightarrow{\circ, \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{est}(R)) \\ \text{est}(R) \\ \exists \exists \end{array}} \\
 \boxed{\begin{array}{c} \epsilon \epsilon P(\text{est}(R)) \\ \text{est}(R) \\ R^\circ \end{array}} \xleftrightarrow{\text{UP of est, } R \text{ transitive}} \boxed{\begin{array}{c} P(\text{est}(R)) \exists \\ \exists \text{est}(R) \end{array}} \\
 \boxed{\begin{array}{c} \epsilon P(\text{est}(R)) \\ \text{est}(R) \epsilon \end{array}}
 \end{array}
 \quad (7.11)$$

,  $R$  transitive

## §7.2. Monotonic algebras

(7.2a)

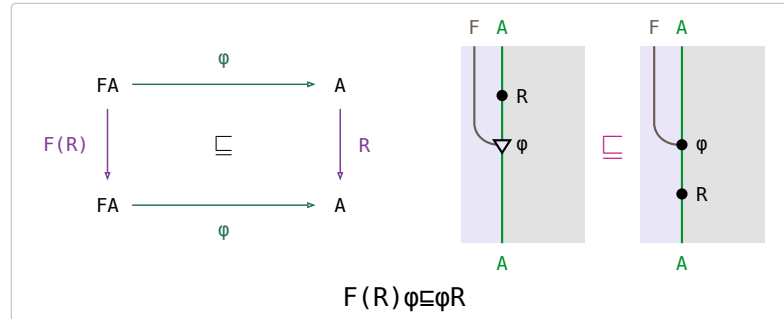
An  $F$ -algebra  $\varphi : FA \rightarrow A$  is **monotonic on**  $R : A \rightarrow A$  when it is lax at  $R$  at  $G := F, F := \text{Id}$ :  $F(R)\varphi \sqsubseteq \varphi R$ .  $R$  is an **endorelation**, so  $B=A$  and the two components  $\varphi_A, \varphi_B$  are the one arrow  $\varphi$  — an algebra, not a family.

For a map  $f : FA \rightarrow A$  that is  $f \circ F(R)f \in \text{Map}(f)$

$\Rightarrow 1 \sqsubseteq f \circ f^\circ, R(S \cup T) = (R S) \cup (R T)$  's  $f^\circ \cdot \dashv f \cdot$  at  $X := F(R)f, Y := R$ , equivalently  $F(R) \sqsubseteq f R f^\circ \cdot f \dashv \cdot f^\circ$  then  $f^\circ \cdot \dashv f \cdot, .$

$(\leq \times \leq) + \sqsubseteq + \leq$  — addition on  $\text{Nat}$  is monotonic on  $\leq$ , which at the point level reads  $c = a + b \wedge a \leq a' \wedge b \leq b' \Rightarrow c \leq a' + b'$ .

(7.2b)



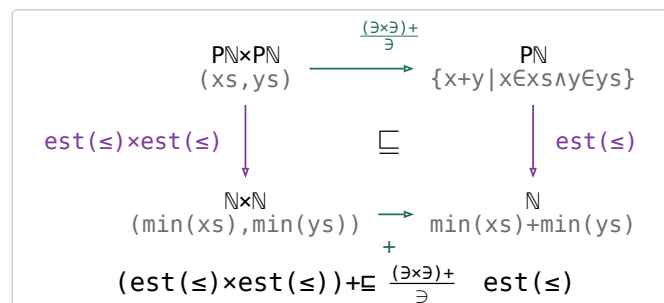
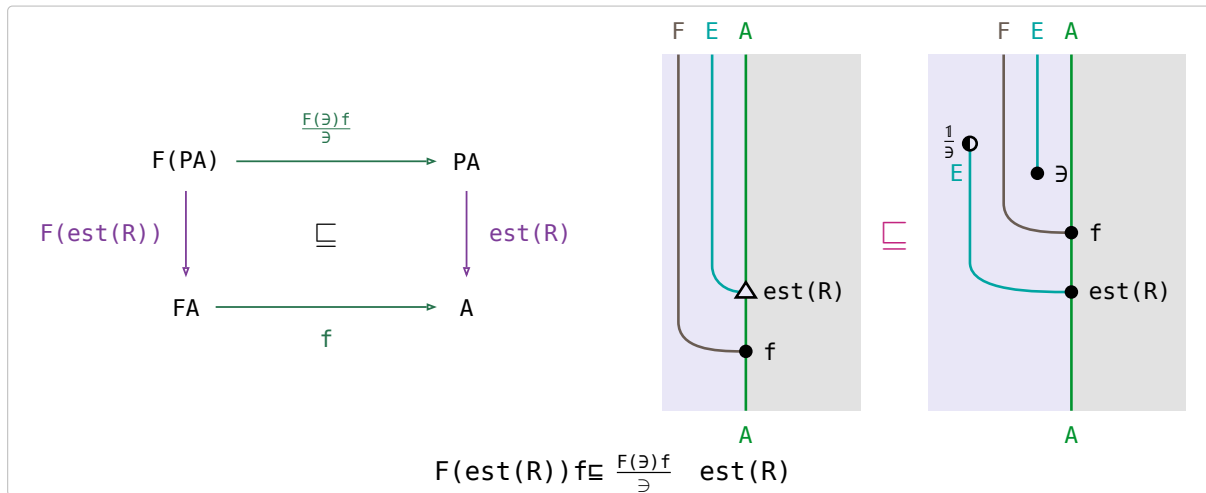
### §7.2.1. Function $f$ is monotonic on $R$ iff it distributes over $R$

(7.2.1a)

$f : FA \rightarrow A$  **distributes over**  $R$  if  $F(\text{est}(R))f \sqsubseteq \frac{F(\exists)f}{\exists} \text{est}(R)$ .

$+$  distributes over  $\leq$ , at the point level  $\min(xs) + \min(ys)$  is a least element of  $\{x+y \mid x \in xs \wedge y \in ys\}$  for  $xs, ys$  non-empty and  $\min \triangleq \text{est}(\leq)$ .

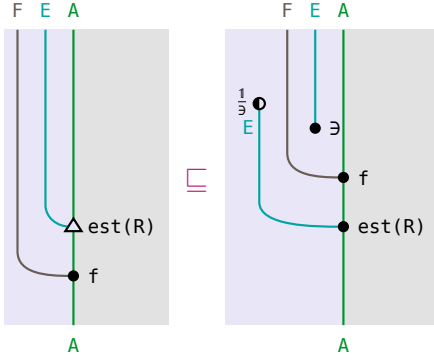
(7.2.1b)



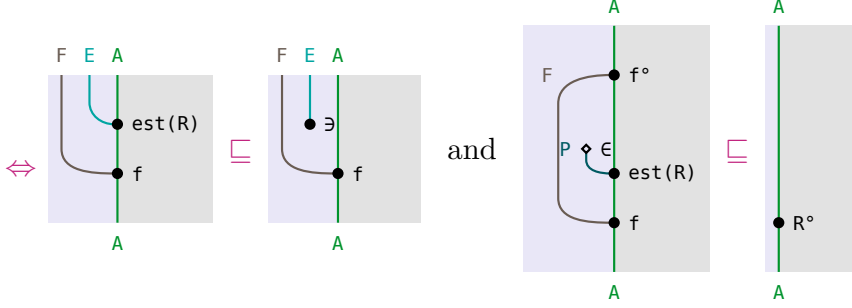
$$f^\circ F(R) f \sqsubseteq R \iff F(\text{est}(R)) f \sqsubseteq \frac{F(\exists)f}{\exists} \text{est}(R)$$

Theorem 7.1

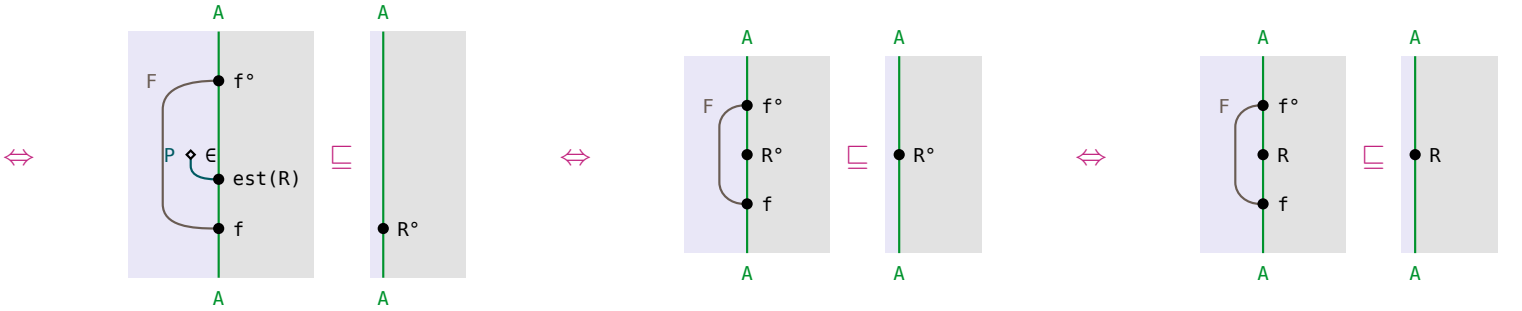
function  $f$  is monotonic over  $R$  if and only if it distributes over  $R$ ;  $f$  a map,  $R$  reflexive



$f$  distributes over  $R \triangleq \text{distributes} \text{ — the fraction bent as } \frac{R}{\exists} \exists = R$



(7.5)  $\frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)$  splits the bound in two, (0.9a.2) moving  $(F(\exists)f)^\circ$  across



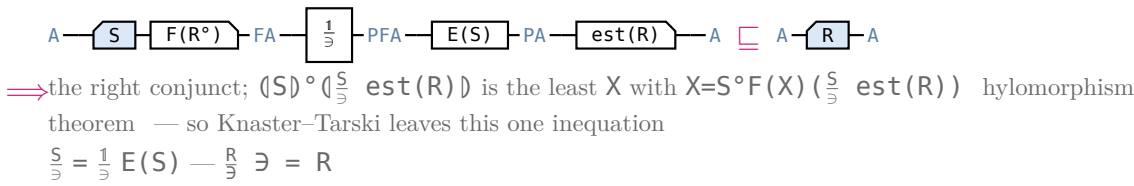
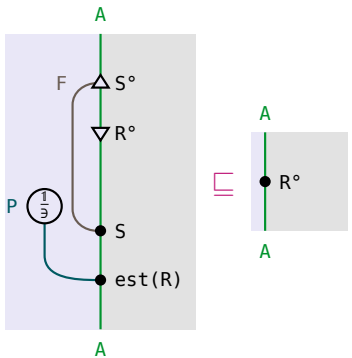
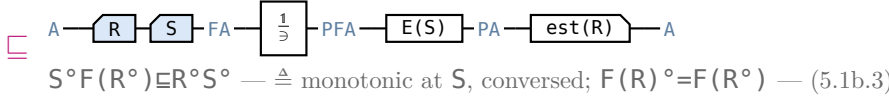
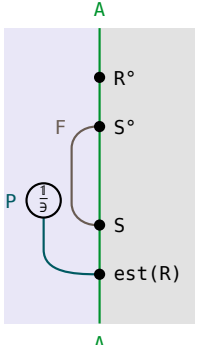
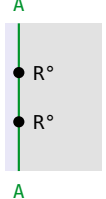
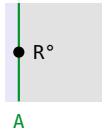
$\text{est}(R) \sqsubseteq \exists \triangleq \text{est}$  — so the first conjunct drops

$(F(\exists)f)^\circ = f^\circ F(\epsilon) \triangleq \epsilon$  — and  $\epsilon \text{est}(R) = R^\circ \triangleq \text{est}, R$  reflexive

both sides conversed —  $F(R^\circ)^\circ = F(R)$ , (5.1b.3)

**§7.2.2.  $R \circ R \subseteq R \wedge \text{MonoAlg}(S, R) \Rightarrow (\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq (\bigcap_{\mathcal{S}} \text{est}(R))$  — taking the best at each step gives a best overall result**

Theorem 7.2

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $(\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq (\bigcap_{\mathcal{S}} \text{est}(R))$<br>the conclusion: one minimum kept at each step is below every result collected and one minimum taken at the end                                                                                                                                                                                                                               |                                                                                       |
| $(\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq (S)$<br>and<br>$\Leftrightarrow (S)^\circ (\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq R^\circ$<br>(7.5) $\bigcap_{\mathcal{S}} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)$ at $(S)$ ; $(\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq (S)^\circ \setminus R^\circ \Leftrightarrow (S)^\circ (\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq R^\circ$<br>— (0.9a.2)        |                                                                                       |
| $\Rightarrow \bigcap_{\mathcal{S}} \text{est}(R) \subseteq S$<br>the left conjunct; (7.5) $\bigcap_{\mathcal{S}} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)$ , and $(-)$ is monotone                                                                                                                                                                                                                                            |                                                                                       |
|  <p>the right conjunct; <math>(S)^\circ (\bigcap_{\mathcal{S}} \text{est}(R))</math> is the least <math>X</math> with <math>X=S^\circ F(X) (\bigcap_{\mathcal{S}} \text{est}(R))</math> hylomorphism theorem — so Knaster–Tarski leaves this one inequation<br/> <math>\bigcap_{\mathcal{S}} = \frac{1}{3} E(S) \text{ — } \frac{R}{3} = R</math></p> |    |
|  <p><math>S^\circ F(R^\circ) \subseteq R^\circ S^\circ \text{ — } \triangle \text{ monotonic at } S, \text{ conversed; } F(R)^\circ = F(R^\circ) \text{ — (5.1b.3)}</math></p>                                                                                                                                                                      |   |
|  <p><math>S^\circ (\bigcap_{\mathcal{S}} \text{est}(R)) \subseteq S^\circ (S^\circ \setminus R^\circ) \subseteq R^\circ \text{ — (7.5) } \bigcap_{\mathcal{S}} \text{est}(R) = S \cap (S^\circ \setminus R^\circ), S (S \setminus R) \subseteq R</math></p>                                                                                         |  |
|  <p>R transitive</p>                                                                                                                                                                                                                                                                                                                                 |  |

$$(7.2.3a)$$

| name                     | definition                                                                                                                                                                                                     | type                                                   | example                                                                                | in words                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1 F                      | $FX = L + E \times X$                                                                                                                                                                                          | $\mathcal{A} \rightarrow \mathcal{A}$                  |                                                                                        | nothing, or a head and a tail                                                                       |
| 2 nil, cons              | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$<br>$\text{list}(A) = [A]$                                                                                                                                        | $1 \rightarrow [A],$<br>$A \times [A] \rightarrow [A]$ | $\text{cons}(3, [1, 2]) = [3, 1, 2]$                                                   | the empty list; a head onto a tail                                                                  |
| 3 $\alpha$               | $[\text{nil}, \text{cons}]$                                                                                                                                                                                    | $F[A] \rightarrow [A]$                                 |                                                                                        | both constructors as one map                                                                        |
| 4 p                      | a coreflexive                                                                                                                                                                                                  | $A \rightarrow A$                                      | $p \triangleq \text{even} \rightarrow 2, p \rightarrow 2,$<br>$3 \notin \text{dom}(p)$ | $\{(a, a) \mid a \text{ passes the test}\}$                                                         |
| 5 R                      | $\text{length} \leq \text{length}^\circ$ , a preorder                                                                                                                                                          | $[A] \rightarrow [A]$                                  | $[1] \ R \ [1, 2]$                                                                     | $xs \ R \ ys \iff \text{length}(xs) \leq \text{length}(ys)$                                         |
| 6 $\rightarrow$ nil      | the constant nil — the second nil of prefix                                                                                                                                                                    | $A \times [A] \rightarrow [A]$                         | $(\rightarrow \text{nil})(3, [1, 2]) = \text{nil}$                                     | drop the pair, return nil                                                                           |
| 7 prefix                 | $\langle [ \text{nil}, \rightarrow \text{nil} \cup \text{cons} ] \rangle$<br>$\text{prefix} = \langle [ \text{nil}, \rightarrow \text{nil} \cup \text{cons} ] \rangle, \text{prefix} = \text{cat}^\circ \pi_1$ | $[A] \rightarrow [A]$                                  | $[3, 1, 2] \ \text{prefix} \ [3, 1]$                                                   | $xs \ \text{prefix} \ ys \iff \exists zs. \ xs = ys \# zs$<br>— at each cons, stop or keep the head |
| 8 S                      | $[\text{nil}, \rightarrow \text{nil} \cup (p \times 1) \text{cons}]$                                                                                                                                           | $F[A] \rightarrow [A]$                                 | $(4, [2]) \ S \ [4, 2],$ and<br>$(3, [2]) \ S \ \text{nil}$ only                       | prefix's algebra with one extra p<br>— stop, or keep a head that passes p                           |
| 9 $(g \rightarrow X, Y)$ | X where g is defined and Y where it is not                                                                                                                                                                     | $A \rightarrow B$                                      |                                                                                        | the test picks the branch                                                                           |

`nil` `R=T`, `nil` `R°=nil` `nil` is the shortest list — below every list, and above only itself, so it loses every `est(R°)`

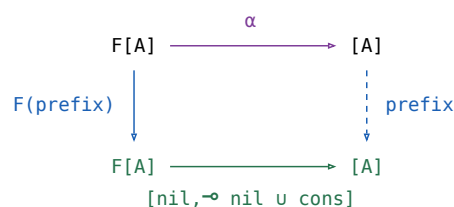
```
prefix = ([nil, nil u cons])
```

$$(7.2.3b)$$

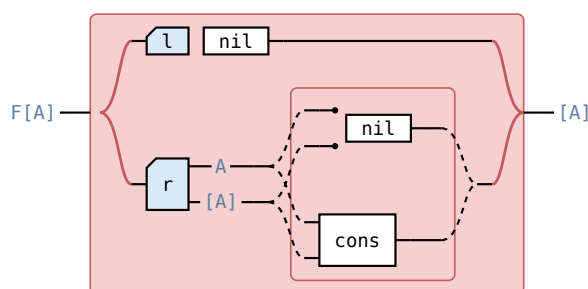
the fold whose algebra, at each **cons**, stops with **nil** or keeps the head:  $xs \text{ prefix } ys \iff \exists zs. xs = ys \# zs$

**the defining square**  $\alpha$   $\text{prefix} = F(\text{prefix})[\text{nil}, \multimap \text{nil} \cup \text{cons}]$  — build the list and then take a prefix, or take a prefix of the tail and then rebuild with the algebra

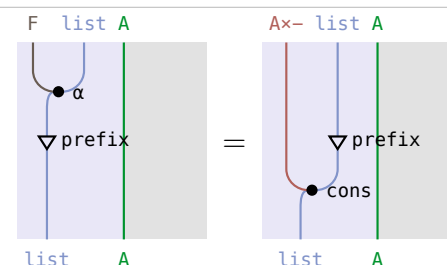
Hinze–Marsden



$F = 1 + A \times -$ , so  $F(\text{prefix}) = 1 + 1 \times \text{prefix}$ : the head passes, the fold recurses on the tail alone



the algebra  $[\text{nil}, \multimap \text{nil} \cup \text{cons}] : F([A]) \rightarrow [A]$  as a circuit: **nil** on the **1** branch; on a pair  $(a, ys')$  two outputs, **nil** and **cons**(**a**, **ys'**)



the `cons` operand of  $\rightarrow$  `nil` u `cons`;  $\rightarrow$  `nil` makes a constant and draws nothing

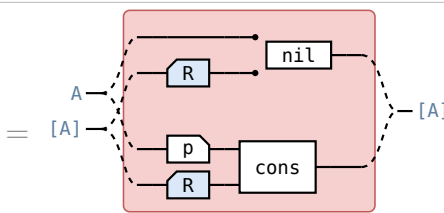
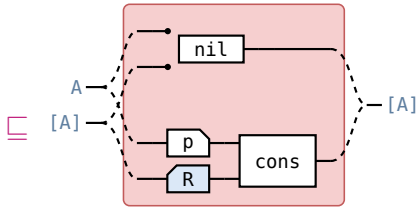
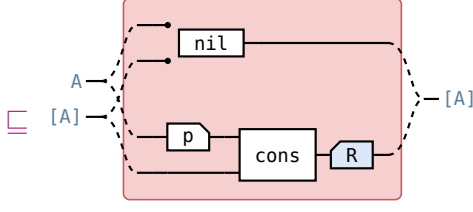
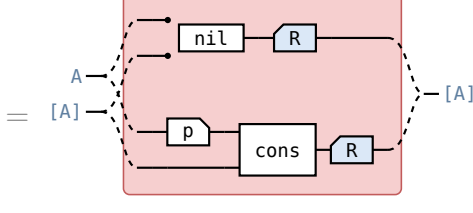
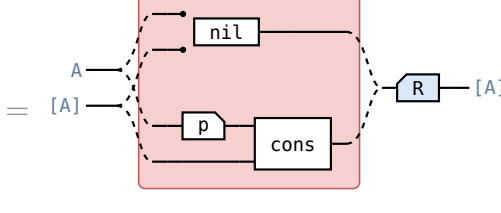
$$\begin{aligned} & \text{nil prefix } ys \iff ys = \text{nil} \\ \text{cons}(a, xs') \text{ prefix } ys & \iff ys = \text{nil} \vee \\ & \exists ys'. xs' \text{ prefix } ys' \wedge \\ & \quad ys = \text{cons}(a, ys') \end{aligned}$$

a right fold:  $\alpha^\circ$  peels,  $F(\text{prefix})$  recurses,  
the algebra runs on the way back;  $\text{init}^*$   
 $(\text{prefix} = \langle [\text{nil}, \neg \text{nil} \cup \text{cons}] \rangle,$   
 $\text{prefix} = \text{cat}^\circ \pi_1)$  is the tail-recursive  
form

lax natural only, **list(R)** **prefix**  $\sqsubseteq$  **prefix**  
**list(R)**: **list(R)** first needs an **R**-image of  
every element, **prefix** first may have dropped  
the ones without

| $\alpha \text{ prefix list}(p) = F(\text{prefix list}(p)) S$ <span style="float: right;">(7.2.3c)</span><br>building the list and then keeping a <b>p</b> -passing prefix of it is keeping one of the tail first, and then building with <b>S</b><br>this same diagram is <b>subseq</b> 's: algebra $[\text{nil}, \pi_2 \cup \text{cons}]$ , type $[A] \rightarrow [A]$ |                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>circuit</b> — the fork is $F([A]) = 1 + A \times [A]$ : nil above, the pair below<br><p><math>\alpha \text{ prefix list}(p)</math></p>                                                                                                                                                                                                                               | <b>Hinze–Marsden</b><br><p>the <b>cons</b> branch alone, without<br/> <math>1 +</math> or <math>\multimap \text{nil}</math></p>                                        |
| <p><math>F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup \text{cons}] \text{list}(p)</math><br/>         defining equation</p>                                                                                                                                                                                                                                  | <p>the <b>cons</b> operand of <math>\multimap \text{nil} \cup</math><br/> <b>cons</b></p>                                                                              |
| <p><math>F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup (p \times \text{list}(p)) \text{cons}]</math><br/>         list(p) through cons</p>                                                                                                                                                                                                                    | <p>the <math>(p \times \text{list}(p))</math> cons<br/>         operand of <math>\multimap \text{nil} \cup</math><br/> <math>(p \times \text{list}(p))</math> cons</p> |
| <p><math>[\text{nil}, \multimap \text{nil} \cup (p \times (\text{prefix list}(p))) \text{cons}]</math><br/>         relator, <b>prefix</b> entire</p>                                                                                                                                                                                                                   |                                                                                                                                                                        |
| <p><math>F(\text{prefix list}(p)) S</math><br/> <b>prefix list(p)</b> entire</p>                                                                                                                                                                                                                                                                                        |                                                                                                                                                                        |

$\triangleq$  **reduce** reads that off as  $\text{prefix list}(p) = (S)$ . fusion cannot:  $\text{list}(p)$  is not entire,  $(1 \times \text{list}(p)) \multimap \text{nil} \sqsubseteq \multimap \text{nil}$ , and no algebra meets the side condition. ,

| $(1 \times R^\circ) (\multimap \text{nil} \cup ((p \times 1) \text{cons})) \sqsubseteq (\multimap \text{nil} \cup ((p \times 1) \text{cons})) R^\circ$ (7.2.3d)<br>shortening the tail and then taking the step lands inside taking the step and then shortening the result |                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| circuit — the cons branch of $F(R^\circ)S \sqsubseteq SR^\circ$                                                                                                                                                                                                             | reason                                                                                                                                                                                                         |
|  $(1 \times R^\circ) (\multimap \text{nil} \cup ((p \times 1) \text{cons}))$                                                                                                               |                                                                                                                                                                                                                |
|  $((1 \times R^\circ) \multimap \text{nil}) \cup ((p \times R^\circ) \text{cons})$                                                                                                         | each operand is reached on its own $(S \cup T) R = (S R) \cup (T R)$ , $(R \cap R') / S = (R / S) \cap (R' / S)$ — and $(1 \times R^\circ)(p \times 1)$ is $p$ and $R^\circ$ on the pair's two strands at once |
|  $\multimap \text{nil} \cup ((p \times R^\circ) \text{cons})$                                                                                                                              | $\multimap$ is the greatest arrow into $1$ , so $(1 \times R^\circ) \multimap \sqsubseteq \multimap$                                                                                                           |
|  $\multimap \text{nil} \cup ((p \times 1) \text{cons} R^\circ)$                                                                                                                          | $\text{cons length} = (1 \times \text{length}) \pi_2$<br>succ with succ monotone — a shorter tail makes a shorter list                                                                                         |
|  $(\multimap \text{nil} R^\circ) \cup ((p \times 1) \text{cons} R^\circ)$                                                                                                                | $\text{nil } R^\circ = \text{nil } R \triangleq \text{length} \leq \text{length}^\circ$ — so the constant branch may carry the $R^\circ$ the other one already has                                             |
|  $(\multimap \text{nil} \cup ((p \times 1) \text{cons})) R^\circ$                                                                                                                        | one $R^\circ$ past the join is the two inside it $(S \cup T) R = (S R) \cup (T R)$ , $(R \cap R') / S = (R / S) \cap (R' / S)$                                                                                 |

the  $\text{nil}$  branch, which no row above draws, is  $\text{nil} \sqsubseteq \text{nil } R^\circ$ .

| $\frac{S}{\text{est}} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ <span style="float: right;">(7.2.3e)</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| the longest of the lists the algebra allows is the <b>cons</b> where the head passes <b>p</b> , and <b>nil</b> where it does not                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                |
| formula                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | reason                         |
| $\begin{array}{c} A \\ [A] \end{array} \xrightarrow{\frac{1}{S}} \text{PF}[A] \rightarrow \text{E}(S) \rightarrow \text{P}[A] \rightarrow \text{est}(R^\circ) \rightarrow [A]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                |
| $= \begin{array}{c} F[A] \\ = \end{array} \left\{ \begin{array}{l} \begin{array}{c} l \\ [A] \end{array} \xrightarrow{\frac{1}{S}} \text{P1} \rightarrow \text{E}(\text{nil}) \rightarrow \text{P}[A] \rightarrow \text{est}(R^\circ) \rightarrow [A] \\ \begin{array}{c} r \\ [A] \end{array} \xrightarrow{A} \xrightarrow{\frac{1}{S}} \text{P}(A \times [A]) \rightarrow \text{E}(\neg \text{nil} \cup ((p \times 1) \text{cons})) \rightarrow \text{P}[A] \rightarrow \text{est}(R^\circ) \rightarrow [A] \end{array} \right.$ $\left[ \frac{\text{nil}}{S} \text{est}(R^\circ), \frac{\neg \text{nil} \cup ((p \times 1) \text{cons})}{S} \text{est}(R^\circ) \right]$ | coproduct of maps              |
| $= \begin{array}{c} F[A] \\ = \end{array} \left\{ \begin{array}{l} \begin{array}{c} l \\ [A] \end{array} \xrightarrow{\text{nil}} \text{est}(R^\circ) \rightarrow [A] \\ \begin{array}{c} r \\ [A] \end{array} \xrightarrow{A} \xrightarrow{\frac{1}{S}} \text{P}(A \times [A]) \rightarrow \text{E}(\neg \text{nil} \cup ((p \times 1) \text{cons})) \rightarrow \text{P}[A] \rightarrow \text{est}(R^\circ) \rightarrow [A] \end{array} \right.$ $[\text{nil}, \frac{\neg \text{nil} \cup ((p \times 1) \text{cons})}{S} \text{est}(R^\circ)]$                                                                                                                            | singleton, $R^\circ$ reflexive |
| $= \begin{array}{c} F[A] \\ = \end{array} \left\{ \begin{array}{l} \text{nil} \\ \begin{array}{c} A \\ [A] \end{array} \xrightarrow{(\pi_1 p \rightarrow \text{cons}, \neg \text{nil})} \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\text{nil } R = \top$         |

the set is  $\{\text{nil}\}$  where  $p$  fails on the head and  $\{\text{nil}, \text{cons}(a, xs)\}$  where it holds, and  $\text{nil}$  loses the second —  $\triangleq \text{est}$  at a two-element set.

(7.2.3f)

| name | definition                                                                                                                                 | type                   | example                                                           | in words                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------|
| S    | $[\text{nil}, \neg \text{nil} \cup (p \times 1) \text{cons}] \ S \triangleq [\text{nil}, \neg \text{nil} \cup ((p \times 1) \text{cons})]$ | $F[A] \rightarrow [A]$ | $(4, [2]) \ S \ [4, 2]$ , and<br>$(3, [2]) \ S \ \text{nil}$ only | prefix's algebra with one extra $p$<br>— stop, or keep a head that passes $p$ |

$$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\exists} \text{est}(R^\circ)$$

Exercise 7.39

$$\text{takewhile}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})])$$

**takewhile:**  $\text{takewhile}(p)(x)$  returns the longest prefix of  $x$  with the property that all its elements satisfy  $p$ ; the catamorphism is the standard implementation.



|     |                                                                    |                                                                                                                     |
|-----|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ | $(7.2.3e) \frac{S}{\exists} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ |
| (b) | $= ([\frac{S}{\exists} \text{est}(R^\circ)])$                      | $[\frac{S}{\exists} \text{est}(R^\circ)] \sqsubseteq ([\frac{S}{\exists} \text{est}(R^\circ)])$                     |
| (c) | $\sqsubseteq ([\frac{S}{\exists} \text{est}(R^\circ)])$            | $\text{prefix list}(p) = (S)$                                                                                       |
| (d) | $= \frac{\text{prefix list}(p)}{\exists} \text{est}(R^\circ)$      |                                                                                                                     |
| (e) | $= \text{takewhile}(p)$                                            |                                                                                                                     |

$$\text{takewhile}(p)^\circ \text{takewhile}(p) \sqsubseteq \text{prefix}^\circ \text{prefix} \sqcap R \sqcap R^\circ \sqsubseteq 1$$

$\text{takewhile}(p) \sqsubseteq \text{prefix list}(p)$  and  $(\text{prefix list}(p))^\circ \text{takewhile}(p) \sqsubseteq R$  — (7.5)  $\frac{S}{\exists} \text{est}(R) = S \sqcap (S^\circ \setminus R^\circ)$  at  $\text{est}(R^\circ)$  — and two prefixes of one list of equal length are equal, so  $\text{takewhile}(p)$  is simple: the longest, not a longest.

$$\frac{\text{prefix list}(p)}{\exists} \text{est}(R^\circ) \text{ entire}$$

$\text{nil}$  is always a  $p$ -prefix and  $R$  is connected on the prefixes of one list, so the longest exists.

$$X \sqsubseteq Y, X \text{ entire}, Y \text{ simple} \implies X = Y$$

$[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$  is a reduce of maps, hence entire — what turns the  $\sqsubseteq$  above into the heading's  $=$ .

**§7.2.4.  $\text{mss} = ([k] \pi_2 \text{est}(\geq))$  —  $\text{mss}$  is a fold followed by taking the largest**

(7.2.4a)

$\text{FX} = 1 + A \times X$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $\text{sum} = ([\text{zero}, \text{plus}])$  and  $\text{segment} = \text{suffix prefix}$  from  $\text{sum} = ([\text{zero}, \text{plus}])$  and  $\text{segment} = \text{suffix prefix}$ .

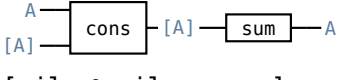
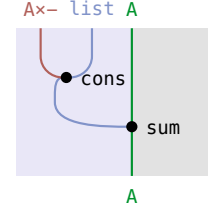
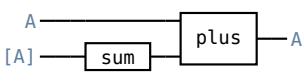
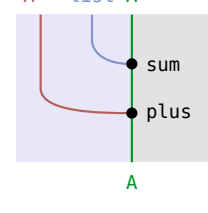
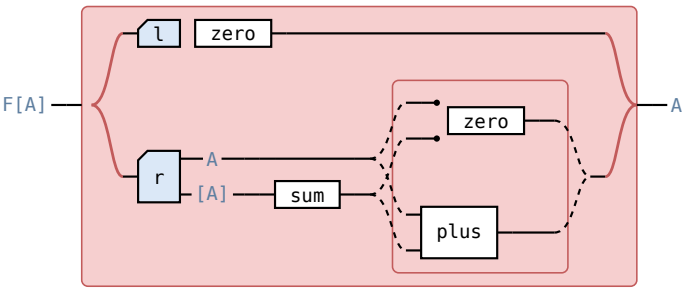
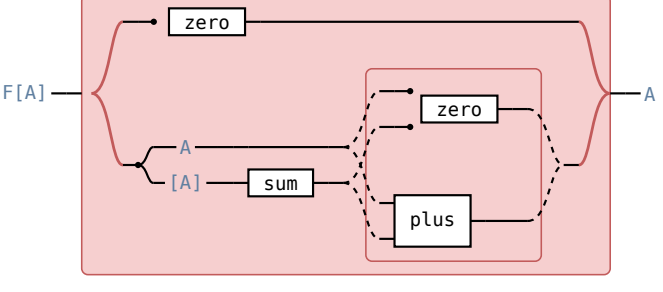
$\text{head} \triangleq \text{cons}^\circ \pi_1$ ,  $\text{wrap} \triangleq (1, \neg \text{nil}) \text{cons}$  the head of a list and the one-element list, beside  $\text{suffix} = \text{cat}^\circ \pi_2$  's  $\text{tail} \triangleq \text{cons}^\circ \pi_2$

$\oplus \triangleq \frac{\neg \text{zero} \cup \text{plus}}{\exists} \text{est}(\geq)$  the set at  $(a, b)$  is  $\{0, a+b\}$ , so  $\oplus$  is the larger of the two,

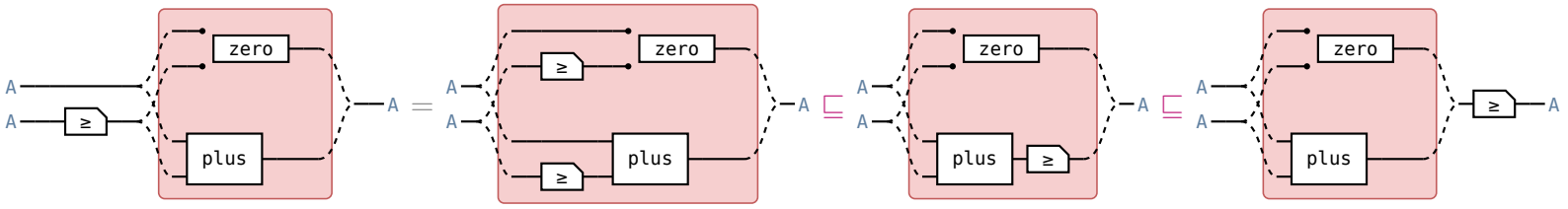
| $\text{mss} = \frac{\text{suffix}}{\exists} E\left(\frac{\text{prefix sum}}{\exists} \text{est}(\geq)\right) \text{est}(\geq) \quad (7.2.4b)$                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| maximum segment sum problem: using <b>segment=suffix prefix</b> , the specification is expressed in this form                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| formula — one wire from [A] to A, its type written along it                                                                                                                                           | reason                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| $[A] \xrightarrow{\frac{1}{\exists}} P[A] \xrightarrow{E(\text{segment sum})} PA \xrightarrow{\text{est}(\geq)} A$                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $= [A] \xrightarrow{\frac{1}{\exists}} P[A] \xrightarrow{E(\text{suffix prefix sum})} PA \xrightarrow{\text{est}(\geq)} A$                                                                            | segment=suffix<br>prefix<br>segment = suffix<br>prefix, $\triangleq$ mss                                                                                                                                                                                                                                                                                                                                                                                 |
| $= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) P[A] \xrightarrow{E(\text{prefix sum})} PA \xrightarrow{\text{est}(\geq)} A$                                                                  | absorption<br>$\frac{S}{\exists} E(R) = \frac{S}{\exists} \frac{R}{\exists}$ —<br>$\text{frac}(S, \exists)$<br>$E(R) = \text{frac}(SR, \exists)$ at<br>$S := \text{suffix}, R := \text{prefix}$<br>sum                                                                                                                                                                                                                                                   |
| $= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) P[A] \xrightarrow{E\left(\frac{\text{prefix sum}}{\exists}\right)} PPA \xrightarrow{\text{union}} PA \xrightarrow{\text{est}(\geq)} A$        | $\frac{R}{\exists} \exists = R, \text{union} = E(\exists)$<br>$\frac{R}{\exists} \exists = R$<br>$\text{frac}(R, \exists) \exists = R$ at<br>$R := \text{prefix sum}$ and<br>$E(R) \triangleq \text{frac}(\exists R, \exists)$ ,<br>$E(R) \triangleq \frac{\exists R}{\exists}$ ; (7.1b.4)<br>$\text{union} \triangleq \text{frac}(\exists \exists, \exists)$ ; the<br>middle equality is $\triangleq$<br>relator's<br>$F(RS) = F(R)F(S)$ at<br>$F := E$ |
| $= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) P[A] \xrightarrow{E\left(\frac{\text{prefix sum}}{\exists}\right)} PPA \xrightarrow{E(\text{est}(\geq))} PA \xrightarrow{\text{est}(\geq)} A$ | $R R \sqsubseteq R$<br>$\Rightarrow P(\text{est}(R)) \text{est}(R)$<br>$= P(\text{dom}(\text{est}(R)))$<br>union $\text{est}(R)$ , the<br>sets non-empty                                                                                                                                                                                                                                                                                                 |
| $= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) P[A] \xrightarrow{E\left(\frac{\text{prefix sum}}{\exists} \text{est}(\geq)\right)} PA \xrightarrow{\text{est}(\geq)} A$                      | relator<br>$\triangleq$ relator —<br>$F(RS) = F(R)F(S)$ at<br>$F := E$                                                                                                                                                                                                                                                                                                                                                                                   |

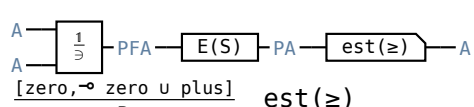
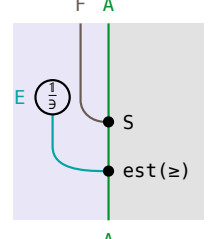
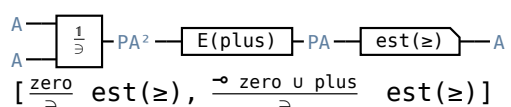
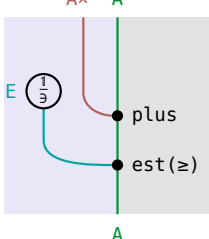
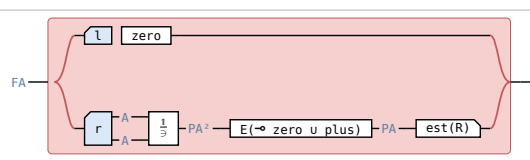
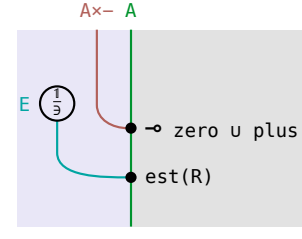
The union step is  $R R \sqsubseteq R$

$\Rightarrow P(\text{est}(R)) \text{est}(R) = P(\text{dom}(\text{est}(R))) \text{union est}(R)$  's  $P(\text{est}(R)) \text{est}(R) = P(\text{dom}(\text{est}(R))) \text{union est}(R)$  — every suffix has the empty prefix, so Dom is 1 here — and  $P(f) = E(f)$  at the map it is applied to (Map(f))  
 $\Rightarrow P(f) = E(f)$ .

| $[nil, \multimap nil \cup cons] \text{ sum} = F(\text{sum}) \text{ S}$ <span style="float: right;">(7.2.4c)</span>                                                                                |                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fusion: <b>prefix</b> expressed as a catamorphism on cons-lists, $([nil, \multimap nil \cup cons])$ , and this is the fusion condition that expresses <b>prefix sum</b> as a catamorphism         |                                                                                                                                                                                                                 |
| <b>circuit</b> — the fork is the bracket's case split $F([A])=1+A \times [A]$ : nil above, the pair and its $u$ below                                                                             | <b>Hinze–Marsden</b>                                                                                                                                                                                            |
|  $[nil, \multimap nil \cup cons] \text{ sum}$                                                                    |  <p>the <b>cons</b> operand of <math>\multimap nil \cup cons</math></p>                                                      |
|  $[nil \text{ sum}, \multimap nil \text{ sum} \cup cons \text{ sum}]$<br>coproduct of maps, composition over $u$ |                                                                                                                                                                                                                 |
|  $[zero, \multimap zero \cup (1 \times sum) \text{ plus}]$<br>sum's defining equation                          |  <p>the <math>(1 \times sum)</math> <b>plus</b> operand of <math>\multimap zero \cup (1 \times sum)</math> <b>plus</b></p> |
|  $[zero, (1 \times sum) (\multimap zero \cup plus)]$<br>$(1 \times sum) \multimap = \multimap$ , sum entire    |                                                                                                                                                                                                                 |
|  $F(\text{sum}) [zero, \multimap zero \cup plus]$<br>relator                                                   |                                                                                                                                                                                                                 |

fusion at  $R := [nil, \multimap nil \cup cons]$ ,  $S := \text{sum}$ : the side condition, so **prefix sum**  $= ([zero, \multimap zero \cup plus])$ . **prefix** is the reduce, **sum** the map fused into it — the intermediate list is gone.

| <b>MonoAlg(S, ≥)</b> <span style="float: right;">(7.2.4d)</span><br>monotonic algebra: an <b>F</b> -algebra <b>S</b> is monotonic on a relation <b>R</b> if $F(R) \text{SES} R$ — the <b>plus</b> branch of $F(\geq) \text{SES} \geq$ ;<br>the <b>zero</b> branch is $\text{zero} \text{E} \text{zero} \geq$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>circuit</b> — the head above, the running sum below; the tape is the <b>u</b><br>$(1 \times \geq) (\neg \text{zero} \cup \text{plus}) \quad ((1 \times \geq) \neg \text{zero}) \cup ((1 \times \neg \text{zero} \cup (\text{plus} \geq) \neg \text{zero} \cup \text{plus})$<br> <p style="text-align: center;">relator, composition over <b>u</b></p> <p style="text-align: right;">(0.8a), <math>(\geq \times \geq) \text{plus} \text{E} \text{plus} \geq</math>; <math>\geq</math> reflexive<br/> <math>(\leq \times \leq) \text{plus} \text{E} \text{plus} \leq</math> is <math>\triangleq</math><br/>         monotonic, written + there,<br/>         and <b>plus</b> is a map, so it is<br/>         monotonic on an order and on<br/>         its opposite together, which<br/>         carries it to <math>\geq</math>.</p> |  |

| $\frac{S}{S} \text{est}(\geq) = [\text{zero}, \oplus]$ <span style="float: right;">(7.2.4e)</span><br>the largest sum the algebra offers is zero from nothing and, from a head and a running sum, the larger of zero<br>and the head added to it                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>circuit</b> — the tape is the coproduct:<br>zero's branch above, plus's below<br> <p style="text-align: center;"><math>\frac{[\text{zero}, \neg \text{zero} \cup \text{plus}]}{S} \text{est}(\geq)</math></p>                                                                                                                                                                                                                      | <b>Hinze–Marsden</b><br> <p style="text-align: center;">the <b>zero</b> arm of the bracket</p>                                                                                                               |
|  <p style="text-align: center;"><math>\frac{[\frac{\text{zero}}{S} \text{est}(\geq), \frac{\neg \text{zero} \cup \text{plus}}{S} \text{est}(\geq)]}{S}</math></p> <p>= coproduct of maps — <math>[R, S] = [\frac{R}{S}, \frac{S}{S}] \ni</math> at<br/> <math>T := [\text{zero}, \neg \text{zero} \cup \text{plus}]</math>, then<br/> <math>[U, V]Z = [UZ, VZ] \text{ — } [R, S] Z = [R Z, S Z]</math>, composition over <b>u</b></p> |  <p style="text-align: center;">the <b>plus</b> operand of the lower arm's <math>\neg \text{zero} \cup \text{plus}</math>, under its <math>\frac{1}{S} E(\dots)</math> and <math>\text{est}(\geq)</math></p> |
|  <p>= singleton, <math>\geq</math> reflexive — <math>\frac{1}{S} \text{est}(R) = 1 \cap R^\circ</math><br/>         's <math>\frac{1}{S} \text{est}(R) = 1 \cap R</math> at <math>R := \geq</math>, <b>zero</b> a map; the<br/>         lower branch is <math>\oplus</math>'s definition, <math>\triangleq \text{mss}</math>, and no<br/>         law</p>                                                                             |                                                                                                                                                                                                              |

with  $\text{MonoAlg}(S, \geq)$  the greedy theorem gives  $([\text{zero}, \oplus]) \text{E} \frac{\text{prefix sum}}{S} \text{est}(\geq)$  — and Exercise 7.40  $\text{mss} = (k) \pi_2 \text{est}(\geq)$  makes it an equality.

$$[\text{nil}, \text{cons}] \langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle) \rangle = F(\langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle) \rangle) k \quad (7.2.4f)$$

$k \triangleq [\text{zero} \langle 1, \frac{1}{\exists} \rangle, \langle w, \langle w \frac{1}{\exists}, \pi_2 \pi_2 \rangle \text{cup} \rangle]$ ,  $w \triangleq (1 \times \pi_1) \oplus$ : the value at the whole list, paired with the set of the values at its suffixes, runs  $k$ 's recursion.

the equation at that branch

$$\text{nil}(g, \frac{\text{suffix}}{\exists} E(g)) = \text{zero} \langle 1, \frac{1}{\exists} \rangle$$

$$\text{cons}(g, \frac{\text{suffix}}{\exists} E(g)) = (1 \times \langle g, \frac{\text{suffix}}{\exists} E(g) \rangle) \langle w, \langle w \frac{1}{\exists}, \pi_2 \pi_2 \rangle \text{cup} \rangle$$

why

$\text{nil}$  has one suffix, itself, and  $g(\text{nil}) = \text{zero}$ , so the set is the singleton  $\{\text{zero}\}$

$g(\text{cons}(a, x)) = a \oplus (g(x))$ , which is  $w$  reading  $g(x)$  off  $\pi_1$ ; and the suffixes of  $\text{cons}(a, x)$  are  $\text{cons}(a, x)$  itself, whose value is that same  $w$ , together with those of  $x$ , which  $\pi_2 \pi_2$  carries — so the two sets meet at  $\text{cup}$

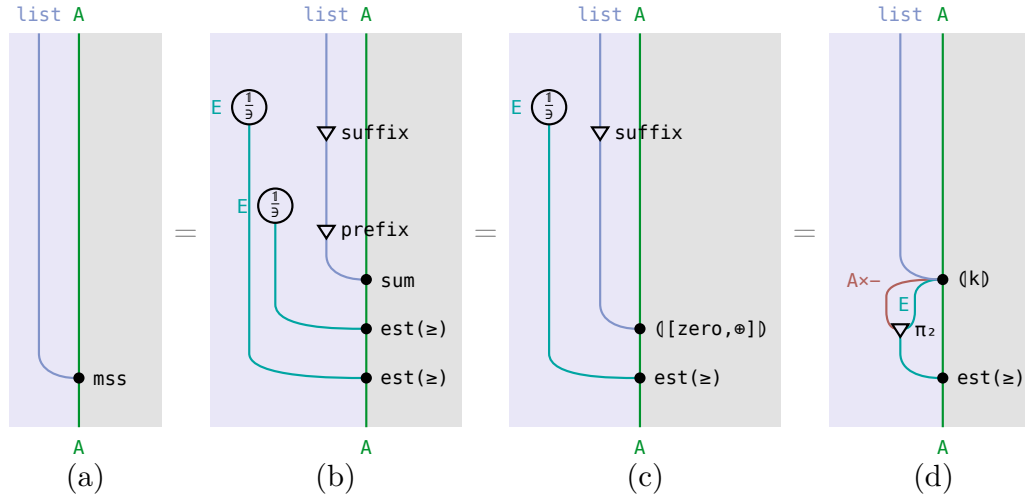
$$\text{mss} = \langle k \rangle \pi_2 \text{est}(\geq)$$

Exercise 7.40

maximum segment sum problem:  $\frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle)$  expressed as the catamorphism  $\langle k \rangle$ , hence

$\text{mss}$  implemented by a linear-time algorithm,  $\oplus \triangleq \text{zero} \vee \text{plus}$   $\text{est}(\geq) \triangleq \text{mss}$ ;  $k$  and  $w$  —

$$[\text{nil}, \text{cons}] \langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle) \rangle = F(\langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle) \rangle) k.$$



|     |                                                                                                          |                                                                                                                              |
|-----|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{mss}$                                                                                             | $(7.2.4b) \text{mss} = \frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists} \text{est}(\geq)) \text{est}(\geq)$ |
| (b) | $= \frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists} \text{est}(\geq)) \text{est}(\geq)$ | $\frac{\text{prefix sum}}{\exists} \text{est}(\geq) = \langle [\text{zero}, \oplus] \rangle$                                 |
| (c) | $= \frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle) \text{est}(\geq)$              | $\langle k \rangle \pi_2 = \frac{\text{suffix}}{\exists} E(\langle [\text{zero}, \oplus] \rangle)$                           |
| (d) | $= \langle k \rangle \pi_2 \text{est}(\geq)$                                                             |                                                                                                                              |

one fold builds the  $n+1$  running maxima and the final  $\text{est}(\geq)$  reads them in one more pass, so  $\text{mss}$  is linear.

§7.2.5.  $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$  —  $\text{filter}(p)$  keeps the items satisfying  $p$ , as one fold

(7.2.5a)

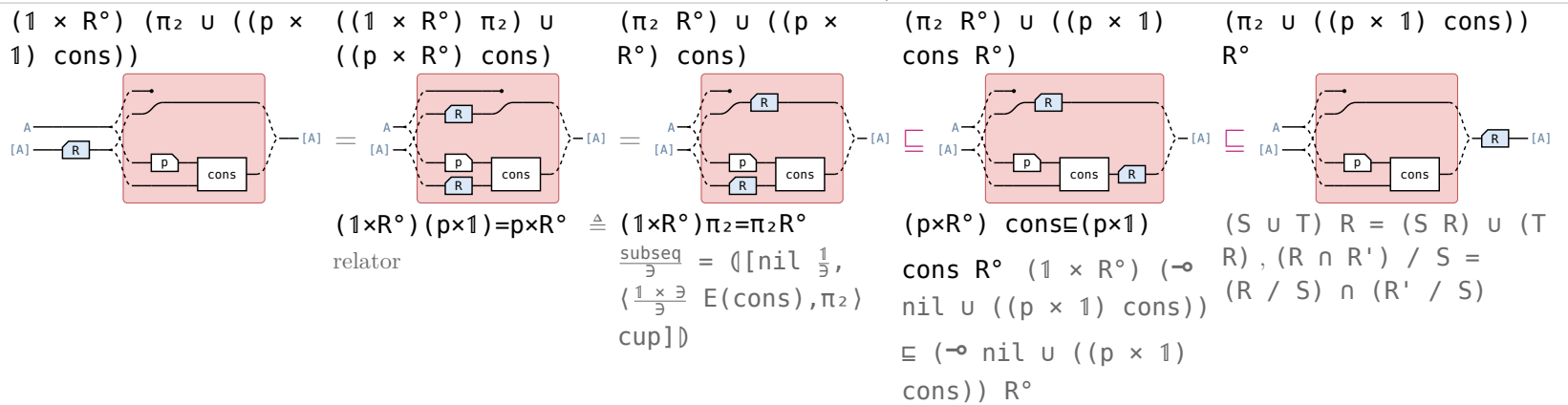
| name                                         | definition                                                                                                                                          | type                           | example                                                              | in words                                                                                                                                                                                    |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $F, \alpha, p, R$                          | as in $FX = L + E \times X, \alpha = [\text{nil}, \text{cons}], p \triangleq \text{corefl}(p), R \triangleq \text{length} \leq \text{length}^\circ$ |                                |                                                                      |                                                                                                                                                                                             |
| 2 $\pi_2$                                    | $\pi_2$ where $\rightarrow \text{nil} \triangleq \rightarrow \text{nil}$ has $\rightarrow \text{nil}$                                               | $A \times [A] \rightarrow [A]$ | $\pi_2(3, [1, 2]) = [1, 2]$                                          | drop the head, keep the tail                                                                                                                                                                |
| 3 $\text{subseq}$                            | $([\text{nil}, \pi_2 \cup \text{cons}])$<br>$\text{subseq} = ([\text{nil}, \text{cons} \cup \pi_2])$                                                | $[A] \rightarrow [A]$          | $[3, 1, 2] \text{ subseq } [3, 2]$                                   | $xs \text{ subseq } ys \iff ys$ is $xs$ with elements dropped — at each $\text{cons}$ , drop the head or keep it                                                                            |
| 4 $S$                                        | $[\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}]$                                                                                                | $F[A] \rightarrow [A]$         | $(4, [2]) S [2]$ and $(4, [2]) S [4, 2]$ , but $(3, [2]) S [2]$ only | $\text{subseq}$ 's algebra with one extra $p$ — drop the head, or keep a head that passes $p$                                                                                               |
| 5 $1 \sqsubseteq \pi_2 R \text{ cons}^\circ$ |                                                                                                                                                     |                                |                                                                      | the tail is one shorter than the cons, so $\pi_2$ loses the $\text{est}(R^\circ)$ at every step — where $\rightarrow \text{nil} \triangleq \rightarrow \text{nil}$ 's loser is $\text{nil}$ |

**MonoAlg( $S, R^\circ$ )**

(7.2.5b)

shortening the tail and then taking the step lands inside taking the step and then shortening the result

**formula** — the  $\text{cons}$  branch of  $F(R^\circ) S \sqsubseteq SR^\circ$ ; **reason** under each circuit



the  $\text{nil}$  branch:  $\text{nil} \sqsubseteq \text{nil } R^\circ$ .

| $\frac{S}{\exists} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]$ (7.2.5c)                      |                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the longest of the lists the algebra allows is the <b>cons</b> where the head passes <b>p</b> , and the tail where it does not |                                                                                                                                                                                                                                                                  |
| formula                                                                                                                        | reason                                                                                                                                                                                                                                                           |
|                                                                                                                                |                                                                                                                                                                                                                                                                  |
|                                                                                                                                | $S = [\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}]$ $S \triangleq [\text{nil}, \pi_2 \cup ((p \times 1) \text{ cons})]$ — and the fraction of a coproduct of maps is the coproduct of their fractions $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \ni$ |
|                                                                                                                                | $\frac{\text{nil}}{\exists}$ is the singleton $\{\text{nil}\}$ , and $\text{est}(R^\circ)$ of a singleton is its element because $R^\circ$ is reflexive $\triangleq \text{est}$                                                                                  |
|                                                                                                                                | $\frac{\pi_2 \cup (p \times 1) \text{ cons}}{\exists} = \left\{ \frac{\pi_2}{\exists}, \frac{(p \times 1) \text{ cons}}{\exists} \right\} \text{cup} \triangleq \text{cup}$                                                                                      |
|                                                                                                                                | $1 \sqsubseteq \pi_2 R \text{ cons}^\circ$ (7.2.5a.5) — $xs \ R \text{ cons}(a, xs)$ , so $\text{est}(R^\circ)$ returns the <b>cons</b> where <b>p</b> a puts it in the set and <b>xs</b> where the set is $\{xs\} \triangleq \text{est}$                        |

the head is dropped, not the whole tail: that is the one place  $\pi_2$  shows against  $\frac{S}{\exists} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \multimap \text{nil})]$  's  $\multimap \text{nil}$ .

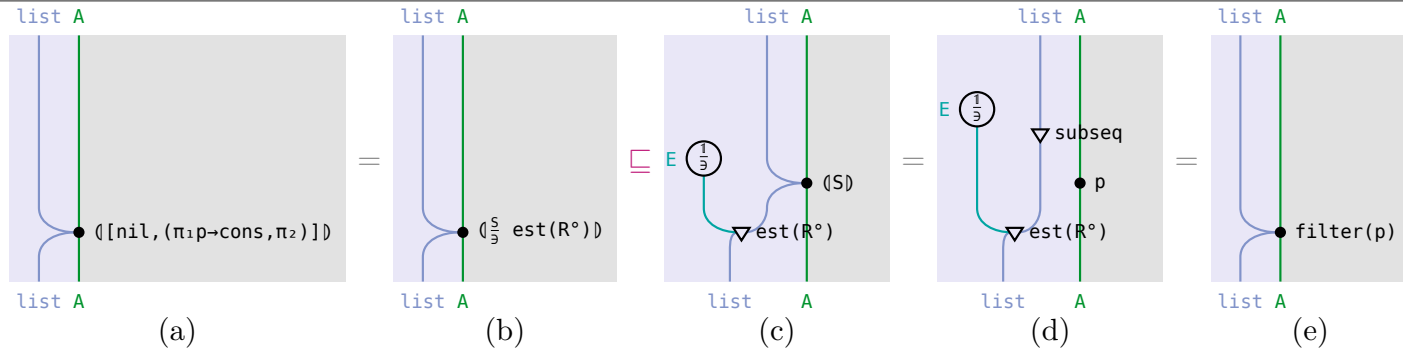
(7.2.5d)

| name | definition                                                                                                               | type                   | example                                                                          | in words                                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| S    | $[\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}]$ $S \triangleq [\text{nil}, \pi_2 \cup ((p \times 1) \text{ cons})]$ | $F[A] \rightarrow [A]$ | $(4, [2]) \ S \ [2]$ and $(4, [2]) \ S \ [4, 2]$ , but $(3, [2]) \ S \ [2]$ only | <b>subseq's</b> algebra with one extra <b>p</b> — drop the head, or keep a head that passes <b>p</b> |

**filter(p) = ([nil, (π<sub>1</sub>p→cons, π<sub>2</sub>)])**

Exercise 7.41

**filter: filter(p)** **x** returns the longest subsequence of **x** with the property that all its elements satisfy **p**; the catamorphism is the standard program; **R** a preorder.



|     |                                                                                |                                                                                                                                       |
|-----|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\llbracket [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)] \rrbracket$ | $(7.2.5c) \frac{S}{\exists} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]$                             |
| (b) | $= \llbracket \frac{S}{\exists} \text{est}(R^\circ) \rrbracket$                | $\llbracket \frac{S}{\exists} \text{est}(R^\circ) \rrbracket \sqsubseteq \llbracket \frac{S}{\exists} \text{est}(R^\circ) \rrbracket$ |
| (c) | $\sqsubseteq \llbracket \frac{S}{\exists} \text{est}(R^\circ) \rrbracket$      | $\text{subseq list}(p) = \llbracket S \rrbracket$                                                                                     |
| (d) | $= \frac{\text{subseq list}(p)}{\exists} \text{est}(R^\circ)$                  |                                                                                                                                       |
| (e) | $= \text{filter}(p)$                                                           |                                                                                                                                       |

the catamorphism is entire and **filter(p)** simple, so  $\sqsubseteq$  is  $=$ .

$(\text{subseq list}(p))^\circ (\text{subseq list}(p)) \cap R \cap R^\circ \sqsubseteq 1$  fails — two **p**-subsequences of one list can be of equal length and different — so §7.2.3's uniqueness argument does not transfer. What survives  $\text{est}(R^\circ)$  is the one keeping **exactly** the passing elements: a subsequence that drops a passing element is beaten by the one that keeps it.

(7.2.5e)

|   | definition                                                                                 | type                  | note                                                                                                                                  |
|---|--------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1 | segment = suffix prefix                                                                    | $[A] \rightarrow [A]$ | A contiguous stretch of <b>xs</b> : a suffix, then a prefix of that.                                                                  |
| 2 | $\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\exists} \text{est}(R^\circ)$    | $[A] \rightarrow [A]$ | The longest subsequence of <b>xs</b> whose every element passes <b>p</b> .<br>$\text{est}(R^\circ)$ is $\triangleq \text{est}$        |
| 3 | $R \triangleq \text{length} \leq \text{length}^\circ$                                      | $[A] \rightarrow [A]$ | The preorder <b>filter</b> and <b>takewhile</b> maximise over: the longer list wins. $\geq \triangleq \leq^\circ$                     |
| 4 | $\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\exists} \text{est}(R^\circ)$ | $[A] \rightarrow [A]$ | The same with <b>prefix</b> for <b>subseq</b> : the longest prefix whose every element passes <b>p</b> .                              |
| 5 | $\text{mss} \triangleq \frac{\text{segment sum}}{\exists} \text{est}(\geq)$                | $[A] \rightarrow A$   | Maximum segment sum. <b>segment=suffix prefix</b> splits it into $\frac{\text{prefix sum}}{\exists} \text{est}(\geq)$ on each suffix. |

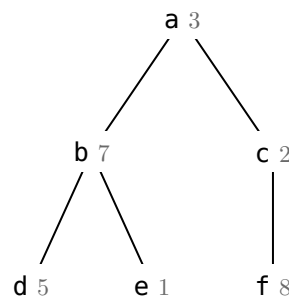
## §7.3. Planning a company party

(7.3a)

|    | definition                                                                      | type                                              | note                                                                                           |
|----|---------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1  | $\text{tree } A ::= \text{node } (A, [\text{tree } A])$                         | $\mathcal{A} \rightarrow \mathcal{A}$             | The company hierarchy: an employee, and the list of subtrees under them.                       |
| 2  | $F(A, B) = A \times [B]$                                                        | $\mathcal{A} \rightarrow \mathcal{A}$             | The base functor <b>tree</b> folds: an employee beside the recursive position, one layer deep. |
| 3  | rating                                                                          | $A \rightarrow \text{Int}$                        | What one employee is worth as a guest.                                                         |
| 4  | $\text{cost} = \text{list}(\text{rating}) \text{ sum}$                          | $[A] \rightarrow \text{Int}$                      | What a guest list is worth.                                                                    |
| 5  | $R = \text{cost} \leq \text{cost}^\circ$                                        | $[A] \rightarrow [A]$                             | The preorder the guest list is maximised over.                                                 |
| 6  | $\text{choose} \triangleq \pi_1 \cup \pi_2$                                     | $[A] \times [A] \rightarrow [A]$                  | Takes one of the two parties a subtree returns.                                                |
| 7  | $\text{include} = (1 \times (\text{list}(\pi_2) \text{ concat})) \text{ cons}$  | $F(A, [A] \times [A]) \rightarrow [A]$            | The party that invites the root, which puts every immediate subtree's root out. A map.         |
| 8  | $\text{exclude} = (1 \times (\text{list}(\text{choose}) \text{ concat})) \pi_2$ | $F(A, [A] \times [A]) \rightarrow [A]$            | The party that leaves the root out, so each subtree is free to choose. Not a map.              |
| 9  | $S \triangleq (\text{include}, \text{exclude})$                                 | $F(A, [A] \times [A]) \rightarrow [A] \times [A]$ | The algebra: one step returns both parties of a subtree at once.                               |
| 10 | $\text{party} = \langle S \rangle \text{ choose}$                               | $\text{tree}(A) \rightarrow [A]$                  | Every guest list the president's ruling allows.                                                |
| 11 | the specification<br>$\frac{\text{party}}{\exists} \text{ est}(R^\circ)$        | $\text{TA} \rightarrow [A]$                       | A guest list of greatest total conviviality.                                                   |

## §7.3.1. include and exclude

(7.3.1a)



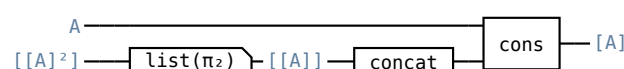
the small grey number is the employee's rating

(7.3.1b)

|   | node    | the algebra's input<br>$A \times [[A] \times [A]]$ | include                                         | exclude                                |
|---|---------|----------------------------------------------------|-------------------------------------------------|----------------------------------------|
| 1 | d, e, f | $(d, [])$                                          | $[d]$                                           | $[]$                                   |
| 2 | b       | $(b, ([d], []), ([e], []))$                        | $[b]$                                           | $[d, e], [d], [e], []$                 |
| 3 | c       | $(c, ([f], []))$                                   | $[c]$                                           | $[f], []$                              |
| 4 | a       | $(a, ([b], \dots), ([c], \dots))$                  | $[a] \# \text{exclude}(b) \# \text{exclude}(c)$ | $\text{choose}(b) \# \text{choose}(c)$ |

with those ratings,  $\text{est}(R^\circ)$  keeps  $([b]=7, [d, e]=6)$  at **b**, where **include** wins, and  $([c]=2, [f]=8)$  at **c**, where **exclude** wins, so at the root  $\text{include}=[a, d, e, f]=17$  beats  $\text{exclude}=[b, f]=15$ , and **choose** takes 17.

(7.3.1c)



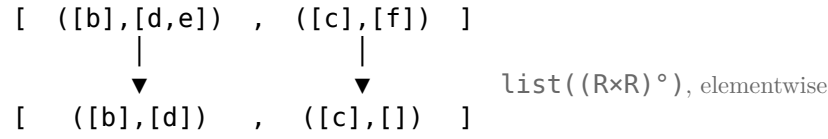


(7.3.1d)

list(choose) [([d],[ ]), ([e],[ ])]  
 1st item ([d],[ ]) choose→[d] or [ ]  
 2nd item ([e],[ ]) choose→[e] or [ ]  
 2×2=4 combinations: [[d],[e]] [[d],[ ]] [[],[e]] [[],[ ]]  
 concat flattens: [d,e] [d] [e] [ ]

(7.3.1e)

### §7.3.2. list((R×R)°)



(7.3.2a)

([b],[d,e])  
 | — exclude: the best party in b's subtree when b does not come  
 | — include: the best party in b's subtree when b comes

(7.3.2b)

| element            | 1st component<br>include         | 2nd component<br>exclude                                              |
|--------------------|----------------------------------|-----------------------------------------------------------------------|
| 1 1st, ([b],[d,e]) | [b] R° [b]<br>7 ≥ 7, reflexivity | [d,e] R° [d]<br>6 ≥ 5, and ([b],[d,e]) is the pair est(R°) keeps at b |
| 2 2nd, ([c],[f])   | [c] R° [c]<br>2 ≥ 2, reflexivity | [f] R° [ ]<br>8 ≥ 0, and ([c],[f]) is the pair est(R°) keeps at c     |

the first component of  $1 \times \text{list}((R \times R)^\circ)$  is the root employee.

$R \triangleq \text{cost} \leq \text{cost}^\circ$  is a preorder, so the components that do not move are related by reflexivity —  $(R \times R)^\circ$  demands a relation in **both** components, it cannot skip one.

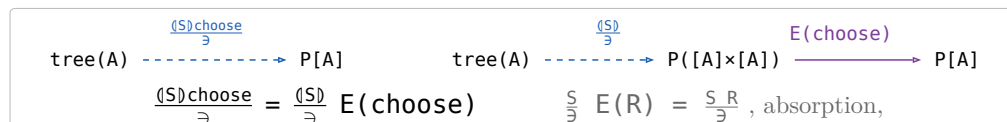
$R$  compares cost only, not contents:  $[f] R^\circ [ ]$  is legal though  $[f]$  has an element  $[ ]$  does not. That is why the three leaves of §7.3.3's proof all reduce to "**cost** is a sum".

### §7.3.3. $(1 \times \text{list}((R \times R)^\circ)) S \sqsubseteq S (R \times R)^\circ$ — $S$ is monotonic on $(R \times R)^\circ$

(7.3.3a)



(7.3.3b)



$$\begin{aligned}
S &\triangleq \{\text{include}, \text{exclude}\} \\
(1 \times \text{list}((R \times R)^\circ)) S &\sqsubseteq S (R \times R)^\circ \\
(1 \times \text{list}((R \times R)^\circ)) \text{include} &\sqsubseteq \text{include } R^\circ \\
(1 \times \text{list}((R \times R)^\circ)) \text{exclude} &\sqsubseteq \text{exclude } R^\circ
\end{aligned}
\tag{7.3.3c}$$

bettering both parties of every subtree before the node's algebra runs gets no further than running it first and bettering the two parties it returns,



|     |                                                                                           |  |
|-----|-------------------------------------------------------------------------------------------|--|
| (a) | $(1 \times \text{list}((R \times R)^\circ)) (1 \times (\text{list}(g) \text{ concat})) h$ |  |
| (b) | $\sqsubseteq (1 \times (\text{list}(g R^\circ) \text{ concat})) h$                        |  |
| (c) | $\sqsubseteq (1 \times (\text{list}(g) \text{ concat } R^\circ)) h$                       |  |
| (d) | $\sqsubseteq (1 \times (\text{list}(g) \text{ concat})) h R^\circ$                        |  |

$g \quad (R \times R)^\circ g \sqsubseteq g R^\circ$   
 $g := \pi_2$  is  $(R \times R)^\circ \pi_2 = (\text{dom}(\pi_1 R^\circ)) \pi_2 R^\circ \sqsubseteq \pi_2 R^\circ$ ,  $g := \pi_1$  its mirror  $(\text{dom}(\pi_2 R^\circ)) \pi_1 R^\circ \sqsubseteq \pi_1 R^\circ$  —  $\triangleq \times$  and  $\langle R, S \rangle \pi_1 = \text{dom}(S) R$ ,  $\langle R, S \rangle \pi_2 = \text{dom}(R) S$ , then  $\text{dom} \sqsubseteq 1$ ;  $g := \text{choose} \triangleq \pi_1 \cup \pi_2$  is the union of the two  $G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$ .  $\text{list}$  monotonic,  $\triangleq$  relator  
 $\text{concat} \quad \text{list}(R^\circ) \text{ concat} \sqsubseteq \text{concat } R^\circ$   
 a LEAF: no law above it.  $\text{cost}$  is a sum, so  $\text{cost}(\text{concat}(xss)) = \text{sum}(\text{list}(\text{cost})(xss))$  and a cheaper part makes a cheaper whole.  
 $h \quad (1 \times R^\circ) h \sqsubseteq h R^\circ$   
 a LEAF:  $\text{cost}(\text{cons}(a, xs)) = \text{rating}(a) + \text{cost}(xs)$ , so a cheaper tail makes a cheaper list. For  $h := \pi_2$  it is an EQUALITY  $(1 \times R^\circ) \pi_2 = (\text{dom}(\pi_1)) \pi_2 R^\circ = \pi_2 R^\circ$ :  $\pi_1$  is a map, hence entire, so  $\text{dom}(\pi_1) = 1$  —  $\text{Entire}(R) \triangleq \text{dom}(R) = 1$

|                                                |                 |               |
|------------------------------------------------|-----------------|---------------|
| $(1 \times (\text{list}(g) \text{ concat})) h$ | $g$             | $h$           |
| $\text{include}$                               | $\pi_2$         | $\text{cons}$ |
| $\text{exclude}$                               | $\text{choose}$ | $\pi_2$       |

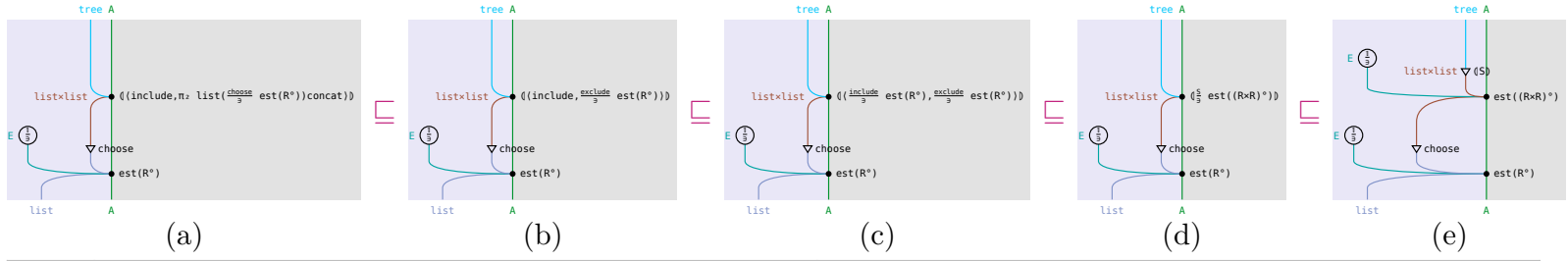
(7.3.3d)

|   |                                                        |                                                                                                                                     |
|---|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $R \triangleq \{(0,0)\} : \{0,1\} \rightarrow \{0,1\}$ | not entire — undefined at 1                                                                                                         |
| 2 | $(R \times R) \text{ choose at } (0,1)$                | nothing — $R \times R$ needs BOTH components related, and $R$ is undefined at 1                                                     |
| 3 | $\text{choose } R \text{ at } (0,1)$                   | 0 — choose picks the first component 0, and $R \ 0 \ 0$ holds                                                                       |
| 4 | $((0,1), 0)$                                           | in choose $R$ , not in $(R \times R) \text{ choose}$                                                                                |
| 5 | $(R \times R) \text{ choose} = \text{choose } R$       | iff $R$ is entire — $\text{dom} \sqsubseteq 1$ is the $g$ row's only inequality step, and it is an equality iff $\text{dom}(R) = 1$ |

#### §7.3.4. The derivation

$$(\langle \text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{3} \text{ est}(R^\circ)) \text{ concat} \rangle) \frac{\text{choose}}{3} \text{ est}(R^\circ) \sqsubseteq \frac{\text{party}}{3} \text{ est}(R^\circ) \quad (7.3.4a)$$

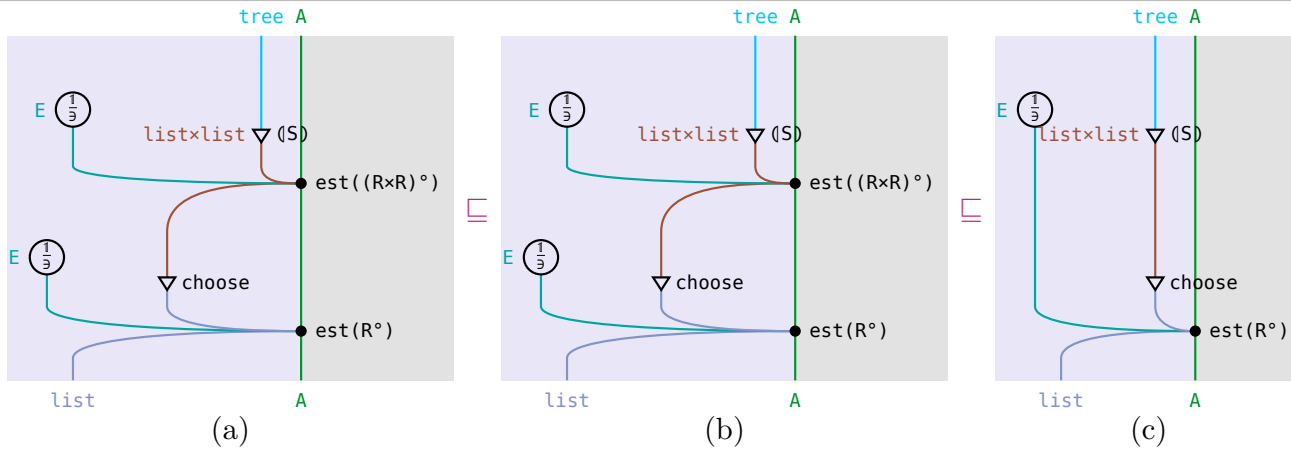
the best of every guest list the president allows is one pass up the tree, each subtree handing up its best party with its boss in and its best with the boss out, and **choose** taking the better of the two at the root



|     |                                                                                                                                                                           |                                                                                                                                                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(\langle \text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{3} \text{ est}(R^\circ)) \text{ concat} \rangle) \frac{\text{choose}}{3} \text{ est}(R^\circ)$          | $\pi_2 \text{ list}(\frac{\text{choose}}{3} \text{ est}(R^\circ)) \text{ concat} \sqsubseteq \frac{\text{exclude}}{3} \text{ est}(R^\circ)$              |
| (b) | $\sqsubseteq (\langle \text{include}, \frac{\text{exclude}}{3} \text{ est}(R^\circ) \rangle) \frac{\text{choose}}{3} \text{ est}(R^\circ)$                                | $\text{include} \sqsubseteq \frac{\text{include}}{3} \text{ est}(R^\circ)$                                                                               |
| (c) | $\sqsubseteq (\langle \frac{\text{include}}{3} \text{ est}(R^\circ), \frac{\text{exclude}}{3} \text{ est}(R^\circ) \rangle) \frac{\text{choose}}{3} \text{ est}(R^\circ)$ | $(\frac{\text{include}}{3} \text{ est}(R^\circ), \frac{\text{exclude}}{3} \text{ est}(R^\circ)) \sqsubseteq \frac{S}{3} \text{ est}((R \times R)^\circ)$ |
| (d) | $\sqsubseteq (\frac{S}{3} \text{ est}((R \times R)^\circ)) \frac{\text{choose}}{3} \text{ est}(R^\circ)$                                                                  | $(\frac{S}{3} \text{ est}((R \times R)^\circ)) \sqsubseteq \frac{(S)}{3} \text{ est}((R \times R)^\circ)$                                                |
| (e) | $\sqsubseteq \frac{(S)}{3} \text{ est}((R \times R)^\circ) \frac{\text{choose}}{3} \text{ est}(R^\circ)$                                                                  |                                                                                                                                                          |

(7.3.4b)

1



|       |                                                                                                          |                                                                                                                                 |
|-------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1 (a) | $\frac{(S)}{3} \text{ est}((R \times R)^\circ) \frac{\text{choose}}{3} \text{ est}(R^\circ)$             | $\text{est}((R \times R)^\circ) \frac{\text{choose}}{3} \text{ est}(R^\circ) \sqsubseteq E(\text{choose}) \text{ est}(R^\circ)$ |
| 2 (b) | $\sqsubseteq \frac{(S)}{3} \text{ est}((R \times R)^\circ) \frac{\text{choose}}{3} \text{ est}(R^\circ)$ | $\frac{(S)}{3} \frac{\text{choose}}{3} = \frac{(S)}{3} E(\text{choose})$                                                        |
| 3 (c) | $\sqsubseteq \frac{(S)}{3} \frac{\text{choose}}{3} \text{ est}(R^\circ)$                                 |                                                                                                                                 |

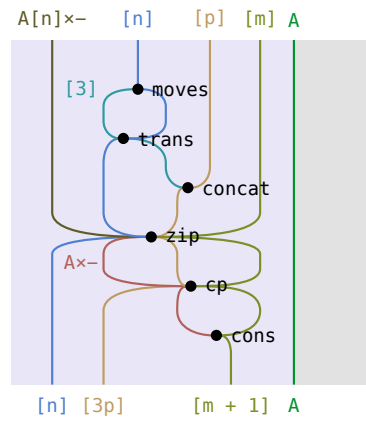
## §7.4. Shortest paths on a cylinder

(7.4a)

|                                                                        | on $L = \text{list}^+$ , 10.3.3                                                                                                     | on $\text{Vec}$                                                                                                                                                                    | note                                                                                                                                           |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 F                                                                    | $F(A, X) = A + A \times X$                                                                                                          | $F(A, A[m]) = A[1] + A[m+1]$                                                                                                                                                       | A square alone, or in front of the path so far.                                                                                                |
| 2 the input                                                            | $L(N\ A)$                                                                                                                           | $A[m][n] \triangleq (A[n])[m]$                                                                                                                                                     | $m$ columns of $n$ squares, outermost index first — the matrix itself.                                                                         |
| 3 a path                                                               | $L\ A$ , a non-empty list                                                                                                           | $A[1] \triangleq A$ , $A[m+1] \triangleq A \times A[m]$                                                                                                                            | Through $m$ columns, $m$ squares: the count is in the type. A one-square path is the square; $\text{Vec}(m)$ is the functor $X \mapsto X[m]$ . |
| 4 the paths into a row                                                 | $P(L\ A)$                                                                                                                           | $A[p][m]$ , $p = 3^{m-1}$                                                                                                                                                          | Three moves per column crossed, so this count is in the type too; a set forgets it.                                                            |
| 5 $\alpha$                                                             | $[\text{wrap}, \text{cons}]$<br>$F(A, LA) \rightarrow LA$                                                                           | $\text{wrap} = 1 : A \rightarrow A[1]$<br>$\text{cons} : A \times A[m] \rightarrow A[m+1]$                                                                                         | $\text{wrap}$ is an identity; $\text{cons}$ is only an iso, so it stays a bead.                                                                |
| 6 moves                                                                | $NA \rightarrow PNA$                                                                                                                | $X[n] \rightarrow X[3][n]$                                                                                                                                                         | $\text{moves}(x) = (\text{up}(x), x, \text{down}(x))$ , a $3 \times n$ matrix.                                                                 |
| 7 trans                                                                | $PNX \rightarrow NPX$                                                                                                               | $X[3][n] \rightarrow X[n][3]$                                                                                                                                                      | The transpose.                                                                                                                                 |
| 8 union                                                                | $PPX \rightarrow PX$                                                                                                                | $\text{concat} : X[j][k] \rightarrow X[jk]$                                                                                                                                        | The $j$ rows laid end to end; $j$ is 3 inside $\text{gen}$ and $n$ inside $\text{paths}$ .                                                     |
| 9 zip                                                                  | $F(NA, NB) \rightarrow NF(A, B)$                                                                                                    | $F(A[n], B[n]) \rightarrow F(A, B)[n]$                                                                                                                                             | Unchanged.                                                                                                                                     |
| 10 cp                                                                  | $F(A, PB) \rightarrow P(F(A, B))$                                                                                                   | $A \times B[p] \rightarrow (A \times B)[p]$                                                                                                                                        | The square paired with each of the $p$ candidates.                                                                                             |
| 11 $\text{est}(R) \triangleq \exists n (\in \text{PPX}) \rightarrow X$ |                                                                                                                                     | $X[k] \rightarrow X$ , $\exists : X[k] \rightarrow X$                                                                                                                              | A component that is $R$ -below every component; $\exists$ is “is some component” and $\in$ its converse.                                       |
| 12 $R \triangleq \text{sum} \leq \text{sum}^\circ$                     | $L\ \text{Nat} \rightarrow L\ \text{Nat}$                                                                                           | $\text{Nat}[m] \rightarrow \text{Nat}[m]$                                                                                                                                          | The cost of a path, which the cheapest minimises — one order per length $m$ .                                                                  |
| 13 $\text{gen}$                                                        | $F(1, \text{moves trans } N(\text{union}))$<br>$\text{zip } N(\text{cp } P(\alpha))$<br>$F(NA, N(P(LA)))$<br>$\rightarrow N(P(LA))$ | $F(1, \text{moves trans } \text{Vec}(n)(\text{concat}))$<br>$\text{zip } \text{Vec}(n)(\text{cp } \text{Vec}(3p)(\text{cons}))$<br>$F(A[n], A[n][p][m]) \rightarrow A[n][3p][m+1]$ | The same composite without $\alpha$ ; the type shown is the $\text{cons}$ side, the $\text{wrap}$ side is $1 : A[n] \rightarrow A[n]$ .        |
| 14 $(\text{gen})$                                                      | $L(N\ A) \rightarrow N(P(L\ A))$                                                                                                    | $A[m][n] \rightarrow A[n][p][m]$                                                                                                                                                   | Columns in, rows out.                                                                                                                          |
| 15 $\text{paths} \triangleq (\text{gen})$<br>$\text{concat}$           | $(\text{gen})$ setify union<br>$L\ N\ \text{Nat} \rightarrow P(L\ \text{Nat})$                                                      | $A[m][n] \rightarrow A[np][m]$                                                                                                                                                     | The $n$ rows of $p$ paths laid end to end: every path of the cylinder, in one row.                                                             |
| 16 $Q$ , 7.4.3                                                         | $F(1, \text{moves trans } N(\text{est}(R)))$<br>$\text{zip } N(\alpha)$                                                             | $F(1, \text{moves trans } \text{Vec}(n)(\text{est}(R)))$<br>$\text{zip}, F(A[n], A[n][m]) \rightarrow A[n][m+1]$                                                                   | The cheapest of three; no $p$ anywhere. $(Q) : A[m][n] \rightarrow A[n][m]$ has the type of a transpose.                                       |
| 17 the specification<br>paths<br>$\text{est}(R)$                       | $L\ N\ \text{Nat} \rightarrow L\ \text{Nat}$                                                                                        | $A[m][n] \rightarrow A[m]$ , $\text{est}(R) : X[np] \rightarrow X$                                                                                                                 | A cheapest path from the entry side to the exit side.                                                                                          |

§7.4.1.  $\text{gen} \triangleq (1 \times (\text{moves trans } \text{Vec}(\text{concat}))) \text{ zip } \text{Vec}(\text{cp } \text{Vec}(\text{cons}))$  — one more column: each square’s paths go to its three neighbours, then the square goes in front

(7.4.1a)



(7.4.1b)

a row is the outer index, a cell one path

$u = ((1,2,3,4), C)$

$C = 5$

6

7

8

$(1 \times (\text{moves trans Vec}(n)(\text{concat})))$

$\text{moves}(C)$

$= 6 \ 7 \ 8 \ 5$

5 6 7 8

8 5 6 7

$\text{trans}(\text{that})$

$= 6 \ 5 \ 8$

7 6 5

8 7 6

5 8 7

$\text{Vec}(n)(\text{concat})(\text{that}) = \text{that}$

$\text{zip}(\text{that})$

$= (1, 6 \ 5 \ 8)$

(2, 7 6 5)

(3, 8 7 6)

(4, 5 8 7)

$\text{Vec}(n)(\text{cp})(\text{that})$

$= 1 \ 6 \ 2 \ 7 \ 3 \ 8 \ 4 \ 5$

1 5 2 6 3 7 4 8

1 8 2 5 3 6 4 7

:  $A[n] \times A[n]$

one candidate per row, one square long:

a column of squares

1 keeps the column, the candidates move down, unmoved, up: a  $3 \times n$  matrix

:  $A[3][n]$

the transpose: row  $k$  gets rows  $k-1, k, k+1$

:  $A[n][3]$

$\text{concat}: X[3][1] \rightarrow X[3]$  is 1

each row: its square, and its candidates

:  $(A \times A[3])[n]$

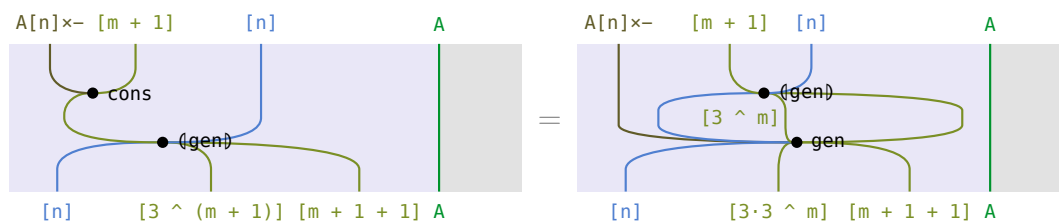
$\text{cp}$  pairs the square with each candidate:

row  $k$  of that is the  $k$ -th  $3 \times 2$  block, left to right; each line a two-square path

:  $A[n][3][2]$

**§7.4.2.  $\text{cons } (\text{gen}) = (1 \times (\text{gen})) \text{ gen}$  — folding a matrix with one more column is folding the rest, then one gen**

(7.4.2a)



the  $[m+1]$  bracket puts the column back on the matrix, then the fold reads all of it

the fold reads the rest under  $F$ , then one  $\text{gen}$  puts the column in front

(7.4.2b)

$xs = 1 \ 2 \ 3 \ 4$

5 6 7 8

:  $A[m][n], m=2$

$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})(5,6,7,8))$

$(\text{gen})(5,6,7,8)$

$= 5$

6

7

8

one column: each square its own one-square path

:  $A[n][1][1]$

both halves are in  $\text{gen} \triangleq (1 \times (\text{moves trans Vec}(\text{concat}))) \text{ zip Vec}(\text{cp Vec}(\text{cons}))$ : the tail folds to the column 5 6 7 8, one path per row and each of them one square long, and  $\text{gen}$  on it is that walk.

**§7.4.3.  $Q \triangleq (1 \times (\text{moves trans Vec}(n)(\text{est}(R)))) \text{ zip Vec}(n)(\text{cons})$  —  $Q$  is  $\text{gen}$  with  $\text{est}(R)$  choosing one path per square**

(7.4.3a)



(7.4.3b)

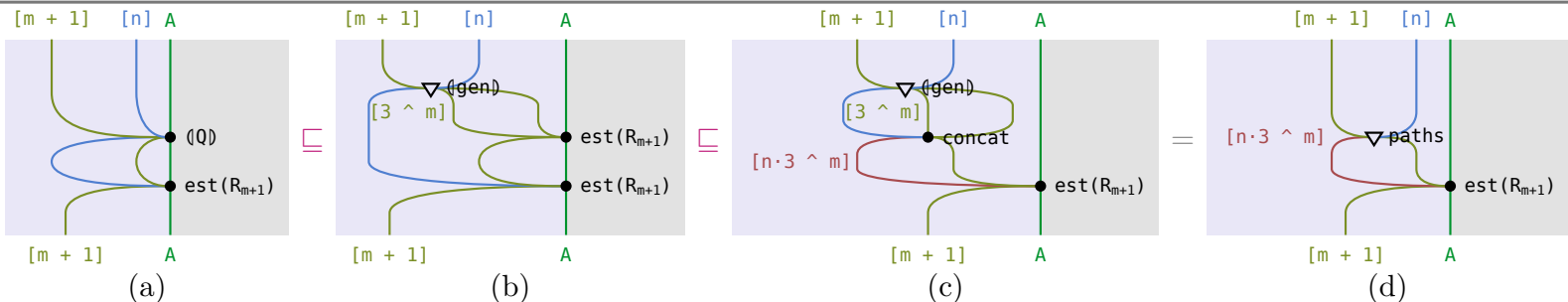
$u = ((1,2,3,4), C)$   
 $C = 5$   
 $6$   
 $7$   
 $8$   
 $(1 \times (\text{moves trans Vec}(n)(\text{est}(R))))$   
 $\text{moves}(C)$   
 $= 6 \ 7 \ 8 \ 5$   
 $5 \ 6 \ 7 \ 8$   
 $8 \ 5 \ 6 \ 7$   
 $\text{trans}(\text{that})$   
 $= 6 \ 5 \ 8$   
 $7 \ 6 \ 5$   
 $8 \ 7 \ 6$   
 $5 \ 8 \ 7$   
 $\text{Vec}(n)(\text{est}(R))(\text{that})$   
 $= 5 \ 5 \ 6 \ 5$   
 $\text{zip}(\text{that})$   
 $= 1 \ 5$   
 $2 \ 5$   
 $3 \ 6$   
 $4 \ 5$

$: A[n] \times A[n][m]$   
 one path per row, one square long  
  
 $1$  keeps the column, the paths move down, unmoved, up: a  $3 \times n$  matrix  
  
 $: A[3][n][m]$   
 the transpose: row  $k$  gets rows  $k-1, k, k+1$   
  
 $: A[n][3][m]$   
 the cheapest into each row – chosen BEFORE  
 $: A[n][m]$  the new square is put in front  
 each row: its square, and the one predecessor that survived; the  $[m+1]$  bracket is cons  
  
 $: A[n][m+1]$

**§7.4.4.**  $(\forall i. 1 \in R_i) \wedge (\forall i. R_i R_i \in R_i) \wedge (\forall i. (1 \times R_i) \text{ cons} \in \text{cons } R_{i+1}) \Rightarrow (Q) \text{ est}(R_{m+1}) \in \text{paths est}(R_{m+1})$   
 — folding  $Q$  gives a smallest of paths

$(\forall i. 1 \in R_i) \wedge (\forall i. R_i R_i \in R_i) \wedge (\forall i. (1 \times R_i) \text{ cons} \in \text{cons } R_{i+1}) \Rightarrow (Q) \text{ est}(R_{m+1}) \in \text{paths est}(R_{m+1})$  (7.4.4a)

a cheapest of all  $np$  paths of the cylinder is beaten by the greedy fold's one path per row and then a cheapest of those  $n$ , which costs  $O(n \times m)$ .



|     |                                                                            |                                                                                                                                                                                                        |
|-----|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(Q) \text{ est}(R_{m+1})$                                                 | $(\forall i. 1 \in R_i) \wedge (\forall i. R_i R_i \in R_i) \wedge (\forall i. (1 \times R_i) \text{ cons} \in \text{cons } R_{i+1})$<br>$\Rightarrow (Q) \in (Q) \text{ Vec}(n)(\text{est}(R_{m+1}))$ |
| (b) | $\sqsubseteq (Q) \text{ Vec}(n)(\text{est}(R_{m+1})) \text{ est}(R_{m+1})$ | $S S \in S$<br>$\Rightarrow \text{Vec}(j)(\text{est}(S)) \text{ est}(S) \in \text{concat est}(S)$                                                                                                      |
| (c) | $\sqsubseteq (Q) \text{ concat est}(R_{m+1})$                              | $f g = f g$                                                                                                                                                                                            |
| (d) | $= \text{paths est}(R_{m+1})$                                              |                                                                                                                                                                                                        |

$S S \in S \Rightarrow \text{Vec}(j)(\text{est}(S)) \text{ est}(S) \in \text{concat est}(S)$ ,  $R$  transitive

a cheapest of each of the  $j$  rows and then a cheapest of those  $j$  is a cheapest of all  $j$  entries laid end to end.

$(\forall i. 1 \in R_i) \wedge (\forall i. R_i R_i \in R_i) \wedge (\forall i. (1 \times R_i) \text{ cons} \in \text{cons } R_{i+1}) \Rightarrow (Q) \in (Q) \text{ Vec}(n)(\text{est}(R_{m+1}))$ ,  $R$  reflexive, transitive and monotonic

the one path per row the greedy fold keeps is one of the  $p$  that  $(Q) \text{ gen}$  generates for that row, and a cheapest of them.

$$R_m \ R_m \sqsubseteq R_m \wedge (1 \times R_m) \text{ cons } \sqsubseteq \text{ cons } R_{m+1} \Rightarrow (1 \times \text{Vec}(n)(\text{est}(R_m))) \ Q \sqsubseteq \text{gen Vec}(n)(\text{est}(R_{m+1})) \quad (7.13)$$

choosing a cheapest of each square's  $\mathbf{p}$  paths before the column is extended is no better than extending first and choosing among the  $3\mathbf{p}$ .



|     |                                                             |                                       |
|-----|-------------------------------------------------------------|---------------------------------------|
| (a) | $\text{gen Vec}(n)(\text{est}(R_{m+1}))$                    | (7.13); zip, trans, moves lax natural |
| (b) | $\sqsubseteq (1 \times \text{Vec}(n)(\text{est}(R_m))) \ Q$ |                                       |

$$(7.13) \text{ on Vec: } (1 \times R_m) \text{ cons } \sqsubseteq \text{ cons } R_{m+1} \Rightarrow (1 \times \text{est}(R_m)) \text{ cons } \sqsubseteq \text{cp Vec}(k)(\text{cons}) \text{ est}(R_{m+1}), R \text{ monotonic}$$

putting the new square in front of every one of the  $\mathbf{p}$  candidates and then choosing a cheapest is beaten by choosing a cheapest first and putting the square in front of that one.

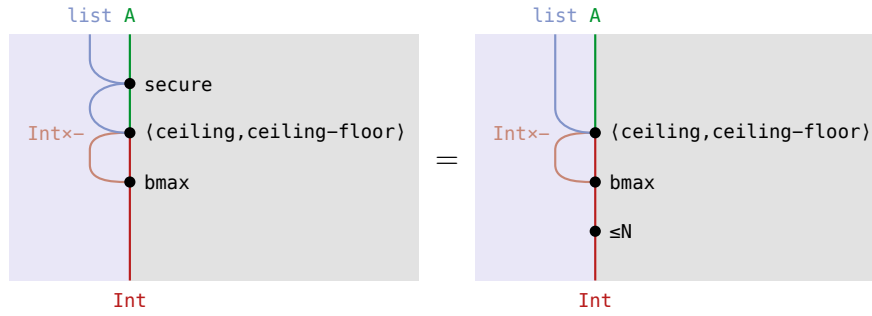
## §7.5. The security van problem

(7.5a)

|    | definition                                                                                                                                                                                                                                                                                                             | type                                        | example                                                                                                       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1  | $R \triangleq \text{length} \leq \text{length}^\circ$                                                                                                                                                                                                                                                                  | $[[A]] \rightarrow [[A]]$                   | $[[a,b,c]]$ to $[[a],[b,c]]$ , and not back: $1 \leq 2$ .                                                     |
| 2  | $\text{ceiling} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\geq)$                                                                                                                                                                                                                                             | $[A] \rightarrow \text{Int}$                | $\text{ceiling}[a,b]$ is the largest of $0, a, a+b$ .                                                         |
| 3  | $\text{floor} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\leq)$                                                                                                                                                                                                                                               | $[A] \rightarrow \text{Int}$                | $\text{floor}[a,b]$ is the smallest of $0, a, a+b$ , so $[a,b]$ carries $\text{ceiling} - \text{floor}$ cash. |
| 4  | <b>secure</b><br>the coreflexive on $x$ with<br>$\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$                                                                                                                                                                                           | $[A] \rightarrow [A]$                       | <b>secure</b> keeps $[a]$ exactly when $-N \leq a \leq N$ .                                                   |
| 5  | <b>ok</b><br>the coreflexive on $(a, xs)$ with $xs$ non-empty and $[a] \# \text{head}(xs)$ secure                                                                                                                                                                                                                      | $A \times [[A]] \rightarrow A \times [[A]]$ | <b>ok</b> keeps $(a, [[b],[c]])$ exactly when $[a,b]$ is secure.                                              |
| 6  | $\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$                                                                                                                                                                                                                                                            | $A \times [[A]] \rightarrow [[A]]$          | $\text{new}(a, [[b],[c]]) = [[a],[b],[c]]$ .                                                                  |
| 7  | $\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$                                                                                                                                                                                                              | $A \times [[A]] \rightarrow [[A]]$          | $\text{glue}(a, [[b],[c]]) = [[a,b],[c]]$ .                                                                   |
| 8  | $\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons secure}) \times 1) \text{ cons}$                                                                                                                                                                                                      | $A \times [[A]] \rightarrow [[A]]$          | $\text{old}(a, [[b],[c]]) = [[a,b],[c]]$ when $[a,b]$ is secure, and nothing otherwise.                       |
| 9  | <b>partition</b> = $([\text{nil}, \text{new} \cup \text{glue}])$<br>cutting the transactions into non-empty segments<br>is one pass along them that either opens a segment<br>for the transaction in hand or puts it on the front<br>of the segment already open                                                       | $[A] \rightarrow [[A]]$                     | <b>partition</b> $[a,b]$ gives $[[a],[b]]$ and $[[a,b]]$ .                                                    |
| 10 | $S \triangleq [\text{nil}, \text{new} \cup \text{old}]$                                                                                                                                                                                                                                                                | $F[[A]] \rightarrow [[A]]$                  | $S(a, [[b],[c]])$ gives $[[a],[b],[c]]$ , and $[[a,b],[c]]$ when $[a,b]$ is secure.                           |
| 11 | $(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow$<br>$\text{partition list}(\text{secure}) = (S)$<br>cutting the transactions into non-empty segments<br>every way and keeping the cuts whose every<br>segment is secure is the one fold whose algebra is<br>$[\text{nil}, \text{new} \cup \text{old}]$ | $[A] \rightarrow [[A]]$                     | Both take $[a,b]$ to $[[a],[b]]$ — each $[a]$ is secure — and to $[[a,b]]$ when $[a,b]$ is.                   |
| 12 | $H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$                                                                                                                                                                                                                       | $[[A]] \rightarrow [[A]]$                   | $[[a],[b,c]]$ to $[[a,b],[c]]$ : $[a]$ is a prefix of $[a,b]$ .                                               |
| 13 | $R; H \triangleq R \circ (R^\circ \Rightarrow H)$                                                                                                                                                                                                                                                                      | $[[A]] \rightarrow [[A]]$                   | $[[a,b,c]]$ to $[[a],[b,c]]$ by $ R $ , and $[[a],[b,c]]$ to $[[a,b],[c]]$ by $R \circ H$ .                   |
| 14 | $ R  \triangleq R \circ \neg R^\circ$                                                                                                                                                                                                                                                                                  | $[[A]] \rightarrow [[A]]$                   | $[[a,b,c]]$ to $[[a],[b,c]]$ : $1 < 2$ . The strict part $R$ splits into: $R; H =  R  \cup (R \circ H)$ .     |
| 15 | the specification<br>$\frac{\text{partition list}(\text{secure})}{\ni} \text{est}(R)$                                                                                                                                                                                                                                  | $[A] \rightarrow [[A]]$                     | $[a,b]$ gives $[[a,b]]$ when $[a,b]$ is secure, and $[[a],[b]]$ otherwise.                                    |

### §7.5.1. secure prefix $\sqsubseteq$ prefix secure — a prefix of a secure segment is secure

(7.5.1a)

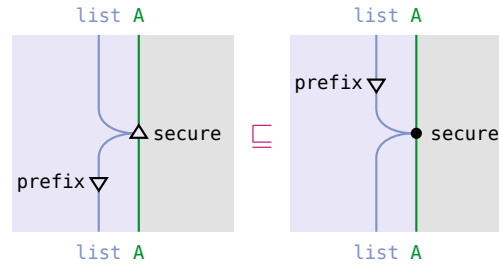


$$\text{secure } \langle \text{ceiling}, \text{ceiling-floor} \rangle \text{ bmax} = \langle \text{ceiling}, \text{ceiling-floor} \rangle \text{ bmax } (\leq N)$$

a stretch passes **secure** before the test exactly where the test's own value passes  $\leq N$  after it, which is  $\text{secure} \triangleq \text{corefl}(\text{secureP}(\text{amount}, N))$

$$\text{'s bmax}(\text{ceiling } x, \text{ceiling } x\text{-floor } x) \leq N$$

Exercise 7.49



$$\text{secure prefix} \sqsubseteq \text{prefix secure}$$

every pair **secure** then **prefix** gives, **prefix** then **secure** gives too — a prefix of a secure stretch is itself secure, which is the prefix-closure

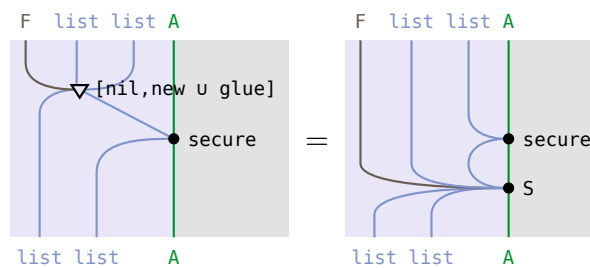
B&dM p.185 names

**§7.5.2.**  $(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow \text{partition list}(\text{secure}) = \{S\}$  — partitioning into secure segments is one fold

(7.5.2a)

| definition                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p>1 <math>\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}</math></p> <p><b>wrap</b> makes the transaction a segment on its own, and <b>cons</b> puts that segment at the front of the schedule: the van is called</p>                                                                                                           |               |
| <p>2 <math>\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}</math></p> <p><math>\text{cons}^\circ</math> splits the schedule into its first segment and the rest, the first <b>cons</b> puts the transaction at the front of that segment, and the second puts the segment back</p> |               |
| <p>3 <math>\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons secure}) \times 1) \text{ cons}</math></p> <p><b>glue</b> with <b>secure</b> on the segment the transaction has just joined: the van stays away only where the longer segment still passes</p>                                                  |               |

(7.5.2b)



$$(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow [\text{nil}, \text{new } u \text{ glue}] \text{ list}(\text{secure}) = F(\text{list}(\text{secure})) S$$

testing every segment for security after the algebra has run is testing it before it ran: a segment **glue** has extended is secure exactly where **old** would have let it be, and the one transaction **new** sets out on its own is secure whatever it is

| $(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow \text{partition list}(\text{secure}) = \langle S \rangle$ <span style="float: right;">Exercise 7.51</span>                                                    |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| cutting the transactions into non-empty segments every way and keeping the cuts whose every segment is secure is the one fold whose algebra is $[\text{nil}, \text{new} \cup \text{old}]$                                     |               |
| step                                                                                                                                                                                                                          | Hinze–Marsden |
| <b>partition list(secure)</b><br>the specification: cut the transactions every way, then keep the cuts whose every segment is secure                                                                                          |               |
| $= \langle [\text{nil}, \text{new} \cup \text{glue}] \rangle \text{list}(\text{secure})$<br>partition is the fold that either opens a segment for the transaction in hand or puts it on the front of the segment already open |               |
| $= \langle [\text{nil}, \text{new} \cup \text{old}] \rangle$<br>fusion: the condition above moves $\text{list}(\text{secure})$ inside that fold, which is what turns $\text{glue}$ into $\text{old}$                          |               |

(7.5.2c)

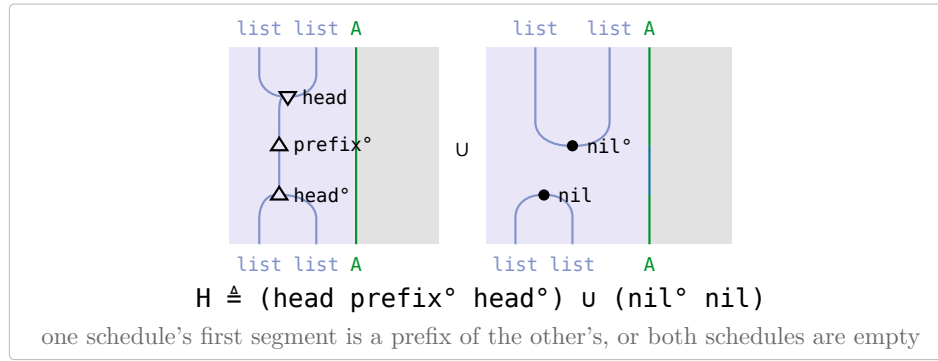
|                                                                                                                                                                                                                                                                                                                            |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                                                                                                                                                                                                                                                                                                            |  |
| $(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow \text{partition list}(\text{secure}) = \langle S \rangle$<br>cutting the transactions every way and then keeping the cuts whose every segment is secure is one pass along them that either calls the van or extends the open segment while it stays secure |  |

### §7.5.3. $(1 \times R) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) R$ — new on an R-related schedule is matched by new or old

| $(1 \times R) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) R$ <span style="float: right;">(7.14)</span>                                                                                                                           |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| calling the van for the transaction on a no-longer schedule gets no further than calling it on this one and shortening the schedule afterwards                                                                                            |               |
| step                                                                                                                                                                                                                                      | Hinze–Marsden |
| <b><math>(1 \times R) \text{ new}</math></b><br>the left side of (7.14) — $R = \text{clen cast} \leq \text{clen cast}^\circ, \text{new} \triangleq \text{new}$                                                                            |               |
| $= (\text{wrap} \times R) \text{ cons}$<br>$\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$ — $(\text{wrap} \times 1) \text{ cons}$ — and $(\text{wrap} \times 1) (1 \times R) = (\text{wrap} \times R)$                       |               |
| $\sqsubseteq \text{new} R$<br>$\text{cons}$ is monotonic on $R$ — $(1 \times R) \text{ cons} \sqsubseteq \text{cons} R$ , consing onto a no-longer schedule leaves it no longer — and $\text{new} \sqsubseteq \text{new} \cup \text{old}$ |               |

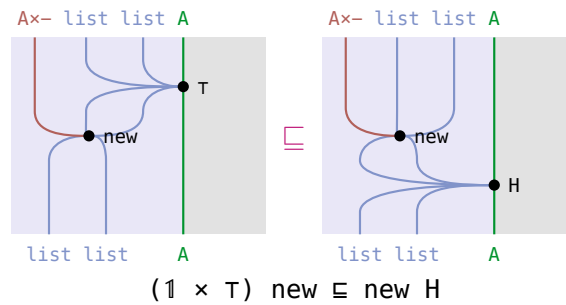
### §7.5.4. $H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$ — the right first segment is a prefix of the left one, or both are empty

(7.5.4a)



**§7.5.5.**  $(1 \times (R;H)) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) (R;H)$  — **new** on an  $R;H$ -related schedule is matched by **new** or **old**

(7.18)



whatever schedule the van is called on, the result's first segment is the one transaction  $[a]$ , so any two results of **new** on that transaction stand in  $H$

$(1 \times (R;H)) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) (R;H)$

(7.16)

calling the van for the transaction on a  $R;H$ -better schedule gets no further than calling it on this one and bettering the whole schedule afterwards

| step                                                                                                                                                                                   | Hinze–Marsden |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $(1 \times (R;H)) \text{ new}$<br>the left side of (7.16)                                                                                                                              |               |
| $\sqsubseteq (1 \times R) \text{ new}$<br>$R;H \sqsubseteq R \text{ — } R;H = R \cap (R^\circ \Rightarrow H)$                                                                          |               |
| $\sqsubseteq \text{new } R \cap \text{new } H$<br>(7.14) as far as its <b>new</b> $R$ line — (7.14) $(1 \times R) \text{ new} \sqsubseteq (\text{new} \cup \text{old}) R$ — and (7.18) |               |
| $= \text{new } (R \cap H)$<br><b>new</b> is a map, and a map distributes over $\cap$                                                                                                   |               |
| $\sqsubseteq \text{new } (R;H)$<br>$X \cap Y \sqsubseteq X;Y$ , and $\text{new} \sqsubseteq \text{new} \cup \text{old}$                                                                |               |

**§7.5.6.**  $(1 \times (R;H)) \text{ old} \sqsubseteq (\text{new} \cup \text{old}) (R;H)$  — **old** on an  $R;H$ -related schedule is matched by **new** or **old**

(7.19)



$$(1 \times T) \text{ old} \sqsubseteq \text{new H}$$

**old** leaves the transaction **[a]** at the front of the first segment, and **[a]** is what **new** makes that segment, so the two first segments are **prefix-related** whatever the schedules were

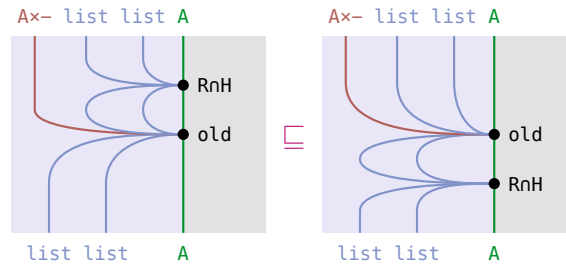
(7.20)



$$(1 \times |R|) \text{ old} \sqsubseteq \text{new R}$$

**old** keeps the schedule's length, so a strictly shorter one still comes out no longer than the one the van's own segment lengthens

(7.21)



$$(1 \times (R \cap H)) \text{ old} \sqsubseteq \text{old } (R \cap H)$$

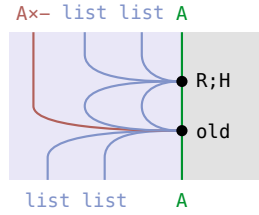
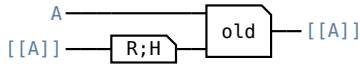
on a tie the one first segment is a prefix of the other, so prefix-closure of **secure** — Exercise 7.49 **secure prefix**  $\sqsubseteq$  **prefix secure** — lets **old** fire on this side too, and it keeps both the length and the prefix

$$(1 \times (R;H)) \text{ old} \sqsubseteq (\text{new} \cup \text{old}) (R;H) \quad (7.17)$$

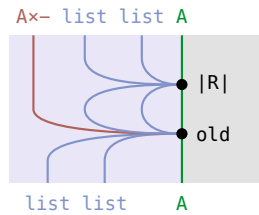
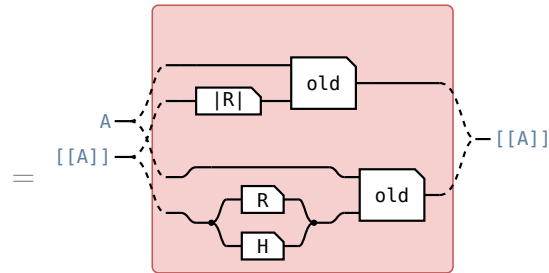
gluing the transaction onto a better schedule for the rest gets no further than gluing it on, or calling the van, and  
bettering the whole schedule after,

**circuit** — the old branch of each union

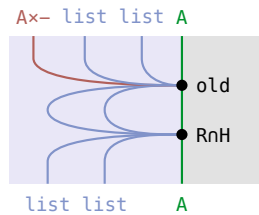
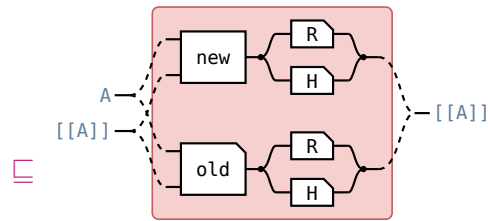
**Hinze–Marsden**



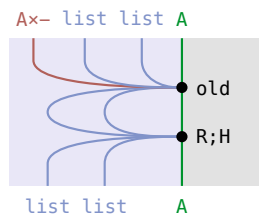
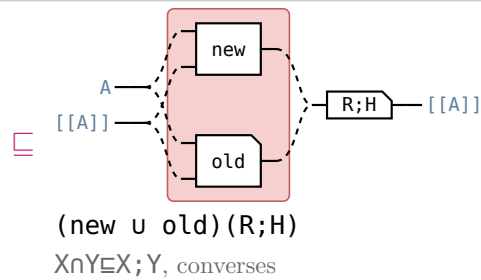
the **old** operand of **new**  $\cup$  **old**, in every row



$(1 \times |R|) \text{ old} \cup (1 \times (R \cap H)) \text{ old}$   
 $R;H = |R| \cup (R \cap H) \text{ — (7.5a.14), } \cup \text{ distributes,}$



$\text{new } (R \cap H) \cup \text{old } (R \cap H)$   
 (7.19) and (7.20) on  $|R|$  — (7.19)  $(1 \times \tau) \text{ old} \sqsubseteq \text{new } H$ , (7.20)  $(1 \times |R|) \text{ old} \sqsubseteq \text{new } R$  — and (7.21) on  $R \cap H$  — (7.21)  $(1 \times (R \cap H)) \text{ old} \sqsubseteq \text{old } (R \cap H)$

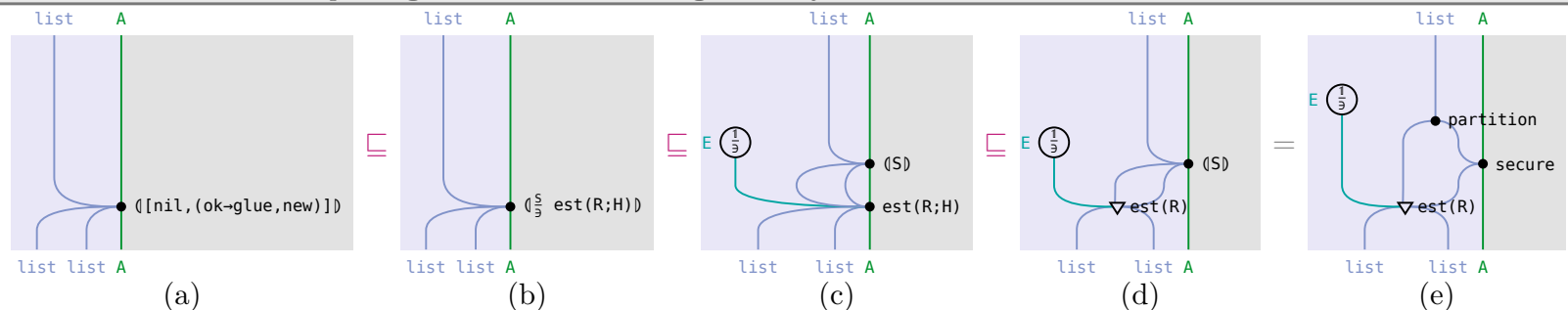


$(\text{new} \cup \text{old}) (R;H)$   
 $X \cap Y \sqsubseteq X; Y$ , converses

### §7.5.7. The derivation

$$(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow ([\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})]) \sqsubseteq \frac{\text{partition list}(\text{secure})}{\exists} \text{est}(R) \quad (7.5.7a)$$

the fewest secure segments the transactions can be cut into are one pass along them, the next transaction glued  
onto the open segment wherever that segment stays secure and the van called where it does not



|     |                                                                        |                                                                                                                             |
|-----|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})]$        | $[\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})] \sqsubseteq \frac{S}{\exists} \text{est}(R;H)$               |
| (b) | $\sqsubseteq ([S] \text{est}(R;H))$                                    | $([S] \text{est}(R;H)) \sqsubseteq \frac{([S])}{\exists} \text{est}(R;H)$                                                   |
| (c) | $\sqsubseteq \frac{([S])}{\exists} \text{est}(R;H)$                    | $R;H \sqsubseteq R$                                                                                                         |
| (d) | $\sqsubseteq \frac{([S])}{\exists} \text{est}(R)$                      | Exercise 7.51 $(\forall a. \text{secureP}(\text{amount})(N, [a])) \Rightarrow \text{partition list}(\text{secure}) = ([S])$ |
| (e) | $= \frac{\text{partition list}(\text{secure})}{\exists} \text{est}(R)$ |                                                                                                                             |

## §8. Thinning Algorithms

### §8.1. Thinning

(8.1)

For  $Q : A \rightarrow A$ ,  $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$  (8.1).

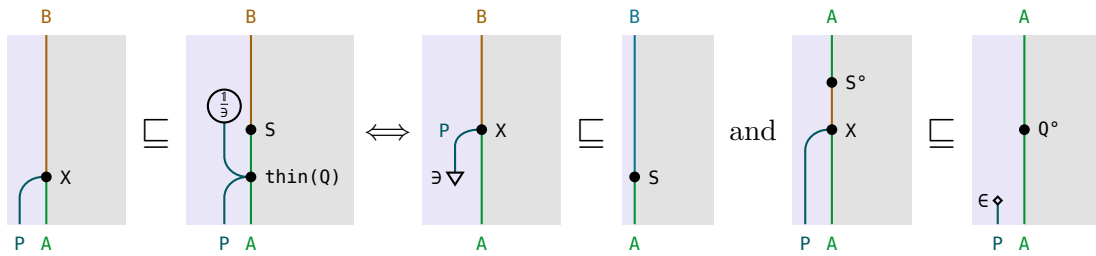
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$  the same  $^\circ$  as  $\triangleq \text{est}$ :  $\epsilon X$  runs  $a \rightarrow ys \rightarrow xs$ , member of  $ys$  first, and  $Q^\circ \epsilon$  runs  $a \rightarrow b \rightarrow xs$ , so  $(a, b) \in Q^\circ$  reads  $b \ Q \ a$ ; at  $Q^\circ \leq$  every  $a \in ys$  keeps some  $b \leq a$  in  $xs$ , the end  $\text{est}(\leq)$  picks

(8.1a)

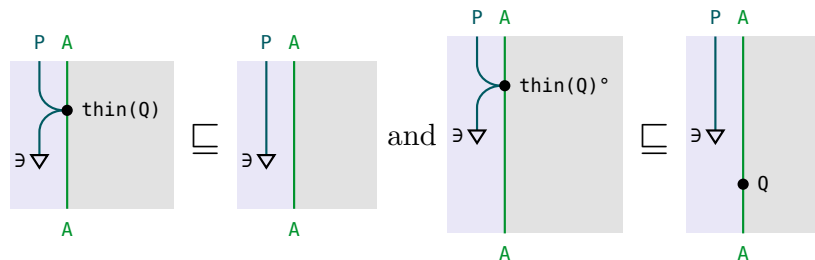
| the law                                                                                                                                                                                                           | what it says                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$                                                                                               | $S$ is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |
| 2 $\text{thin}(Q) \ni \subseteq \ni$                                                                                                                                                                              | everything a thinning keeps was in the set                                                                                                                                           |
| 3 $\text{thin}(Q)^\circ \ni \subseteq \ni Q$                                                                                                                                                                      | every element of the set has a $Q$ -lower bound among the kept ones                                                                                                                  |
| 4 $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$                                                                                                                                                | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                                                                                    |
| 5 $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$<br>$Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$                                                                          | keeping everything is always a legal thinning                                                                                                                                        |
| 6 $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{est}(R) \text{preorder}(Q)$<br>$\implies \text{est}(R) \subseteq \text{thin}(Q) \text{est}(R)$ | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                                                                                |
| 7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists} \quad (8.2) - (8.2) \text{est}(Q) \frac{1}{\exists} \subseteq \text{thin}(Q)$                                                                         | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on                                                                |
| 8 $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant                                            | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                                                                                   |
| 9 $P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q) \quad (8.4) - (8.4)$<br>$P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$                                                          | thinning each member set is a thinning of the union                                                                                                                                  |

#### §8.1.1. $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)



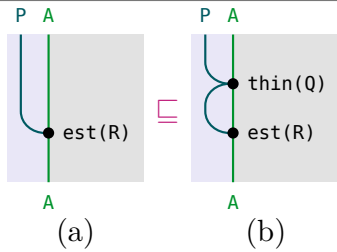
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$  at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \ni$ : the left side is  $\text{thin}(Q) \subseteq \text{thin}(Q)$ , since  $\frac{\exists}{\exists} = \wedge(\exists) = 1$

#### §8.1.2. $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$\text{preorder}(Q) \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \quad (8.1.2a)$$

**est(R)**

the  $\sqsubseteq$  half: keeping everything is a  
thinning —  $1 \sqsubseteq Q$



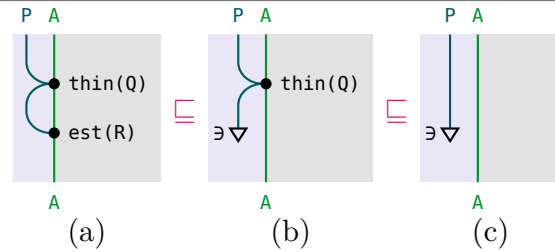
|     |                                             |                                                                                                                                                                                                              |
|-----|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | est(R)                                      | $1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$<br>$\Rightarrow 1 \sqsubseteq \text{thin}(Q),$<br>$Q \sqsubseteq Q$<br>$\Rightarrow \text{thin}(Q)$<br>$\text{thin}(Q) \sqsubseteq \text{thin}(Q)$ |
| (b) | $\sqsubseteq \text{thin}(Q) \text{ est}(R)$ |                                                                                                                                                                                                              |

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

the  $\sqsubseteq$  half, first condition of the UP of

**est** at  $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq$

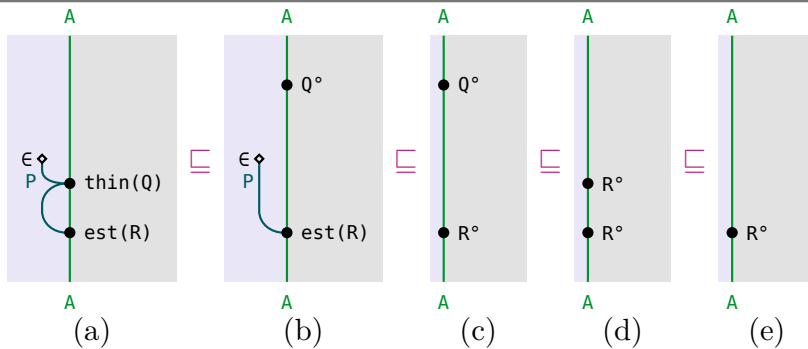
**est(R)  $\Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$**



|     |                                      |                                              |
|-----|--------------------------------------|----------------------------------------------|
| (a) | thin(Q) est(R)                       | $R \cap S \sqsubseteq R$                     |
| (b) | $\sqsubseteq \text{thin}(Q) \exists$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| (c) | $\sqsubseteq \exists$                |                                              |

$$Q \sqsubseteq R \wedge \text{preorder}(R) \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the  $\sqsubseteq$  half, second condition



|     |                                         |                                              |
|-----|-----------------------------------------|----------------------------------------------|
| (a) | $\in \text{thin}(Q) \text{ est}(R)$     | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ |
| (b) | $\sqsubseteq Q^\circ \in \text{est}(R)$ | $\in \text{est}(R) \sqsubseteq R^\circ$      |
| (c) | $\sqsubseteq Q^\circ R^\circ$           | $Q \sqsubseteq R$                            |
| (d) | $\sqsubseteq R^\circ R^\circ$           |                                              |
| (e) | $\sqsubseteq R^\circ$                   |                                              |

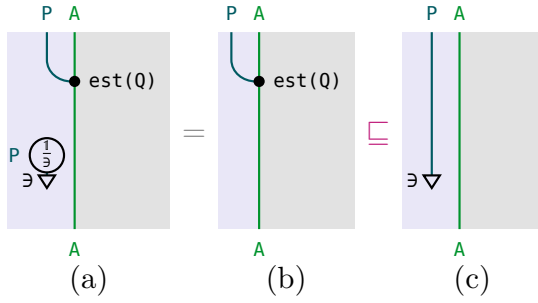
$$\text{est}(Q) \stackrel{1}{\exists} \sqsubseteq \text{thin}(Q)$$

(8.2)

the singleton holding a  $Q$ -least member of a set is a thinning of that set

$$\text{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

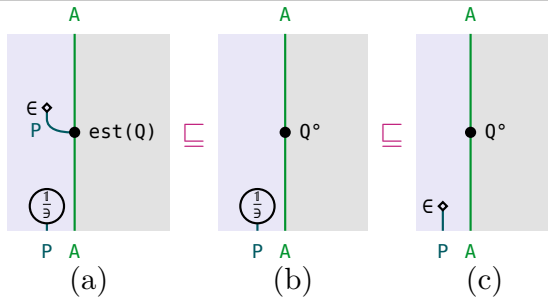
every member of the singleton is a member of the set —  $\triangleq \text{thin}, \ni/\ni$  half



|     |                                           |                                 |
|-----|-------------------------------------------|---------------------------------|
| (a) | $\text{est}(Q) \stackrel{1}{\exists} \ni$ | $\stackrel{1}{\exists} \ni = 1$ |
| (b) | $= \text{est}(Q)$                         | $R \cap S \sqsubseteq R$        |
| (c) | $\sqsubseteq \ni$                         |                                 |

$$\in \text{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every member of the set —  $\triangleq \text{thin}, \in/(Q^\circ \in)$  half

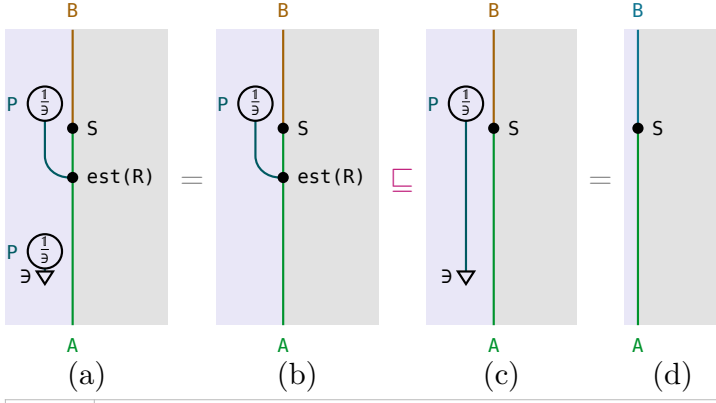


|     |                                             |                                         |
|-----|---------------------------------------------|-----------------------------------------|
| (a) | $\in \text{est}(Q) \stackrel{1}{\exists}$   | $\in \text{est}(R) \sqsubseteq R^\circ$ |
| (b) | $\sqsubseteq Q^\circ \stackrel{1}{\exists}$ | $\stackrel{1}{\exists} \sqsubseteq \in$ |
| (c) | $\sqsubseteq Q^\circ \in$                   |                                         |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \in S$$

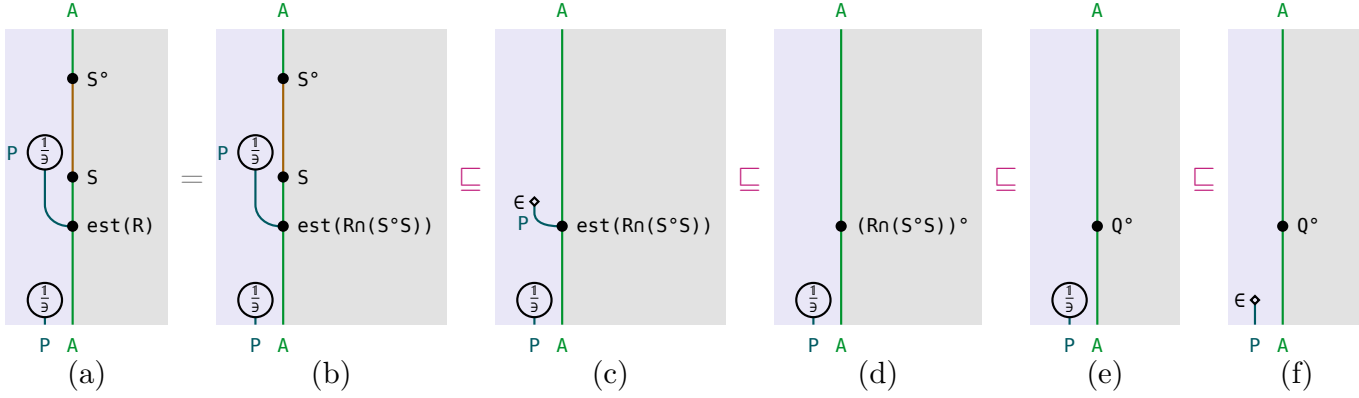
every member of the singleton is a value  $S$  returns —  $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$ , first condition



|     |                                                         |                             |
|-----|---------------------------------------------------------|-----------------------------|
| (a) | $\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$ | $\frac{1}{\exists} \ni = 1$ |
| (b) | $= \frac{S}{\exists} \text{est}(R)$                     | $R \cap S \subseteq R$      |
| (c) | $\subseteq \frac{S}{\exists} \ni$                       | $\frac{R}{\exists} \ni = R$ |
| (d) | $= S$                                                   |                             |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument —  $X \subseteq \frac{S}{\exists} \text{thin}(Q) \Leftrightarrow X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$ , second condition



|     |                                                                                |                                                                                      |
|-----|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$                    | $\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$ |
| (b) | $= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$ | $T^\circ \frac{1}{\exists} \subseteq \in$                                            |
| (c) | $\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$               | $\in \text{est}(R) \subseteq R^\circ$                                                |
| (d) | $\subseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$                       | $R \cap (S^\circ S) \subseteq Q$                                                     |
| (e) | $\subseteq Q^\circ \frac{1}{\exists}$                                          | $\frac{1}{\exists} \subseteq \in$                                                    |
| (f) | $\subseteq Q^\circ \in$                                                        |                                                                                      |

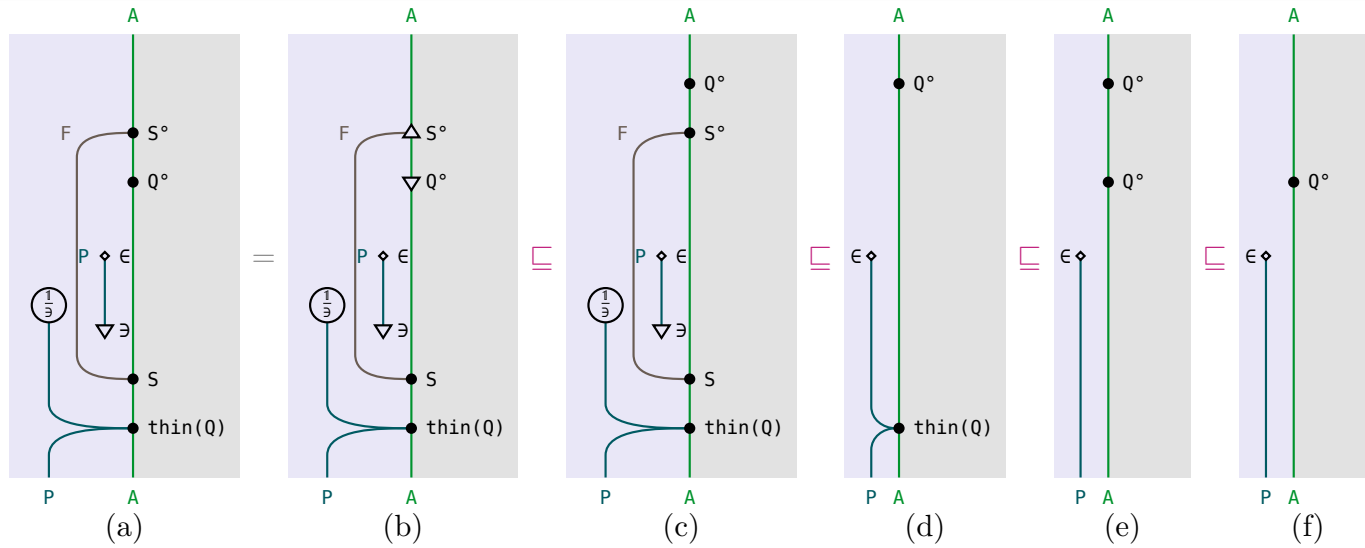
$$\text{preorder}(Q) \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \frac{\langle S \rangle}{\exists} \text{thin}(Q)$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle S \rangle \quad \text{and} \quad \langle S \rangle \circ \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \in$$

$$X \sqsubseteq \frac{\langle S \rangle}{\exists} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \sqsubseteq Q^\circ \in \text{at } X \triangle \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle, \langle S \rangle \text{ for its } S$$

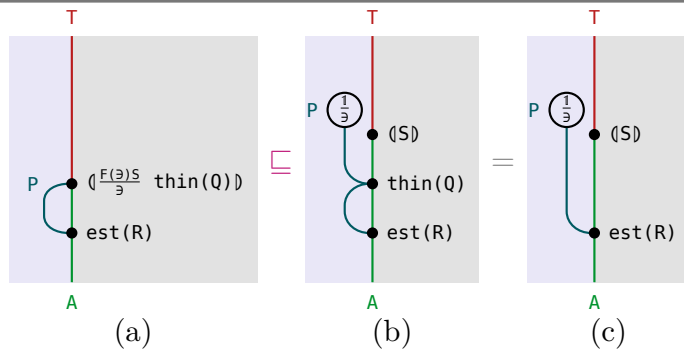
 $\Rightarrow$  fusion, hylomorphism theorem

|     |                                                                                  |                                                               |
|-----|----------------------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$             |                                                               |
| (b) | $= S^\circ F(Q^\circ) F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$        | $\text{MonoAlg}(S, Q)$                                        |
| (c) | $\sqsubseteq Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $\Rightarrow S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$  |
| (d) | $\sqsubseteq Q^\circ \in \text{thin}(Q)$                                         | $S^\circ F(\in) \frac{F(\exists) S}{\exists} \sqsubseteq \in$ |
| (e) | $\sqsubseteq Q^\circ Q^\circ \in$                                                | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$                  |
| (f) | $\sqsubseteq Q^\circ \in$                                                        |                                                               |

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \frac{\langle S \rangle}{\exists} \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself



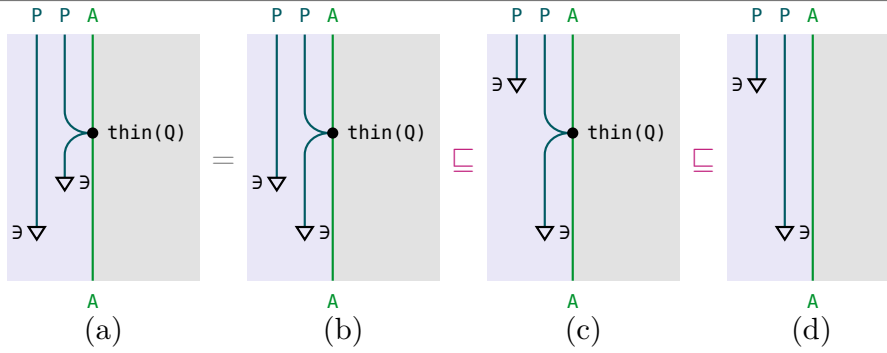
|     |                                                                              |                                                                                                                                        |
|-----|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R)$  | $\text{Theorem 8.1 } \text{preorder}(Q) \wedge \text{MonoAlg}(S, Q)$                                                                   |
| (b) | $\sqsubseteq \frac{\langle S \rangle}{\exists} \text{thin}(Q) \text{est}(R)$ | $\Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \frac{\langle S \rangle}{\exists} \text{thin}(Q)$ |
| (c) | $= \frac{\langle S \rangle}{\exists} \text{est}(R)$                          | $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R)$                                                                  |
|     |                                                                              | $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$                                                                             |

$$P(\text{thin}(Q)) \text{ union } \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union } \ni \sqsubseteq \ni$$

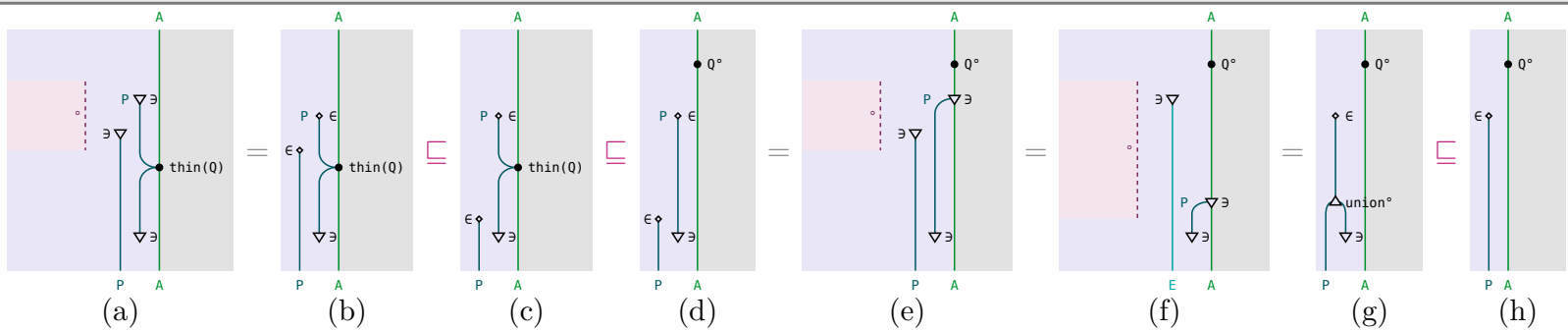
every member of the result is a member of a member set —  $X \sqsubseteq \bigcup \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , first condition



|     |                                        |                                      |
|-----|----------------------------------------|--------------------------------------|
| (a) | $P(\text{thin}(Q)) \text{ union } \ni$ | $\text{union } \ni = \ni \ni$        |
| (b) | $= P(\text{thin}(Q)) \ni \ni$          | $P(R) \ni \sqsubseteq \ni R$         |
| (c) | $\sqsubseteq \ni \text{thin}(Q) \ni$   | $\text{thin}(Q) \ni \sqsubseteq \ni$ |
| (d) | $\sqsubseteq \ni \ni$                  |                                      |

$$(\ni \ni)^\circ P(\text{thin}(Q)) \text{ union } \sqsubseteq Q^\circ \in$$

every member of a member set has a  $Q$ -lower bound in the result —  $X \sqsubseteq \bigcup \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , second condition



|     |                                                    |                                              |
|-----|----------------------------------------------------|----------------------------------------------|
| (a) | $(\ni \ni)^\circ P(\text{thin}(Q)) \text{ union}$  | $(R S)^\circ = S^\circ R^\circ$              |
| (b) | $= \in \in P(\text{thin}(Q)) \text{ union}$        | $\in P(R) \sqsubseteq R \in$                 |
| (c) | $\sqsubseteq \in \text{thin}(Q) \in \text{union}$  | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ |
| (d) | $\sqsubseteq Q^\circ \in \in \text{union}$         | $(R S)^\circ = S^\circ R^\circ$              |
| (e) | $= Q^\circ (\ni \ni)^\circ \text{union}$           | $\text{union } \ni = \ni \ni$                |
| (f) | $= Q^\circ (\text{union } \ni)^\circ \text{union}$ | $(R S)^\circ = S^\circ R^\circ$              |
| (g) | $= Q^\circ \in \text{union}^\circ \text{union}$    | $\text{union} = E(\ni)$                      |
| (h) | $\sqsubseteq Q^\circ \in$                          |                                              |

## §8.2. Paths in a layered network

(8.2a)

|    | name                            | type                                                               | definition                                                                             | meaning                                                                                                                                                          |
|----|---------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | F                               | $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$           | $F(B, X) = B + (B \times X)$                                                           | a path is either one vertex $a$ , or a vertex $a$ followed by a path $xs$                                                                                        |
| 2  | $\alpha$                        | $F(A, LA) \rightarrow LA$                                          | $\alpha = [\text{wrap}, \text{cons}]$                                                  | build a path: wrap one vertex as $[a]$ , or put $a$ in front of $xs$                                                                                             |
| 3  | wrapz                           | $A \rightarrow LA \times \mathbb{N}$                               | $\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$ | a path of one vertex, no edge, cost 0                                                                                                                            |
| 4  | consw                           | $A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$ | $(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$                  | put $a$ in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$                                                           |
| 5  | cost                            | $LA \rightarrow \mathbb{N}$                                        | $\text{cost} \triangleq \llbracket [\text{wrapz}, \text{consw}] \rrbracket \pi_2$      | the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$                                                                       |
| 6  | R                               | $LA \rightarrow LA$                                                | $R \triangleq \text{cost} \leq \text{cost}^\circ$                                      | $xs$ is cheaper than $ys$                                                                                                                                        |
| 7  | L(R)                            | $LA \rightarrow LB$                                                | $\llbracket F(\exists, 1) \alpha \rrbracket = L(\exists)$                              | $L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer                                                                      |
| 8  | minpath                         | $L(PA) \rightarrow LA$                                             | $\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$                  | the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one |
| 9  | Q                               | $LA \rightarrow LA$                                                | $Q \triangleq R \cap (\text{head} \text{ head}^\circ)$                                 | like $R$ , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$                                                                  |
| 10 | l                               | $B \rightarrow B + C$                                              | $a \mapsto l(a)$                                                                       | mark a value as the left alternative                                                                                                                             |
| 11 | r                               | $C \rightarrow B + C$                                              | $b \mapsto r(b)$                                                                       | mark a value as the right alternative                                                                                                                            |
| 12 | cpl                             | $PB \times A \rightarrow P(B \times A)$                            | $\text{cpl} \triangleq \text{cp}(1 \times A)$                                          | pair every element of the set with the value beside it                                                                                                           |
| 13 | cpr                             | $A \times PB \rightarrow P(A \times B)$                            | $\text{cpr} \triangleq \text{cp}(A \times 1)$                                          | pair the value beside the set with every element of the set                                                                                                      |
| 14 | $\frac{F(\exists, 1)}{\exists}$ | $F(PB, X) \rightarrow PF(B, X)$                                    | $\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$                             | cpl transpose: from layer $x$ and a set of paths $ps$ , choose a vertex $a \in x$                                                                                |
| 15 | $\frac{F(1, \exists)}{\exists}$ | $F(B, PX) \rightarrow PF(B, X)$                                    | $\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$           | cpr transpose: from a vertex $a$ and a set of paths $ps$ , choose a tail $xs \in ps$                                                                             |
| 16 | step                            | $A \times P(LA) \rightarrow LA$                                    | $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$                      | put $a$ in front of every path in $ps$ and keep a cheapest one                                                                                                   |

(8.2b)

|   | the law                                                                         | what it says                                                                                                                                                             |
|---|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                             |
| 2 | $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

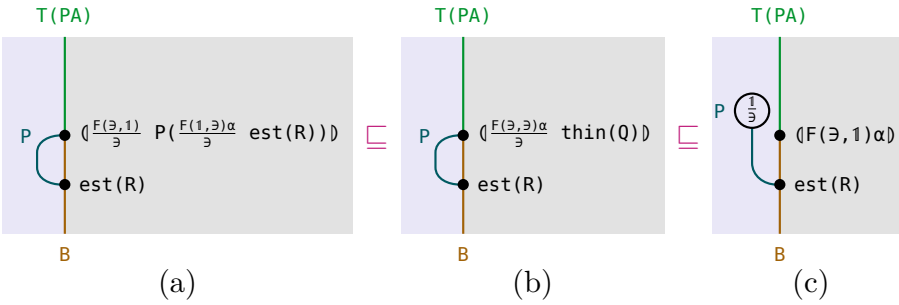
(8.2a)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge R \sqcap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \sqsubseteq Q \Rightarrow \quad (8.2c)$$

$$\left( \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R) \sqsubseteq \left( \frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

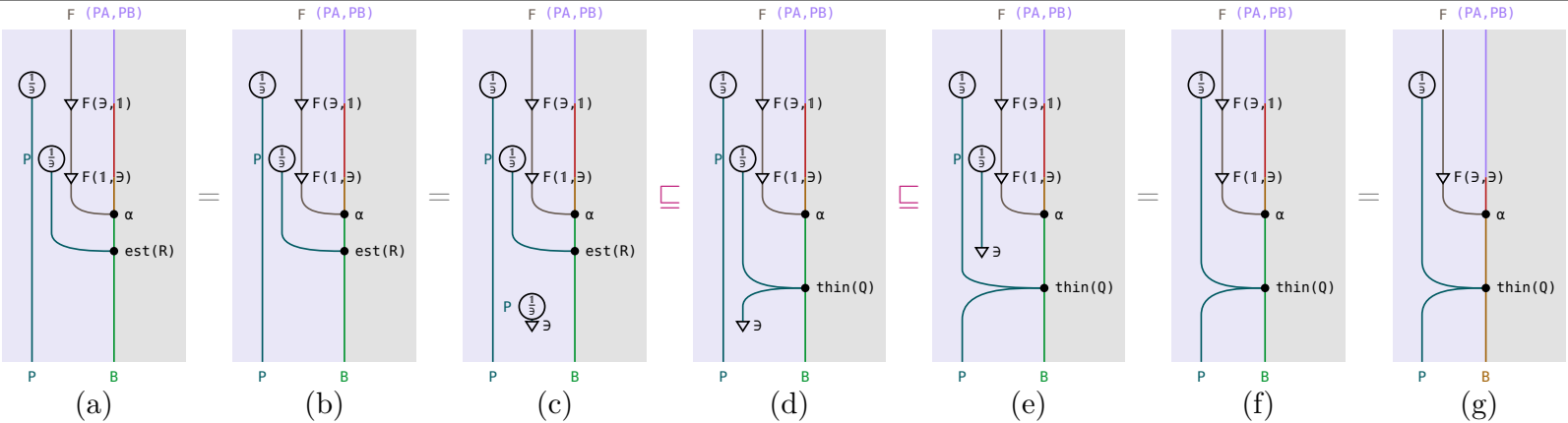
- Why it is needed:  $\text{est}(R)$  keeps one cheapest  $w$  of the set  $\Lambda(S)(x)$ ;  $\text{thin}(Q)$  needs every dropped  $z$  to be  $Q$ -beaten by a kept one. For  $\{w\}$  alone to be a thinning we need  $w \sqsubseteq z$  for every  $z$  in  $\Lambda(S)(x)$ .
- We have  $w \sqsubseteq z$  ( $w$  is cheapest) and  $w \sqsubseteq (S^\circ S) z$  (both are built by  $S$  from the same argument  $x$ ); the hypothesis turns the two into  $w \sqsubseteq z$ : among candidates from one argument, cheaper already means  $Q$ -better.



|     |                                                                                                                                  |                                                                                                                                                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\left( \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R)$ | (8.2d) $R \sqcap (S^\circ S) \sqsubseteq Q$<br>$\Rightarrow \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$ |
| (b) | $\sqsubseteq \left( \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R)$                             | $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge$<br>$\text{MonoAlg}(F(\exists, 1) \alpha, Q)$                                                                                                            |
| (c) | $\sqsubseteq \left( \frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$                                                  | $\Rightarrow \left( \frac{F(\exists, 1) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R) \sqsubseteq \left( \frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$                                                         |

$$S \triangleq F(1, \exists) \alpha, R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



|     |                                                                                                                                    |                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$                                |                                                                                                                                                            |
| (b) | $= \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$                              | $P \left( \frac{1}{\exists} \right) E(\exists) = 1$                                                                                                        |
| (c) | $= \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \frac{1}{\exists} \right) E(\exists)$ | (8.3) $R \sqcap (S^\circ S) \sqsubseteq Q$<br>$\Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{thin}(Q)$ |
| (d) | $\sqsubseteq \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \right) P(\text{thin}(Q)) E(\exists)$     | (8.4) $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$                                                                                   |
| (e) | $\sqsubseteq \frac{F(\exists, 1)}{\exists} P \left( \frac{F(1, \exists) \alpha}{\exists} \right) E(\exists) \text{thin}(Q)$        | $\frac{V}{\exists} S = \frac{V}{\exists} P \left( \frac{S}{\exists} \right) \text{ union}$                                                                 |
| (f) | $= \frac{F(\exists, 1)}{\exists} \frac{F(1, \exists) \alpha}{\exists} \text{thin}(Q)$                                              | $F(R, 1) F(1, S) = F(R, S)$                                                                                                                                |
| (g) | $= \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$                                                                      |                                                                                                                                                            |

$$= \text{cp}(F) P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$$

at the layered network  $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}])$ :  $\frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$  —  $\text{step} \triangleq \text{cpr}$   
 $P(\text{cons}) \text{est}(R)$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \times | \\ & & R & & R \quad S \\ & & \downarrow & & \downarrow \quad \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u  $F(R, S)$  v holds iff one of:

- u, v are both  $\text{inl}$  and  $d \sqsubseteq R \sqsubseteq d'$ : the leaf moves by R;
- u, v are both  $\text{inr}$  and  $p_1 \sqsubseteq R \sqsubseteq q_1 \wedge p_2 \sqsubseteq S \sqsubseteq q_2$ : the vertex moves by R, the rest by S.
- An  $\text{inl}$  never relates to an  $\text{inr}$ :  $R + S \triangleq [R \text{ inl}, S \text{ inr}]$  returns to the summand it came from, and  $F(1) = 1$  allows no crossing.
- $F(\exists, \exists)$  relates  $\text{inr}(\{a, b\}, \{p, q\})$  to  $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$ , and  $\text{inl}(\{a, b\})$  to  $\text{inl}(a), \text{inl}(b)$ .

## §8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow PA$ ,  $\text{cup} : PA \times PA \rightarrow PA$ ,  $\text{cp}(F) \text{ cp}(F) \triangleq \frac{F(\exists)}{3}$ ,  $\text{listcp} : F(L) \rightarrow LF$ ,  $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$  for  $\leq$  a connected preorder.

$\text{thinlist}(Q)$  is any  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$  with  $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ ; one is  $\llbracket \text{nil}, \text{bump } Q \rrbracket$ ,  $\text{bump } Q (a, []) = [a]$ ,  $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

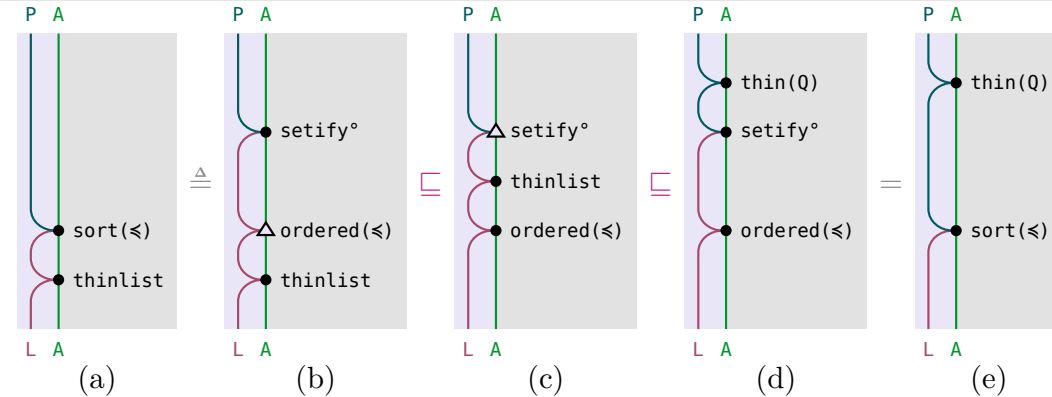
**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $\leq$  a connected preorder with both  $f_1, f_2$  monotonic on  $\leq$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

(8.3b)

| the law                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | what it says                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$<br>(8.5), $Q$ connected, $xs$ non-empty                                                                                                                                                                                                                                                                                                                                                                            | what thinning should come to when it can: one element                                                              |
| 2 $\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6)<br>$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ | thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys |
| 3 $\text{sort}(\leq) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)                                                                                                                                                                                                                                                                                                                                                                                                              | a minimum of the sorted list is a minimum of the set                                                               |
| 4 $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ (8.8)                                                                                                                                                                                                                                                                                                                                                                                             | shunt a function through a sort                                                                                    |
| 5 $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), $p$ coreflexive                                                                                                                                                                                                                                                                                                                                                                                    | filtering a sorted list sorts the restricted set                                                                   |
| 6 $(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)                                                                                                                                                                                                                                                                                                                                                                         | merging two sorted lists sorts their union                                                                         |
| 7 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$<br>(8.11), $F$ linear                                                                                                                                                                                                                                                                                                                                                                          | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$                                 |

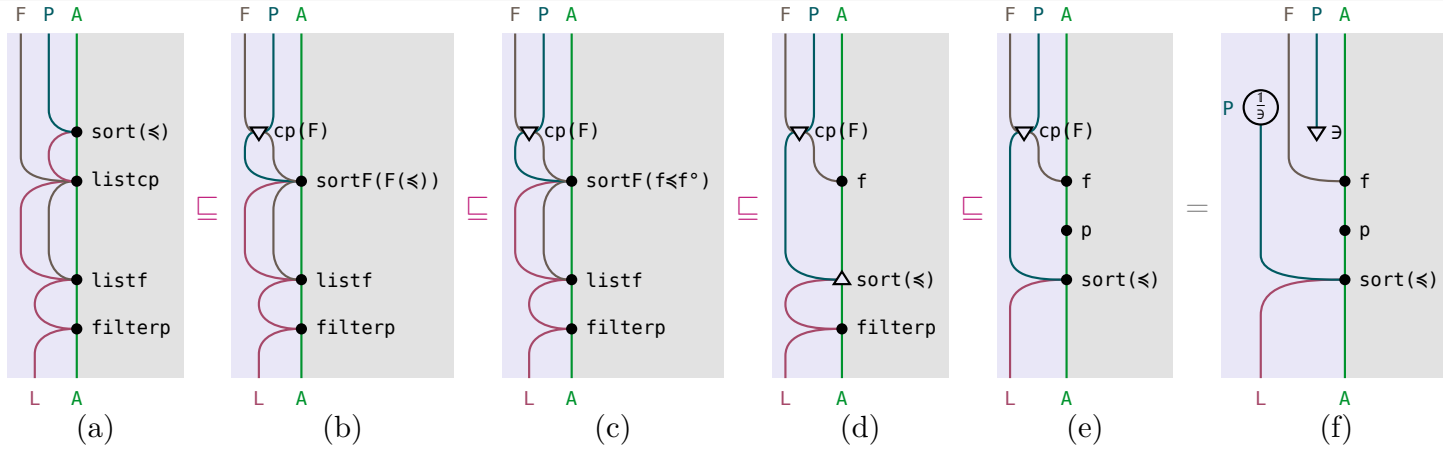
$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$   
a thinning of the sorted list lists a thinning of the set —  $\leq$  a connected preorder,  $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ .

(8.6)



|     |                                                                         |                                                                                                                                                                                                                                                                                  |
|-----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{sort}(\leq) \text{ thinlist}$                                    | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$                                                                                                                                                                                                         |
| (b) | $\triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$ | $\text{Coreflexive}(\text{ordered}(\leq)) \wedge g \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \Rightarrow \text{setify}^\circ \text{ ordered}(\leq) g \sqsubseteq \text{setify}^\circ g \text{ ordered}(\leq)$ |
| (c) | $\sqsubseteq \text{setify}^\circ \text{ thinlist ordered}(\leq)$        | $\text{Map}(\text{setify}) \wedge g \text{ setify} \sqsubseteq \text{setify } T \Rightarrow \text{setify}^\circ g \text{ ordered}(\leq) \sqsubseteq T \text{ setify}^\circ \text{ ordered}(\leq)$                                                                                |
| (d) | $\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$ |                                                                                                                                                                                                                                                                                  |
| (e) | $= \text{thin}(Q) \text{ sort}(\leq)$                                   |                                                                                                                                                                                                                                                                                  |

**Map(f)  $\wedge$  ( $\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$ )  $\wedge$  MonoAlg(f,  $\leq$ )  $\wedge$  sortF(f  $\leq$  f $^\circ$ ) listf  $\sqsubseteq$  Lemma 8.1**  
**P(f) sort( $\leq$ )  $\wedge$  sort( $\leq$ ) filterp  $\sqsubseteq$  E(p) sort( $\leq$ )  $\wedge$  F(sort( $\leq$ )) listcp  $\sqsubseteq$  cp(F) sortF(F( $\leq$ ))**  
 **$\Rightarrow$  F(sort( $\leq$ )) listcp listf filterp  $\sqsubseteq$   $\frac{F(\exists) \ f \ p}{\exists}$  sort( $\leq$ )**  
 one sorted list built from sorted arguments, instead of a set built and then sorted — f : FA $\rightarrow$ A monotonic  
 on  $\leq$ , p coreflexive, F linear.



|     |                                                               |                                                                                                 |
|-----|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| (a) | F(sort( $\leq$ )) listcp listf filterp                        | F(sort( $\leq$ )) listcp $\sqsubseteq$ cp(F) sortF(F( $\leq$ ))                                 |
| (b) | $\sqsubseteq$ cp(F) sortF(F( $\leq$ )) listf filterp          | Map(f)<br>$\Rightarrow$ MonoAlg(f, R) $\iff$ F(R) $\sqsubseteq$ f R f $^\circ$                  |
| (c) | $\sqsubseteq$ cp(F) sortF(f $\leq$ f $^\circ$ ) listf filterp | sortF(f $\leq$ f $^\circ$ ) listf $\sqsubseteq$ P(f) sort( $\leq$ )                             |
| (d) | $\sqsubseteq$ cp(F) P(f) sort( $\leq$ ) filterp               | sort( $\leq$ ) filterp $\sqsubseteq$ E(p) sort( $\leq$ )<br>Map(f)<br>$\Rightarrow$ P(f) = E(f) |
| (e) | $\sqsubseteq$ cp(F) E(f p) sort( $\leq$ )                     | $\frac{S}{\exists} E(R) = \frac{S \ R}{\exists}$                                                |
| (f) | $= \frac{F(\exists) \ f \ p}{\exists}$ sort( $\leq$ )         |                                                                                                 |

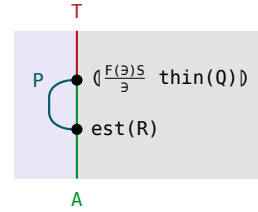
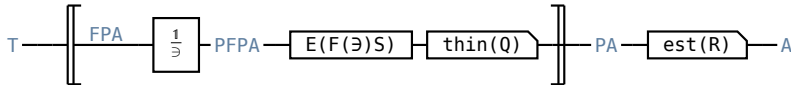
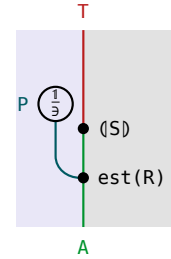
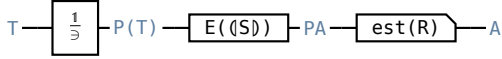
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(f_1 \text{ p}_1, Q) \wedge$   
 $\text{MonoAlg}(f_2 \text{ p}_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge$   
 $\text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq$   
 $P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq)$   
 $\wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$   
 $\wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$   
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = (f_1 \text{ p}_1) \cup (f_2 \text{ p}_2) \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq))$   
 $\text{thinlist}(Q) \sqsubseteq \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$

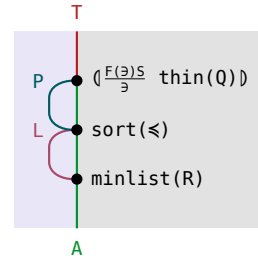
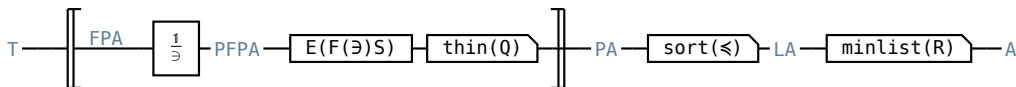
a fold on sorted lists of partial solutions, thinned at every step — at  $\triangleq \text{thinlist}$ 's binary thinning data.

circuit

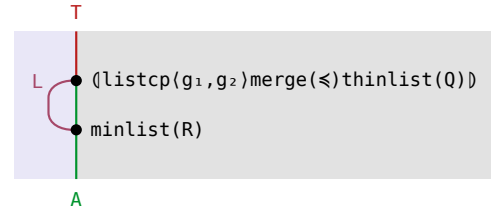
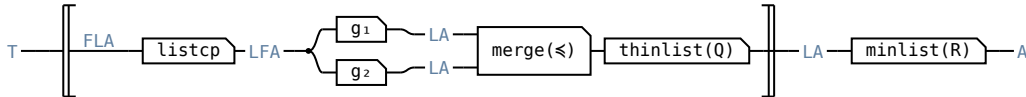
Hinze–Marsden



$\sqsubseteq$  Corollary 8.1  $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge \text{MonoAlg}(S, Q)$   
 $\Rightarrow (\frac{F(3)S}{3} \text{ thin}(Q)) \text{ est}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$  at  $f_1 p_1$  and  $f_2 p_2$  monotonic on  $Q$  —  
 $\triangleq \text{thinlist}$



$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$  — (8.3b.3) at its  $Q \triangleq R$



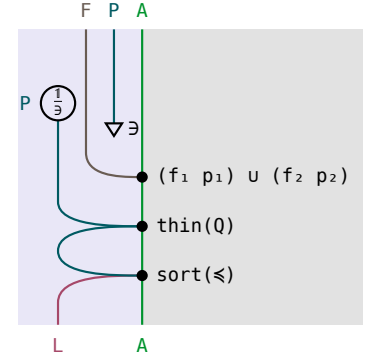
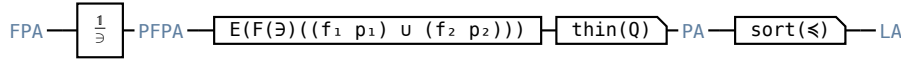
fusion at  $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y))$   
 $\wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1)$   
 $\text{sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq)$   
 $\text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge$   
 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq))$   
 $\text{merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$   
 $\Rightarrow F(\text{sort}(\leq)) \text{ listcp} \langle \text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2 \rangle \text{ merge}(\leq)$   
 $\text{thinlist}(Q) \sqsubseteq \frac{F(3) ((f_1 \text{ p}_1) \cup (f_2 \text{ p}_2))}{3} \text{ thin}(Q) \text{ sort}(\leq)$

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{ listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)
\end{aligned}$$

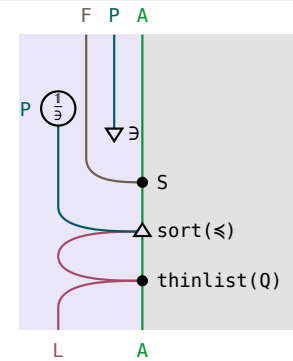
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FPA} \rightarrow \left[ \frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)S) \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \quad \text{--- (8.3b.2)}
\end{aligned}$$



$$\begin{aligned}
& \text{FPA} \rightarrow \left[ \frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)f_1 p_1) \rightarrow \text{PA} \rightarrow \text{cup} \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left( \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \right) \text{ cup} \text{ sort}(\leq) \text{ thinlist}(Q) \\
& S = (f_1 p_1) \cup (f_2 p_2) \quad \text{---} \triangleq \text{thinlist, then} \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[ \frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)f_1 p_1) \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left( \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \right) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \quad \text{--- (8.3b.6)}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[ \frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)f_1 p_1) \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{LA} \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left( \frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(\leq) \right) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& \langle X, Y \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) = \langle X \text{ sort}(\leq), Y \text{ sort}(\leq) \rangle \quad \text{--- absorption}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow F(\text{sort}(\leq)) \rightarrow \text{FLA} \rightarrow \text{listcp} \rightarrow \text{LFA} \rightarrow \left( \begin{array}{c} \text{listf}_1 \quad \text{filterp}_1 \\ \text{listf}_2 \quad \text{filterp}_2 \end{array} \right) \rightarrow \text{LA} \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& F(\text{sort}(\leq)) \text{ listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq) \text{ thinlist}(Q) \\
& \text{Lemma 8.1 } \text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, \leq) \wedge \\
& \text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq P(f) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp} \sqsubseteq E(p) \text{ sort}(\leq) \wedge \\
& F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \\
& \Rightarrow F(\text{sort}(\leq)) \text{ listcp} \text{ listf} \text{ filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then} \\
& X \langle g_1, g_2 \rangle \sqsubseteq \langle X g_1, X g_2 \rangle \quad \text{--- } X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle ; g_i \triangleq \text{list}(f_i) \text{ filter}(p_i) \quad \text{---} \triangleq \text{thinlist}
\end{aligned}$$

## §8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$ ,  $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$ ,  $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$ .

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])$ , within  $w$  the coreflexive on  $\text{xs}$  with weight  $\text{xs} \leq w$ ,  $0 \leq w$ .

$R \triangleq \text{value} \geq \text{value}^\circ$ ,  $Q \triangleq \text{weight} \leq \text{weight}^\circ$ ,  $P \triangleq R$ ,.

$\text{FA} = 1 + \text{Item} \times A$ ,  $\text{listcp} = \text{wrap} + \text{cpr}$ ,  $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$   $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$ ,  
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$ .

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$ ,  $h_2 \triangleq \text{list}(\pi_2)$ .

(8.4b)

| the law                                                                                                                                      | what it says                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) (\text{cons} (\text{within } w)) \sqsubseteq \text{cons} (\text{within } w) R$<br>FALSE                                      | not monotonic on $R$ : a selection of greater value need not still fit once one more item goes in |
| 2 $(1 \times Q) (\text{cons} (\text{within } w)) \sqsubseteq \text{cons} (\text{within } w) Q$<br>$(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$ , | both halves are monotonic on $Q$ once ties in value are broken by weight                          |

$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R$  (8.4c)  
 $[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq$   
 $P([\text{nil}, \pi_2]) \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F)$   
 $\text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup} \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt}))$   
 $\text{sort}(R) \wedge \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp} (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{merge}(R) \text{thinlist})$   
 $\text{minlist} \sqsubseteq \frac{\text{subseq within}(w)}{\supset} \text{est}(R)$

the knapsack problem, as a fold that thins the packings kept at each item

| circuit                                                                                                                                                                                                                                    | Hinze–Marsden |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <p><math>\text{subseq} (\text{within } w) \supset \text{est}(R)</math></p>                                                                                                                                                                 |               |
| <p><math>\frac{[(\text{nil}, \text{cons}) (\text{within } w) \cup [(\text{nil}, \pi_2)]]}{\supset} \text{est}(R)</math><br/> fusion, weights non-negative.</p>                                                                             |               |
| <p><math>[\text{listcp} (g_1, g_2) \text{merge } R \text{thinlist}(Q)] \text{minlist}(R)</math><br/> binary thinning theorem, at <math>P \triangleq R</math>, <math>F</math> linear, <math>Q</math> from (8.4b.2)</p>                      |               |
| <p><math>\text{listcp} = \text{wrap} + \text{cpr}</math>, <math>g_i = [\text{list}(\text{nil}), h_i] \rightarrow \triangleq \text{within}</math>;<br/> <math>\text{minlist}(R)</math> is head, packings coming out in descending value</p> |               |

## §8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[a]#xs, glue(a,xs)=[a]#head(xs)#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

| the law                                                                                       | what it says                                                                                                             |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1 (1×R) glue⊆glue R FALSE                                                                     | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| 2 (1×Q) new⊆new Q<br>(1×Q) (glue (ok w))⊆glue (ok w)Q<br>cons monotonic on collect≤collect°., | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| 3 merge T=cat; P≐head prefix head° also serves                                                | T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                        |

(8.5c)

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\
& \text{wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \in P([\text{wrap} \text{ wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau \\
& [\text{wrap} \text{ wrap}, \text{glue}]^\circ) \text{listf}_2 \in P([\text{wrap} \text{ wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \in E(\text{ok}(w)) \text{sort}(\tau) \\
& \wedge F(\text{sort}(\tau)) \text{listcp} \in \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \in \text{cup} \text{sort}(\tau) \wedge \\
& \text{sort}(\tau) \text{thinlist} \in \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \in \text{est}(R) \Rightarrow ([\text{listcp} \\
& (\text{listf}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\tau) \text{thinlist}) \text{minlist} \in \frac{\text{partition fits}(w)}{\supset} \text{est}(R) \\
& \text{a paragraph laid out as a fold that thins the layouts kept at each word}
\end{aligned}$$

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Hinze–Marsden |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(L(\text{Word})) \xrightarrow{E(\text{partition fits}(w))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & \frac{\text{partition list}^*(\text{fits } w)}{\supset} \text{est}(R) \end{aligned} $                                                                                                                                                                                                                                                        |               |
| $ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(L(\text{Word})) \xrightarrow{E([\text{wrap} \text{ wrap}, \text{new}] \cup ([\text{wrap} \text{ wrap}, \text{glue}] \text{ok}(w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap} \text{ wrap}, \text{new}] \cup ([\text{wrap} \text{ wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(R) \\ & \text{fusion, every word fits on a line by itself.} \end{aligned} $                                                      |               |
| $ \begin{aligned} & L(\text{Word}) \xrightarrow{\frac{1}{\supset}} P(L(\text{Word})) \xrightarrow{E([\text{wrap} \text{ wrap}, \text{new}] \cup ([\text{wrap} \text{ wrap}, \text{glue}] \text{ok}(w)))} P(L(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} L(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap} \text{ wrap}, \text{new}] \cup ([\text{wrap} \text{ wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(R) \\ & \text{the algebra as } (f_1 p_1) \cup (f_2 p_2), p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist.} \end{aligned} $ |               |
| $ \begin{aligned} & \sqsubseteq ([\text{listcp} \langle g_1, g_2 \rangle \text{cat thinlist}(Q)] \text{minlist}(R)) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } T = \text{cat} \text{ ((8.5b.3)), } Q \\ & \text{from (8.5b.2)} \end{aligned} $                                                                                                                                                                                                                                                                                             |               |
| $ \begin{aligned} & = ([\text{start}, \text{cpr} \langle h_1, h_2 \rangle \text{cat thinlist}(Q)] \text{minlist}(R)) \\ & \text{listcp=wrap+cpr, } g_i \text{ along the coproduct — } \triangleq \text{partition} \end{aligned} $                                                                                                                                                                                                                                                                                                                                           |               |

## §8.6. Bitonic tours

(8.6a)

$FA = (City \times City) + (City \times A)$ , the base functor of cons-lists of length at least two;  $listcp = wrap + cpr$ ;  $tc : City \times City \rightarrow Real$ , neither positive nor symmetric.

$start(a, b) = ([a, b], [a, b])$ ,  $dropl(a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$ ,  $dropr(a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$ ,  $tour \triangleq ([start, dropl \cup dropr])$ .

$cost(xs, ys) = outcost\ xs + incost\ ys$ ,  $outcost[a_0, \dots, a_n] = tc(a_0, a_1) + \dots + tc(a_{n-1}, a_n)$ ,  $incost[a_0, \dots, a_n] = tc(a_1, a_0) + \dots + tc(a_n, a_{n-1})$ .

$next \triangleq tail\ head$ ,  $next2 \triangleq next \times next$ ,  $head2 \triangleq head \times head$ ,  $R \triangleq cost \leq cost^\circ$ ,  $Q \triangleq Rn(next2\ next2^\circ) \cap (head2\ head2^\circ)$ ,  $P \triangleq T$ ,  $g_1 \triangleq list([start, dropl])$ ,  $g_2 \triangleq list([start, dropr])$ .

(8.6b)

| the law                                                                                                                                                      | what it says                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ dropl} \sqsubseteq \text{dropl}\ R \quad \text{FALSE}$<br>$(1 \times R) \text{ dropr} \sqsubseteq \text{dropr}\ R \quad \text{FALSE}$ | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| 2 $(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl}\ Q$<br>$(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr}\ Q$ ,                                     | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |

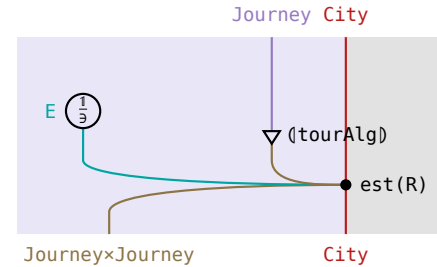
(8.6c)

$(\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([start, droplFn] \tau [start, droplFn]^\circ) \ g_1 \sqsubseteq$   
 $P([start, droplFn]) \text{ sort}(\tau) \wedge \text{sortF}([start, droprFn] \tau [start, droprFn]^\circ) \ g_2 \sqsubseteq$   
 $P([start, droprFn]) \text{ sort}(\tau) \wedge F(\text{sort}(\tau)) \text{ listcp} \sqsubseteq cp(F) \text{ sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{ cat}$   
 $\sqsubseteq \text{cup sort}(\tau) \wedge \text{sort}(\tau) \text{ thinlist} \sqsubseteq \text{thin}(Qc(tc)) \text{ sort}(\tau) \wedge \text{sort}(\tau) \text{ minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}$   
 $\langle g_1, g_2 \rangle \text{ cat thinlist}) \text{ minlist} \sqsubseteq \frac{(\text{tourAlg})}{\exists} \text{ est}(R)$   
 a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit

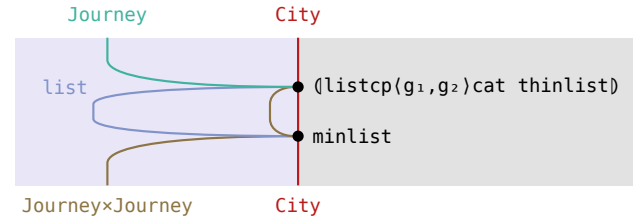
Hinze–Marsden

$dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} P(dCL(City \times City, City)) \xrightarrow{E(\langle \text{tourAlg} \rangle)} P(Journey(City)^2) \xrightarrow{\text{est}(R)} Journey(City)$   
 $\text{tour} \xrightarrow{\exists} \text{est}(R)$



$dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} P(dCL(City \times City, City)) \xrightarrow{E(\langle [start, droplFn] \cup [start, droprFn] \rangle)} P(Journey(City)^2) \xrightarrow{\text{est}(R)} Journey(City)$   
 $= \frac{([start, dropl \cup dropr])}{\exists} \text{est}(R)$   
 $\text{tour} \triangleq ([start, dropl \cup dropr]) \triangleq \text{tour}$

$dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} P(dCL(City \times City, City)) \xrightarrow{E(\langle [start, droplFn] \cup [start, droprFn] \rangle)} P(Journey(City)^2) \xrightarrow{\text{est}(R)} Journey(City)$   
 $\sqsubseteq ([\text{listcp} \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)] \text{ minlist}(R))$   
 binary thinning theorem, at  $P \triangleq T$  with merge  $\tau = \text{cat}$ ,  $Q$  from (8.6b.2)



$([start\ wrap, cpr\ (list(dropl), list(dropr)) \text{ cat}$   
 $\text{thinlist}(Q)] \text{ minlist}(R))$   
 $\text{listcp} = \text{wrap} + \text{cpr}$ ,  $g_i = [list(start), list(drop_i)] \triangleq \text{tour}$ ;  
 quadratic, two tours added per step

## §9. Dynamic Programming

### §9.1. Theory

#### §9.1.1. Example: segmenting a list

- $(\mathsf{T}) = \text{concat}$   
 $\mathsf{T} = [\text{nil}, \text{cat}]$  takes a non-empty first list, so  $\mathsf{T}^\circ$  cuts a non-empty prefix off a list every way, and folding with  $\mathsf{T}$  flattens a list of segments
- $(\mathsf{h}) = \text{list}(\text{sum})$   
 $\mathsf{h} = [\text{nil}, \text{cons}(\text{sum} \times 1)]$  puts a segment's sum in front, so folding with  $\mathsf{h}$  sums every segment
- $\mathsf{H} = \text{partition list}(\text{sum})$   
 $\mathsf{H}$  segments a list every way, then sums each segment; with  $\mathsf{R}$  comparing the largest entry,  $\mathsf{M}$  cuts the list so that the largest segment sum is as small as possible

(9.1.1b)

| $\mathbf{x}$                                            | $\mathbf{h(x)}$                         |
|---------------------------------------------------------|-----------------------------------------|
| 1 $\mathsf{l}(() )$                                     | $[\ ]$                                  |
| 2 $\mathsf{r}([\mathsf{c}], [\ ])$                      | $[\mathsf{c}]$                          |
| 3 $\mathsf{r}([\mathsf{a}, \mathsf{b}], [\mathsf{c}])$  | $[\mathsf{a} + \mathsf{b}, \mathsf{c}]$ |
| 4 $\mathsf{r}([\mathsf{a}], [\mathsf{b} + \mathsf{c}])$ | $[\mathsf{a}, \mathsf{b} + \mathsf{c}]$ |

- $\mathsf{H}([\mathsf{a}, \mathsf{b}, \mathsf{c}], \mathsf{ys}) \iff \mathsf{ys} = [\mathsf{a} + \mathsf{b} + \mathsf{c}] \vee \mathsf{ys} = [\mathsf{a}, \mathsf{b} + \mathsf{c}] \vee \mathsf{ys} = [\mathsf{a} + \mathsf{b}, \mathsf{c}] \vee \mathsf{ys} = [\mathsf{a}, \mathsf{b}, \mathsf{c}]$   
 $\mathsf{H}$  sends  $[\mathsf{a}, \mathsf{b}, \mathsf{c}]$  to the sums of its four segmentations and nothing else
- $\mathsf{T}^\circ \mathsf{F}(\mathsf{H}) \mathsf{h} = \mathsf{H}$   
cut once, solve the rest, put the pieces back together

(9.1.1c)

(9.1.1d)

$\mathsf{h} : \mathsf{FB} \rightarrow \mathsf{B}$  a map,  $\mathsf{T} : \mathsf{FA} \rightarrow \mathsf{A}$  an  $\mathsf{F}$ -algebra,  $\mathsf{R} : \mathsf{B} \rightarrow \mathsf{B}$ .

$\mathsf{H} \triangleq (\mathsf{T})^\circ (\mathsf{h}) : \mathsf{A} \rightarrow \mathsf{B}$ ,  $\mathsf{M} \triangleq \frac{\mathsf{H}}{\mathsf{S}} \text{est}(\mathsf{R})$  the problem to be solved,  $(\mu \mathsf{X} : \mathsf{G}(\mathsf{X}))$  as in  $\triangleq \mu$ .

$\mathsf{DPSetting}(\mathsf{h}, \mathsf{R}) \triangleq \text{Map}(\mathsf{h}) \wedge \text{MonoAlg}(\mathsf{h}, \mathsf{R}^\circ) \wedge \mathsf{R}^\circ \mathsf{R}^\circ \subseteq \mathsf{R}^\circ$

$\mathsf{h}$  is a map,  $\mathsf{h}$  is monotonic on  $\mathsf{R}^\circ$ , and  $\mathsf{R}^\circ$  is transitive — assumed by every theorem below

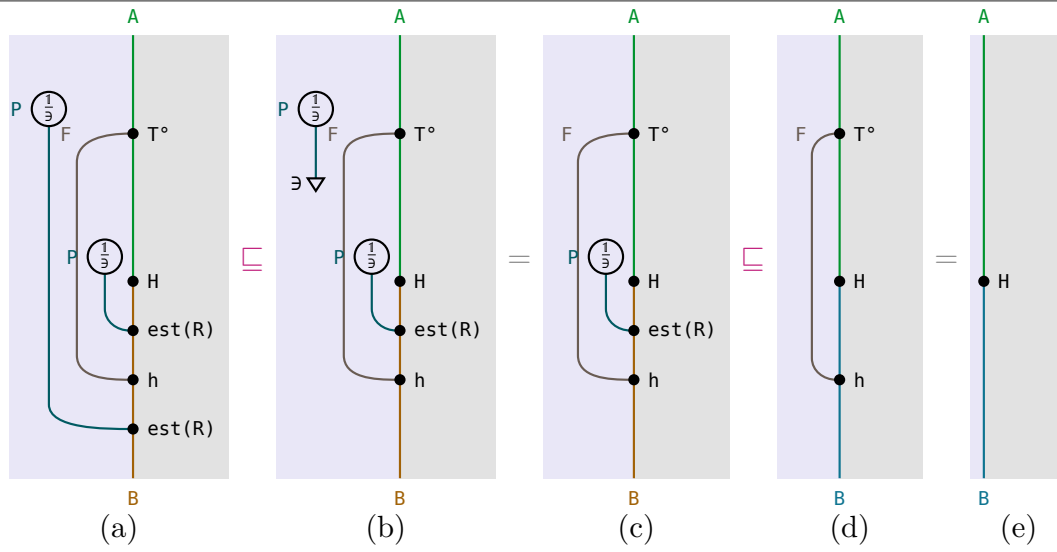
**$(\mu \mathsf{X} : \frac{\mathsf{T}^\circ}{\mathsf{S}} \mathsf{P}(\mathsf{F}(\mathsf{X}) \mathsf{h}) \text{est}(\mathsf{R})) \subseteq \frac{\mathsf{H}}{\mathsf{S}} \text{est}(\mathsf{R})$**   
every answer the recursion returns is an optimal one

Theorem 9.1

- $\frac{\mathsf{T}^\circ}{\mathsf{S}}$  takes the input apart one step every way;  $\mathsf{F}(\mathsf{X})$  solves each part by the recursion  $\mathsf{X}$ ;  $\mathsf{h}$  assembles each candidate;  $\text{est}(\mathsf{R})$  keeps a best one
- $\sqsubseteq \mathsf{M}$ : that never misses the optimum, though all candidates are never listed as  $\mathsf{H}$  would
- $\mathsf{h}$  monotonic on  $\mathsf{R}$ : a better part never makes the whole worse
- $\sqsubseteq$ , not  $=$ : some optima may not be returned; that the recursion returns anything at all needs  $\mathsf{T}^\circ$  to stop taking apart — dynamic programming theorem

$$T^\circ F(H) h = H \implies \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \sqsubseteq H \quad (9.2)$$

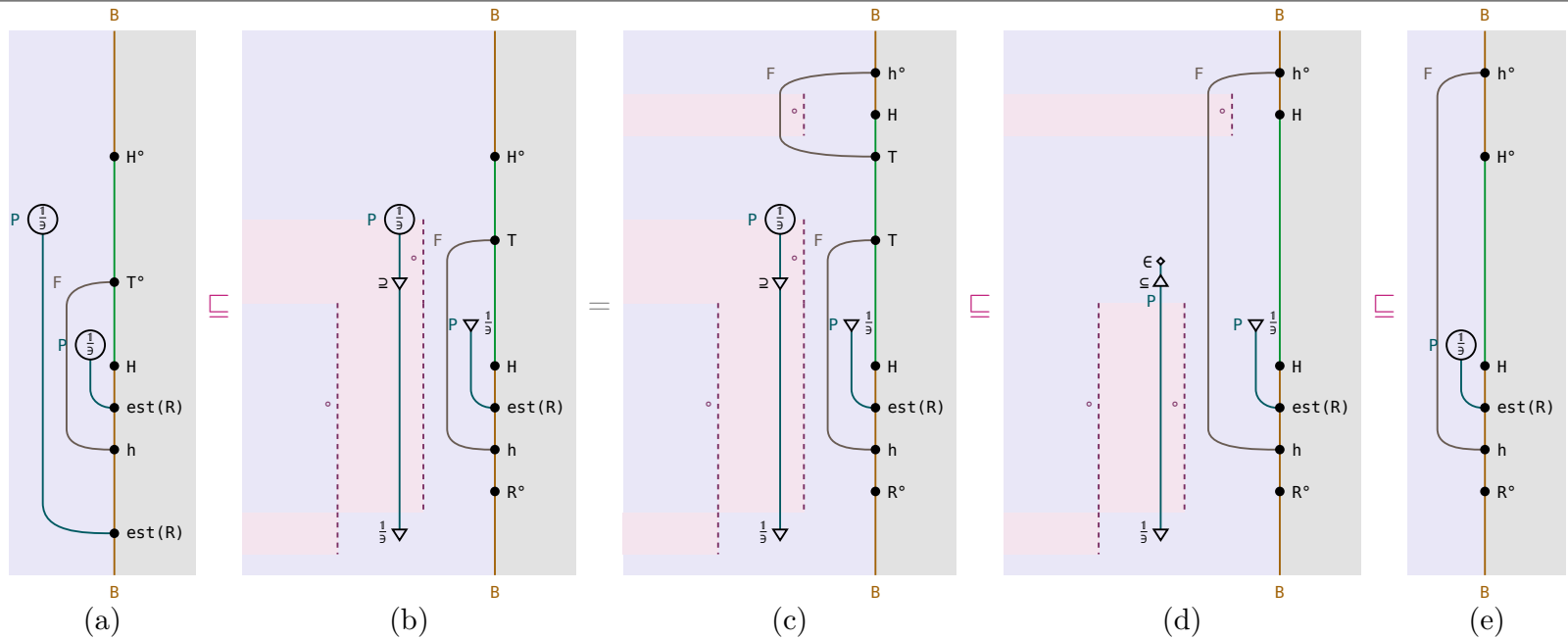
taking the input apart every way  $T$  allows ( $\frac{T^\circ}{\exists}$ ), solving each part by  $M$  and keeping an optimum ( $P(F(M)h) \text{est}(R)$ )  
returns only what  $H$  returns



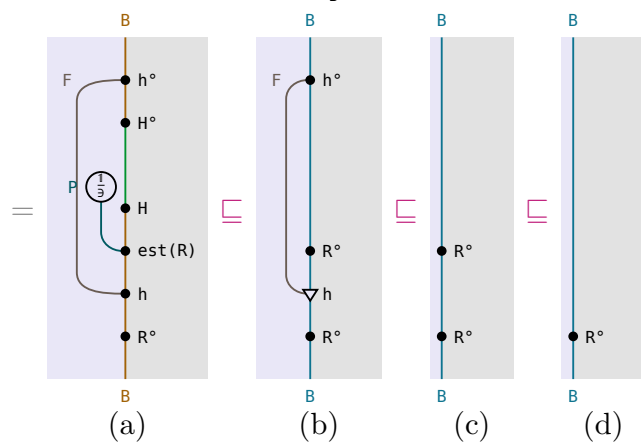
|     |                                                                                    |                                                 |
|-----|------------------------------------------------------------------------------------|-------------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$    | $P(S) \text{est}(R) \sqsubseteq \exists S$      |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{est}(R)) h$ | $\frac{R}{\exists} \exists = R$                 |
| (c) | $= T^\circ F(\frac{H}{\exists} \text{est}(R)) h$                                   | $\frac{R}{\exists} \text{est}(R) \sqsubseteq S$ |
| (d) | $\sqsubseteq T^\circ F(H) h$                                                       | $T^\circ F(H) h = H$                            |
| (e) | $= H$                                                                              |                                                 |

$$T^\circ F(H) h = H \implies H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \sqsubseteq R^\circ \quad (9.3)$$

for every  $\mathbf{b}$  that  $\mathbf{H}$  returns from an input, the step  $\frac{T^\circ}{\exists} P(F(\mathbf{M})h) \text{est}(R)$  returns from that input only  $\mathbf{b}'$  with  $\mathbf{R}$  relating  $\mathbf{b}'$  to  $\mathbf{b}$



|     |                                                                                                                      |                                                                   |
|-----|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$                              | $P(S) \text{est}(R) \sqsubseteq \epsilon \setminus (S R^\circ)$   |
| (b) | $\sqsubseteq H^\circ \frac{T^\circ}{\exists} (\epsilon \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$    | $T^\circ F(H) h = H$<br>$\implies H^\circ = h^\circ F(H^\circ) T$ |
| (c) | $= h^\circ F(H^\circ) T \frac{T^\circ}{\exists} (\epsilon \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$ | $T \frac{T^\circ}{\exists} \sqsubseteq \epsilon$                  |
| (d) | $\sqsubseteq h^\circ F(H^\circ) \epsilon (\epsilon \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$        | $S (S \setminus R) \sqsubseteq R$                                 |
| (e) | $\sqsubseteq h^\circ F(H^\circ) F(\frac{H}{\exists} \text{est}(R)) h R^\circ$                                        |                                                                   |



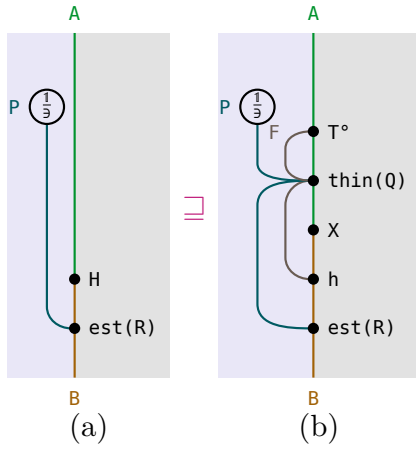
|     |                                                                  |                                                               |
|-----|------------------------------------------------------------------|---------------------------------------------------------------|
| (a) | $= h^\circ F(H^\circ \frac{H}{\exists} \text{est}(R)) h R^\circ$ | $S^\circ \frac{S}{\exists} \text{est}(R) \sqsubseteq R^\circ$ |
| (b) | $\sqsubseteq h^\circ F(R^\circ) h R^\circ$                       | $\text{MonoAlg}(h, R^\circ)$                                  |
| (c) | $\sqsubseteq R^\circ R^\circ$                                    | $R^\circ R^\circ \sqsubseteq R^\circ$                         |
| (d) | $\sqsubseteq R^\circ$                                            |                                                               |

$$\text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{thin}(Q) P(F(X) h) \text{est}(R)) \sqsubseteq \frac{H}{\exists} \text{est}(R)$$

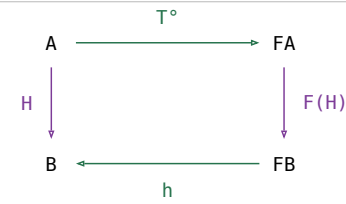
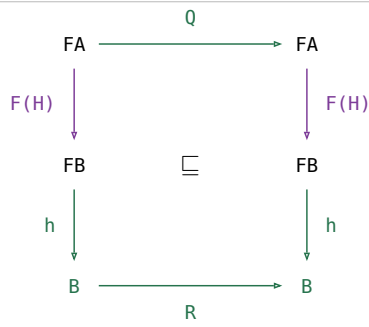
Theorem 9.2

an optimum over everything  $H$  returns is reached by taking the input apart every way  $T$  allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

$$H \triangleq (T)^\circ (h) \triangleq H$$



|     |                                                                              |                                                                                                                                                                                                                                                                                                    |
|-----|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{H}{\exists} \text{est}(R)$                                            | $QF(H)h \sqsubseteq F(H)hR, h \text{ monotonic on } R \text{ — } Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R), X \sqsubseteq \text{est}(R) \Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$ |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \text{thin}(Q) P(F(X) h) \text{est}(R)$ |                                                                                                                                                                                                                                                                                                    |



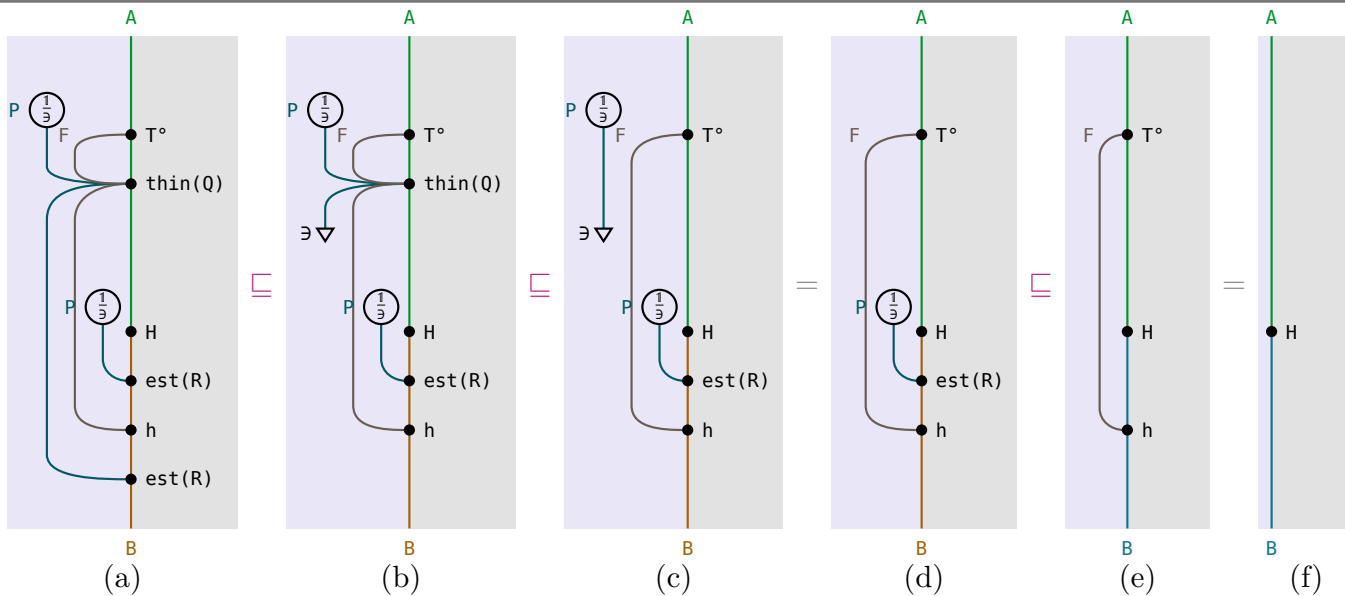
$T^\circ F(H)h = H$ : taking the input apart once by  $T^\circ$ , solving the parts by  $F(H)$  and assembling by  $h$  returns exactly what  $H$  returns

$QF(H)h \sqsubseteq F(H)hR$ : every result of thinning by  $Q$ , solving the parts by  $F(H)$  and assembling by  $h$  is reached by  $R$  from a result of solving by  $F(H)$  and assembling by  $h$  without  $Q$

$$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \sqsubseteq H$$

(9.1.1e)

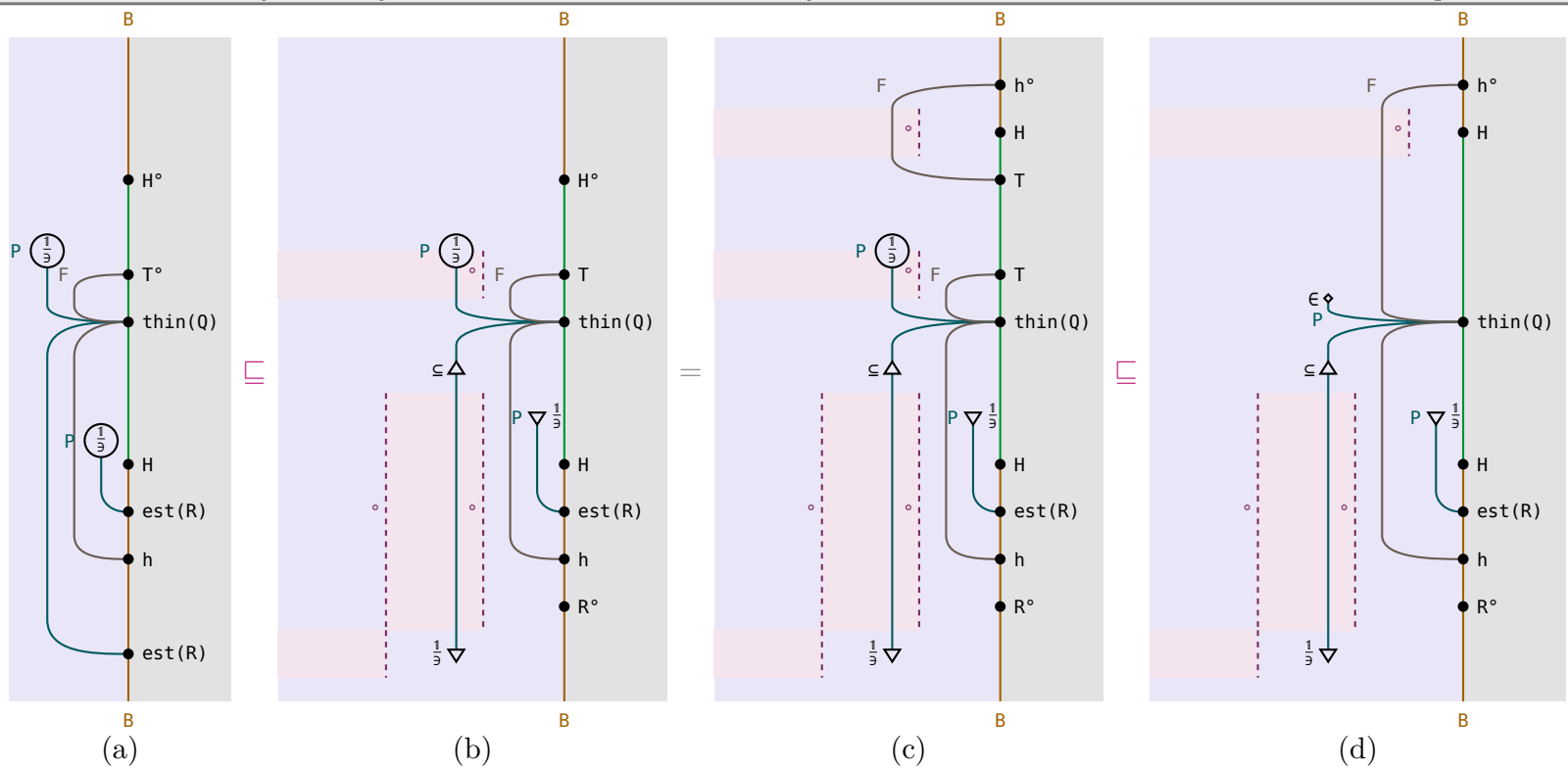
taking the input apart every way  $T$  allows ( $\frac{T^\circ}{\exists}$ ), dropping the parts  $\text{thin}(Q)$  rejects, solving each rest by  $M$  and keeping an optimum returns only what  $H$  returns



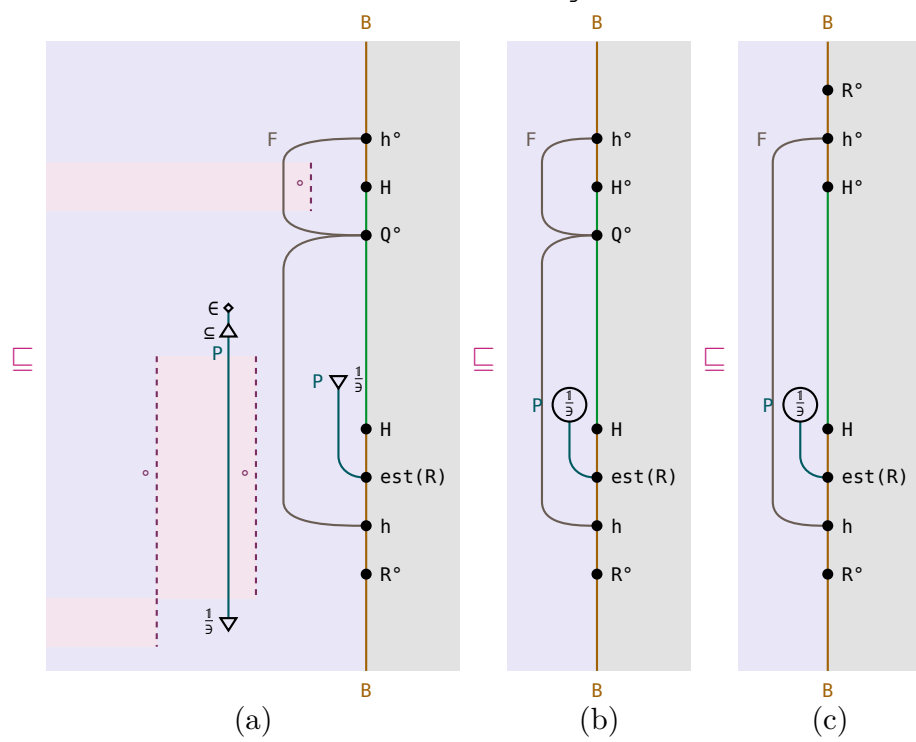
|     |                                                                                                   |                                                 |
|-----|---------------------------------------------------------------------------------------------------|-------------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$    | $P(S) \text{est}(R) \sqsubseteq \exists S$      |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \text{thin}(Q) \exists F(\frac{H}{\exists} \text{est}(R)) h$ | $\text{thin}(Q) \exists \sqsubseteq \exists$    |
| (c) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{est}(R)) h$                | $\frac{R}{\exists} \exists = R$                 |
| (d) | $= T^\circ F(\frac{H}{\exists} \text{est}(R)) h$                                                  | $\frac{H}{\exists} \text{est}(R) \sqsubseteq S$ |
| (e) | $\sqsubseteq T^\circ F(H) h$                                                                      | $T^\circ F(H) h = H$                            |
| (f) | $= H$                                                                                             |                                                 |

$$T^\circ F(H) h = H \wedge Q F(H) h \subseteq F(H) h R \implies H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R) \subseteq R^\circ \quad (9.1.1f)$$

$H^\circ$  followed by the body at  $M$  is  $\mathbb{E}R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input

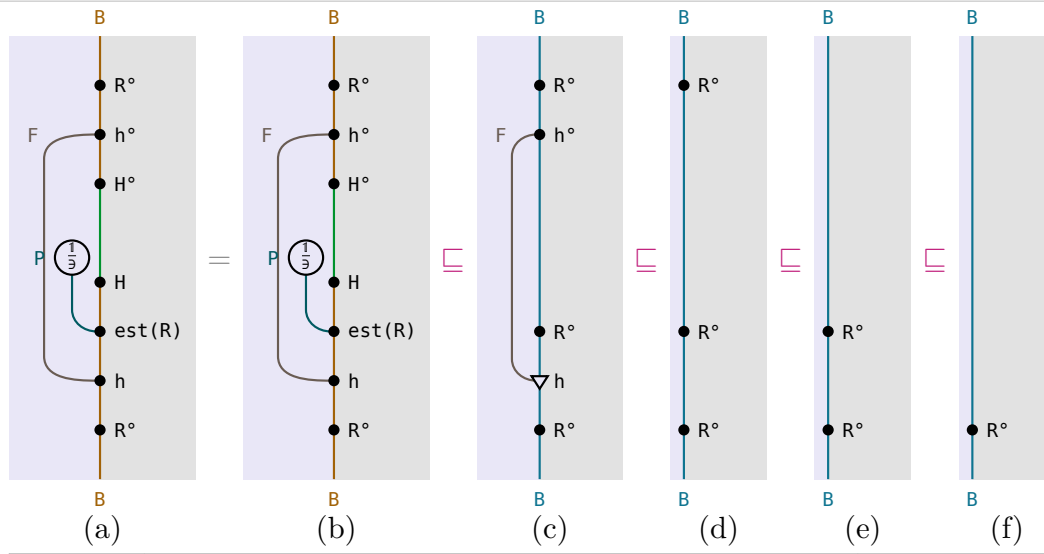


|     |                                                                                                                                       |                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R)) h) \text{est}(R)$                                | $P(S) \text{est}(R) \subseteq \mathbb{E} \setminus (S R^\circ)$ |
| (b) | $\sqsubseteq H^\circ \frac{T^\circ}{\exists} \text{thin}(Q) (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$    | $T^\circ F(H) h = H$                                            |
| (c) | $= h^\circ F(H^\circ) T \frac{T^\circ}{\exists} \text{thin}(Q) (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$ | $\implies H^\circ = h^\circ F(H^\circ) T$                       |
| (d) | $\sqsubseteq h^\circ F(H^\circ) \mathbb{E} \text{thin}(Q) (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$      | $T \frac{T^\circ}{\exists} \subseteq \mathbb{E}$                |
|     |                                                                                                                                       | $\mathbb{E} \text{thin}(Q) \subseteq Q^\circ \mathbb{E}$        |



|     |                                                                                                                           |                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| (a) | $\sqsubseteq h^\circ F(H^\circ) Q^\circ \mathbb{E} (\mathbb{E} \setminus (F(\frac{H}{\exists} \text{est}(R)) h R^\circ))$ | $S (S \setminus R) \subseteq R$                                            |
| (b) | $\sqsubseteq h^\circ F(H^\circ) Q^\circ F(\frac{H}{\exists} \text{est}(R)) h R^\circ$                                     | $Q F(H) h \subseteq F(H) h R$                                              |
| (c) | $\sqsubseteq R^\circ h^\circ F(H^\circ) F(\frac{H}{\exists} \text{est}(R)) h R^\circ$                                     | $\implies h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$ |

1

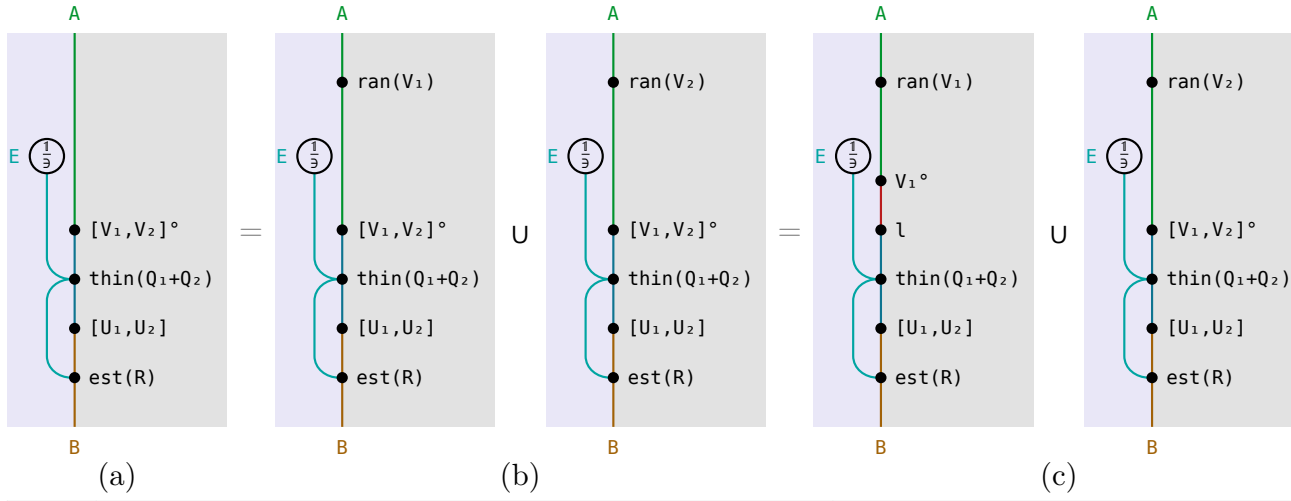


|       |                                                                                |                                                            |
|-------|--------------------------------------------------------------------------------|------------------------------------------------------------|
| 1 (a) | $R^\circ \ h^\circ \ F(H^\circ) \ F(\frac{H}{\exists} \ est(R)) \ h \ R^\circ$ |                                                            |
| 2 (b) | $= R^\circ \ h^\circ \ F(H^\circ \ \frac{H}{\exists} \ est(R)) \ h \ R^\circ$  | $S^\circ \ \frac{S}{\exists} \ est(R) \sqsubseteq R^\circ$ |
| 3 (c) | $\sqsubseteq R^\circ \ h^\circ \ F(R^\circ) \ h \ R^\circ$                     | $\text{MonoAlg}(h, R^\circ)$                               |
| 4 (d) | $\sqsubseteq R^\circ \ R^\circ \ R^\circ$                                      | $R^\circ \ R^\circ \sqsubseteq R^\circ$                    |
| 5 (e) | $\sqsubseteq R^\circ \ R^\circ$                                                | $R^\circ \ R^\circ \sqsubseteq R^\circ$                    |
| 6 (f) | $\sqsubseteq R^\circ$                                                          |                                                            |

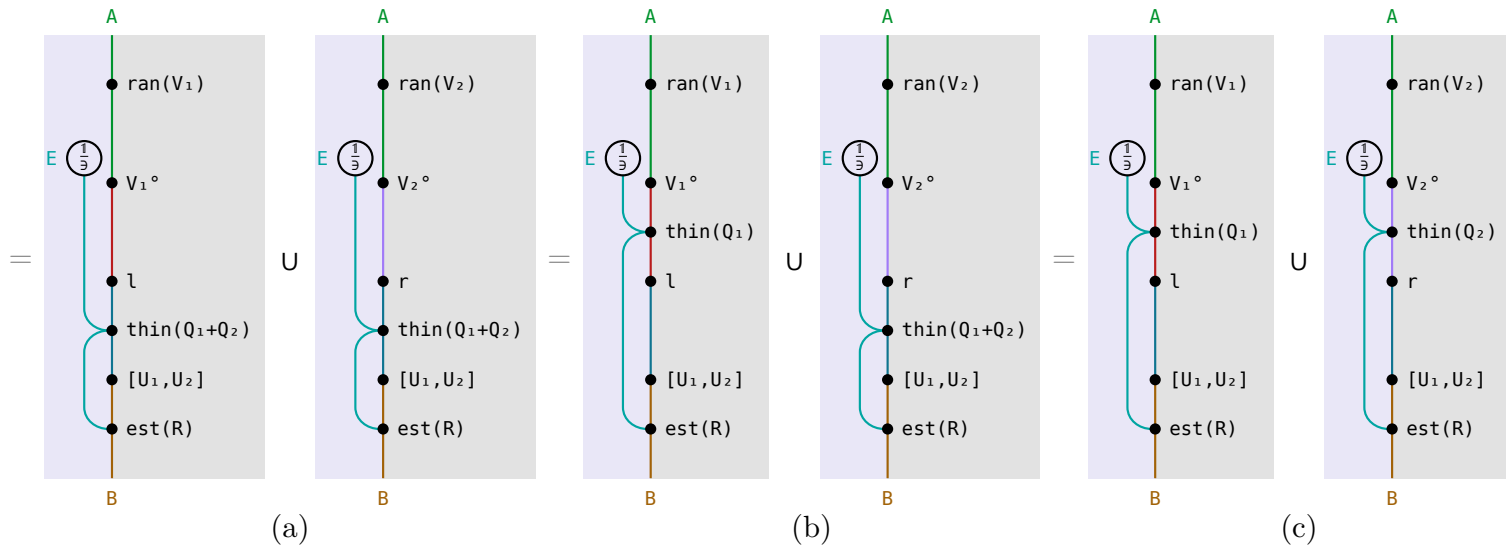
Proposition 9.1

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R) = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(U_2) \ \text{est}(R))$$

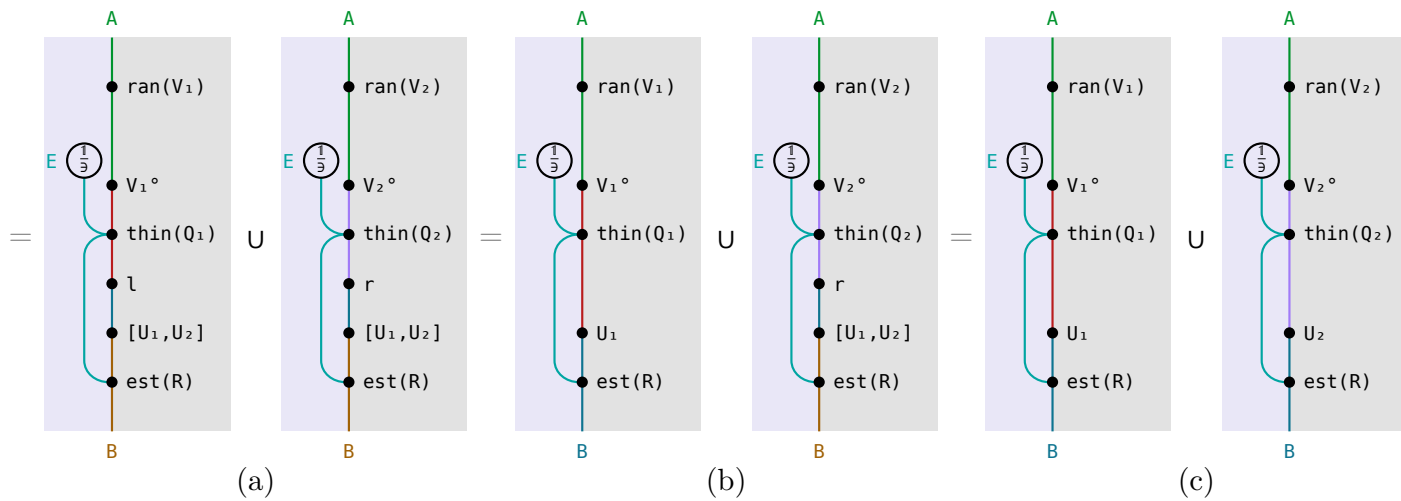
when  $V_1$  and  $V_2$  have disjoint ranges, the body over  $[V_1, V_2]$  runs the  $V_1$  problem on inputs  $V_1$  reaches and the  $V_2$  problem on inputs  $V_2$  reaches



|     |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)$                                                                                                                                        | $\frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R) = (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ |
| (b) | $= (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$ | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \ P(l)$                                                                                                                                                                           |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$        | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \ P(r)$                                                                                                                                                                           |



|     |                                                                                                                                                                                                                      |                                                           |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_1 + Q_2) \ P([U_1, U_2]) \ \text{est}(R))$   | $P(l) \ \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) \ P(l)$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ | $P(r) \ \text{thin}(Q_1 + Q_2) = \text{thin}(Q_2) \ P(r)$ |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ |                                                           |



|     |                                                                                                                                                                                                                      |                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(l) \ P([U_1, U_2]) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$ | $l \ [R, S] = R$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(r) \ P([U_1, U_2]) \ \text{est}(R))$               | $r \ [R, S] = S$ |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{thin}(Q_1) \ P(U_1) \ \text{est}(R)) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{thin}(Q_2) \ P(U_2) \ \text{est}(R))$                             |                  |

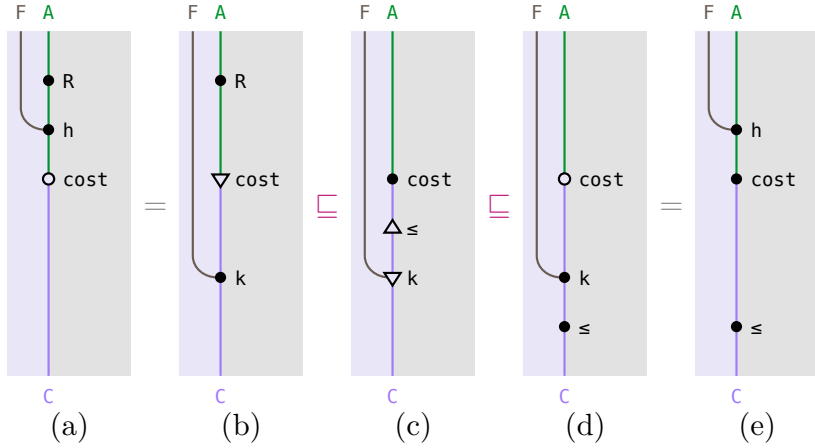
**Map(cost)  $\wedge$  R = cost  $\leq$  cost $^\circ$   $\wedge$  h cost = F(cost) k  $\wedge$  F( $\leq$ ) k  $\sqsubseteq$  k  $\leq \implies$  MonoAlg(h,R)** Proposition 9.2

if R compares two values by comparing their **costs** under  $\leq$ , and h then **cost** equals F(**cost**) then

a k monotonic on  $\leq$ , then h is monotonic on R

**Map(cost)  $\wedge$  R = cost  $\leq$  cost $^\circ$   $\wedge$  h cost = F(cost) k  $\wedge$  F( $\leq$ ) k  $\sqsubseteq$  k  $\leq \implies$  F(R) h cost  $\sqsubseteq$  h cost  $\leq$**

by the definition of R and shunting at the map **cost**, F(R)h  $\sqsubseteq$  hR is F(R)h cost  $\sqsubseteq$  h cost  $\leq$ , which the chain proves



|     |                                     |                                                                        |
|-----|-------------------------------------|------------------------------------------------------------------------|
| (a) | F(R) h cost                         | h cost = F(cost) k                                                     |
| (b) | = F(R) F(cost) k                    | Map(f) $\wedge$ R = f S f $^\circ$<br>$\implies$ R f $\sqsubseteq$ f S |
| (c) | $\sqsubseteq$ F(cost) F( $\leq$ ) k | F( $\leq$ ) k $\sqsubseteq$ k $\leq$                                   |
| (d) | $\sqsubseteq$ F(cost) k $\leq$      | h cost = F(cost) k                                                     |
| (e) | = h cost $\leq$                     |                                                                        |

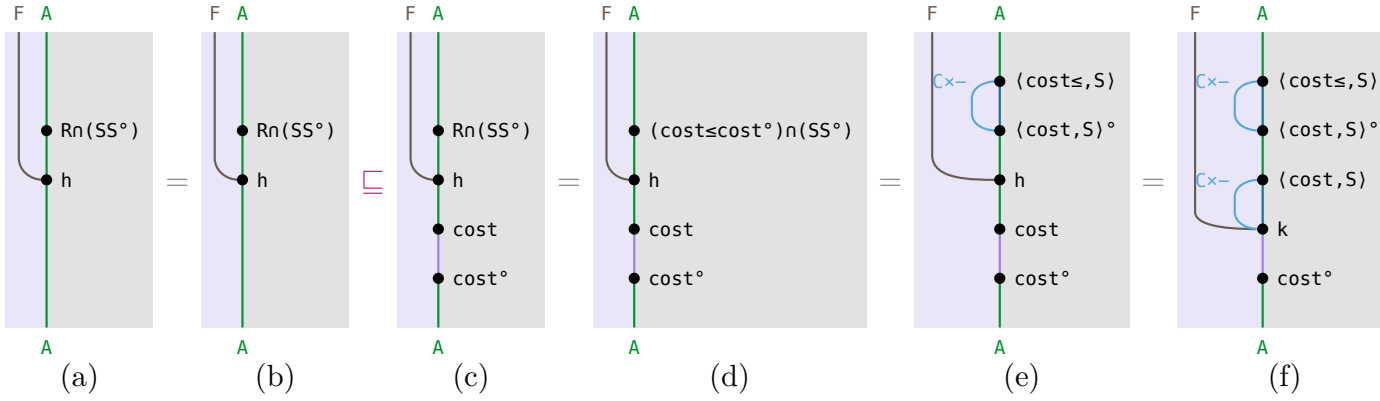
(9.1.1h)

|   | the condition                                                                                                                                                                                                | how it is discharged                                                               |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1 | F(R)h $\sqsubseteq$ hR<br>Proposition 9.2, R $\triangleq$ cost $\leq$ cost $^\circ$ , h cost = F(cost)k, F( $\leq$ ) k $\sqsubseteq$ k $\leq$                                                                | monotonicity when the cost is itself a fold with a step k<br>monotonic on $\leq$   |
| 2 | F(Rn(H $^\circ$ H))h $\sqsubseteq$ hR<br>Proposition 9.3, R $\triangleq$ cost $\leq$ cost $^\circ$ , h cost = F((cost, H $^\circ$ ))k,<br>F( $\leq \times 1$ ) k $\sqsubseteq$ k $\leq$ , H $^\circ$ simple; | monotonicity <b>in context</b> : k may also read the input the part was built from |
| 3 | QF(H)h $\sqsubseteq$ F(H)hR at Q $\triangleq$ F(U,V)<br>Proposition 9.4, U, V preorders, F(U,R)h $\sqsubseteq$ hR, Vh $\sqsubseteq$ hR                                                                       | both conditions at once, split along the two arguments of a bifunctor              |

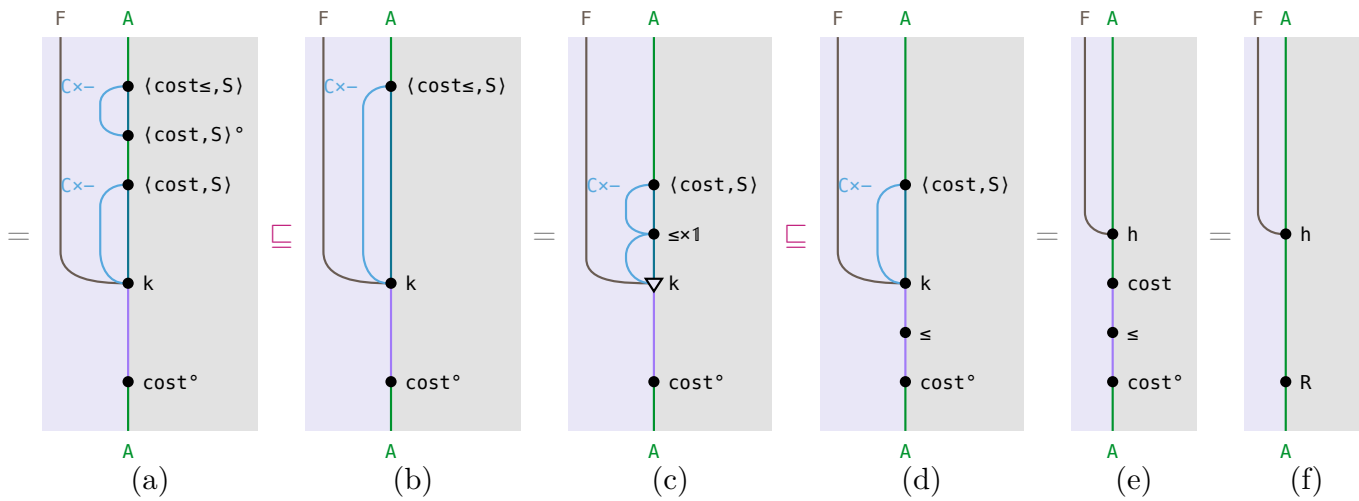
$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge h \text{ cost} = F(\langle \text{cost}, S \rangle) k \wedge F(\leq \times 1) k \sqsubseteq$  Proposition 9.3

$k \leq \implies F(R \cap (S S^\circ)) h \sqsubseteq h R$

improving each part by  $R$  within its  $S$ -context, then assembling by  $h$ , is below  $hR$



|     |                                                                                                                                        |                                                                                                              |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| (a) | $F(R \cap (S S^\circ)) h$                                                                                                              |                                                                                                              |
| (b) | $= F(R \cap (S S^\circ)) h$                                                                                                            | $\text{Map}(f)$<br>$\implies 1 \sqsubseteq f f^\circ$                                                        |
| (c) | $\sqsubseteq F(R \cap (S S^\circ)) h \text{ cost cost}^\circ$                                                                          | $R = \text{cost} \leq \text{cost}^\circ$                                                                     |
| (d) | $= F((\text{cost} \leq \text{cost}^\circ) \cap (S S^\circ)) h \text{ cost cost}^\circ$                                                 | $\langle X, Y \rangle \langle R, S \rangle^\circ = \langle X R^\circ \rangle \cap \langle Y S^\circ \rangle$ |
| (e) | $= F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ) h \text{ cost cost}^\circ$                              | $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$                                                       |
| (f) | $= F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ) F(\langle \text{cost}, S \rangle) k \text{ cost}^\circ$ |                                                                                                              |



|     |                                                                                                                                     |                                                                            |
|-----|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| (a) | $= F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle) k \text{ cost}^\circ$ |                                                                            |
| (b) | $\sqsubseteq F(\langle \text{cost} \leq, S \rangle) k \text{ cost}^\circ$                                                           | $\langle X, Y \rangle \langle R \times 1 \rangle = \langle X R, Y \rangle$ |
| (c) | $= F(\langle \text{cost}, S \rangle) F(\leq \times 1) k \text{ cost}^\circ$                                                         | $F(\leq \times 1) k \sqsubseteq k \leq$                                    |
| (d) | $\sqsubseteq F(\langle \text{cost}, S \rangle) k \leq \text{cost}^\circ$                                                            | $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$                     |
| (e) | $= h \text{ cost} \leq \text{cost}^\circ$                                                                                           | $R = \text{cost} \leq \text{cost}^\circ$                                   |
| (f) | $= h R$                                                                                                                             |                                                                            |

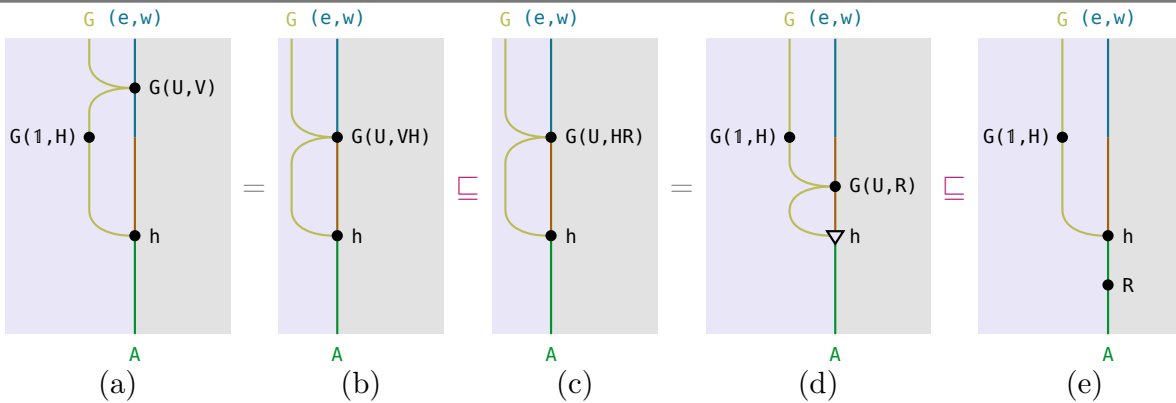
$G(U, R) h \sqsubseteq h R \wedge V H \sqsubseteq H R \implies G(U, V) G(1, H) h \sqsubseteq G(1, H) h R$

Proposition 9.4

thinning the parts by  $U$  in the first argument and by  $V$  in the second ( $G(U, V)$ ), then solving by  $H$

and assembling by  $h$ , gives only what solving and assembling and then improving by  $R$  gives

$(G(1, H) h R)$



|     |                           |                             |
|-----|---------------------------|-----------------------------|
| (a) | $G(U, V) G(1, H) h$       |                             |
| (b) | $= G(U, V H) h$           | $V H \sqsubseteq H R$       |
| (c) | $\sqsubseteq G(U, H R) h$ |                             |
| (d) | $= G(1, H) G(U, R) h$     | $G(U, R) h \sqsubseteq h R$ |
| (e) | $\sqsubseteq G(1, H) h R$ |                             |

(9.2a)

- $(\exists a. o = \text{cpy}(a)) \vee (\exists a. o = \text{del}(a)) \vee (\exists a. o = \text{ins}(a))$   
an edit operation copies, deletes or inserts one character
- $\text{FX} = 1 + (0p \times X)$   
 $\text{F}(0p, X)$  is the empty sequence or one operation paired with an  $X$
- $\text{edit} = ([\text{base}, \text{step}])$   
 $\text{edit}$  is the fold of  $[\text{base}, \text{step}]$  over an edit sequence, returning the two strings it reconstitutes
- $\text{length} = ([\text{zero}, \pi_2 \text{ succ}])$   
the fold of  $[\text{zero}, \pi_2 \text{ succ}]$  counts the operations of a sequence
- $\text{R} = \text{length} \leq \text{length}^\circ$   
 $\text{R}$  relates  $\text{es}$  to  $\text{fs}$  when  $\text{es}$  is no longer than  $\text{fs}$
- $\text{V} = \text{suffix}^\circ \times \text{suffix}^\circ$   
 $\text{V}$  relates two pairs of strings when each string of the first is a suffix of the matching string of the second
- $\text{Q}(\text{l}(u), \text{l}(u))$   
 $\text{Q}$  is the identity on the **base** summand
- $\text{Q}(\text{r}(p), \text{r}(q)) \iff \text{V}(\pi_2(p), \pi_2(q))$   
on the **step** summand  $\text{Q}$  compares any two operations ( $\text{U} \triangleq \text{T}$ ) and the two pairs of strings by  $\text{V}$
- $\text{empty}(p, q) \iff p = q \wedge p = ([], [])$   
 $\text{empty}$  is the coreflexive holding only at the pair of two empty strings
- $q \in \text{unstep}(p) \implies \text{step}(\text{r}(q)) = p$   
every decomposition  $\text{unstep}$  returns is sent back to its input by  $\text{step}$

(9.2b)

|   | <b>x</b>                   | <b><math>\alpha(x)</math></b> |
|---|----------------------------|-------------------------------|
| 1 | $\text{l}(u)$              | $[\ ]$                        |
| 2 | $\text{r}((o, \text{es}))$ | $[o]\#\text{es}$              |

(9.2c)

|   | <b>x</b>                              | <b><math>[\text{base}, \text{step}](x)</math></b> |
|---|---------------------------------------|---------------------------------------------------|
| 1 | $\text{l}(u)$                         | $([\ ], [\ ])$                                    |
| 2 | $\text{r}((\text{cpy}(a), (xs, ys)))$ | $([a]\#xs, [a]\#ys)$                              |
| 3 | $\text{r}((\text{del}(a), (xs, ys)))$ | $([a]\#xs, ys)$                                   |
| 4 | $\text{r}((\text{ins}(a), (xs, ys)))$ | $(xs, [a]\#ys)$                                   |

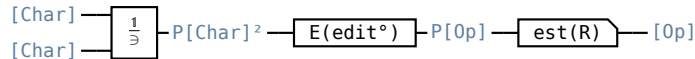
(9.2d)

|   | <b>p</b>             | <b><math>\text{unstep}(p)</math></b>                                                                   |
|---|----------------------|--------------------------------------------------------------------------------------------------------|
| 1 | $([\ ], [\ ])$       | $[\ ]$                                                                                                 |
| 2 | $([a]\#xs, [\ ])$    | $[(\text{del}(a), (xs, [\ ]))]$                                                                        |
| 3 | $([\ ], [b]\#ys)$    | $[(\text{ins}(b), ([\ ], ys))]$                                                                        |
| 4 | $([a]\#xs, [b]\#ys)$ | if $a = b$ then $[(\text{cpy}, xs, ys)]$ else $[(\text{del}, xs, [b]\#ys), (\text{ins}, [a]\#xs, ys)]$ |

$$\begin{aligned} \text{mle} = & (\text{empty} \rightarrow \text{nil}, \text{unstep } \text{list}((1 \times \text{mle}) \text{cons}) \text{minlist}(\text{R})), (\mu X : \frac{[\text{base}, \text{step}]^\circ}{3} \text{thin}(\text{Q}) \text{P}(\text{F}(\text{X}) \\ & [\text{nil}, \text{cons}]) \text{est}(\text{R})) \sqsubseteq \frac{\text{edit}^\circ}{3} \text{est}(\text{R}) \end{aligned} \quad (9.2e)$$

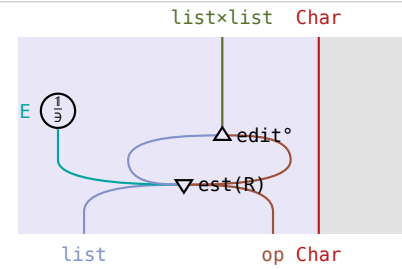
a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

## circuit



the specification —  $\triangleq$  edit

## Hinze–Marsden

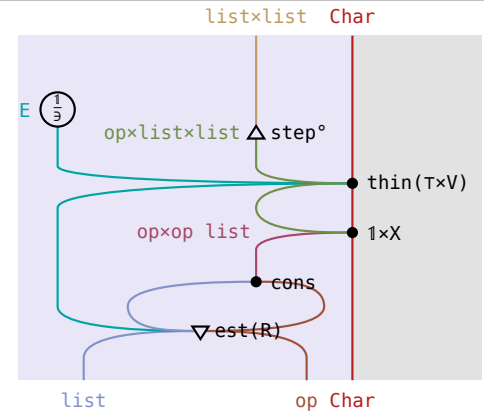


at  $Q \triangleq 1 + (U \times V)$ . Monotonicity  $F(R) \alpha \sqsubseteq \alpha R$  is Proposition 9.2 at  $\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])$  with  $\text{succ}$  monotonic on  $\leq$ , so  $\text{cons}$  is monotonic on  $R$ . The thinning condition  $\sqsubseteq QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$  over  $F(0p, [\text{Char}] \times [\text{Char}])$  is Proposition 9.4 at  $U \triangleq T$  —  $F(T, R) \alpha \sqsubseteq \alpha R$ , left as an exercise, so any two operations may be compared — and  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ , —  $\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}, \text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$ .  $\text{edit} \cdot \text{suffix} = \text{tail}^*$  and  $BA \sqsubseteq CB \Rightarrow BA^* \sqsubseteq C^*B$  cut those to one step each: drop the operation that produced the head, or weaken its  $\text{cpy}$  to a  $\text{del}$ , never lengthening the sequence

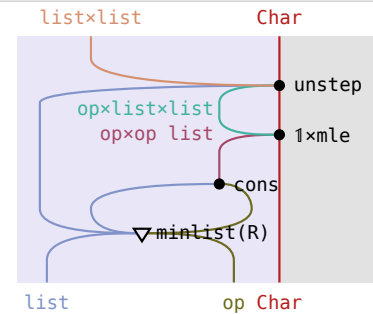
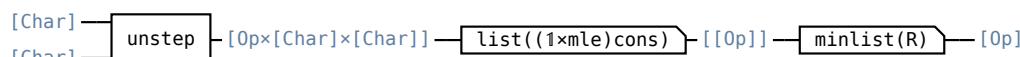
$\sqsubseteq$  Proposition 9.1:  $\text{base}$  and  $\text{step}$  have disjoint ranges, so the fixed point splits into one branch per summand —  $\text{empty}$ , the coreflexive on  $(xs, ys)$  with both lists empty, is where  $\text{base}$  returns



$\sqsubseteq$  Proposition 9.1:  $\text{base}$  and  $\text{step}$  have disjoint ranges, so the fixed point splits into one branch per summand —  $\text{empty}$ , the coreflexive on  $(xs, ys)$  with both lists empty, is where  $\text{base}$  returns



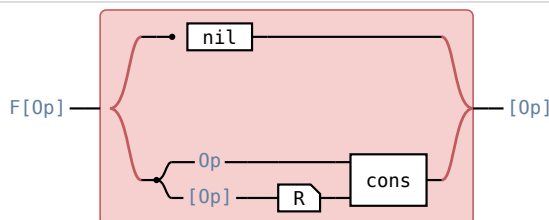
$\sqsubseteq$   $\text{unstep}$  implements  $\frac{\text{step}^\circ}{3} \text{thin}(U \times V)$  — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and  $\text{minlist}(R)$  implements  $\text{est}(R)$ . The same subproblem is solved many times over, so the running time is exponential in the two lengths



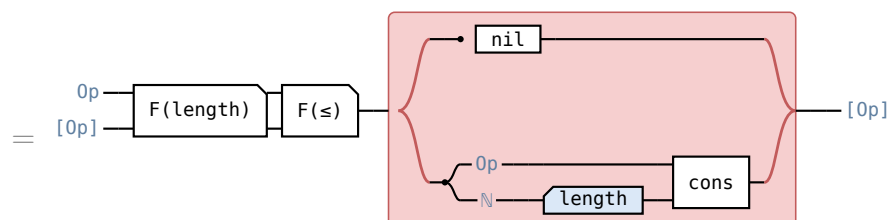
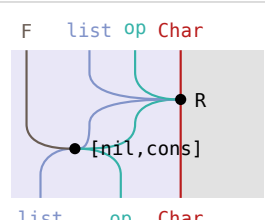
$$F(R) [\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}] R \quad (9.2f)$$

comparing the tails by  $R$  and then putting an operation in front ( $F(R) \alpha$ ) relates only sequences that  $R$  also relates once the operation is in front ( $\alpha R$ )

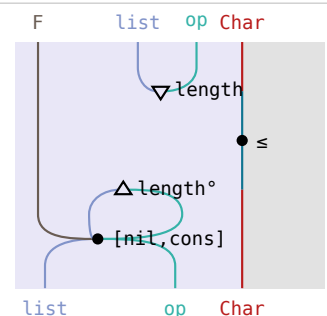
## circuit

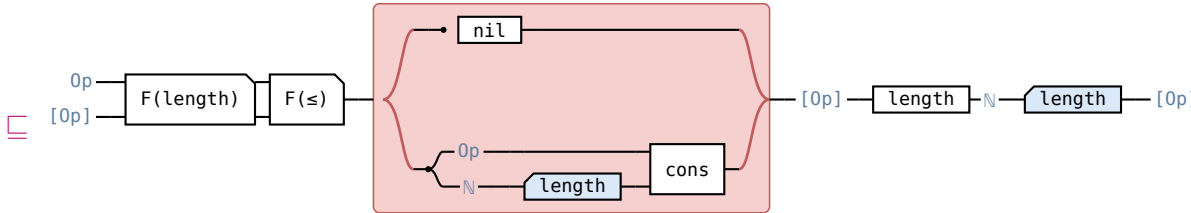
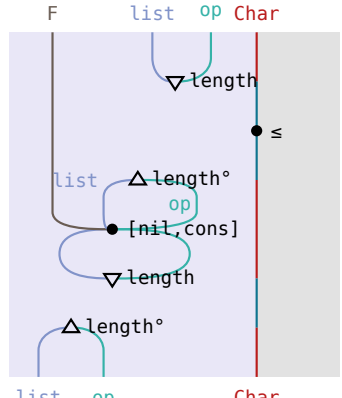
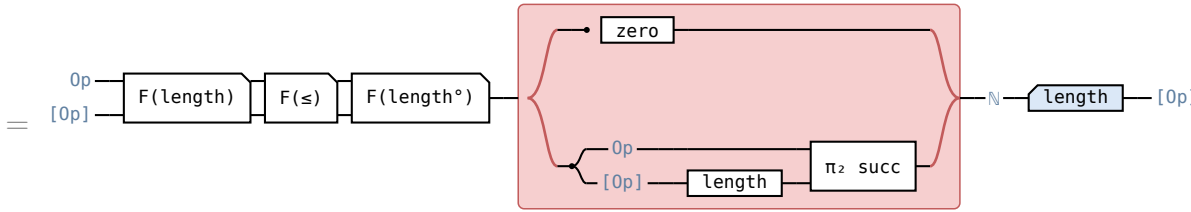
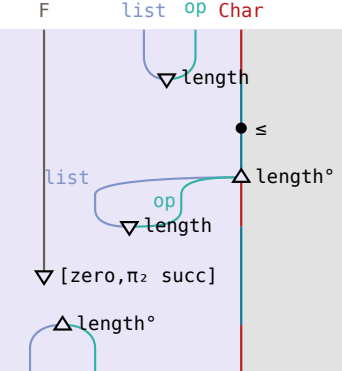
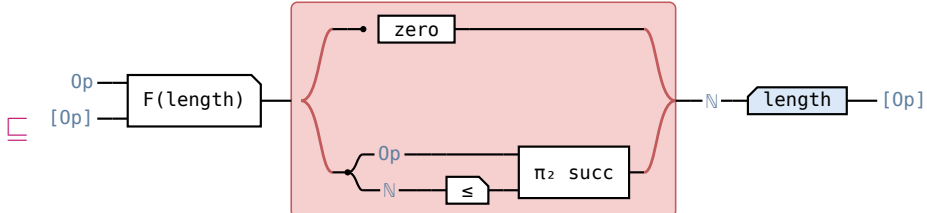
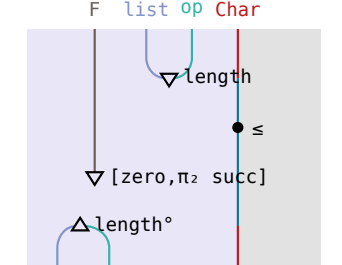
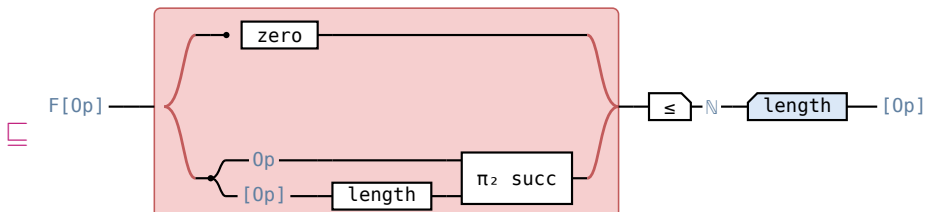
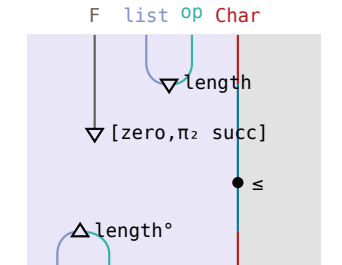
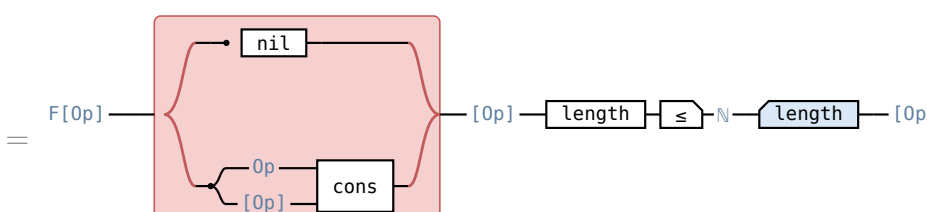
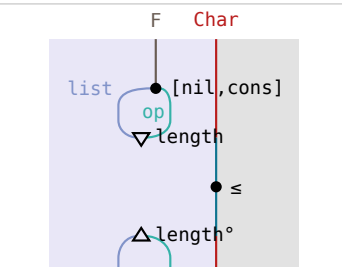
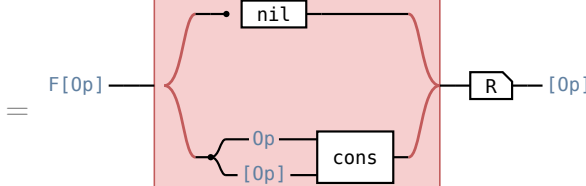
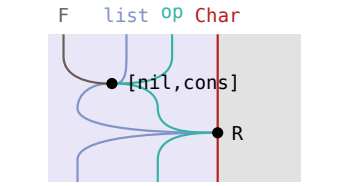


## Hinze–Marsden



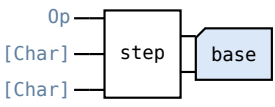
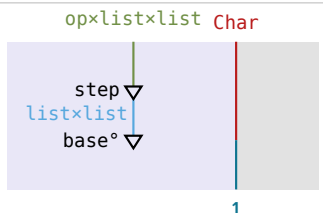
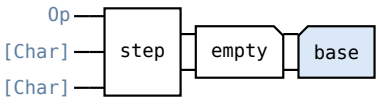
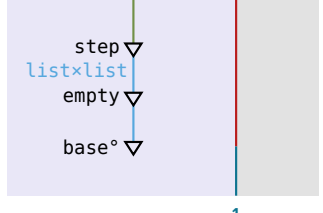
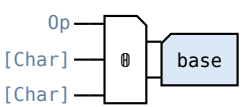
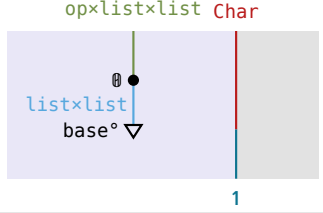
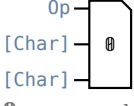
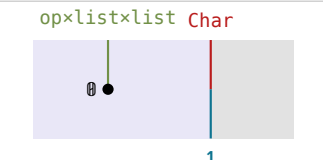
$R \triangleq \text{length} \leq \text{length}^\circ$  —  $\triangleq$  edit;  $F$  preserves composition



| circuit                                                                                                                                                                                                                                                                   | Hinze–Marsden                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <p><math>\text{length}</math> is a map, so entire: <math>1 \sqsubseteq \text{length} \text{ length}^\circ</math></p>                                                                   |    |
|  <p><math>\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}] : \text{length}^\triangleleft ([\text{zero}, \pi_2 \text{ succ}]) \triangleq \text{edit}</math></p> |    |
|  <p><math>\text{length}</math> is a map, so simple, and <math>F</math> is monotonic: <math>F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1</math></p>                     |  |
|  <p><math>\text{succ}</math> is monotonic on <math>\leq</math>: <math>F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq</math></p>          |  |
|  <p><math>F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}</math> again</p>                                                                                  |  |
|  <p><math>R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}</math></p>                                                                                        |  |

| $Q \text{ F(edit}^\circ) [\text{nil}, \text{cons}] \sqsubseteq \text{F(edit}^\circ) [\text{nil}, \text{cons}] R$<br>every edit sequence built from a decomposition $Q$ puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                  | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                          |               |
| $=$ <p><math>Q \triangleq 1 + (U \times V)</math> is <math>F(U, V)</math> at <math>U \triangleq \tau</math> — <math>\triangleq \text{edit}</math>; and the bifunctor <math>F</math> preserves composition:<br/> <math>F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)</math></p>             |               |
| $\sqsubseteq$ <p>Proposition 9.4's second condition <math>V \text{ edit}^\circ \sqsubseteq \text{edit}^\circ R</math> — <math>\text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}</math></p>                                                                                                           |               |
| $=$ <p><math>F</math> preserves composition: <math>F(U, \text{edit}^\circ R) = F(1, \text{edit}^\circ)F(U, R)</math></p>                                                                                                                                                                                 |               |
| $\sqsubseteq$ <p>Proposition 9.4's first condition <math>F(\tau, R) \alpha \alpha R</math>, left as an exercise in the book: <b>cons</b> adds one to both lengths whatever the two operations are</p>                                                                                                    |               |

| $\text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}$ (9.2h)<br>cutting a front off either string an edit sequence produces $(\text{edit } V^\circ)$ leaves a pair that a sequence no longer produces $(R^\circ \text{ edit})$                                                                                                                      |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                               | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                                       |               |
| $=$<br>$V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ , and $\times$ preserves composition                                                                                                                                                                                                                                             |               |
| $\sqsubseteq$<br>$\text{edit } (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit}$ : drop the operation that produced the head, or weaken its <b>cpy</b> to an <b>ins</b> . Proved by induction at <b>suffix</b> directly, not through $\text{suffix} = \text{tail}^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ as the book does |               |
| $\sqsubseteq$<br>$\text{edit } (1 \times \text{suffix}) \sqsubseteq R^\circ \text{ edit}$ , the mirror image: <b>cpy</b> weakens to a <b>del</b>                                                                                                                                                                                                      |               |
| $\sqsubseteq$<br>$R^\circ$ is transitive, $\leq$ being so                                                                                                                                                                                                                                                                                             |               |

| $\text{step } \text{base}^\circ = \emptyset$ <span style="float: right;">(9.2i)</span><br>no pair of strings is both a result of <b>step</b> and the result $([], [])$ of <b>base</b>                                                  |                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                | Hinze–Marsden                                                                        |
|                                                                                                                                                       |   |
|  <p><math>\text{base}</math> returns only <math>([], [])</math>, so <math>\text{base}^\circ = \text{empty } \text{base}^\circ</math></p>              |   |
|  <p><math>\text{step } \text{empty} = \emptyset</math>: <b>cpy</b> and <b>del</b> put a character on the left string, <b>ins</b> one on the right</p> |   |
|  <p><math>\emptyset</math> composed with anything is <math>\emptyset</math></p>                                                                      |  |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$ , $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$<br>$= \text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)])$ , $\text{fstcol} = \text{list}(\text{del}) \text{ tails}$<br>the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail $u$ of $xs$ and $v$ of $ys$ , so the columns are built right to left                          |
| 2 | $\text{column}(xs, [b] \# ys) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$<br>$= \text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)])(xus)$ , $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$<br>each column is a fold built bottom to top, over $xs$ zipped with the adjacent pairs of the column to its right                                                                                      |
| 3 | $\text{base}(b, u) = [[\text{ins}(b)] \# u]$<br>$= \text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$<br>$w = \text{head}(ws)$ ; these <b>base</b> , <b>step</b> are not <b>edit</b> 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths |

### §9.3. Optimal bracketing

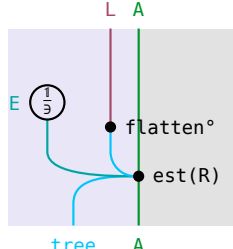
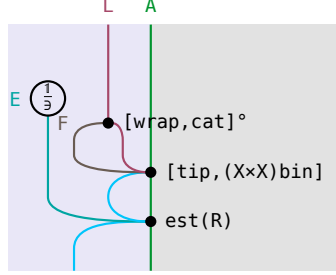
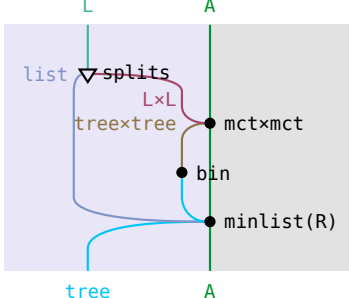
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <p>tree <math>A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)</math>, <math>FX = A + X^2</math>, so <math>F(R) = 1 + R^2</math>; <math>h \triangleq [\text{tip}, \text{bin}]</math>, <math>\text{flatten} \triangleq ([\text{wrap}, \text{cat}]) : \text{tree } A \rightarrow \text{list}^+ A</math>, <math>H = \text{flatten}^\circ</math>.</p> <p><math>\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])</math>, <math>\text{opt} \triangleq (\text{zero}, \text{st})</math>, <math>\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))</math>.</p> <p><b>sb</b> associative, so <math>\text{size} = \text{flatten } \text{sz}</math> for a map <math>\text{sz}</math> <math>R \triangleq \text{cost} \leq \text{cost}^\circ</math>, <math>g \triangleq [\text{zero}, (1 \times \text{sz})^2 \text{ opb } \pi_1]</math>, single the coreflexive on singleton lists.</p> <p><math>\text{splits} \triangleq (\text{inits}^+, \text{tails}^+)</math> <b>zip</b>, an implementation of <math>\frac{\text{cat}^\circ}{\text{zip}}</math>; <math>\text{array} \triangleq \text{inits } \text{list}(\text{row})</math>, <math>\text{row} \triangleq \text{tails } \text{list}(\text{mct})</math>, <math>\text{col} \triangleq \text{inits } \text{list}(\text{mct})</math>.</p> <p><math>\text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)</math>, <math>\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}</math>, <math>\text{process} \triangleq ((\text{tip } \text{wrap}) \times 1) \text{ loop}(\text{next})</math>.</p> | (9.3a) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|

| $\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \text{ g} = [\text{tip}, \text{bin}] \text{ cost}$ (9.5)<br>pairing each subtree with its cost and its flattening and then applying $\text{g}$ gives what building the tree by $[\text{tip}, \text{bin}]$<br>and taking its <b>cost</b> gives                                                                                                                                                                                                                                      |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Hinze–Marsden |
| $F(\text{tree}(A)) \rightarrow \boxed{F(\langle \text{cost}, \text{flatten} \rangle)} \rightarrow F(\text{Int} \times \text{LA}) \rightarrow \boxed{\text{g}} \rightarrow \text{Int}$                                                                                                                                                                                                                                                                                                                                                                                  |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{zero} \\ \text{r} \text{ tree}(A) \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{flatten} \text{ LA} \text{ sz} \\ \text{cost} \\ \text{flatten} \text{ LA} \text{ sz} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ <p>definition of <math>\text{g}</math>; coproducts and products <math>\triangleq \text{flatten}</math></p>                                                                                                                 |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{zero} \\ \text{r} \text{ tree}(A) \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{size} \\ \text{cost} \\ \text{size} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ <p><math>\text{flatten sz} = \text{size}</math>, since <math>\text{sb}</math> is associative <math>\triangleq \text{flatten}</math></p>                                                                                                                                      |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{zero} \\ \text{r} \text{ tree}(A) \end{array} \right\} \rightarrow \text{bin} \text{ tree}(A) \rightarrow \left\{ \begin{array}{l} \text{cost} \\ \text{size} \end{array} \right\} \rightarrow \text{Int} \times \text{S} \rightarrow \text{opb} \rightarrow \text{Int}$ <p><math>\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])</math>, at a node <math>\text{bin}(\text{cost}, \text{size})</math><br/> <math>= (\text{cost}, \text{size})^2 \text{ opb}</math></p> |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{tip} \text{ tree}(A) \text{ cost} \\ \text{r} \text{ tree}(A) \text{ bin} \text{ tree}(A) \text{ cost} \end{array} \right\} \rightarrow \text{Int}$ <p><math>\text{tip cost} = \text{zero}</math>; products, <math>\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}</math></p>                                                                                                                                                                                                            |               |
| $A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} \text{tip} \\ \text{r} \text{ tree}(A) \text{ bin} \end{array} \right\} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ <p>coproducts, <math>[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{ cost}</math></p>                                                                                                                                                                                                                                               |               |

| $F(\leq \times 1) \quad g \sqsubseteq g \leq$ <span style="float: right;">(9.6)</span>                                                                                       |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| raising the cost of either subtree ( $\leq \times 1$ in both slots) and then applying $g$ returns a value at least what $g$ returns on the costs before raising ( $g \leq$ ) |               |
| circuit                                                                                                                                                                      | Hinze–Marsden |
| $F(\text{Int} \times \text{LA}) \rightarrow \boxed{F(\leq \times 1)} \rightarrow \boxed{g} \rightarrow \text{Int}$                                                           |               |
| $A + \text{Int} \times \text{LA} \times \text{Int} \times \text{LA} \rightarrow$<br>definition of $g \triangleq \text{flatten}$                                              |               |
| $A + \text{Int} \times \text{LA} \times \text{Int} \times \text{LA} \rightarrow$<br>definition of $\text{opb}$ , and $+$ monotonic                                           |               |
| $A + \text{Int} \times \text{LA} \times \text{Int} \times \text{LA} \rightarrow$<br>$\leq$ reflexive, so $\text{zero} \sqsubseteq \text{zero} \leq$ ; coproducts             |               |
| $F(\text{Int} \times \text{LA}) \rightarrow \boxed{g} \rightarrow \boxed{\leq} \rightarrow \text{Int}$<br>definition of $g \triangleq \text{flatten}$                        |               |

$$\text{mct}=(\text{single} \rightarrow \text{head tip}, \langle \text{init col}, \text{tail row} \rangle \text{ mix}), \text{ Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\mathfrak{g}} \text{ P}([\text{tip}, (X \times X) \text{ bin}]) \text{ est}(\text{R})) \sqsubseteq \frac{\text{flatten}^\circ}{\mathfrak{g}} \text{ est}(\text{R}) \quad (9.3b)$$

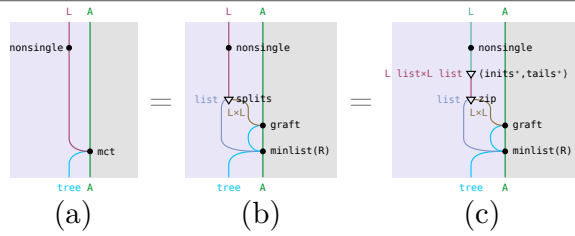
a least-cost bracketing of  $a_1 \oplus \dots \oplus a_n$  is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hinze–Marsden                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $\text{LA} \rightarrow \boxed{\frac{1}{\mathfrak{g}}} \rightarrow \text{P}(\text{LA}) \rightarrow \boxed{\text{E}(\text{flatten}^\circ)} \rightarrow \text{P}(\text{tree}(\text{A})) \rightarrow \boxed{\text{est}(\text{R})} \rightarrow \text{tree}(\text{A})$ <p>the specification <math>\dashv \triangleq \text{flatten}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| $\text{LA} \rightarrow \boxed{\frac{1}{\mathfrak{g}}} \rightarrow \text{P}(\text{LA}) \rightarrow \boxed{\text{E}([\text{wrap}, \text{cat}]^\circ)} \rightarrow \text{PF}(\text{LA}) \rightarrow \boxed{\text{P}([\text{tip}, (X \times X) \text{ bin}])} \rightarrow \text{P}(\text{tree}(\text{A})) \rightarrow \boxed{\text{est}(\text{R})} \rightarrow \text{tree}(\text{A})$ <p>split the list in every way, bracket both halves, join. The condition is monotonicity <b>in context</b>,<br/> <math>\sqsubseteq \text{F}(\text{Rn}(\text{flatten} \text{ flatten}^\circ)) \text{ h} \sqsubseteq \text{hR}</math> — only trees with the same flattening are compared — which is Proposition 9.3 at <math>\text{H}^\circ = \text{flatten}</math>, a map, with (9.5) <math>[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) \text{g}</math> (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6) <math>(1 + (\leq \times 1)^2) \text{g} \sqsubseteq \text{g} \leq (\text{g} \text{ monotonic on } \leq \text{ in its two cost arguments})</math></p> |    |
| $\text{LA} \rightarrow \boxed{\frac{1}{\mathfrak{g}}} \rightarrow \text{P}(\text{LA}) \rightarrow \boxed{\text{E}(\text{cat}^\circ)} \rightarrow \text{P}(\text{LA}^2) \rightarrow \boxed{\text{P}((X \times X) \text{ bin})} \rightarrow \text{P}(\text{tree}(\text{A})) \rightarrow \boxed{\text{est}(\text{R})} \rightarrow \text{tree}(\text{A})$ <p><math>\equiv</math> Proposition 9.1: <b>wrap</b> and <b>cat</b> have disjoint ranges, and <b>single</b> is the coreflexive on singleton lists, where <b>wrap</b> returns</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |
| $\text{LA} \rightarrow \boxed{\text{splits}} \rightarrow [\text{LA}^2] \rightarrow \boxed{\text{list}((\text{mct} \times \text{mct}) \text{ bin})} \rightarrow [\text{tree}(\text{A})] \rightarrow \boxed{\text{minlist}(\text{R})} \rightarrow \text{tree}(\text{A})$ <p><math>\sqsubseteq \text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle</math> zip implements <math>\frac{\text{cat}^\circ}{\mathfrak{g}}</math> and <math>\text{minlist}(\text{R})</math> implements <math>\text{est}(\text{R})</math>.<br/> Exponential, since the segments of one list overlap</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| $\text{mct}=(\text{single} \rightarrow \text{head tip}, \langle \text{init col}, \text{tail row} \rangle \text{ mix}) \quad (9.7)$ <p><math>\equiv</math> the tabulation: <math>\text{mct } \text{xs}</math> is needed for every non-empty segment <math>\text{xs}</math>, so the values are held as an array of rows, <math>\text{array} \triangleq \text{inits list}(\text{row}), \text{row} \triangleq \text{tails list}(\text{mct}), \text{col} \triangleq \text{inits list}(\text{mct}), \text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(\text{R})</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                       |
| $\text{col}=(\text{single} \rightarrow \text{head tip wrap}, \langle \text{init col}, \text{tail row} \rangle \text{ next}) \quad (9.8)$ $\text{cons col}=(1 \times \text{array}) \text{ process} \quad (9.9)$ <p><math>\equiv \text{row}=(\text{single} \rightarrow \text{head tip wrap}, \langle \text{mct}, \text{tail row} \rangle \text{ cons}) \quad (9.10)</math><br/> a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop,<br/> <math>\text{next} \triangleq \langle \pi_1, \text{mix} \rangle \text{ snoc}, \text{ process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                       |
| $\text{array}=(\text{fstcol}, \text{addcol}), \text{ fstcol} \triangleq \text{tip wrap wrap}$ <p><math>\equiv \text{addcol} \triangleq \langle \pi_1 \text{ tip wrap}, \text{step} \rangle \text{ cons}, \text{ step} \triangleq \langle \text{process tail}, \pi_2 \rangle \text{ zip list}(\text{cons})</math><br/> the program: one fold building the array column by column, cubic in the length of the input</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                       |

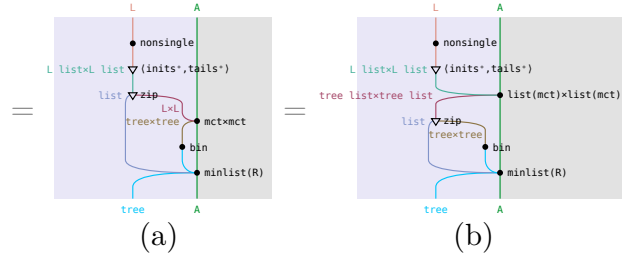
**nonsingle mct = nonsingle (init col,tail row) mix**

(9.7)

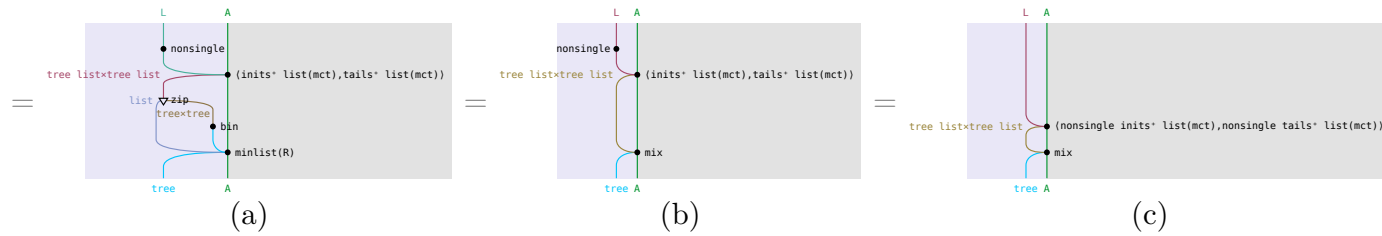
on a list of two or more elements, **mct** gives what **mix** gives on the column of its **init** and the row of its **tail**



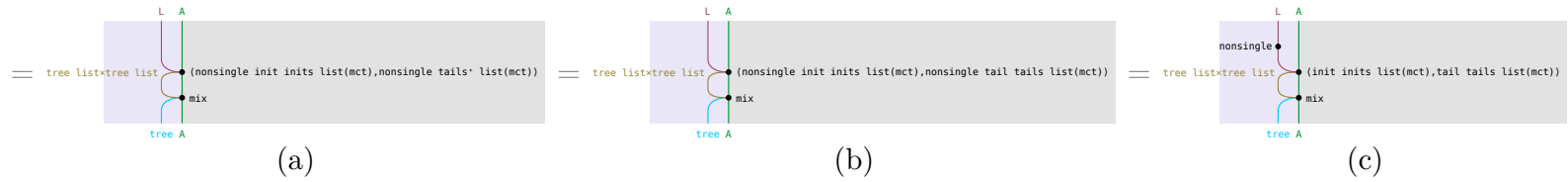
|     |                                                        |                                           |
|-----|--------------------------------------------------------|-------------------------------------------|
| (a) | nonsingle mct                                          | nonsingle splits list(graft) minlist(R)   |
| (b) | = nonsingle splits list(graft) minlist(R)              | splits = (inits+,tails+) zip              |
| (c) | = nonsingle (inits+,tails+) zip list(graft) minlist(R) | graft $\triangleq$ (mct $\times$ mct) bin |



|     |                                                                                     |                                                             |
|-----|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
| (a) | = nonsingle (inits+,tails+) zip list(mct $\times$ mct) list(bin) minlist(R)         | zip list(f $\times$ g) M = (list(f) $\times$ list(g)) zip M |
| (b) | = nonsingle (inits+,tails+) (list(mct) $\times$ list(mct)) zip list(bin) minlist(R) | (R,S) (U $\times$ V) = (R U,S V)                            |



|     |                                                                          |                                             |
|-----|--------------------------------------------------------------------------|---------------------------------------------|
| (a) | = nonsingle (inits+ list(mct),tails+ list(mct)) zip list(bin) minlist(R) | mix = zip list(bin) minlist(R)              |
| (b) | = nonsingle (inits+ list(mct),tails+ list(mct)) mix                      | nonsingle (U,V) = (nonsingle U,nonsingle V) |
| (c) | = (nonsingle inits+ list(mct),nonsingle tails+ list(mct)) mix            | nonsingle inits+ M = nonsingle init inits M |



|     |                                                                       |                                             |
|-----|-----------------------------------------------------------------------|---------------------------------------------|
| (a) | = (nonsingle init inits list(mct),nonsingle tails+ list(mct)) mix     | nonsingle tails+ M = nonsingle tail tails M |
| (b) | = (nonsingle init inits list(mct),nonsingle tail tails list(mct)) mix | nonsingle (U,V) = (nonsingle U,nonsingle V) |
| (c) | = nonsingle (init inits list(mct),tail tails list(mct)) mix           | col = inits list(mct)                       |

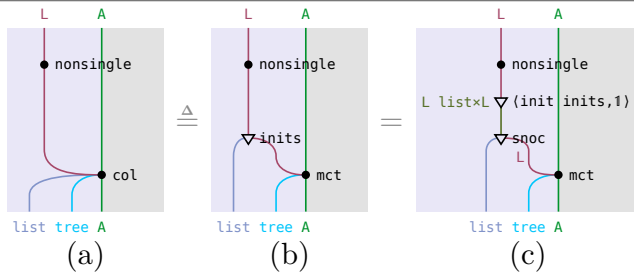


|     |                                                 |                       |
|-----|-------------------------------------------------|-----------------------|
| (a) | = nonsingle (init col,tail tails list(mct)) mix | row = tails list(mct) |
| (b) | = nonsingle (init col,tail row) mix             |                       |

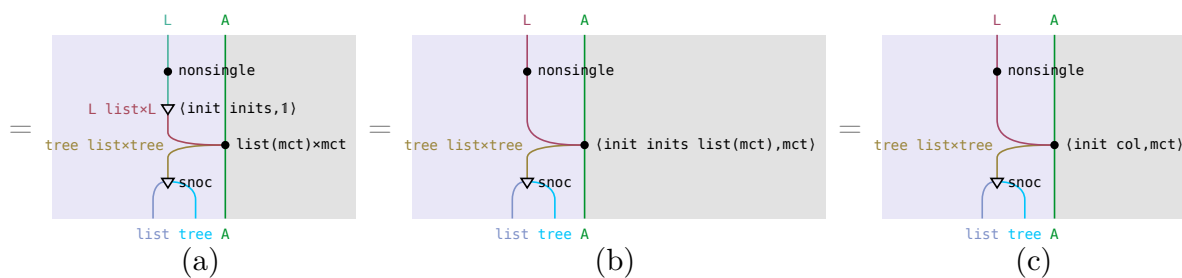
**nonsingle col = nonsingle (init col,tail row) next**

(9.8)

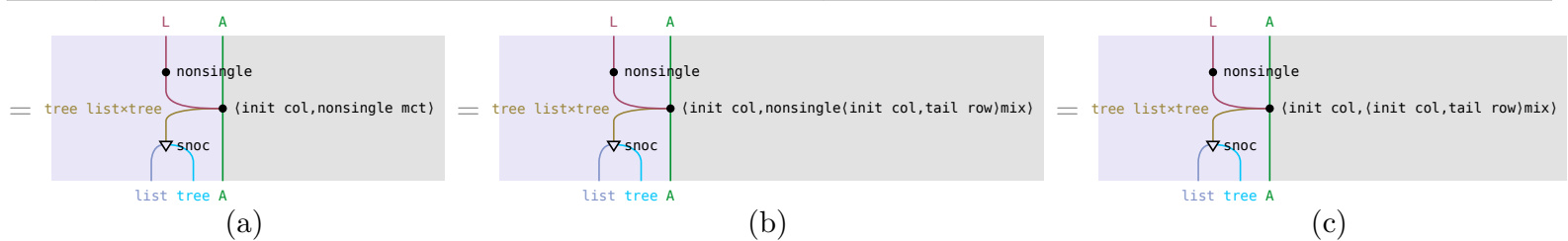
on a list of two or more elements, the column is what **next** makes of the column of its **init** and the row of its **tail**



|     |                                           |                                                     |
|-----|-------------------------------------------|-----------------------------------------------------|
| (a) | nonsingle col                             | col $\triangleq$ inits list(mct)                    |
| (b) | $\triangleq$ nonsingle inits list(mct)    | nonsingle inits M = nonsingle (init inits,1) snoc M |
| (c) | = nonsingle (init inits,1) snoc list(mct) | snoc list(f) = (list(f) × f) snoc                   |



|     |                                                   |                                                                                       |
|-----|---------------------------------------------------|---------------------------------------------------------------------------------------|
| (a) | = nonsingle (init inits,1) (list(mct) × mct) snoc | $\langle R, S \rangle (U \times V) = \langle R U, S V \rangle$                        |
| (b) | = nonsingle (init inits list(mct), mct) snoc      | col = inits list(mct)                                                                 |
| (c) | = nonsingle (init col, mct) snoc                  | nonsingle $\langle U, V \rangle =$ nonsingle $\langle U, \text{nonsingle } V \rangle$ |



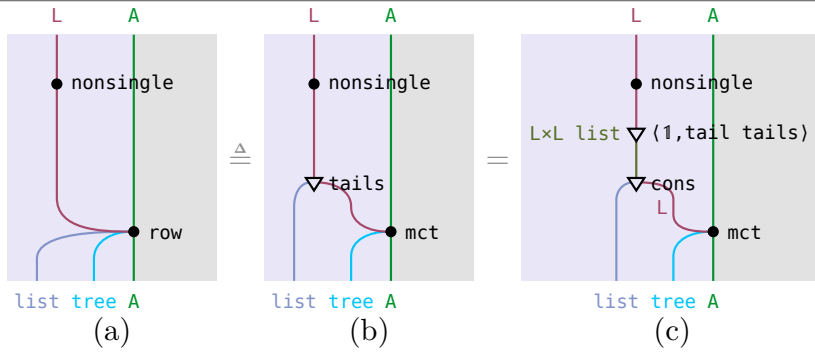
|     |                                                                 |                                                                                                                                       |
|-----|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| (a) | = nonsingle (init col, nonsingle mct) snoc                      | (9.7) nonsingle mct = nonsingle (init col, tail row) mix                                                                              |
| (b) | = nonsingle (init col, nonsingle (init col, tail row) mix) snoc | nonsingle $\langle U, V \rangle =$ nonsingle $\langle U, \text{nonsingle } V \rangle$                                                 |
| (c) | = nonsingle (init col, (init col, tail row) mix) snoc           | $\langle \text{init col}, V \rangle \langle \pi_1, M \rangle = \langle \text{init col}, \langle \text{init col}, V \rangle M \rangle$ |



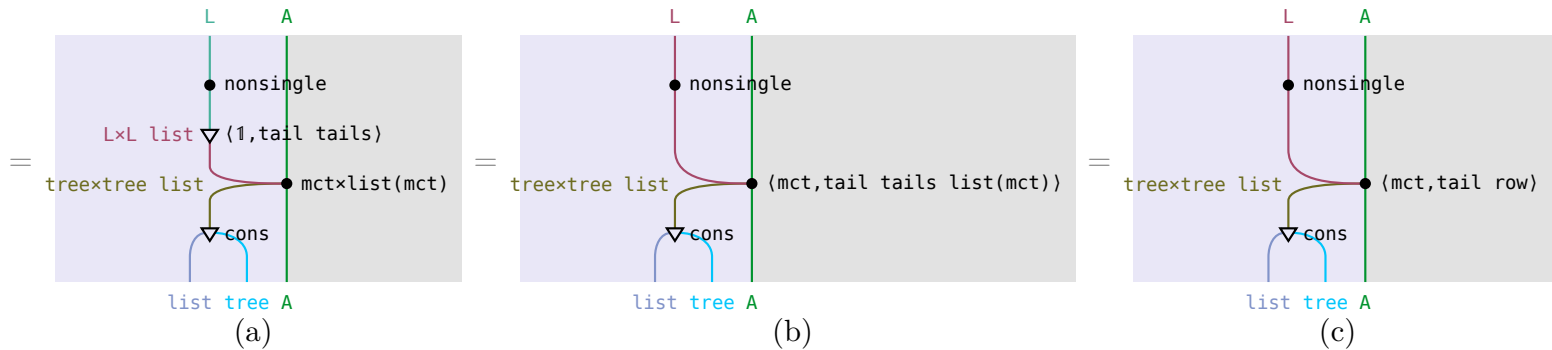
|     |                                                                           |                                                 |
|-----|---------------------------------------------------------------------------|-------------------------------------------------|
| (a) | = nonsingle (init col, tail row) $\langle \pi_1, \text{mix} \rangle$ snoc | next = $\langle \pi_1, \text{mix} \rangle$ snoc |
| (b) | = nonsingle (init col, tail row) next                                     |                                                 |

**nonsingle row = nonsingle (mct,tail row) cons**

(9.10)

on a list of two or more elements, the row is **mct** of the whole list consed onto the row of its **tail**

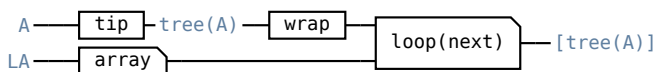
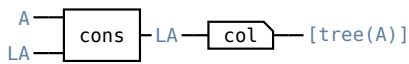
|     |                                                                     |                                                                                                      |
|-----|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| (a) | nonsingle row                                                       | $\text{row} \triangleq \text{tails list(mct)}$                                                       |
| (b) | $\triangleq$ nonsingle tails list(mct)                              | $\text{nonsingle tails } M = \text{nonsingle } \langle 1, \text{tail tails} \rangle \text{ cons } M$ |
| (c) | $=$ nonsingle $\langle 1, \text{tail tails} \rangle$ cons list(mct) | $\text{cons list}(f) = (f \times \text{list}(f)) \text{ cons}$                                       |



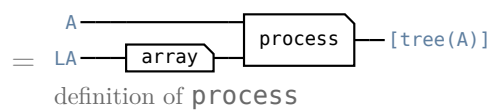
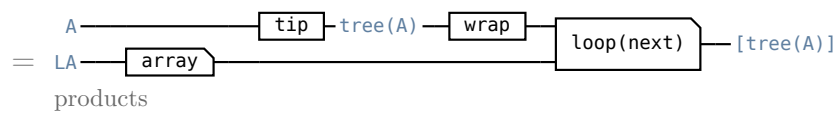
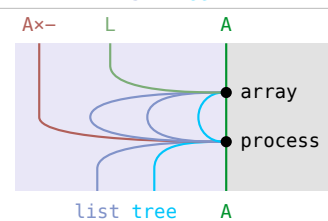
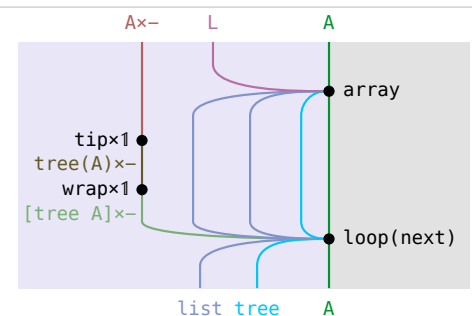
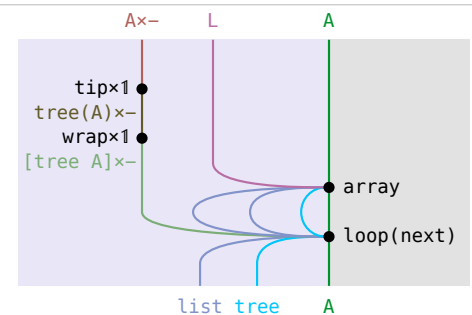
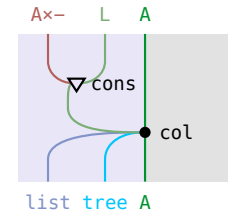
|     |                                                                                    |                                                                |
|-----|------------------------------------------------------------------------------------|----------------------------------------------------------------|
| (a) | $=$ nonsingle $\langle 1, \text{tail tails} \rangle$ (mct $\times$ list(mct)) cons | $\langle R, S \rangle (U \times V) = \langle R U, S V \rangle$ |
| (b) | $=$ nonsingle $\langle \text{mct}, \text{tail tails list(mct)} \rangle$ cons       | $\text{row} = \text{tails list(mct)}$                          |
| (c) | $=$ nonsingle $\langle \text{mct}, \text{tail row} \rangle$ cons                   |                                                                |

**cons col = (1  $\times$  array) process**

(9.9)

the column of **a** consed onto **x** is what **process** makes of **a** and the array of **x****circuit**

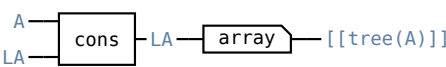
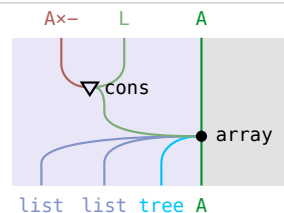
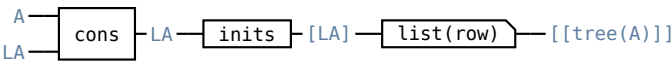
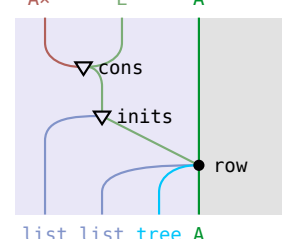
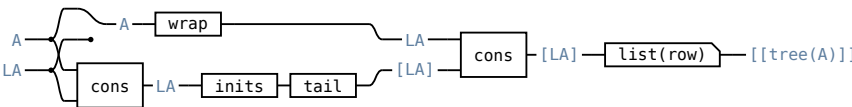
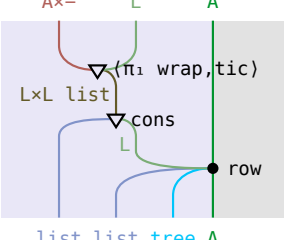
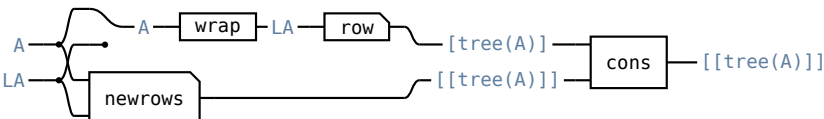
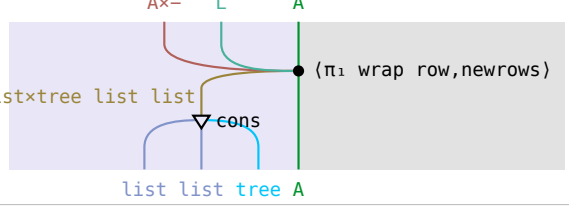
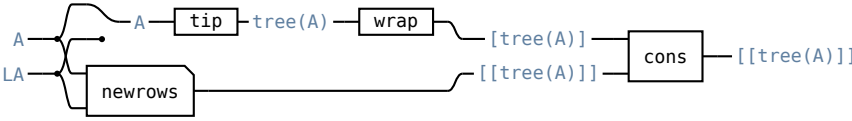
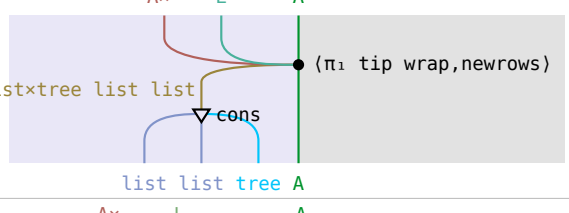
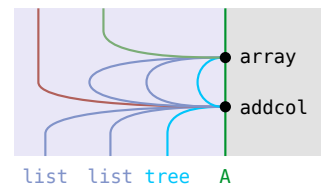
$=$  Exercise 9.13 with  $k \triangleq \text{cons col}$ ,  $g \triangleq \text{tip wrap}$ ,  $h \triangleq \text{row}$ ,  $f \triangleq \text{next}$ : its two equations are (9.8)  
 — (9.8)  $\text{nonsingle col} = \text{nonsingle } \langle \text{init col}, \text{tail row} \rangle \text{ next}$  — at  $a:[b]$   
 and at  $a:(u++[b])$ , with  $[a] \text{ col} = [\text{tip}(a)]$

**Hinze–Marsden**

| <b>tops = (1 × array) process tail</b> <span style="float: right;">(9.3c)</span><br>the trees at the top of the new rows are what <b>process</b> makes of <b>a</b> and the array of <b>x</b> , less its first |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                       | Hinze–Marsden |
|                                                                                                                                                                                                               |               |
| $= \begin{array}{c} A \\ LA \end{array} \text{cons} \text{LA} \text{inits} \text{tail} \text{LA} \text{list(mct)} \text{tree(A)}$ <p>definition of tops<math>\triangleq</math>tic list(mct), our name</p>     |               |
| $= \begin{array}{c} A \\ LA \end{array} \text{cons} \text{LA} \text{col} \text{tail} \text{tree(A)}$ <p>tail list(f)=list(f) tail; definition of col</p>                                                      |               |
| $= \begin{array}{c} A \\ LA \end{array} \text{array} \text{process} \text{tail} \text{tree(A)}$ <p>(9.9) — (9.9) cons col = (1 × array) process</p>                                                           |               |

| <b>rests = <math>\pi_2</math> array</b> <span style="float: right;">(9.3d)</span><br>the rest of each new row is the array of <b>x</b>                                                                               |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                              | Hinze–Marsden |
|                                                                                                                                                                                                                      |               |
| $= \begin{array}{c} A \\ LA \end{array} \text{cons} \text{LA} \text{inits} \text{tail} \text{LA} \text{list(tail row)} \text{tree(A)}$ <p>definition of rests<math>\triangleq</math>tic list(tail row), our name</p> |               |
| $= \begin{array}{c} A \\ LA \end{array} \text{array} \text{tree(A)}$ <p>tic list(tail)=<math>\pi_2</math> inits; definition of array</p>                                                                             |               |

| $\text{newrows} = (1 \times \text{array}) \text{ step}$ <span style="float: right;">(9.3e)</span>                                                                                                                                                                                                                                  |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| the rows of the array of $\mathbf{a}:\mathbf{x}$ below its first are what <b>step</b> makes of $\mathbf{a}$ and the array of $\mathbf{x}$                                                                                                                                                                                          |               |
| circuit                                                                                                                                                                                                                                                                                                                            | Hinze–Marsden |
|                                                                                                                                                                                                                                                                                                                                    |               |
| $= \text{LA} \rightarrow \text{cons} \rightarrow \text{LA} \rightarrow \text{inits} \rightarrow \text{tail} \rightarrow \text{LA} \rightarrow \text{list}(\text{row}) \rightarrow [[\text{tree}(\text{A})]]$ definition of $\text{newrows} \triangleq \text{tic list}(\text{row})$ , our name                                      |               |
| $= \text{LA} \rightarrow \text{cons} \rightarrow \text{LA} \rightarrow \text{inits} \rightarrow \text{tail} \rightarrow \text{LA} \rightarrow \text{list}(\text{mct, tail row cons}) \rightarrow [[\text{tree}(\text{A})]]$ (9.10) — (9.10) nonsingle row = nonsingle (mct, tail row)<br>cons — on the non-singleton tic lists     |               |
| $= \text{LA} \rightarrow \text{tops} \rightarrow \text{rests} \rightarrow \text{zip} \rightarrow \text{tree}(\text{A}) \times \text{tree}(\text{A}) \rightarrow \text{list}(\text{cons}) \rightarrow [[\text{tree}(\text{A})]]$ $\text{list}(f,g) = (\text{list}(f), \text{list}(g))$ zip; products; definitions of tops and rests |               |
| $= \text{LA} \rightarrow \text{array} \rightarrow \text{step} \rightarrow [[\text{tree}(\text{A})]]$ $\text{tops} = (1 \times \text{array})$ process tail and rests = $\pi_2$ array;<br>products; definition of step                                                                                                               |               |

| $\text{cons array} = (1 \times \text{array}) \text{ addcol}$ <span style="float: right;">(9.3f)</span><br>the array of <b>a</b> consed onto <b>x</b> is what <b>addcol</b> makes of <b>a</b> and the array of <b>x</b> |                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                | Hinze–Marsden                                                                         |
|                                                                                                                                       |    |
| $=$ <br>definition of <b>array</b>                                                                                                    |    |
| $=$ <br>cons inits= $\langle \pi_1 \text{ wrap, tic} \rangle$ cons, abbreviating cons inits tail by tic                              |    |
| $=$ <br>cons list(f)=(f $\times$ list(f)) cons; definition of <b>newrows</b>                                                       |   |
| $=$ <br>wrap row=tip wrap                                                                                                          |  |
| $=$ <br>newrows = (1 $\times$ array) step ; products; definition of <b>addcol</b>                                                   |  |

## §9.4. Data compression

(9.4a)

$[A] ::= \text{nil} \mid \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A \mid \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym } \text{Char} \mid \text{ptr } (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$

$\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}])$  with  $c, p$  the constant costs of a symbol and a pointer;  $R \triangleq \text{size} \leq \text{size}^\circ$ ,  $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$ , the two  $\tau$  on symbols and on pointers.

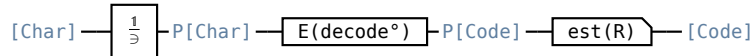
$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{(xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws\}$ , the longest repeated tail;  $\text{reduce } (ws \# [a]) = (zs \neq [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)])$  with  $(xs, (ys, zs)) = \text{lrt } (ws \# [a]).$

$$0 \leq c \wedge 0 \leq p \Rightarrow (\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{3} \text{thin}(Q) P([\text{nil}, (X \times 1) \text{snoc}]) \text{est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{3} \text{est}(R), \quad (9.4b)$$

$$\text{encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1) \text{snoc}) \text{minlist}(R))$$

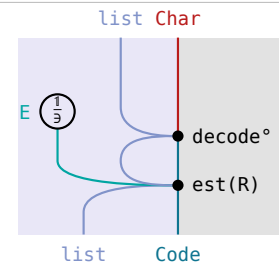
a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

### circuit



the specification —  $\triangleq$  decode

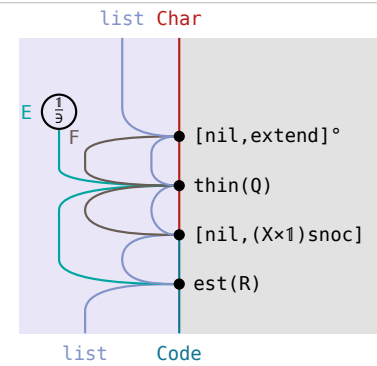
### Hinze–Marsden



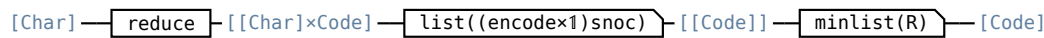
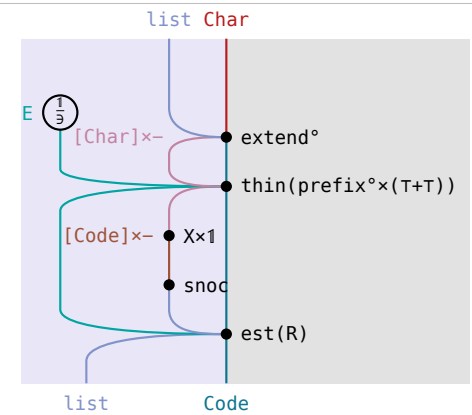
at  $Q \triangleq F(T+T, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (T+T))$ , the two  $T$  on symbols and on pointers.

Monotonicity  $F(R) \alpha \sqsubseteq \alpha R$  is routine, the two costs being constants:  $(T+T)[c, p] = [c, p]$ .

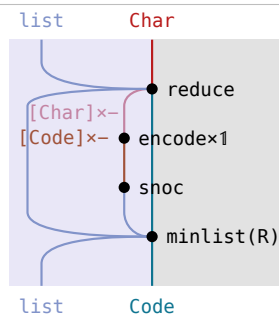
Proposition 9.4 splits the thinning condition in two —  $F(T+T, R) \alpha \sqsubseteq \alpha R$ , and  $\text{decode prefix} \sqsubseteq R^\circ \text{decode}$ , for which  $\text{decode init} \sqsubseteq R^\circ \text{decode}$  is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins



Proposition 9.1:  $\text{nil}$  and  $\text{extend}$  have disjoint ranges. The decompositions of one string are  $\frac{\text{extend}^\circ}{3} (ws\#[a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs\#zs = ws\#[a], ys\#zs \text{ a proper prefix of } ws\}$  — take the last character as a symbol, or end with a pointer



$\text{reduce}$  implements  $\frac{\text{extend}^\circ}{3} \text{thin}(\text{prefix}^\circ \times (T+T))$ : thinning leaves at most two, the symbol and the pointer of the longest repeated tail  $\text{lrt}$ . Again exponential; the book gives no tabulation for it

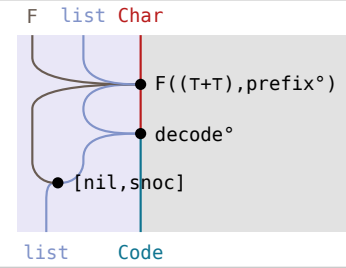
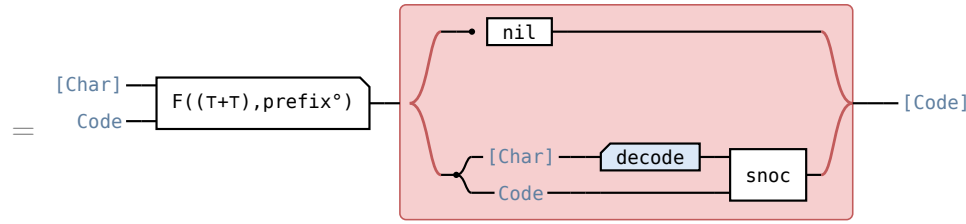
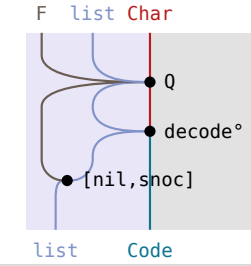
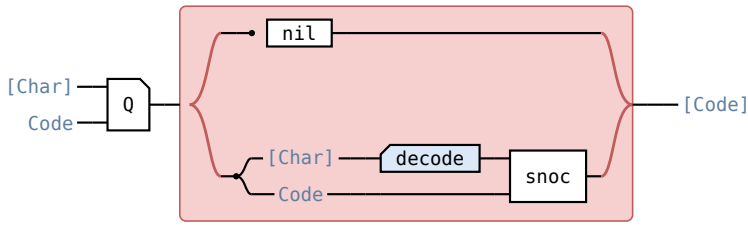


$$\theta \leq c \wedge \theta \leq p \Rightarrow Q \text{ F}(\text{decode}^\circ) [\text{nil}, \text{snoc}] \sqsubseteq \text{F}(\text{decode}^\circ) [\text{nil}, \text{snoc}] R \quad (9.4c)$$

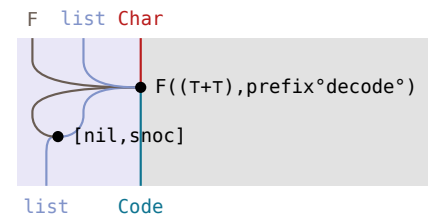
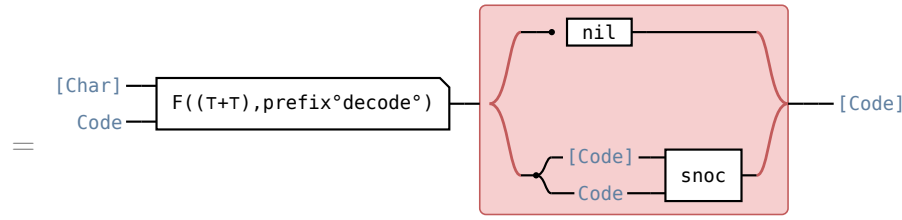
every code sequence built from a decomposition  $Q$  puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

**circuit**

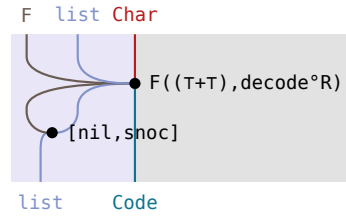
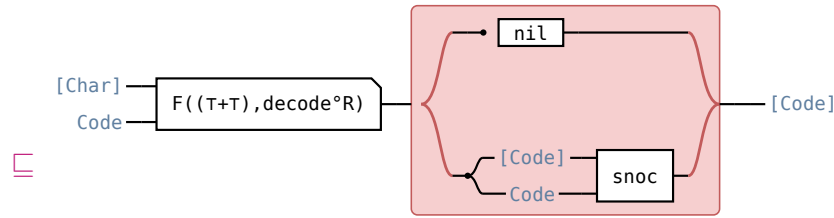
**Hinze–Marsden**



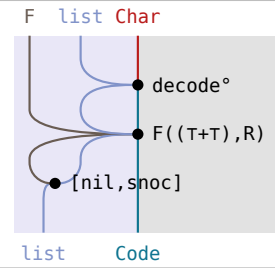
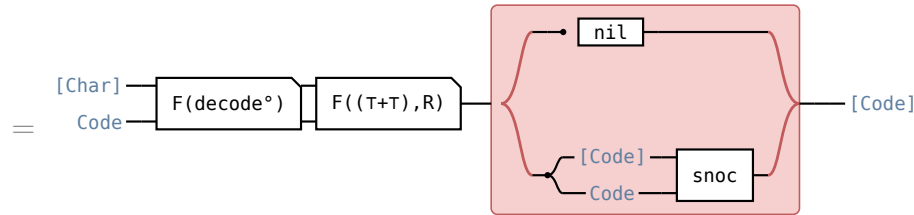
$Q \triangleq 1 + (\text{prefix}^\circ \times (T+T))$  is  $F(T+T, \text{prefix}^\circ) \triangleq \text{decode}$



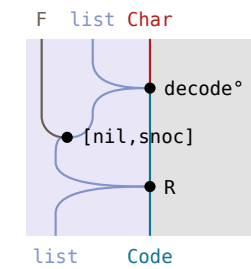
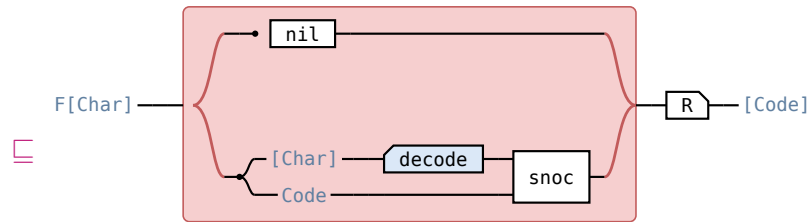
the bifunctor  $F$  preserves composition:  $F(U, \text{prefix}^\circ) F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)$



Proposition 9.4's second condition  $\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R$ : dropping the last **sym**, or shortens or drops the last pointer. Proved at **prefix** directly by induction, not through **init**



$F$  preserves composition:  $F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ) F(U, R)$



Proposition 9.4's first condition  $F(T+T, R) \alpha \sqsubseteq \alpha R$ , left as an exercise in the book: **snoc** adds the cost of the last element to both sides, and  $[c, p](T+T) = [c, p]$

| <div> <div>extend nil<sup>°</sup> = 0</div> <div>(9.4d)</div> </div> <div>no string is both a result of <b>extend</b> and the result [] of <b>nil</b></div> |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                     | Hinze–Marsden |
|                                                                                                                                                             |               |
| <p>nil returns only [], so nil<sup>°</sup>=null nil<sup>°</sup></p>                                                                                         |               |
| <p>extend null=0: a symbol ends the string with a character, a pointer with its non-empty ZS</p>                                                            |               |
| <p>0 composed with anything is 0</p>                                                                                                                        |               |

## §10. Greedy Algorithms

### §10.1. Theory

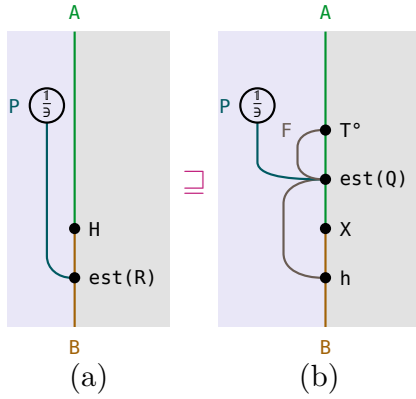
(10.1a)

$h, T, R, H, M$  as in  $\triangle H$ ; additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\exists}$  returns, so that  $\frac{T^\circ}{\exists} \text{ est}(Q)$  is entire.

**Map(h)  $\wedge$  MonoAlg(h,R)  $\wedge$  R R  $\subseteq$  R  $\wedge$  ThinCondition(T,h,R,Q)  $\Rightarrow$  ( $\mu X : \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h) \subseteq \frac{H}{\exists} \text{ est}(R)$**  Theorem 10.1

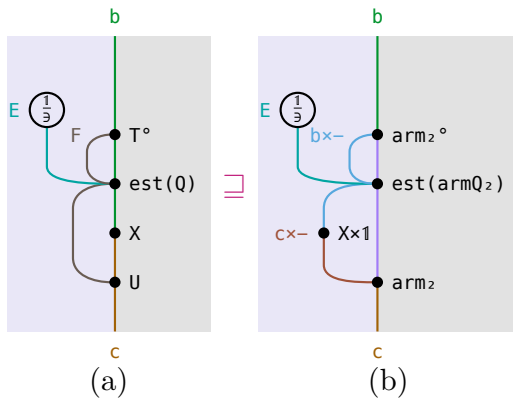
the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

$H \triangle (T)^\circ (h) \triangle H, Q$



|     |                                                             |  |
|-----|-------------------------------------------------------------|--|
| (a) | $\frac{H}{\exists} \text{ est}(R)$                          |  |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h$ |  |

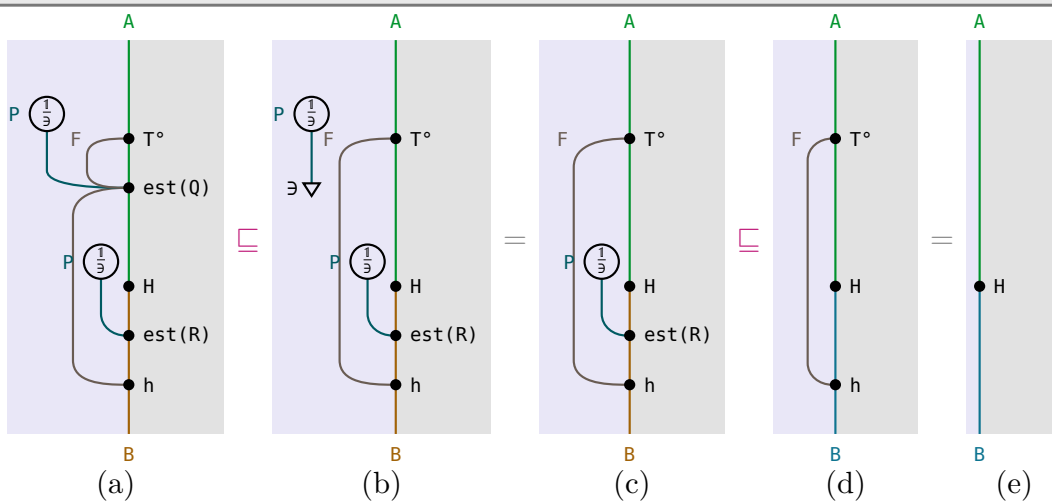
the body at  $T$  a coproduct of two arms



|     |                                                                                                       |                                               |
|-----|-------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{ est}(Q) F(X) U$                                                       | Proposition 10.1, $V_2 V_1^\circ = \emptyset$ |
| (b) | $\sqsubseteq \frac{\text{arm}_2^\circ}{\exists} \text{ est}(\text{arm}_2) (X \times 1) \text{ arm}_2$ |                                               |

**$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h \subseteq H$**  (10.1b)

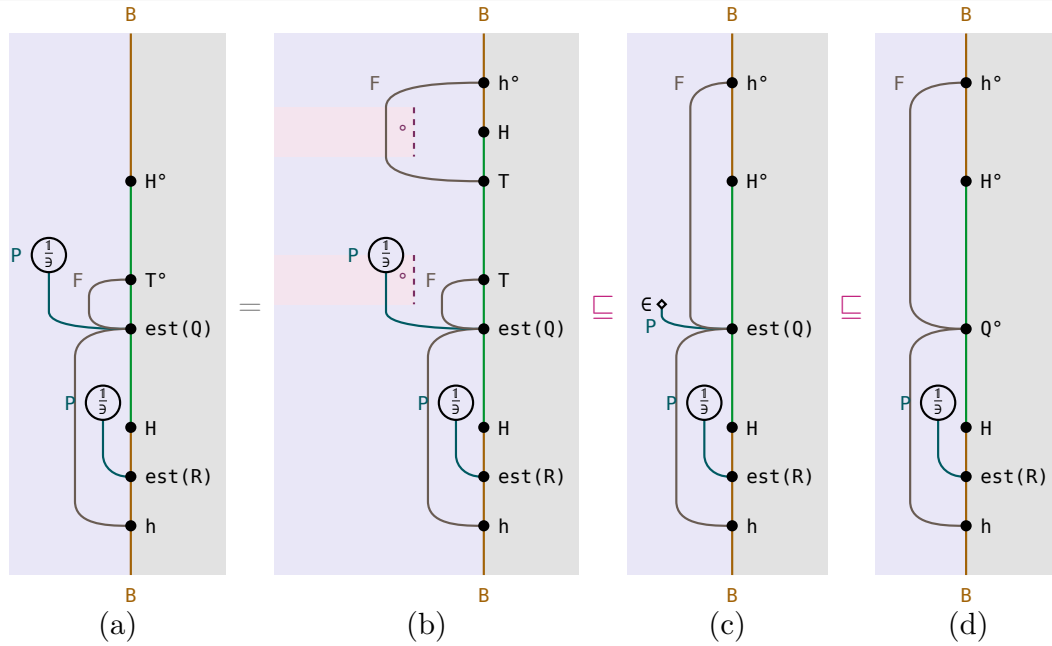
taking the input apart in one  $Q$ -extreme way, solving the parts by  $M$  and assembling by  $h$  returns only what  $H$  returns



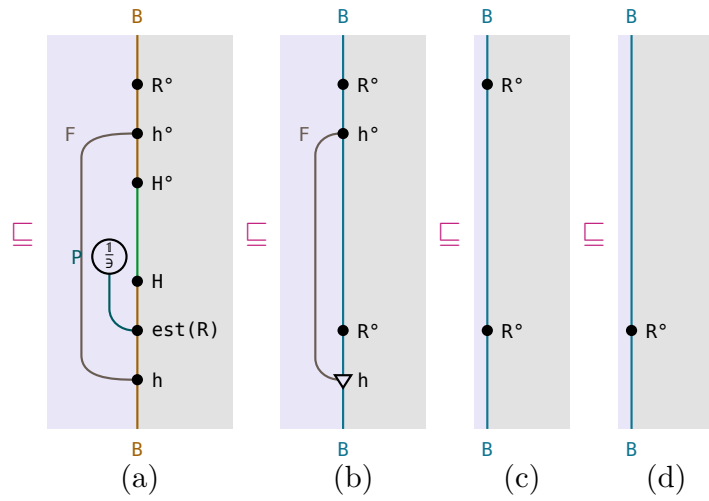
|     |                                                                                     |                                                |
|-----|-------------------------------------------------------------------------------------|------------------------------------------------|
| (a) | $\frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h$      | $\text{est}(Q) \subseteq \exists$              |
| (b) | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{ est}(R)) h$ | $\frac{R}{\exists} \exists = R$                |
| (c) | $= T^\circ F(\frac{H}{\exists} \text{ est}(R)) h$                                   | $\frac{R}{\exists} \text{ est}(R) \subseteq S$ |
| (d) | $\sqsubseteq T^\circ F(H) h$                                                        | $T^\circ F(H) h = H$                           |
| (e) | $= H$                                                                               |                                                |

$$\text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H) h = H \wedge Q F(H) h \subseteq F(H) h R \implies H^\circ \frac{T^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h \subseteq R^\circ \quad (10.1c)$$

$H^\circ$  followed by the greedy body at  $M$  is  $\subseteq R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input



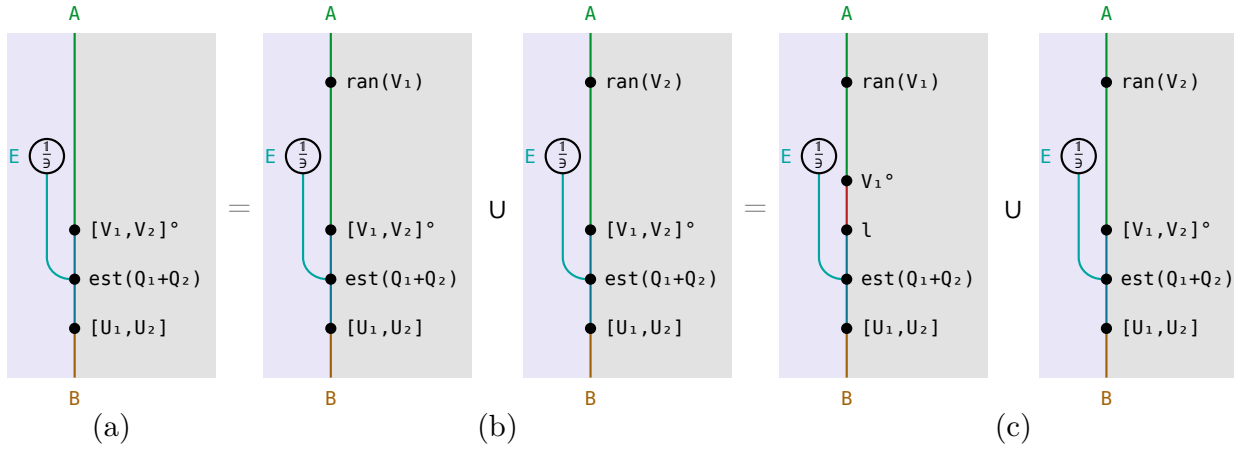
|     |                                                                                         |                                                                                                             |
|-----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| (a) | $H^\circ \frac{T^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h$                | $T^\circ F(H) h = H$<br>$\implies H^\circ = h^\circ F(H^\circ) T$                                           |
| (b) | $= h^\circ F(H^\circ) T \frac{T^\circ}{3} \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h$ | $T \frac{T^\circ}{3} \subseteq \in$                                                                         |
| (c) | $\subseteq h^\circ F(H^\circ) \in \text{est}(Q) F(\frac{H}{3} \text{est}(R)) h$         | $\in \text{est}(R) \subseteq R^\circ$                                                                       |
| (d) | $\subseteq h^\circ F(H^\circ) Q^\circ F(\frac{H}{3} \text{est}(R)) h$                   | $Q F(H) h \subseteq F(H) h R$<br>$\implies h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$ |



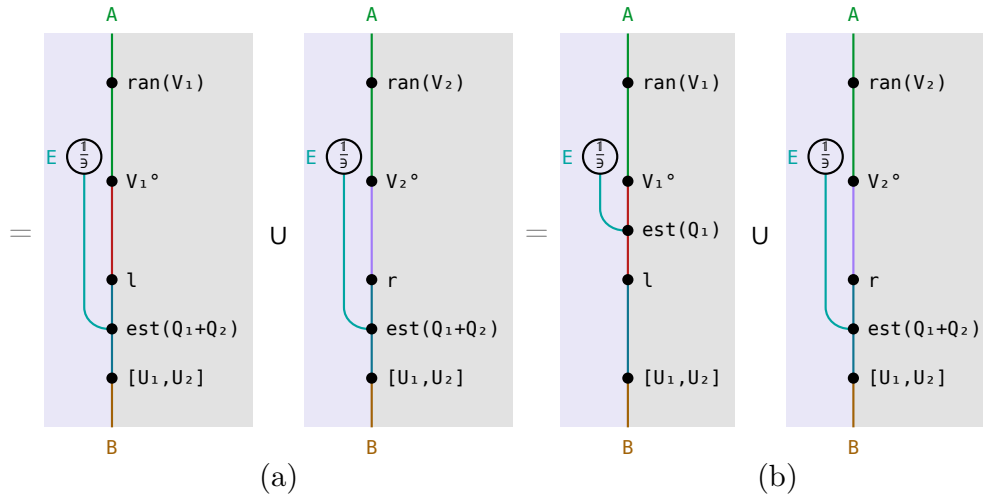
|     |                                                                    |                                                                                    |
|-----|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| (a) | $\subseteq R^\circ h^\circ F(H^\circ \frac{H}{3} \text{est}(R)) h$ | $S^\circ \frac{S}{3} \text{est}(R) \subseteq R^\circ$                              |
| (b) | $\subseteq R^\circ h^\circ F(R^\circ) h$                           | $\text{Map}(f)$<br>$\implies \text{MonoAlg}(f, R) \iff f^\circ F(R) f \subseteq R$ |
| (c) | $\subseteq R^\circ R^\circ$                                        | $R^\circ R^\circ \subseteq R^\circ$                                                |
| (d) | $\subseteq R^\circ$                                                |                                                                                    |

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ U_2) \quad \text{Proposition 10.1}$$

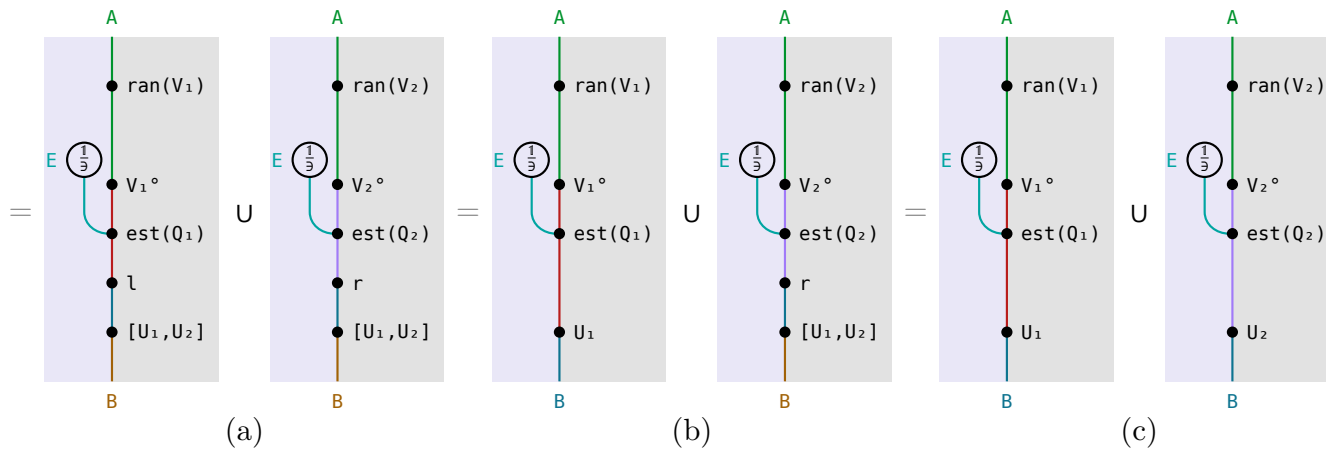
when  $V_1$  and  $V_2$  have disjoint ranges, the greedy step over  $[V_1, V_2]$  runs the  $V_1$  step on inputs  $V_1$  reaches and the  $V_2$  step on inputs  $V_2$  reaches



|     |                                                                                                                                                                             |                                                                                                                                                                                                                                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2]$                                                                                                         | $\frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2])$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2])$      | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_1) \ \frac{V_1^\circ}{\exists} P(l)$                                                                                             |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{\exists} = \text{ran}(V_2) \ \frac{V_2^\circ}{\exists} P(r)$                                                                                             |



|     |                                                                                                                                                                                  |                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $P(l) \text{est}(Q_1 + Q_2) = \text{est}(Q_1) \ l$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$        | $P(r) \text{est}(Q_1 + Q_2) = \text{est}(Q_2) \ r$ |



|     |                                                                                                                                                                    |                |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| (a) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ r [U_1, U_2])$ | $l [R, S] = R$ |
| (b) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ r [U_1, U_2])$          | $r [R, S] = S$ |
| (c) | $= (\text{ran}(V_1) \ \frac{V_1^\circ}{\exists} \text{est}(Q_1) \ U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{\exists} \text{est}(Q_2) \ U_2)$                   |                |

## §10.2. The detab-entab problem

(10.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over  $\text{snoc}$ -lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ .

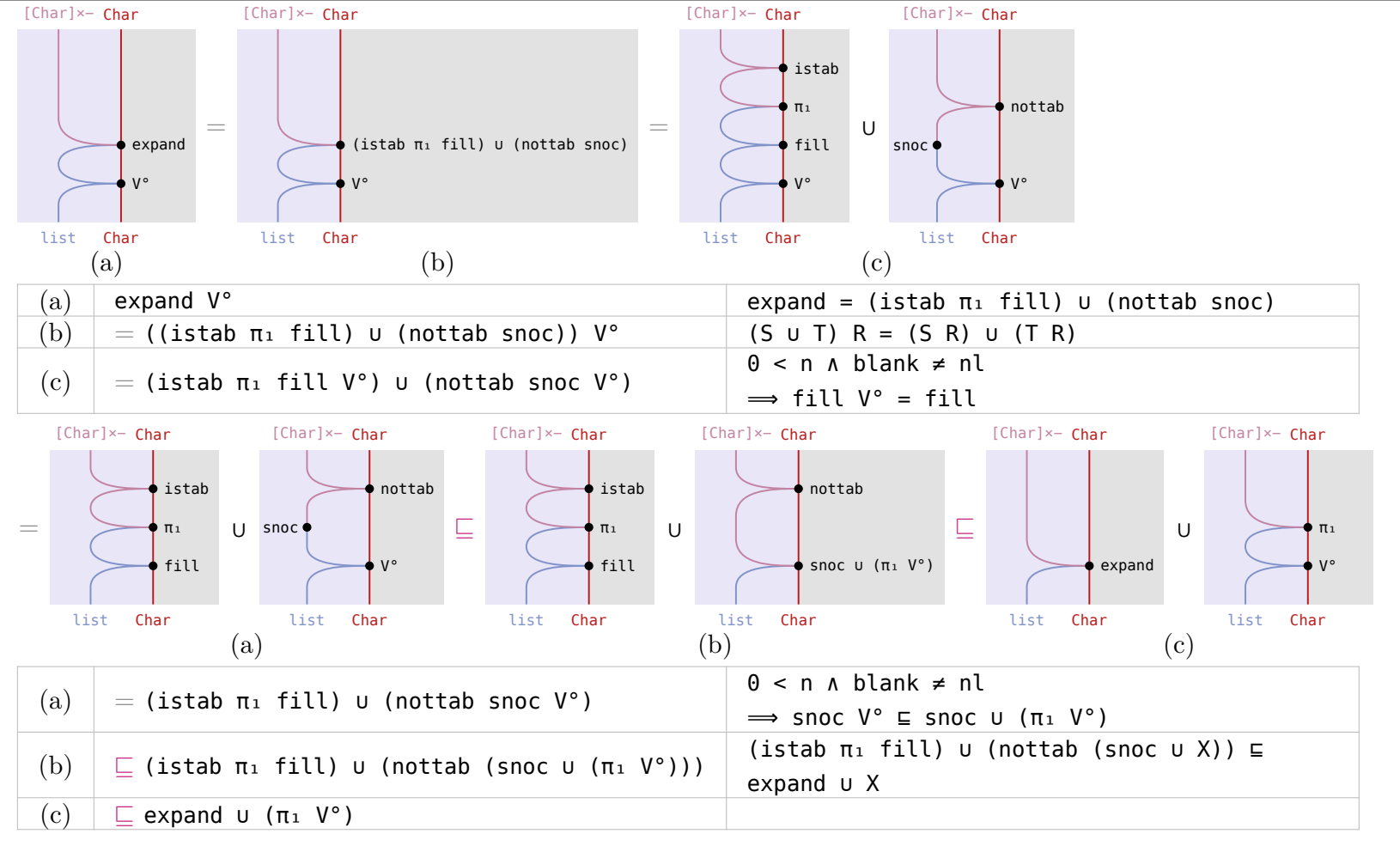
$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a \ U \ b \iff a = \text{TB} \vee a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill}$  ( $\text{unfill } xs = \text{fill } xs$ );  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$ .

$$0 < n \wedge \text{blank} \neq \text{nl} \implies \text{expand } V^\circ \sqsubseteq \text{expand } u \ (\pi_1 V^\circ) \quad (10.2b)$$

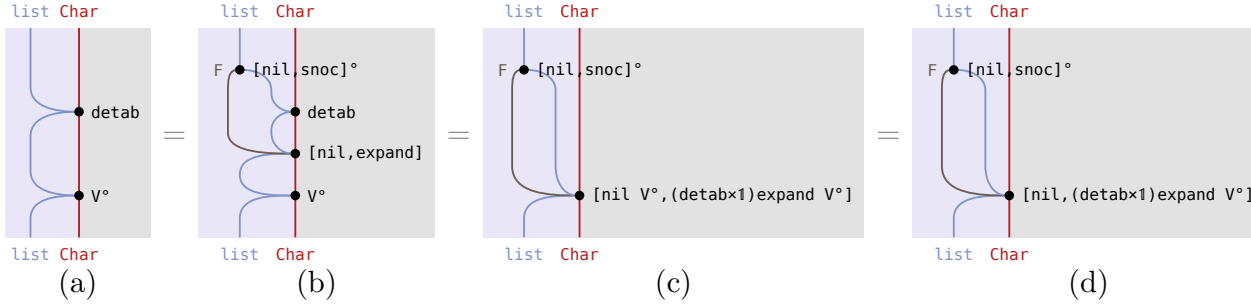
shortening the output of one **expand** step to a  $V$ -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ( $\pi_1 V^\circ$ )



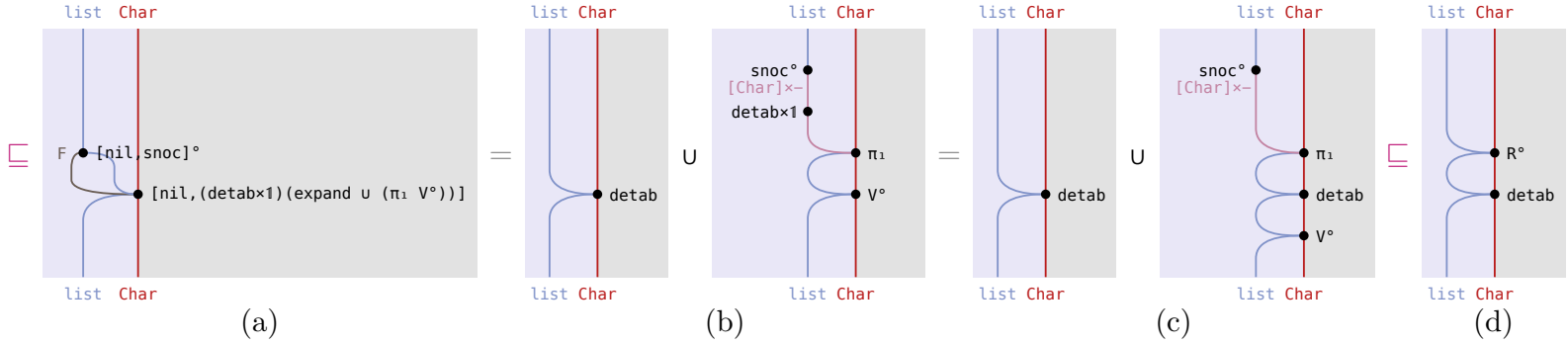
(10.2a)

$$0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{ detab} \quad (10.2c)$$

any  $V$ -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one



|     |                                                                                                   |                                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{detab } V^\circ$                                                                           | $\text{detab} = [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}]$                                           |
| (b) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$           | $[R, S] Z = [R Z, S Z]$                                                                                                                |
| (c) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil } V^\circ, \text{expand } V^\circ]$ | $\text{nil } V^\circ = \text{nil}$                                                                                                     |
| (d) | $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } V^\circ]$          | $(10.2b) \ 0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } u (\pi_1 V^\circ)$ |

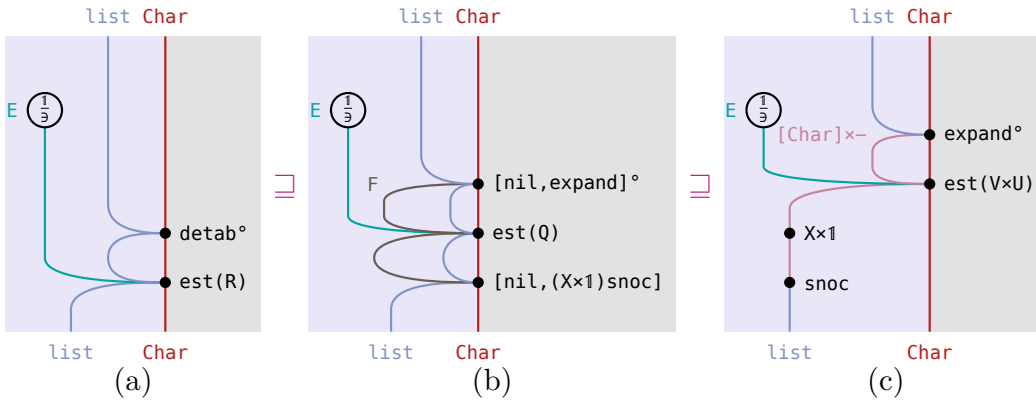


|     |                                                                                                              |                                                                                                                                                           |
|-----|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u (\pi_1 V^\circ)]$ | $[\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u X] = \text{detab } u (\text{snoc}^\circ (\text{detab} \times 1) X)$        |
| (b) | $= \text{detab } u (\text{snoc}^\circ (\text{detab} \times 1) \pi_1 V^\circ)$                                | $\text{snoc}^\circ (\text{detab} \times 1) \pi_1 Y = \text{snoc}^\circ \pi_1 \text{detab } Y$                                                             |
| (c) | $= \text{detab } u (\text{snoc}^\circ \pi_1 \text{detab } V^\circ)$                                          | $0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow \text{detab } u (\text{snoc}^\circ \pi_1 \text{detab } V^\circ) \sqsubseteq R^\circ \text{detab}$ |
| (d) | $\sqsubseteq R^\circ \text{detab}$                                                                           |                                                                                                                                                           |

$$0 < n \wedge \text{blank} \neq n\mathbf{l} \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R) \quad (10.2d)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

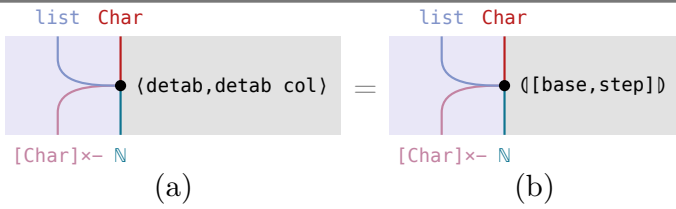
the specification —  $\triangleq \text{detab}$ ;  $\text{detab entab} = 1$  and nothing shorter does



|     |                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{detab}^\circ}{3} \text{est}(R)$                                                           | at $Q \triangleq 1 + (V \times U)$ : one character of input is decided at each step. $F(\tau, R) \alpha \alpha R$ . $\text{detab prefix} \sqsubseteq R^\circ \text{detab}$ is FALSE, — at $n=8$ , $\text{detab}$ $[a, b, c, d, e, \text{TB}] = [a, b, c, d, e, \text{BL}, \text{BL}, \text{BL}]$ , whose prefix $[a, b, c, d, e, \text{BL}, \text{BL}]$ is longer than any input giving it, and $\text{detab } V^\circ \sqsubseteq R^\circ \text{detab}$ holds. $\text{expand } V^\circ \sqsubseteq \text{expand } u (\pi_1 V^\circ)$ — shortening the output either leaves the last step alone or discards it |
| (b) | $\sqsubseteq \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>expand</b> have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (c) | $\sqsubseteq \frac{\text{expand}^\circ}{3} \text{est}(V \times U) (X \times 1) \text{snoc}$            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

$$\text{blank} \neq \text{nl} \wedge \text{tb} \neq \text{nl} \Rightarrow \langle \text{detab}, \text{detab col} \rangle = \langle \llbracket \text{base}, \text{step} \rrbracket \rangle \quad (10.2e)$$

one pass over the input carries the output so far together with its column



|     |                                                                    |                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \text{detab}, \text{detab col} \rangle$                   | both components are folds over the same input; <b>step</b> on $(\text{detab } x, \text{col}(\text{detab } x))$ is $(\text{detab}, \text{col detab})$ one character further — not a tabulated row |
| (b) | $= \langle \llbracket \text{base}, \text{step} \rrbracket \rangle$ |                                                                                                                                                                                                  |

Exercise 10.1

- $\text{blank} \neq \text{nl} \wedge \text{tb} \neq \text{nl} \Rightarrow \langle \llbracket \text{base}, \text{step} \rrbracket \rangle (\text{convert}(l), r) \Leftrightarrow r = \text{loop}(\text{step}, (\llbracket \cdot \rrbracket, 0), l)$   
the fold of  $\llbracket \text{base}, \text{step} \rrbracket$  over the input converted to a snoc-list is the left loop of **step** from **base**

- $\pi_1(\text{loop}(\text{step}, (\llbracket \cdot \rrbracket, c_0), x)) = \text{loop}'((f, g), (c_0, x))$

Exercise 10.1: when **step** only appends  $f(c, a)$  to the output and moves the state by  $g$ , the output of the loop is  $\text{loop}'(f, g)$ , which never carries the output

- $a \neq \text{tb} \Rightarrow \text{expand}(y, b) = \text{snoc}(x, a) \Leftrightarrow \text{fill}(y) = \text{snoc}(x, a) \wedge b = \text{tb} \vee (y, b) = (x, a)$   
**expand** produces  $x\#a$  from a tab after a string that fills to it, or from  $x$  and  $a$

(10.2f)

- $0 < n \wedge \text{blank} \neq \text{nl} \Rightarrow (\exists y. \text{fill}(y) = \text{snoc}(x, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n = 0$   
a string is a **fill** exactly when it ends in a blank on a tab stop

(10.1)

- $\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x)$

(10.1): the output of **entab** is that of **unfill**  $x$  followed by the trailing blanks held back

- $\text{tbc}[] = 0$

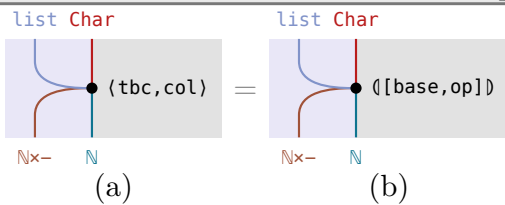
the empty string has no trailing blanks

- $\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$

a blank off a tab stop adds one held blank; anything else releases them all

$$\text{blank} \neq \text{nl} \Rightarrow \langle \text{tbc}, \text{col} \rangle = \langle \llbracket \text{base}, \text{op} \rrbracket \rangle \quad (10.2g)$$

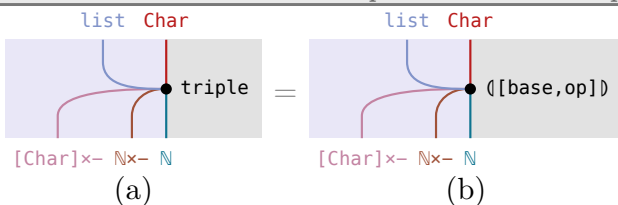
one pass counts the held blanks and the column together



|     |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle \text{tbc}, \text{col} \rangle$                         | <b>op</b> on $(\text{tbc } x, \text{col } x)$ is $(\text{tbc}, \text{col})$ one character further, from the equations for <b>tbc</b> — (10.1) $\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x)$ , $\text{tbc}[] = 0$ , $\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$ |
| (b) | $= \langle \llbracket \text{base}, \text{op} \rrbracket \rangle$ |                                                                                                                                                                                                                                                                                                                                                                                                                                |

$$\text{blank} \neq \text{nl} \Rightarrow \text{triple} = \langle \llbracket \text{base}, \text{op} \rrbracket \rangle \quad (10.2h)$$

one pass carries the output up to the held blanks, their count and the column



|     |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | <b>triple</b>                                                    | <b>op</b> on <b>triple</b> $x$ is <b>triple</b> one character further: (10.1) where the blanks are released — (10.1) $\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x)$ , $\text{tbc}[] = 0$ , $\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$ |
| (b) | $= \langle \llbracket \text{base}, \text{op} \rrbracket \rangle$ |                                                                                                                                                                                                                                                                                                                                                                                                       |

| $\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$ <span style="float: right;">(10.2i)</span>                                                                 |                                                                      |                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| entab is triple with the held blanks appended to its output                                                                                                                                  |                                                                      |                                                                                                                                                                                                                                                                                         |
| <div style="display: flex; justify-content: space-around; margin-top: 10px;"> <div style="text-align: center;"> <p>(a)</p> </div> <div style="text-align: center;"> <p>(b)</p> </div> </div> |                                                                      |                                                                                                                                                                                                                                                                                         |
| (a)                                                                                                                                                                                          | entab                                                                | (10.1) — (10.1) $\text{entab}(x) = \text{entab} (\text{unfill } x) ++ \text{blanks } (\text{tbc } x), \text{tbc}[] = 0, \text{tbc}(\text{snoc}(x,a)) = \text{if } a = \text{blank} \wedge \text{col } (\text{snoc } x \ a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$ |
| (b)                                                                                                                                                                                          | $= \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$ |                                                                                                                                                                                                                                                                                         |

### §10.3. The minimum tardiness problem

(10.3a)

$\text{FX} = 1 + (X \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags, snag putting a job into a bag;  $\text{bagify} \triangleq (\beta) : [\text{Job}] \rightarrow \text{Bag Job}$ ,  $\text{H} = \text{bagify}^\circ$ .

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$  the completion, due and weighting quantities of a job;  $\text{penalty}(xs, j) = (\text{sum}(\text{list}(\text{ct})(xs)) + \text{ct}(j) - \text{dt}(j)) \times \text{wt}(j)$ .

$\text{cost} \triangleq \frac{\text{prefix}}{\sup} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ est}(\geq)$ ,  $\text{cost} [] = 0$ ,  $\text{cost } (xs \# [j]) = \text{bmax} (\text{cost } xs, \text{penalty } (xs, j))$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$\text{perm} \triangleq \text{bagify } \text{bagify}^\circ = ([\text{nil}, \text{add}])$ ,  $\text{add } (xs, j) = ys \# [j] \# zs$  for some  $xs = ys \# zs$ .

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$ ,  $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$ ,  $Q \triangleq f \leq f^\circ$ ,  $Q' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$ .

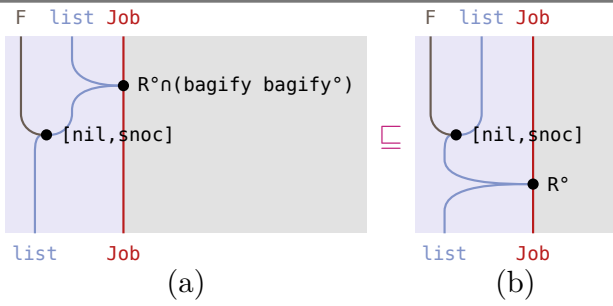
$g \triangleq [\text{zero}, \text{penalty}]$  (10.5),  $m \triangleq [\text{zero}, \pi_1 \text{ cost}]$  (B&dM's  $h$ , renamed as in  $(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j))$ )  
 $\Rightarrow (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{\sup} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{bagify}^\circ}{\sup} \text{ est}(R) )$  (10.6).

| $[\text{nil}, \text{snoc}] \text{ cost} = F((\text{cost}, \text{bagify})) k$ <span style="float: right;">(10.3b)</span>                                                                      |                                          |                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job                                                                                    |                                          |                                                                                                                                                                              |
| <div style="display: flex; justify-content: space-around; margin-top: 10px;"> <div style="text-align: center;"> <p>(a)</p> </div> <div style="text-align: center;"> <p>(b)</p> </div> </div> |                                          |                                                                                                                                                                              |
| (a)                                                                                                                                                                                          | $[\text{nil}, \text{snoc}] \text{ cost}$ | $\text{cost } (xs \# [j]) = \text{bmax} (\text{cost } xs, \text{penalty } (xs, j))$ ,<br>and $\text{penalty}(xs, j)$ reads only the bag of $xs$ — $\triangleq \text{bagify}$ |
| (b)                                                                                                                                                                                          | $= F((\text{cost}, \text{bagify})) k$    |                                                                                                                                                                              |

| $F(\geq \times 1) k \sqsubseteq k \geq$ <span style="float: right;">Exercise 10.5</span>                                                                                                     |                      |                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------|
| a larger cost before the last job gives a larger cost after it                                                                                                                               |                      |                                                 |
| <div style="display: flex; justify-content: space-around; margin-top: 10px;"> <div style="text-align: center;"> <p>(a)</p> </div> <div style="text-align: center;"> <p>(b)</p> </div> </div> |                      |                                                 |
| (a)                                                                                                                                                                                          | $F(\geq \times 1) k$ | $\text{bmax}$ is monotone in its first argument |
| (b)                                                                                                                                                                                          | $\sqsubseteq k \geq$ |                                                 |

$$F(R^\circ \cap (\text{bagify bagify}^\circ)) [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R^\circ \quad (10.2)$$

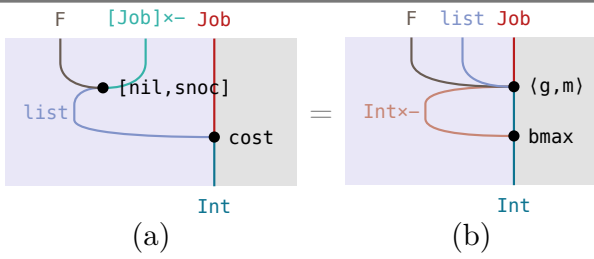
improving the schedule before the last job, among schedules of the same bag, improves the whole schedule



|     |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $F(R^\circ \cap (\text{bagify bagify}^\circ)) [\text{nil}, \text{snoc}]$ | Proposition 9.3 at $S \triangleq \text{bagify}$ , $\leq \triangleq$ — Proposition 9.3<br>$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge h$<br>$\text{cost} = F((\text{cost}, S)) \ k \wedge F(\leq \times 1) \ k \sqsubseteq k \leq$<br>$\Rightarrow F(R \cap (S \ S^\circ)) \ h \sqsubseteq h \ R$ , with $[\text{nil}, \text{snoc}] \text{cost}$<br>$= F((\text{cost}, \text{bagify})) \ k$ and Exercise 10.5 $F(\geq \times 1) \ k$<br>$\sqsubseteq k \geq$ |
| (b) | $\sqsubseteq [\text{nil}, \text{snoc}] R^\circ$                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

$$[\text{nil}, \text{snoc}] \text{cost} = \langle g, m \rangle \text{bmax} \quad (10.4)$$

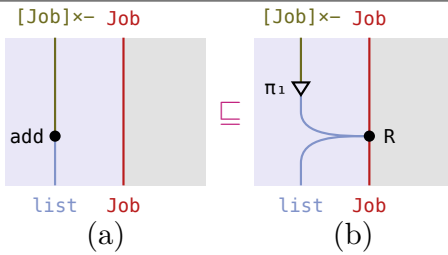
the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it



|     |                                         |                                                                                                                                                                                 |
|-----|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{snoc}] \text{cost}$ | $\text{cost} (\text{xs} \# [\text{j}]) = \text{bmax} (\text{cost } \text{xs}, \text{penalty} (\text{xs}, \text{j}))$<br>and $\text{bmax}$ commutes — $\triangleq \text{bagify}$ |
| (b) | $= \langle g, m \rangle \text{bmax}$    |                                                                                                                                                                                 |

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 R \quad (10.7)$$

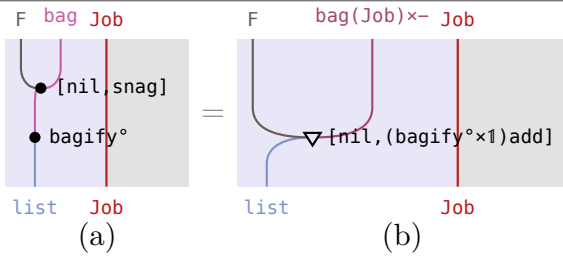
putting one more job anywhere into a schedule never lowers its cost



|     |                       |                                                                                   |
|-----|-----------------------|-----------------------------------------------------------------------------------|
| (a) | $\text{add}$          | induction on where $j$ goes: every job after it starts later, so no penalty falls |
| (b) | $\sqsubseteq \pi_1 R$ |                                                                                   |

$$[\text{nil}, \text{snag}] \text{bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{add}] \quad (10.8)$$

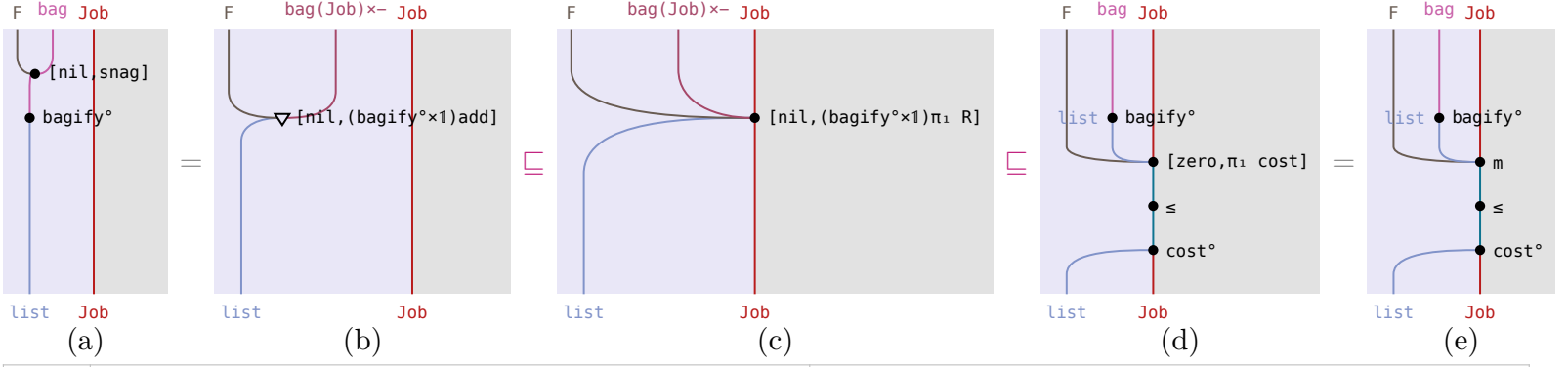
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



|     |                                                             |                                                                            |
|-----|-------------------------------------------------------------|----------------------------------------------------------------------------|
| (a) | $[\text{nil}, \text{snag}] \text{bagify}^\circ$             | take out the last copy of $j$ and put it back — $\triangleq \text{bagify}$ |
| (b) | $= [\text{nil}, (\text{bagify}^\circ \times 1) \text{add}]$ |                                                                            |

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow [nil, snag] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job

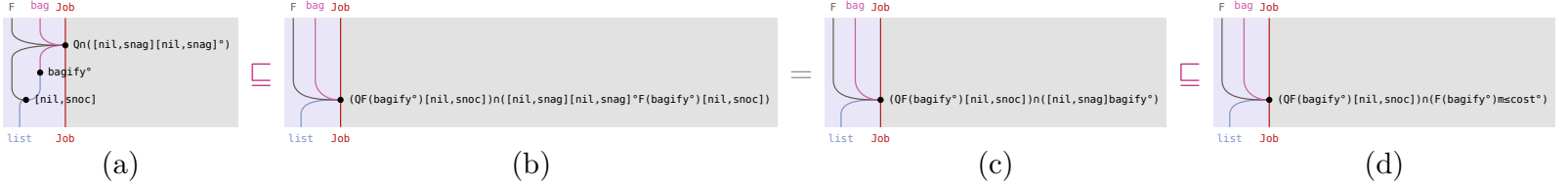


|     |                                                                                        |                                                                                                                |
|-----|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| (a) | $[nil, snag] \text{ bagify}^\circ$                                                     | (10.8) $[nil, snag] \text{ bagify}^\circ = [nil, (bagify^\circ \times 1) \text{ add}]$                         |
| (b) | $= [nil, (bagify^\circ \times 1) \text{ add}]$                                         | (10.7) $(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 R$ |
| (c) | $\sqsubseteq [nil, (bagify^\circ \times 1) \pi_1 R]$                                   | $[nil, \pi_1 R] \sqsubseteq [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$                                 |
| (d) | $\sqsubseteq F(\text{bagify}^\circ) [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$ |                                                                                                                |
| (e) | $= F(\text{bagify}^\circ) m \leq \text{cost}^\circ$                                    |                                                                                                                |

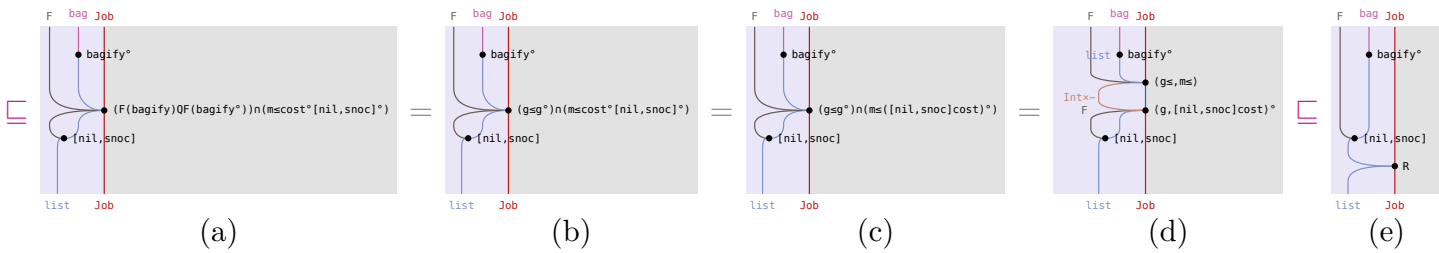
$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow (Q n ([nil, snag] [nil, snag]^\circ)) F(\text{bagify}^\circ) [nil, snoc] \sqsubseteq \quad (10.3)$$

$$F(\text{bagify}^\circ) [nil, snoc] R$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



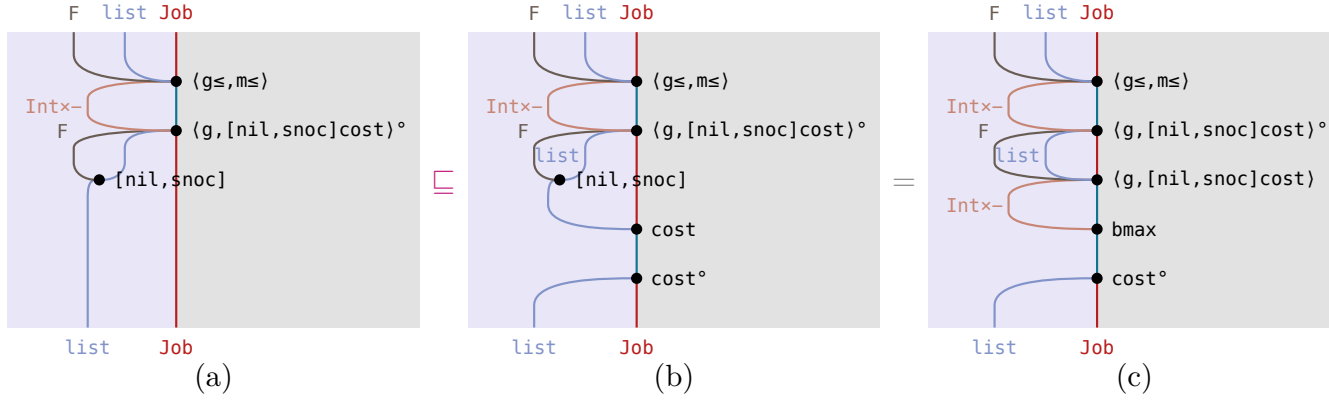
|     |                                                                                                                           |                                                                                                                                                                                                  |
|-----|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $(Q n ([nil, snag] [nil, snag]^\circ)) F(\text{bagify}^\circ) [nil, snoc]$                                                | $(S n T) R \sqsubseteq (S R) n (T R)$                                                                                                                                                            |
| (b) | $\sqsubseteq (Q F(\text{bagify}^\circ) [nil, snoc]) n ([nil, snag] [nil, snag]^\circ F(\text{bagify}^\circ) [nil, snoc])$ | $[nil, snag]^\circ F(\text{bagify}^\circ) [nil, snoc] = \text{bagify}^\circ$                                                                                                                     |
| (c) | $= (Q F(\text{bagify}^\circ) [nil, snoc]) n ([nil, snag] \text{ bagify}^\circ)$                                           | (10.3c) $(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow [nil, snag] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ$                    |
| (d) | $\sqsubseteq (Q F(\text{bagify}^\circ) [nil, snoc]) n (F(\text{bagify}^\circ) m \leq \text{cost}^\circ)$                  | $(Q F(\text{bagify}^\circ) [nil, snoc]) n (F(\text{bagify}^\circ) M) \sqsubseteq F(\text{bagify}^\circ) ((F(\text{bagify}^\circ) Q F(\text{bagify}^\circ)) n (M [nil, snoc]^\circ)) [nil, snoc]$ |



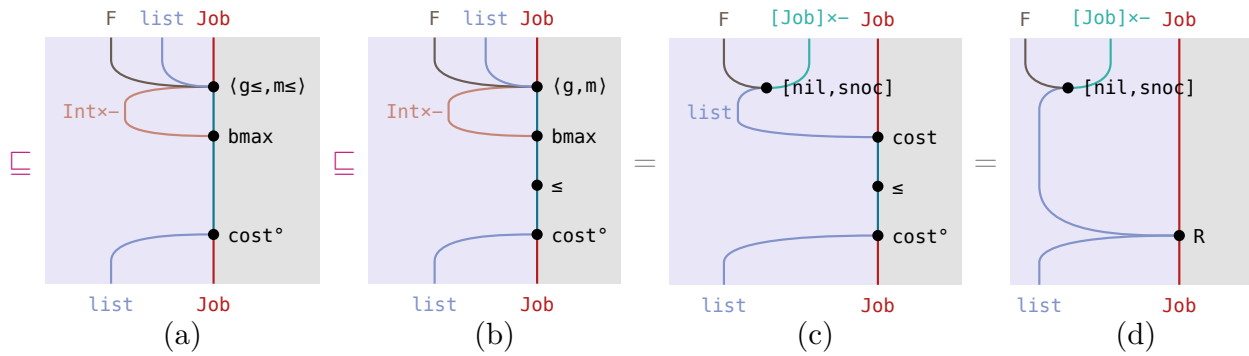
|     |                                                                                                                                               |                                                                                                                    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq F(\text{bagify}^\circ) ((F(\text{bagify}) Q F(\text{bagify}^\circ)) n (m \leq \text{cost}^\circ [nil, snoc]^\circ)) [nil, snoc]$ | $F(\text{bagify}) Q F(\text{bagify}^\circ) = g \leq g^\circ$                                                       |
| (b) | $= F(\text{bagify}^\circ) ((g \leq g^\circ) n (m \leq \text{cost}^\circ [nil, snoc]^\circ)) [nil, snoc]$                                      | $(R S)^\circ = S^\circ R^\circ$                                                                                    |
| (c) | $= F(\text{bagify}^\circ) ((g \leq g^\circ) n (m \leq ([nil, snoc] \text{ cost})^\circ)) [nil, snoc]$                                         | $\langle X, Y \rangle \langle R, S \rangle^\circ = (X R^\circ) n (Y S^\circ)$                                      |
| (d) | $= F(\text{bagify}^\circ) (g \leq, m \leq) (g, [nil, snoc] \text{ cost})^\circ [nil, snoc]$                                                   | (10.3d) $\langle g \leq, m \leq \rangle (g, [nil, snoc] \text{ cost})^\circ [nil, snoc] \sqsubseteq [nil, snoc] R$ |
| (e) | $\sqsubseteq F(\text{bagify}^\circ) [nil, snoc] R$                                                                                            |                                                                                                                    |

$$\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R \quad (10.3d)$$

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job,  
gives a schedule costing at most the whole



|     |                                                                                                                                                                                      |                                                                                                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}]$                                                           | $1 \sqsubseteq \text{cost cost}^\circ$                                                                                                    |
| (b) | $\sqsubseteq \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}] \text{ cost cost}^\circ$                       | $[\text{nil}, \text{snoc}] \text{ cost} = \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \text{ bmax}$                         |
| (c) | $= \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \text{ bmax cost}^\circ$ | $\langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \sqsubseteq 1$ |



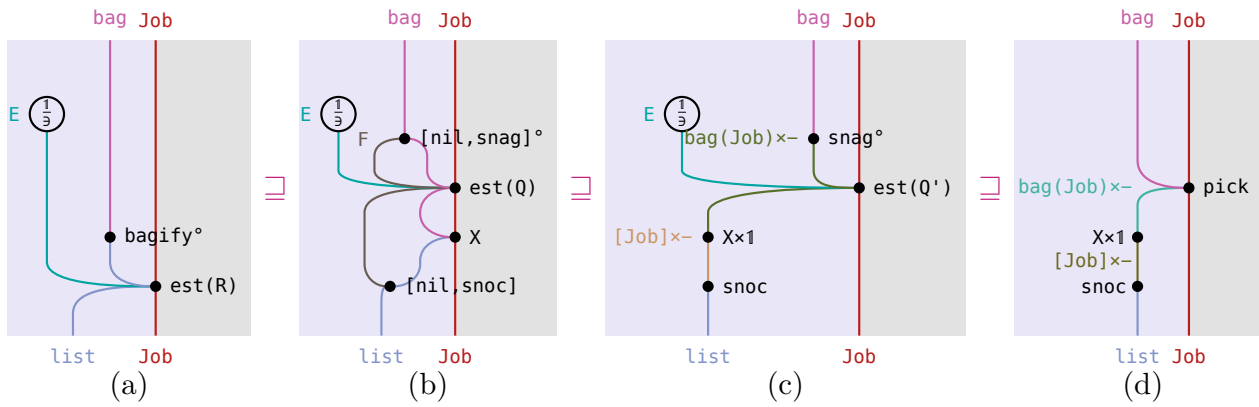
|     |                                                                        |                                                                                                  |
|-----|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| (a) | $\sqsubseteq \langle g \leq, m \leq \rangle \text{ bmax cost}^\circ$   | $\langle g \leq, m \leq \rangle \text{ bmax} \sqsubseteq \langle g, m \rangle \text{ bmax} \leq$ |
| (b) | $\sqsubseteq \langle g, m \rangle \text{ bmax} \leq \text{cost}^\circ$ | $(10.4) [\text{nil}, \text{snoc}] \text{ cost} = \langle g, m \rangle \text{ bmax}$              |
| (c) | $= [\text{nil}, \text{snoc}] \text{ cost} \leq \text{cost}^\circ$      | $R = \text{cost} \leq \text{cost}^\circ$                                                         |
| (d) | $= [\text{nil}, \text{snoc}] R$                                        |                                                                                                  |

$$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick} (\text{schedule} \times 1) \text{ snoc}), (\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \quad (10.3e)$$

$$\frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$$

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag,  
putting it last, and scheduling what is left the same way

the specification —  $\triangleq \text{bagify}$



|     |                                                                                                             |                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$                                                        | No greedy <b>reduce</b> exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole |
| (b) | $\sqsubseteq \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>snag</b> have disjoint ranges                                                                                                  |
| (c) | $\sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q') (X \times 1) \text{ snoc}$                   | $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q')$ , a partial function, quadratic in the number of jobs                                  |
| (d) | $\sqsubseteq \text{pick} (X \times 1) \text{ snoc}$                                                         |                                                                                                                                                                    |

## §10.4. The TeX problem

(10.4a)

$\text{intern} \triangleq \text{val round} : \text{Decimal} \rightarrow [0, 2^{16})$ ,  $\text{val} \triangleq ([\text{zero}, \text{shift}])$ ,  $\text{shift } (d, r) = (d+r)/10$ ,  $\text{round } r$  rounds  $2^{16}r$  to the nearest integer:  $\text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1$ .

$\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17})$ ,  $r \text{ inrange } (a, b) \iff a < r < b$ ,  $\text{round}^\circ = \text{interval inrange}$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ .

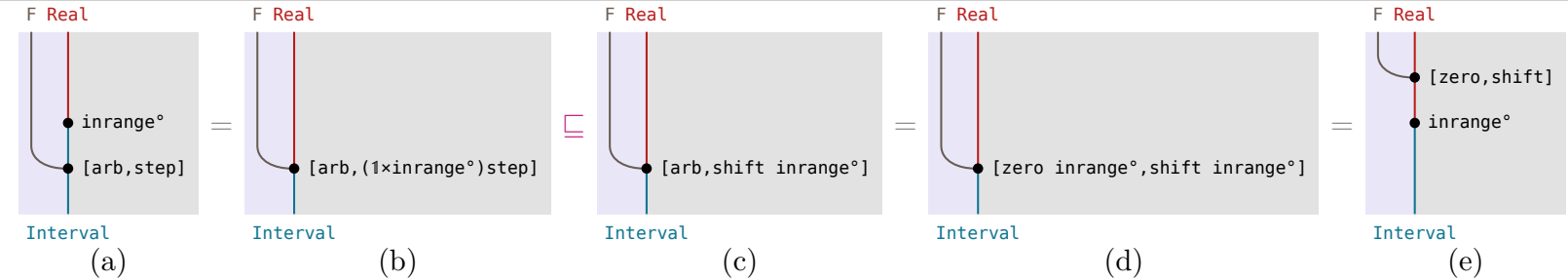
Interval the pairs  $(a, b)$  with  $\text{Legal}(a, b) \iff 0 < b < 1$  and  $a < b$  (10.9);  $\text{step\_legal}$  is what types  $\text{step } [\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$ ,  $\text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10)$ .

$\text{FX} = 1 + (\text{Digit} \times X)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $H \triangleq ([\text{arb}, \text{step}])^\circ$ ,  $! : \text{Digit} \times \text{Interval} \rightarrow 1$ ,  $Q \triangleq (l^\circ !^\circ r) \cup 1$   $l, r$  are  $[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$  's injections into  $\text{FX} = 1 + (\text{Digit} \times X)$ , so  $l : 1 \rightarrow \text{FX}$  and  $r : \text{Digit} \times X \rightarrow \text{FX}$ ,  $w \triangleq 2^{17}$ .

$([\text{arb}, \text{step}])^\circ \sqsubseteq \text{val inrange}^\circ$

(10.4b)

every interval got by folding a decimal's digits with  $[\text{arb}, \text{step}]$  has the decimal's value strictly inside it

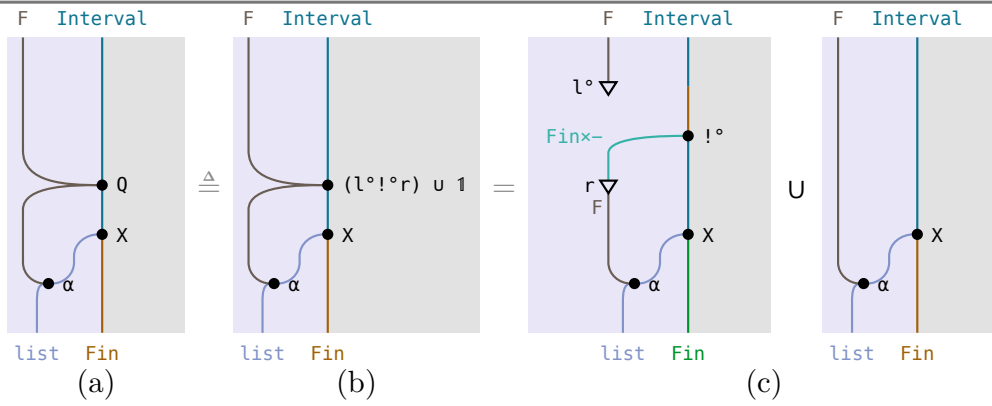


|     |                                                               |                                                                                      |
|-----|---------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (a) | $F(\text{inrange}^\circ) [\text{arb}, \text{step}]$           | $F(S) [T, U] = [T, (1 \times S) U]$                                                  |
| (b) | $= [\text{arb}, (1 \times \text{inrange}^\circ) \text{step}]$ | $(1 \times \text{inrange}^\circ) \text{step} \sqsubseteq \text{shift inrange}^\circ$ |
| (c) | $\sqsubseteq [\text{arb}, \text{shift inrange}^\circ]$        | $\text{zero inrange}^\circ = \text{arb}$                                             |
| (d) | $= [\text{zero inrange}^\circ, \text{shift inrange}^\circ]$   | $[R, S] Z = [R Z, S Z]$                                                              |
| (e) | $= [\text{zero}, \text{shift}] \text{inrange}^\circ$          |                                                                                      |

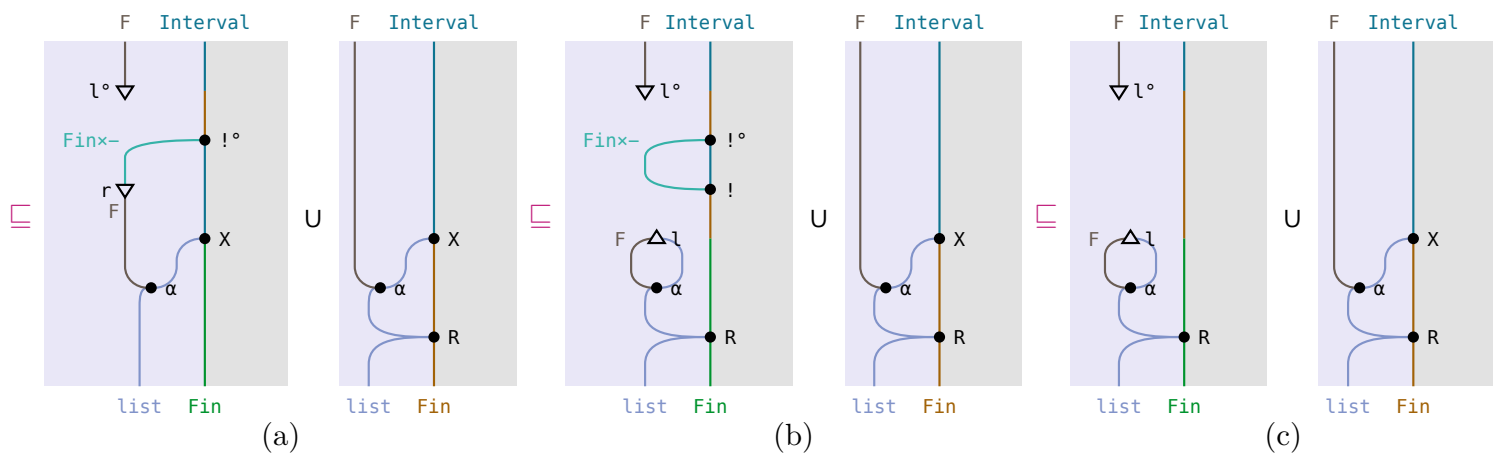
$$Q \ F(X) \ \alpha \sqsubseteq F(X) \ \alpha \ R$$

(10.4c)

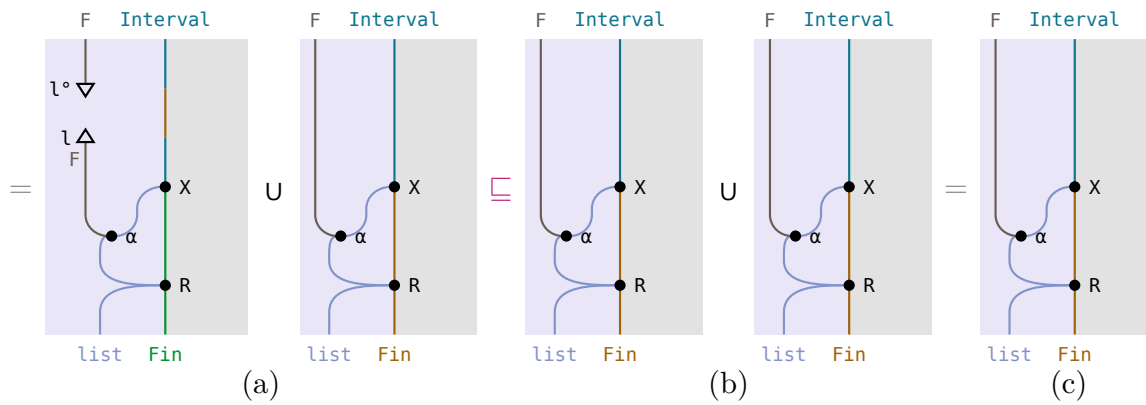
choosing by  $Q$  among the one-step decompositions before building with  $F(X)$  and  $\alpha$  yields only decimals that are  $R$ -related to one built without choosing



|     |                                                          |                                         |
|-----|----------------------------------------------------------|-----------------------------------------|
| (a) | $Q \ F(X) \ \alpha$                                      | $Q \triangleq (l^o \ !^o \ r) \cup 1$   |
| (b) | $\triangleq ((l^o \ !^o \ r) \cup 1) \ F(X) \ \alpha$    | $(S \cup T) \ R = (S \ R) \cup (T \ R)$ |
| (c) | $= (l^o \ !^o \ r \ F(X) \ \alpha) \cup (F(X) \ \alpha)$ | $1 \sqsubseteq R$                       |



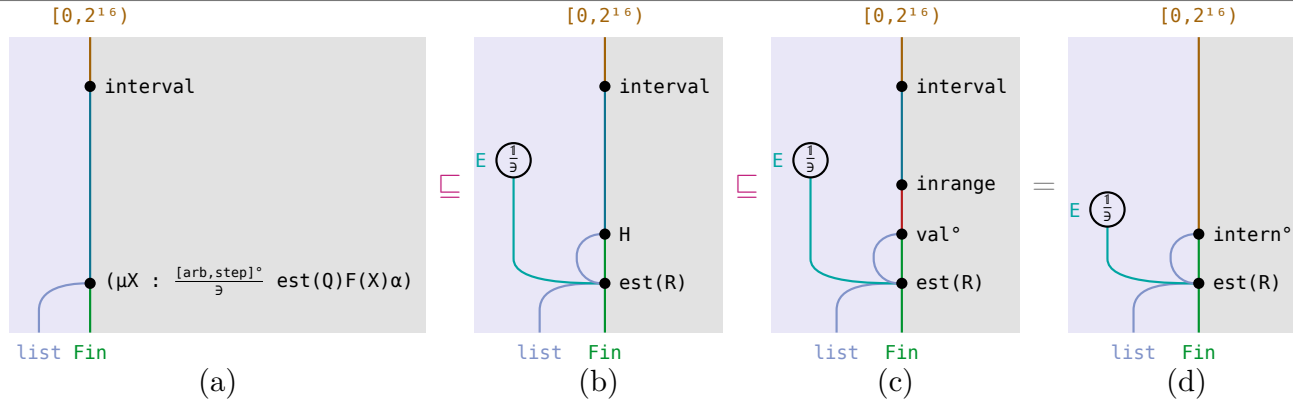
|     |                                                                         |                                                    |
|-----|-------------------------------------------------------------------------|----------------------------------------------------|
| (a) | $\sqsubseteq (l^o \ !^o \ r \ F(X) \ \alpha) \cup (F(X) \ \alpha \ R)$  | $r \ F(X) \ \alpha \sqsubseteq ! \ l \ \alpha \ R$ |
| (b) | $\sqsubseteq (l^o \ !^o \ ! \ l \ \alpha \ R) \cup (F(X) \ \alpha \ R)$ | $!^o \ ! \sqsubseteq 1$                            |
| (c) | $\sqsubseteq (l^o \ l \ \alpha \ R) \cup (F(X) \ \alpha \ R)$           | $l \ F(X) = l$                                     |



|     |                                                            |                         |
|-----|------------------------------------------------------------|-------------------------|
| (a) | $= (l^o \ l \ F(X) \ \alpha \ R) \cup (F(X) \ \alpha \ R)$ | $l^o \ l \sqsubseteq 1$ |
| (b) | $\sqsubseteq (F(X) \ \alpha \ R) \cup (F(X) \ \alpha \ R)$ | $R \cup R = R$          |
| (c) | $= F(X) \ \alpha \ R$                                      |                         |

$$\text{extern}(n)=f(2n-1,2n+1), \quad \text{interval} \left( \mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha \right) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of  $2^{-16}$  is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



|     |                                                                                                              |                                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{interval} \left( \mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha \right)$ | $\frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F\left(\frac{H}{3} \text{est}(R)\right) \alpha \sqsubseteq \frac{H}{3} \text{est}(R)$ |
| (b) | $\sqsubseteq \text{interval} \frac{H}{3} \text{est}(R)$                                                      | $\frac{H}{3} \text{est}(R) \sqsubseteq \frac{\text{inrange val}^\circ}{3} \text{est}(R)$                                                       |
| (c) | $\sqsubseteq \text{interval} \frac{\text{inrange val}^\circ}{3} \text{est}(R)$                               | $\frac{\text{intern}^\circ}{3} = \text{interval} \frac{\text{inrange val}^\circ}{3}$                                                           |
| (d) | $= \frac{\text{intern}^\circ}{3} \text{est}(R)$                                                              |                                                                                                                                                |

- $f(a,b) = x \iff a < 0 \wedge x = [] \vee \neg a < 0 \wedge (\exists d. \exists y. d = [10b] \wedge f(10a-d, 10b-d) = y \wedge x = [d] \# y)$   
 $f$  gives  $(a,b)$  the empty decimal when  $a < 0$ , and otherwise the digit  $d = [10b]$  followed by what it gives  $(10a-d, 10b-d)$  (10.4e)
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha) = f$   
the least solution of the greedy recursion of  $\text{interval} \left( \mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q) F(X) \alpha \right) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R)$  is  $f$
- $d = [10b] \wedge e = [10b] \implies d = e$   
Exercise 10.12: two digits  $d$  and  $e$  with  $0 < 10b-d < 1$  and  $0 < 10b-e < 1$  are equal
- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q) = []$   
the program's first clause:  $f(p,q) = []$  when  $p < 0$
- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \wedge d = 10q/w \implies f(p,q) = [d] \# f(10p-wd, 10q-wd)$   
the program's second clause: with  $d = (10q) \text{ div } w$ ,  $f(p,q)$  is  $d$  followed by  $f(10p-w \cdot d, 10q-w \cdot d)$
- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$   
on the pairs the program reaches from  $\text{interval } n$ , the integer  $f$  at  $(p,q)$  is a value of the rational  $f$  at  $(p/w, q/w)$
- $\text{interval } f = \text{extern}$   
 $f$  after  $\text{interval}$  is the program  $\text{extern}$ , whose value at  $n$  is  $f(2n-1, 2n+1)$

## §10. Appendix

### §10.1. $P(S) \text{ est}(R) = (\exists S) n(\in \setminus (SR^\circ))$

(10.1a)

$(p, q)$  tabulates  $W \triangleq (\exists S) n(\in \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$ , a map.

(10.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(10.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f \cdot, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(10.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \in \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

### §10.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(10.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{(7.11) R \sqsubseteq R \sqsubseteq R} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \Rightarrow P(\text{est}(R)) \text{ est}(R) \sqsubseteq \text{union est}(R) \end{array}$$

(10.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(10.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \xleftarrow{T(U \neg V) \sqsubseteq T U \neg T V, \cdot \exists \neg \exists, (7.3) \text{ union } (\in \setminus R) = \in \setminus (\in \setminus R)} \boxed{\begin{array}{c} (\exists \exists) n(\in \setminus (\in \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists n(\in \setminus (\in \setminus R^\circ))) \\ \exists (\exists n(\in \setminus R^\circ)) \end{array}} \quad \text{count of } T \cdot \neg T \setminus \\ \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\in \text{ lax natural}} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \\ \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{est}(R)^\circ \sqsubseteq \epsilon, \epsilon \epsilon \text{union} \sqsubseteq \epsilon} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est} \end{array}$$

### §10.3. Shortest paths on a cylinder, on lists

(10.3a)

|    | definition                                                                                                         | type                                             | note                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 1  | $F(A,X)=A+A \times X$                                                                                              | $\mathcal{A} \rightarrow \mathcal{A}$            | The base functor: one square of the new column, alone or in front of the path so far.                                           |
| 2  | $L = \text{list}^+$                                                                                                | $\mathcal{A} \rightarrow \mathcal{A}$            | A path is a non-empty list of squares, one per column crossed.                                                                  |
| 3  | $\alpha$ the initial algebra                                                                                       | $F(\text{dCL}(L,E)) \rightarrow \text{dCL}(L,E)$ | $\alpha(5)=[5]$ and $\alpha(1,[5])=[1,5]$ : a path is started by one square, or extended by one.                                |
| 4  | $N$<br>the n-tuple relator                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$            | One component per row of a column, so the fold carries n answers at once.                                                       |
| 5  | $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                                    | $[\mathbb{N}] \rightarrow [\mathbb{N}]$          | The cost of a path, which the cheapest minimises.                                                                               |
| 6  | setify                                                                                                             | $A[n] \rightarrow PA$                            | $\text{setify}(1,2,3,4)=\{1,2,3,4\}$ — which row a component came from is forgotten.                                            |
| 7  | moves                                                                                                              | $A[n] \rightarrow P(A[n])$                       | $\text{moves}(x)=\{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.                                    |
| 8  | trans                                                                                                              | $P(A[n]) \rightarrow (P\ A)[n]$                  | $\text{trans}\{(a,b,c), (x,y,z)\} = (\{a,x\}, \{b,y\}, \{c,z\})$ — component k of the result is the set of the k-th components. |
| 9  | zip                                                                                                                | $F(a[n]) \rightarrow (F\ a)[n]$                  | $\text{zip}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$ .      |
| 10 | $\text{cp} \triangleq \frac{F(1,\exists)}{\exists}$                                                                | $F(PA) \rightarrow PFA$                          | $\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$ .                                                          |
| 11 | $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union}))$<br>$\text{zip } N(\text{cp}(F(A,-))\ E(\alpha))$ | $F(NA, N(P(TA))) \rightarrow N(P(TA))$           | $\text{gen}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (10.3.1b).                                       |
| 12 | $\text{paths} \triangleq (\text{gen})\ \text{setify}\ \text{union}$                                                | $T(NA) \rightarrow P(TA)$                        | $\text{paths}[(1,2,3,4), (5,6,7,8)]$ is the union of (10.3.1b)'s four sets: 12 paths, 3 from each entry row.                    |
| 13 | the specification<br>$\text{paths est}(R)$                                                                         | $[\mathbb{N}[n]] \rightarrow [\mathbb{N}]$       | A cheapest path from the entry side to the exit side.                                                                           |

§10.3.1.  $N(E(L\ A))$ 

1 5  
 2 6  
 3 7       $[(1,2,3,4), (5,6,7,8)] : L\ N\ \text{Nat}$   
 4 8

(10.3.1a)

$(\text{gen})[(5,6,7,8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$

(10.3.1b)

$\text{gen}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) =$   
 $(\{[1,5], [1,6], [1,8]\},$   
 $\{[2,5], [2,6], [2,7]\},$   
 $\{[3,6], [3,7], [3,8]\},$   
 $\{[4,5], [4,7], [4,8]\})$

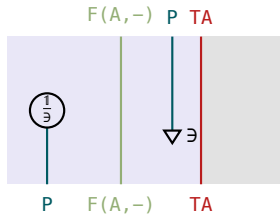
(10.3.1c)

|   |   |                                                        |
|---|---|--------------------------------------------------------|
| 1 | N | the 4-tuple, one component per row of the entry column |
| 2 | E | the set of paths that can start in that row            |
| 3 | L | a path is the non-empty list of the squares it crosses |

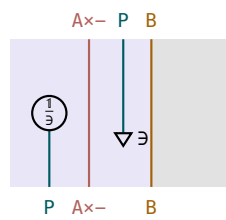
row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is  $\text{moves trans } N(\text{union})$  of  $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$  : component  $k$  collects the paths of rows  $k-1, k, k+1$ .

### §10.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

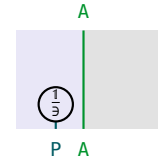
(10.3.2a)



cp on all of  $F(A, E(L A))$



the  $A \times -$  summand:  $\exists$  picks one path,  $1\% \exists$  collects the results



the  $A$  summand, no  $E$  to distribute:  $1\% \exists$  alone,  $a \mapsto \{a\}$

(10.3.2b)

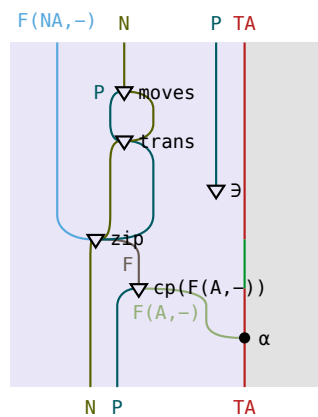
| type                                                                | note                                                                                                                        |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 $F(A, -) = A + A \times -$                                        | one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$                        |
| 2 $\exists : E(L A) \rightarrow L A$                                | the $\exists$ inside $F(1, \exists)$ : a set of paths, one of them — $L A$ is one path, $A$ one square                      |
| 3 $F(1, \exists) = 1 + 1 \times \exists$                            | $: A + A \times E(L A) \rightarrow A + A \times L A$ , the relator acting on each summand                                   |
| 4 $\exists : E(A + A \times L A) \rightarrow A + A \times L A$      | the $\exists$ under the bar: a set of cells, one of them — a different $\exists$                                            |
| 5 $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$ | $\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$                                                    |
| 6 $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$                  | what the picture draws: the unit makes the outer $E$ , leaving one $\exists$ — $\frac{R}{\exists} = \frac{1}{\exists} E(R)$ |

(10.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$  the right summand  
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$   $1$  keeps the square 1,  $\exists$  picks one path  
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$  one output per path in  $w$   
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$  the three of them, collected

### §10.3.3. $\text{gen} = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$

(10.3.3a)

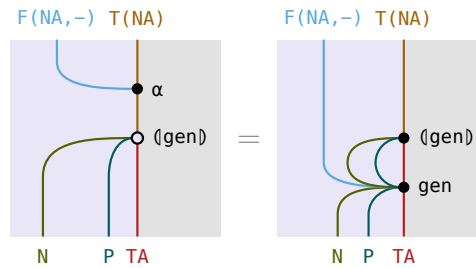


(10.3.3b)

$u = ((1,2,3,4),(\{[5]\},\{[6]\},\{[7]\},\{[8]\}))$  :  $F(N A, N(E(L A)))$   
 $F(1, \text{moves trans } N(\text{union}))$  : 1 keeps the column, the path SETS move  
 $\text{moves}(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$   
 $= (\{[6]\},\{[7]\},\{[8]\},\{[5]\}),$   
 $(\{[5]\},\{[6]\},\{[7]\},\{[8]\}),$   
 $(\{[8]\},\{[5]\},\{[6]\},\{[7]\})$  :  $E(N(E(L A)))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6]\},\{[5]\},\{[8]\},\{[7]\},\{[6]\},\{[5]\}),$   
 $\{[8]\},\{[7]\},\{[6]\},\{[5]\},\{[8]\},\{[7]\})$  :  $N(E(E(L A)))$  row k gets rows k-1, k, k+1  
 $N(\text{union})(\text{that})$   
 $= (\{[5],[6],[8]\},\{[5],[6],[7]\},$   
 $\{[6],[7],[8]\},\{[5],[7],[8]\})$  :  $N(E(L A))$  every path into row k, none dropped  
 $\text{zip}(\text{that})$   
 $= ((1,\{[5],[6],[8]\}), (2,\{[5],[6],[7]\}),$   
 $(3,\{[6],[7],[8]\}), (4,\{[5],[7],[8]\}))$  :  $N(F(A, E(L A)))$  each row: its square, and the paths it may be put in front of  
 $N(\text{cp } P(\alpha))(\text{that})$   
 $= (\{[1,5],[1,6],[1,8]\},\{[2,5],[2,6],[2,7]\},$   
 $\{[3,6],[3,7],[3,8]\},\{[4,5],[4,7],[4,8]\})$  :  $N(E(L A))$  cp pairs the square with each path,  $\alpha$  prefixes it:  $\alpha(1,[5])=[1,5]$

#### §10.3.4. $\text{gen}$ is an F-algebra; $(\text{gen}): \alpha(\text{gen})=F(1, (\text{gen}))\text{gen}$

(10.3.4a)

 $\alpha$  puts the column back on the list, then the fold reads all of itthe fold reads the rest under  $F$ , then one  $\text{gen}$  puts the column in front

(10.3.4b)

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N A)$$

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (10.3.1b): the tail folds to  $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$ , one path per row and each of them one square long, and  $\text{gen}$  on it is (10.3.3b)'s walk.

#### §10.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$ , an $F(N A, -)$ -algebra: $Q=F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(10.3.5a)

$Q : F(N A, N(L A)) \rightarrow N(L A)$  : the fold's algebra: a new column, one path per row  
 $u = ((1,2,3,4),([5],[6],[7],[8]))$  :  $F(N A, N(L A))$   
 $F(1, \text{moves trans } N(\text{est}(R)))$  : 1 keeps the column, the paths move  
 $\text{moves}([5],[6],[7],[8])$   
 $= (\{[6],[7],[8],[5]\},$   
 $\{[5],[6],[7],[8]\},$   
 $\{[8],[5],[6],[7]\})$  :  $E(N(L A))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6],[5],[8]\},\{[7],[6],[5]\},$   
 $\{[8],[7],[6]\},\{[5],[8],[7]\})$  :  $N(E(L A))$  row k gets rows k-1, k, k+1  
 $N(\text{est}(R))(\text{that})$   
 $= ([5],[5],[6],[5])$  :  $N(L A)$  the cheapest into each row – chosen BEFORE the new square  
 $\text{zip}(\text{that})$   
 $= ((1,[5]), (2,[5]), (3,[6]), (4,[5]))$  :  $N(F(A, L A))$  each row: its square, and the one predecessor that survived  
 $N(\alpha)(\text{that})$   
 $= ([1,5],[2,5],[3,6],[4,5])$  :  $N(L A)$   $\alpha(1,[5])=[1,5]$

$$F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge zip = [N(inl), zip' \ N(inr)] \wedge inl \ \alpha = wrap \wedge inr \ \alpha \quad (10.3.5b)$$

$$= cons \Rightarrow Q = [N(wrap), pm \ zip' \ N(cons)]$$

at the last column  $Q$  starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra  $R \ R \sqsubseteq R \wedge F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$

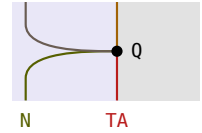
$\Rightarrow (Q)$  setify  $\text{est}(R) \sqsubseteq \text{paths } \text{est}(R)$  's last step folds

**circuit** — the fork is  $F(NA, N(LA)) = NA + NA \times N(LA)$

**Hinze–Marsden**

$$F(N \ A) \ (N \ (T \ A)) \rightarrow \boxed{Q} \rightarrow N(T \ A)$$

$$F(N \ A, N \ (T \ A))$$

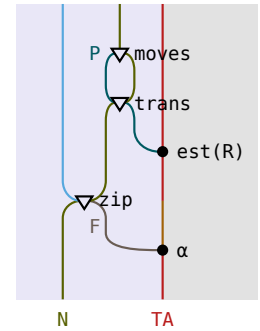


$$FN(T \ A) \rightarrow \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \rightarrow FN(T \ A) \rightarrow \boxed{zip} \rightarrow N(F \ A \ (T \ A)) \rightarrow \boxed{N(\alpha)} \rightarrow N(T \ A)$$

= the fusion condition  $R \ R \sqsubseteq R \wedge F(1, \text{est}(R)) \ \alpha \sqsubseteq cp(F(A, -)) \ E(\alpha) \ \text{est}(R)$

$\Rightarrow F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$  read as a definition,

$$F(NA, -) \ N \ TA$$



$$F(N \ A) \ (N \ (T \ A)) \rightarrow \left( \begin{array}{l} \boxed{l} \rightarrow N(p_0) \rightarrow \boxed{N(wrap)} \\ \boxed{r} \rightarrow q \rightarrow \boxed{pm} \rightarrow q \rightarrow \boxed{zip'} \rightarrow N(q_0) \rightarrow \boxed{N(cons)} \end{array} \right) \rightarrow N(T \ A)$$

$$zip = 1 + zip', \ \alpha = [wrap, cons]$$

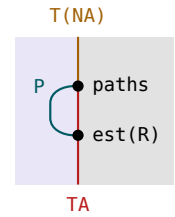
$$R \ R \sqsubseteq R \wedge F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R)) \Rightarrow (Q) \text{ setify } \text{est}(R) \sqsubseteq \text{paths } \text{est}(R) \quad (10.3.5c)$$

shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in  $O(n \times m)$  time;  $Q$  is  $F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge \text{zip} = [N(\text{inl}), \text{zip}' \ N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha = \text{cons}$   
 $\Rightarrow Q = [N(\text{wrap}), pm \ \text{zip}' \ N(\text{cons})]$  's algebra.

circuit

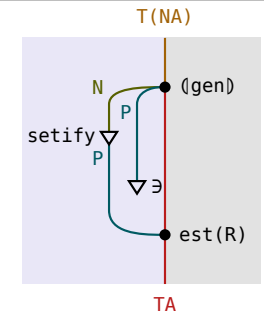
Hinze–Marsden

$$T(N \ A) \text{---} \boxed{\text{paths}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$



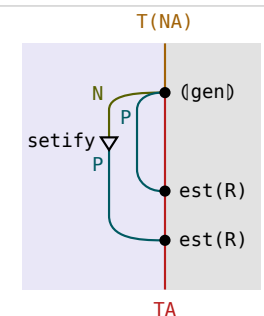
$$= T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{\text{union}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

paths  $\triangleq$  (gen) setify union at paths



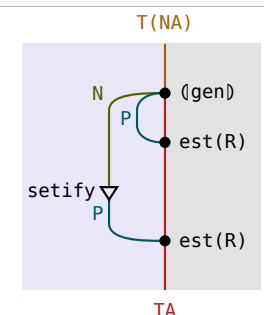
$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{P(\text{est}(R))} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$P(\text{est}(R)) \ \text{est}(R) \sqsubseteq \text{union } \text{est}(R)$  — a minimum in each set, then a minimum of those;  $R$  transitive



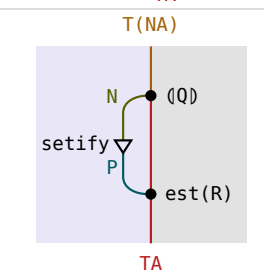
$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

setify lax natural



$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (T \ A))]} \text{---} \boxed{Q} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

fusion at  $R \ R \sqsubseteq R \wedge F(1, \text{est}(R)) \ \alpha \sqsubseteq \text{cp}(F(A, -)) \ E(\alpha) \ \text{est}(R)$   
 $\Rightarrow F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$



$$R \ R \sqsubseteq R \wedge F(1, \text{est}(R)) \ \alpha \sqsubseteq \text{cp}(F(A, -)) \ E(\alpha) \ \text{est}(R) \Rightarrow F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R)) \quad (10.3.5d)$$

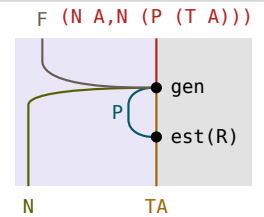
fusion: the condition for fusion in  $R \ R \sqsubseteq R \wedge F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$

$\Rightarrow (Q) \text{ setify } \text{est}(R) \sqsubseteq \text{paths } \text{est}(R)$  's last step, used to derive a definition of  $Q$ .

circuit

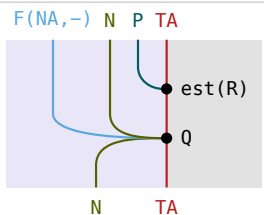
Hinze–Marsden

$$F(N \ A) \ (N \ (P \ (T \ A))) \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T \ A)$$



$$\sqsubseteq FNP(T \ A) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T \ A) \text{---} \boxed{Q} \text{---} N(T \ A)$$

(7.13), then zip, trans, moves lax natural



(7.13) is  $F(1, \text{est}(R)) \ \alpha \sqsubseteq \text{cp } P(\alpha) \ \text{est}(R)$ , Theorem 7.1 at the map  $\alpha$  with  $\frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.